Working Notes on Modular Koszul Duality

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Abstract

One local form $\eta_{ij} = d\log(z_i - z_j)$, Arnold's relation, and the Borcherds–Jacobi identity. Together they build the genus-zero chiral bar differential. At higher genus the prime form or theta propagator introduces curvature, and the correct ambient category is coderived. The proved output is more sharply scoped: bar-cobar inversion, Ran Verdier/Koszul intertwining, the scalar genus tower $F_g = \kappa\lambda_g^{\mathrm{FP}}$ on the uniform-weight lane, cross-channel corrections for multi-weight algebras, and the shadow Postnikov tower on verified primary lines.

These notes are about what happens when you compute with a 1-form, what it teaches you when you fail, and why the failures contain more information than the successes.

Contents

1 A 1-form

$$\eta_{ij} \;=\; d\log(z_i - z_j).$$

This is the simplest local meromorphic 1-form on the configuration space of two distinct points in a coordinate disk. It has a first-order pole along $z_i = z_j$, with residue 1. On a general curve the global propagator is the prime form or theta-normalised analogue; the local residue is still 1.

A double pole $dz/(z-w)^2$ carries no coordinate-independent residue. A regular form extracts nothing at the collision locus. The logarithmic derivative threads the needle: strong enough to detect the collision, mild enough to be canonical.

For three points:

$$\eta_{12} \wedge \eta_{23} \;+\; \eta_{23} \wedge \eta_{31} \;+\; \eta_{31} \wedge \eta_{12} \;=\; 0.$$

This is Arnold's relation. It holds because

$$\frac{1}{(z_1 - z_2)(z_2 - z_3)} + \frac{1}{(z_2 - z_3)(z_3 - z_1)} + \frac{1}{(z_3 - z_1)(z_1 - z_2)} \;=\; 0,$$

the partial-fractions identity: three terms, each a product of two simple poles, summing to zero. The identity is elementary (clear the denominator and observe that the numerator vanishes). It is the only relation in $H^*(\mathrm{Conf}_n(\mathbb{C}), \mathbb{C})$, replicated in all higher degrees by the same mechanism.

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Place elements $\phi_1, \ldots, \phi_n$ of a chiral algebra $\mathcal{A}$ at n points on X. Dress them in logarithmic forms. Define a differential by extracting residues at collisions. Arnold's relation kills the Orlik–Solomon triple-collision piece, while Borcherds/Jacobi kills the algebraic bracket piece. Together they give $d^2 = 0$ at genus 0: the **chiral bar complex** $\bar{B}_X(\mathcal{A})$. At genus $g \geq 1$ the same local residue calculation is accompanied by the Hodge curvature term $m_0 = \kappa(\mathcal{A})\omega_g$.

Under a coordinate change $z \mapsto w(z)$, the propagator transforms as $\eta_{ij} \mapsto \eta_{ij} + d\log\frac{w(z_i)-w(z_j)}{z_i-z_j}$. Near the diagonal this becomes $d\log w'(z_i)$: the standard cocycle for $\mathrm{Diff}(X)$. The propagator is a BRST ghost for diffeomorphisms. One does not postulate this. It follows from the form of η.

2 Two poles

The Heisenberg OPE $\alpha(z)\alpha(w) = k/(z-w)^2$ has one pole. The bar differential extracts a scalar and stops. Everything works.

Kac–Moody has two:

$$J^a(z)J^b(w) \;=\; \frac{k\kappa^{ab}}{(z-w)^2} \;+\; \frac{f^{ab}{}_c J^c(w)}{z-w} \;+\; \text{reg.}$$

The simple pole is the Lie bracket. The double pole is the invariant form. The differential splits: $d = d_{\mathrm{br}} + d_{\mathrm{lev}}$.

$d_{\mathrm{br}}^2 \neq 0$ whenever both poles are present. This was verified across all 2048 sign conventions arising from eleven independent binary choices (grading, suspension, contraction, Koszul signs, ordering, …). No convention saves it.

What happens: iterated residues at a triple collision produce cross-terms between bracket and level. Jacobi cancels the bracket-bracket part. The bracket-level cross-terms survive. The identity is

$$d_{\mathrm{br}}^2 \;=\; -\{d_{\mathrm{br}},\, d_{\mathrm{lev}}\}.$$

The pair $(d_{\mathrm{br}}, d_{\mathrm{lev}})$ does not form a bicomplex. The total d squares to zero not by Jacobi but by the Borcherds identity, which governs all poles at once. This identity is the algebraic shadow of Stokes' theorem on the Fulton–MacPherson compactification $\overline{C}_n(X)$: at the corner where three points collide, the two adjacent codimension-1 faces carry opposite orientations. The cancellation is topological.

This is the conformal anomaly. The classical BRST operator does not square to zero. The quantum correction restores nilpotence. The BV quantum master equation $\hbar\Delta S + \frac{1}{2}\{S, S\} = 0$ is the same equation.

The obstruction to $d_{\mathrm{br}}^2 = 0$ is not an obstruction to the theory.

3 Three logarithms

On $\mathbb{P}^1$ the propagator is single-valued, and $d^2 = 0$ by Arnold. On an elliptic curve E_τ, the function $z_1 - z_2$ is not single-valued. Replace it by the odd theta function, with the correction forced by double-periodicity:

$$\eta_{12}^{(1)} = d\log\vartheta_1(z_1 - z_2 \mid \tau) + \frac{2\pi i\ \mathrm{Im}(z_1 - z_2)}{\mathrm{Im}\,\tau}\, d\bar{z}_1.$$

Arnold's relation fails. What remains is the Arakelov (1, 1)-form:

$$d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g.$$

One number, $\kappa(\mathcal{A})$, extracted from the leading OPE coefficient, controls the curvature.

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Three differentials act on the genus-g bar complex.

(i) d_0: genus-0 collision differential. Built from η_{ij} on $\mathbb{P}^1$. Squares to zero by Arnold. This is classical Koszul duality.

(ii) d_{fib}: fiberwise differential on a fixed curve Σ_g. $d^2_{\text{fib}} = \kappa(\mathcal{A}) \cdot \omega_g \neq 0$.

(iii) D_g: total corrected differential, with A-period parameters $t_k = \oint_{A_k} \omega_k$. $D_g^2 = 0$.

The correction that restores nilpotence is a Lagrangian splitting. The A-cycles form a Lagrangian subspace of $H^1(\Sigma_g, \mathbb{C})$ for the intersection pairing; the curvature lies in the symplectic complement; the A-period corrections project it to zero. Nilpotence comes from the symplectic topology of the surface.

In the curved A_∞ framework: $m_0 = \kappa \cdot \omega_g$, and $m_1^2(a) = [m_0, a]$ (the commutator, with its sign). The condition $\kappa = 0$ is simultaneously: $d^2_{\text{fib}} = 0$; the bar complex being an honest (uncurved) derived object. For the bosonic string, the relevant vanishing is $\kappa_{\text{eff}} = \kappa(\text{matter}) + \kappa(\text{ghost}) = 0$ at $c = 26$: the curvature of the *combined* system vanishes, not that of a single algebra. When $\kappa \neq 0$ the bar complex is curved, and the curvature is the invariant.

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These are three levels of a single logarithm.

Single-valued on $\mathbb{P}^1$	$d_0^2 = 0$ at genus 0
Multi-valued on Σ_g; monodromy $2\pi i$	$d^2_{\text{fib}} = \kappa \cdot \omega_g$ at genus $g \geq 1$
Trivialised on universal cover	$D_g^2 = 0$ (Gauss–Manin correction)

At genus 1 the tadpole amplitude is $\kappa \cdot E_2(\tau)$. The Eisenstein series transforms as $E_2(-1/\tau) = \tau^2 E_2(\tau) + 6\tau/(\pi i)$, and the anomalous term is controlled by κ.

For uniform-weight algebras (all generators of the same conformal weight), the obstruction classes at all genera are controlled by a single generating function:

$$\sum_{g \geq 1} \kappa(\mathcal{A}) \cdot \lambda_g \cdot \hbar^{2g} \;=\; \kappa(\mathcal{A}) \cdot \Big(\hat{A}(i\hbar) - 1\Big).$$

The $\hat{A}$-genus, from index theory. The Bernoulli numbers encode the failure of Arnold's relation at higher genus, and $\hat{A}(ix) = (x/2)/\sin(x/2)$ has all positive coefficients. Each F_g is positive. At genus 1 the formula $F_1 = \kappa/24$ holds unconditionally for all families; at $g \geq 2$ for multi-weight algebras it remains open (op:multi-generator-universality).

Nothing in the OPE data predicts the $\hat{A}$-genus. It lives in the geometry of $\overline{C}_n(X)$.

4 Calls from the future

The 1-form contains structures it has no right to know about. Each computation reveals one: the $\hat{A}$-genus, the Feigin–Frenkel involution, the quantum group, the Langlands dual. Each revelation opens onto what has not yet been computed.

The temptation is to start with the full OPE, with representation theory, with ∞-categories, to import machinery. A better approach is to start with η_{ij} and follow where it leads, retaining awareness of all the directions one has not yet moved in. The five theorems that follow are not the content of the 1-form. They are the first five things it was willing to say.

5 The Fourier dictionary

The bar construction behaves like a Fourier transform on the proved abelian and Verdier-dual lanes. In general this is a dictionary, not a theorem that identifies every row with a Fourier kernel.

Definition 5.1. $\log^{\mathrm{cat}}(\mathcal{A}) := \bar{B}_X(\mathcal{A})$ and $\exp^{\mathrm{cat}}(C) := \Omega_X(C)$.

Classical Fourier	**Chiral bar/Koszul surface**
Kernel: $e^{-2\pi i x\xi}$	Kernel: $\eta_{ij} = d\log(z_i - z_j)$
$\widehat{f * g} = \hat{f} \cdot \hat{g}$	$\mathbb{D}_{\mathrm{Ran}} \circ \bar{B} \simeq \bar{B} \circ (-)^!$ (Thm A)
$\hat{\hat{f}}(x) = f(-x)$	$\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (Thm B)
$\lVert f\rVert^2 = \lVert\hat{f}\rVert^2$	$Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ (Thm C)
Char. fn. $\to$ moments	$\kappa(\mathcal{A}) \to$ genus tower via $\hat{A}$ (Thm D)
Parseval (completeness)	$\mathrm{ChirHoch}^*(\mathcal{A})$ polynomial, Koszul-functorial (Thm H)

When X is a point, η vanishes, d_{res} reduces to the binary bracket, and we recover the classical bar construction of Loday–Vallette. Operadic Koszul duality is the zero-dimensional shadow.

For Heisenberg on E_τ, the bar complex is the Fourier–Mukai kernel on $\mathrm{Jac}(E_\tau) \times \mathrm{Jac}(E_\tau)^\vee$. For non-abelian algebras this abelian theorem becomes a guiding Fourier-type dictionary; the exact universal statements remain bar-cobar inversion, Ran Verdier duality, and Koszul duality.

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The logarithm converts multiplication to addition. The bar construction converts the OPE (a product, with poles) to the bar differential (nilpotent, additive). The cobar reverses: it inserts delta-function currents at boundary strata. Verdier duality on $\overline{C}_n(X)$ exchanges residues with inclusions. Theorems A–D are four faces of this one duality.

$\exp(\log(z)) = z$ on $\lvert z-1\rvert < 1$	$\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ on the Koszul locus
$\log(1+z) = z - z^2/2 + \cdots$	PBW spectral sequence: Taylor expansion of the log
Convergence radius	E_2 degeneration (Koszulness)
Analytic continuation past radius	Coderived category
$\log(ab) = \log(a) + \log(b)$	$Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ (Thm C)
$\log(1) = 0$	$\bar{B}(k)$ is acyclic
Single-valued on simply connected domain	$d_0^2 = 0$ at genus 0
Multi-valued on $\mathbb{C}^\times$; monodromy $2\pi i$	$d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ at genus $g \geq 1$
Residue at branch point	Curvature $\kappa(\mathcal{A})$
Leading coefficient of $\log(1+z) = z + \cdots$	Modular characteristic $\kappa(\mathcal{A})$ (Thm D)

The rows marked by Theorems A–D are theorem surfaces with their stated hypotheses. The Fourier language is a comparison principle unless an abelian Fourier–Mukai or Ran Verdier theorem is explicitly invoked.

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At $n = 4$, the Fulton–MacPherson compactification $\overline{M}_{0,5}$ is the del Pezzo surface $\mathrm{Bl}_4(\mathbb{P}^2)$, with 10 boundary divisors. The relation $\partial^2 = 0$ on this surface yields the A_∞ pentagon identity. The coherence relations of homotopical algebra come from blowing up $\mathbb{P}^2$ four times.

For Koszul algebras: $m_k = 0$ for $k \geq 3$. For interacting algebras: $m_3 \neq 0$, and the pentagon is the first obstruction to formality. Whether a chiral algebra is Koszul is a question about the geometry of a del Pezzo surface.

6 The frame atom

The Heisenberg algebra $\mathcal{H}_k$ sees all five theorems and encounters none of the pitfalls. Not because it is simple, but because its OPE has one pole.

Bar cohomology at weight n has dimension $p(n-2)$ for $n \geq 2$ (the leading entry at weight 1 is the generator itself): $1, 1, 1, 2, 3, 5, 7, 11, \ldots$ –-the generating function $\prod_{n\geq 2}(1-q^n)^{-1}$, a piece of the reciprocal Dedekind eta. A modular form, at the simplest example.

$\mathcal{H}_k$ is not self-dual: $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$. This is a critical pitfall. Heisenberg non-self-duality is the prototype: $\mathcal{H}_k^! \neq \mathcal{H}_{-k}$.

The Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$ on Kac–Moody levels is Verdier duality. The $-k$ dualises the form; the $-2h^\vee$ comes from the canonical class of X. For a Koszul pair $(\widehat{\mathfrak{g}}_k, \widehat{\mathfrak{g}}_{-k-2h^\vee})$:

$$c + c' \;=\; 2\dim(\mathfrak{g}),$$

independent of k. This is Theorem C at the scalar level. The Freudenthal–de Vries formula ensures the sum is always an integer. For $\mathfrak{sl}_2$: $c + c' = 6$ at every non-critical level.

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At the critical level $k = -h^\vee$: the Sugawara tensor is undefined (not "c diverges"), the bar complex is uncurved ($\kappa = 0$), and

$$H^0\big(\bar{B}(\widehat{\mathfrak{g}}_{-h^\vee})\big) \;\cong\; \mathrm{Fun}\big(\mathrm{Op}_{\mathfrak{g}^\vee}(D)\big) \;:$$

functions on the space of Langlands-dual opers.

H^1 gives 1-forms on opers; H^2 gives 2-forms. At H^3 the Cartan transgression kills the top Lie cohomology class via a nonzero d_4 differential.

Four subjects–-representation theory, geometric Langlands, bar-cobar duality, the classical limit of the BV complex–-converge on a single cochain complex with zero curvature.

At admissible levels $k = -h^\vee + p/q$, the bar complex acquires periodic CDG structure indexed by q-th roots of unity. This is the door to logarithmic CFT.

7 The bestiary

What does the bar complex count?

Algebra	**Dims (1–6)**	**Growth**	**Sequence**
Heisenberg	1, 1, 1, 2, 3, 5	$e^{C\sqrt{n}}$	partitions $p(n{-}2)$
Free fermion	1, 1, 2, 3, 5, 7	$e^{C\sqrt{n}}$	partitions $p(n{-}1)$
bc ghosts	2, 3, 6, 13, 28, 59	2^n	–-
$\beta\gamma$	2, 4, 10, 26, 70, 192	$3^n/\sqrt{n}$	A025565
$\widehat{\mathfrak{sl}}_2$	3, 5, 15, 36, 91, 232	$3^n/n^{3/2}$	modified Riordan
Virasoro	1, 2, 5, 12, 30, 76	$3^n/n^{3/2}$	Motzkin differences
$\widehat{\mathfrak{sl}}_3$	8, 36, 204, **1352**	8^n	(conjectured)
$\mathcal{W}_3$	2, 5, 16, **52, 171**	3.3^n	(conjectured)
$Y(\mathfrak{sl}_2)$	4, 10, 28, **82**	3^n	$3^n + 1$ (conjectured)

Warning 7.1. The classical Riordan generating function

$$R(x) \;=\; \frac{1 + x - \sqrt{(1-3x)(1+x)}}{2x(1+x)}$$

yields 3, 6, 15, 36, 91, 232, . . . for the Riordan numbers. But $H^2(\bar{B}(\widehat{\mathfrak{sl}}_2)) = 5$, not 6. This is proved by explicit Chevalley–Eilenberg computation: at weight 3, ker $= 8$, im $= 3$, $H^2_3 = 5$. The generating function that governs the bar cohomology of $\widehat{\mathfrak{sl}}_2$ shares the discriminant $(1-3x)(1+x)$ with the Riordan numbers but differs at the second coefficient. The naive formula is wrong at $n = 2$ and correct for $n \geq 3$.

After Drinfeld–Sokolov reduction to the Virasoro algebra, the generating function rearranges into Motzkin differences $M(n{+}1) - M(n)$: lattice paths in the half-plane. Both come from the same discriminant $\Delta = (1-3x)(1+x)$; the DS reduction preserves the branch cut.

The $\mathfrak{sl}_3$ family has discriminant $1 - 3x - x^2$, with roots involving $\sqrt{13}$ –-and $13 = 3^2 + 4$. The recurrence depth of the generating function equals $\mathrm{rank}(\mathfrak{g}) + 1$ in all known cases.

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Free fields have one OPE pole. Their bar cohomology grows sub-exponentially: partitions, $e^{C\sqrt{n}}$. Interacting fields have two poles, and their bar cohomology grows exponentially: 3^n, 8^n. The jump from one pole to two is an asymptotic phase transition.

8 Five invariants

The proved and conjectural surfaces around a Koszul chiral algebra $\mathcal{A}$ are organised by a 5-tuple:

$$C_{\mathcal{A}} = (\Theta_{\mathcal{A}},\ \kappa(\mathcal{A}),\ \Delta_{\mathcal{A}},\ Q_{\mathcal{A}},\ H_{\mathcal{A}}).$$

$\Theta_{\mathcal{A}}$: universal Maurer–Cartan class. $\kappa(\mathcal{A})$: scalar modular characteristic. $\Delta_{\mathcal{A}}$: spectral discriminant. $Q_{\mathcal{A}}$: shadow metric (the quadratic form on each primary line controlling the entire infinite tower). $H_{\mathcal{A}}$: chiral Hochschild duality on the generic/PBW/Koszul locus, with boundary cases carrying their stated hypotheses.

This is not an assertion that ordinary Hochschild cohomology satisfies a universal duality for all levels. Theorem H is a chiral Hochschild statement; at critical, admissible, or non-PBW inputs the relevant claim is conditional or conjectural unless the local theorem surface says otherwise.

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For every simple $\mathfrak{g}$, cyclic cohomology forces the Maurer–Cartan class to lie on a one-dimensional line: $\Theta^{\min}_{\mathcal{A}} = \eta \otimes \Gamma_{\mathcal{A}}$ for some tautological coefficient $\Gamma_{\mathcal{A}}$. On the uniform-weight lane, the identification $\Gamma_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \Lambda$ holds at all genera. One number, one kernel, one class.

Why: $H^2_{\mathrm{cyc}}(\mathfrak{g}, \mathfrak{g}) = H^3(\mathfrak{g}) = \mathbb{C}$ for every simple $\mathfrak{g}$. One-dimensional. The cubic Casimir of $\mathfrak{sl}_3$ lives in $H^5(\mathfrak{g})$: degree 5 in cyclic cohomology, while the Maurer–Cartan equation lives at degree 2. The degrees do not meet. Higher Casimirs cannot contribute, at any rank.

Genera 1 and 2 are rigid: no freedom beyond κ. At genus 3:

$$l_1(\theta_3 - \kappa \cdot \mu \otimes \lambda_3) = -\tfrac{1}{6}\, l_3(\theta_1, \theta_1, \theta_1),$$

the first correction, purely Killing. The l_2 term vanishes by Jacobi: the first nontrivial partition of 3 as a sum of smaller genera is $1 + 1 + 1$, and l_2 needs only two inputs.

The modular operator $\kappa(\lambda)$ on $\overline{\mathcal{M}}_g$ is nilpotent: $\kappa(\lambda)^{g-1} = 0$ on $\mathcal{M}_g$ (Graber–Vakil). At $g = 2$: zero already. At $g = 3$: nilpotency index ≤ 2. The operator does not cycle. It dies.

9 The BV identification

The BV bracket $\{-, -\}$ of degree $+1$ is modelled by the simple-pole OPE residue, and the BV Laplacian Δ is the genus-raising double-pole operator. The quantum master equation is the BV expression of the same curved compatibility that the bar complex sees through the coderived category. This is proved on the stated BV/bar theorem surfaces, not as an unqualified identity in the ordinary derived category.

The conformal anomaly is one object seen from three windows:

- the curvature $m_0 = \kappa \cdot \omega_g$ of the curved A_∞ (the commutator $m_1^2(a) = [m_0, a]$);
- the obstruction class in $R^*(\overline{\mathcal{M}}_g)$;

- the failure $\hbar\Delta S + \frac{1}{2}\{S, S\} \neq 0$ when $c \neq 26$.

For the bosonic string, the anomaly cancellation is measured on the Virasoro stress-tensor lane: $c_{\text{matter}} = 26$ and $c_{\text{ghost}} = -26$, hence $\kappa_{\text{Vir}} = c/2$ gives $\kappa_{\text{matter}} + \kappa_{\text{ghost}} = 13 - 13 = 0$. This is distinct from the Heisenberg level convention, where a rank-one level-one free boson has $\kappa(\mathcal{H}_1) = 1$. The QME statement belongs to the Virasoro/conformal-anomaly scalar lane and to the coderived BV/bar comparison.

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The mode sums converge for every standard family at every $0 < q < 1$: polynomial OPE growth and subexponential sector growth suffice. For Heisenberg, the genus-g partition function collapses to a one-particle Bergman kernel: a Fredholm determinant, computed from the prime form.

10 Koszulness

$U(\mathfrak{g})$ is Koszul. This says nothing about $\widehat{\mathfrak{g}}_k$. The classical bar complex uses one product. The chiral bar complex uses the full OPE, all poles at once. Higher poles are not perturbations. The Sugawara tensor $T = \frac{1}{2(k+h^\vee)} \sum_a {:}J^a J^a{:}$ produces quartic poles in $T(z)T(w)$ via double Wick contractions.

There is a circularity. To compute bar cohomology: assume Koszulness (spectral sequence collapses at E_2). Bar cohomology would then verify Koszulness (via $H_{\mathcal{A}}(t) \cdot H_{\mathcal{A}^!}(-t) = 1$). Breaking this circle requires an independent collapse proof, and every such proof is specific to its family.

Kac–Moody. Whitehead's theorem ($H^n(\mathfrak{g}, V) = 0$ for $n = 1, 2$) handles the off-diagonal, and Killing contraction the invariant piece. The PBW enrichment at E_1 is $\mathfrak{g} \otimes \mathfrak{g}$ (dimension 9 for $\mathfrak{sl}_2$), not $\Lambda^2(\mathfrak{g})$ (dimension 3).

Virasoro. The subalgebra $\{L_{-1}, L_0, L_1\} \cong \mathfrak{sl}_2$ acts independently of the central charge, because $m^3 - m = 0$ for $m \in \{-1, 0, 1\}$. The Casimir eigenvalues at every weight $h \geq 2$ are positive.

Principal $\mathcal{W}_N$. Unique weight-2 mechanism from the stress tensor; triangular d_2 argument.

There is no universal collapse theorem. Each algebra must be convicted on its own evidence.

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The meta-theorem assembles twelve characterisations of chiral Koszulness. Among conditions (i)–(x), eight are genuinely independent bidirectional equivalences; condition (vi) (Barr–Beck–Lurie) is a proved consequence of (v); condition (viii) (chiral Hochschild concentration in degrees $\{0, 1, 2\}$) is a proved one-way consequence on the Koszul locus. One further characterisation–-the Lagrangian criterion–-requires perfectness hypotheses. One–-D-module purity–-is proved in one direction.

The eight genuinely independent bidirectional equivalences plus the two proved consequences: PBW degeneration; A_∞-formality; Ext diagonal vanishing; bar-cobar counit being a quasi-isomorphism; Barr–Beck–Lurie comparison being an equivalence; factorisation homology concentrated in degree 0; chiral Hochschild vanishing outside degrees $\{0, 1, 2\}$; Kac–Shapovalov nondegeneracy in the bar-relevant range; FM boundary acyclicity; shadow-formality at arities 2, 3, 4.

$V_k(\mathfrak{g})$ is Koszul for *all* k. $L_k(\mathfrak{g})$ may fail at admissible levels. The PBW criterion sees the universal algebra. Failure for simple quotients comes from vacuum null vectors.

11 Two things called E_1

On a smooth curve X:

$$\mathsf{E}_\infty\text{-chiral} \;\subset\; \mathsf{P}_\infty\text{-chiral} \;\subset\; \mathsf{E}_1\text{-chiral}.$$

E_∞-chiral: vertex algebra. Skew-symmetric bracket, unordered configuration spaces. Heisenberg, Kac–Moody, Virasoro, $\mathcal{W}$-algebras.

E_1-chiral: drop skew-symmetry. R-twisted locality: $J_a(z)J_b(w) - R^{ab}_{cd}(z-w)\,J_d(w)J_c(z) = 0$. When $R = P$ (permutation): E_∞. When $R \neq P$: ordered configuration spaces, Yangians, quantum lattice algebras.

Meanwhile, separately: the little n-disks operad E_n lives on real n-manifolds. A complex curve is real 2-dimensional: the chiral bar complex lives at $n = 2$ (Arnold relations). Chern–Simons on a 3-manifold lives at $n = 3$ (Totaro relations).

An E_1-*chiral* algebra lives on a curve. A topological E_1-algebra lives on an interval. These are different objects with an overloaded name. On an interval there are no configuration spaces of points on a complex curve, no Arnold relation in the chiral sense, and no chiral logarithmic propagator. The chiral collision mechanism ceases there; the topological E_1 lane persists as a separate subject.

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The Kazhdan–Lusztig equivalence sends $\widehat{\mathfrak{g}}_k$-modules to $U_q(\mathfrak{g})$-modules, with $q = e^{\pi i/(k+h^\vee)}$ and braiding by the universal R-matrix. Verdier duality on $\overline{C}_n(X)$ reverses the braiding: $R \mapsto R^{-1}$.

Without ordering: $k \mapsto -k - 2h^\vee$ (Feigin–Frenkel). With ordering: $R \mapsto R^{-1}$ (Drinfeld–Kohno).

$$Y_\hbar(\mathfrak{sl}_N)^{!,\mathrm{ch}} \;\cong\; Y_{R^{-1},\hbar}(\mathfrak{sl}_N)^{\mathrm{ch}}.$$

This single identity unifies three theorems. Feigin–Frenkel is its E_∞ shadow. Drinfeld–Kohno is its monodromy. $\mathsf{Ass}^! = \mathsf{Ass}$ is its shadow at a point.

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Lattice vertex algebras make the $\mathsf{E}_\infty/\mathsf{E}_1$ transition visible. A 2-cocycle ε on a lattice Λ produces V_Λ^ε. When ε is symmetric: E_∞. Otherwise: strictly E_1. Koszul duality inverts the cocycle: $\varepsilon \mapsto \varepsilon^{-1}$. For even unimodular Λ: self-duality, $V_\Lambda^! \cong V_\Lambda$. The E_8 lattice vertex algebra is Koszul self-dual.

A Coisson algebra (Beilinson–Drinfeld) is a Poisson vertex algebra: a commutative $\mathcal{D}_X$-algebra with λ-bracket. The quantisation tower runs:

$$\text{Coisson} \xrightarrow{\text{star product}} \mathsf{E}_\infty\text{-chiral} \xrightarrow{R\text{-matrix}} \mathsf{E}_1\text{-chiral}.$$

Holomorphic-topological gauge theories in 3d produce Coisson algebras. Their quantisation lands at E_∞, not E_1. Reaching E_1 requires the KZ equations on a curve: a 2-dimensional phenomenon. Three dimensions is not enough.

12 What the 1-form becomes

The genus-completed bar differential is $D_{\mathcal{A}} = d_0 + \sum_{g\geq 1} \hbar^g d_{\mathcal{A}}^{(g)}$.

Define

$$\Theta_{\mathcal{A}} \;:=\; D_{\mathcal{A}} - d_0.$$

Since $d_0^2 = 0$ and $D_{\mathcal{A}}^2 = 0$:

$$0 = (d_0 + \Theta_{\mathcal{A}})^2 = [d_0, \Theta_{\mathcal{A}}] + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}].$$

The Maurer–Cartan equation. $\Theta_{\mathcal{A}}$ is MC because $D_{\mathcal{A}}^2 = 0$. No construction was needed.

Everything in these notes is a shadow of this element. The five theorems. The curvature. The shadow metric. The sewing amplitudes. The Riordan numbers. The $\hat{A}$-genus. The Polyakov effective action. The Fredholm determinant. The Koszul sign. The 6960.

The 1-form $\eta = d\log(z_i - z_j)$ at two points became a differential d at n points. The differential became a genus-expanded $D_{\mathcal{A}}$. The genus expansion became the MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$. From one 1-form, one relation, and one extraction of residues: a universal invariant of chiral algebras that organises the entire nonlinear structure of the modular operad.

$D_{\mathcal{A}}^2 = 0$ at the convolution level follows from $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$. At the ambient level, with planted-forest corrections from the log FM compactification, $D_{\mathcal{A}}^2 = 0$ is proved via Mok's normal-crossings theorem on $\mathrm{FM}_n(X|D)$ for snc pairs. The planted forests are $\mathrm{Trop}(\mathrm{FM}_n(C|D))$: tropicalisation of the collision geometry.

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The shadow Postnikov obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ projects $\Theta_{\mathcal{A}}$ to arity $\leq r$. At each finite order, κ, $\mathfrak{C}$, $\mathfrak{Q}$ emerge as the arity-2, 3, 4 projections. The inverse limit converges: all obstruction classes $[\mathfrak{o}_{r+1}]$ vanish.

The depth of the tower classifies the algebra:

Class	**Name**	$r_{\max}$	**Examples**
G	Gaussian	2	Heisenberg
L	Lie/tree	3	affine KM
C	contact	4	$\beta\gamma$
M	mixed	∞	Virasoro, $\mathcal{W}_N$

On a 1-dimensional primary slice, $r_{\max} \in \{2, 3, \infty\}$. The contact class escapes by stratum separation. The quintic is forced: $\mathfrak{o}_{\mathrm{Vir}}^{(5)} \neq 0$.

The shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$, with critical discriminant $\Delta = 8\kappa S_4$, controls the entire tower: the Riccati-type shadow equation $H^2 = t^4 Q_L$ determines the generating function. The metric defines a logarithmic connection $\nabla^{\mathrm{sh}} = d - Q_L'/(2Q_L)\,dt$ with residue 1/2 at each zero of Q_L and monodromy -1: the Koszul sign.

On the Virasoro rank-one lane, Koszul duality $c \mapsto 26 - c$ gives

$$\Delta_{\mathrm{Vir}}(c) + \Delta_{\mathrm{Vir}}(26 - c) = \frac{6960}{(5c + 22)(152 - 5c)}.$$

This is not a universal formula for arbitrary $\mathcal{A}$; it is the Virasoro discriminant complementarity. The self-dual point is $c = 13$.

13 The weight-4 Gram matrix

Basis of the weight-4 vacuum Verma: $|1\rangle = L_{-4}|0\rangle$, $|2\rangle = L_{-2}^2|0\rangle$.

G_{11}: $\langle 0|L_4L_{-4}|0\rangle = (8\cdot 0 + \frac{c}{12}\cdot 4\cdot 15)\langle 0|0\rangle = 5c$.

G_{22}: $L_2L_{-2}|0\rangle = \frac{c}{2}|0\rangle$, then $L_2L_{-2}^2|0\rangle = (8+c)L_{-2}|0\rangle$, so $L_2^2L_{-2}^2|0\rangle = \frac{c(c+8)}{2}|0\rangle$.

G_{12}: $L_4L_{-2}^2|0\rangle = 6\cdot\frac{c}{2}|0\rangle = 3c$.

$$G = \begin{pmatrix} 5c & 3c \\ 3c & \frac{c(c+8)}{2} \end{pmatrix}, \qquad \det G = \frac{c^2(5c+22)}{2}.$$

Quasi-primary: $|\Lambda\rangle = L_{-2}^2|0\rangle - \frac{3}{5}L_{-4}|0\rangle$. The coefficient $-3/5$ (not $-3/10$) because the state corresponding to $\partial^2 T$ is $2L_{-4}|0\rangle$. Norm: $\langle\Lambda|\Lambda\rangle = c(5c+22)/10$.

*

The $\mathcal{W}_4$ structure constant:

$$c_{334}^2 = \frac{42\,c^2(5c+22)}{(c+24)(7c+68)(3c+46)}.$$

Numerator zeros = Gram determinant zeros = $c^2(5c+22)$: where a null vector collapses Λ onto $\partial^2 T$, eliminating room for $W^{(4)}$.

Denominator zeros = higher Kac determinant zeros: $c = -24$, $c = -68/7$, $c = -46/3$.

Overall coefficient 42: fixed by the classical $c \to \infty$ limit.

The quartic contact invariant is $Q_{\mathrm{Vir}}^{\mathrm{ct}} = 10/[c(5c+22)]$, with poles at $c = 0$ and $c = -22/5$–-precisely the zeros of $\det G$. The shadow obstruction tower's poles are the representation theory's zeros.

14 The $\widehat{\mathfrak{sl}}_3$ barrier

$H^4(\bar{B}(\widehat{\mathfrak{sl}}_3)) = 1352$. Conjectured.

The rational generating function

$$H_{\bar{B}}(x) \;=\; \frac{4x(2-13x-2x^2)}{(1-8x)(1-3x-x^2)}$$

fits H^1 through H^3 and predicts $H^4 = 1352$.

What is proved:

(i) **Koszul dual from series inversion.** $b_n = [1, 8, 28, 140, 392, 2884, 2716, 75488]$. All positive through degree 7. Chiral excess $\varepsilon_n = b_n - \binom{8}{n}$: $[0, 0, 0, 84, 322, \ldots]$.

(ii) **No quadratic model.** The chiral OPE is not a quadratic algebra. The gap $204 - 120 = 84$ equals $\varepsilon_3 = 140 - 56$.

(iii) $d_{\mathrm{br}}^2 \neq 0$**.** All 2048 sign conventions. Rank of d_{br}^2 is 8.

(iv) **Casimir decomposition.** At $n = 4$ (dim 4096): the three highest $3C_2$-eigenspaces (108, 120, 144, dimensions $112 + 486 + 125 = 723$) have bracket rank 0. Entirely in the kernel. Invariant tensor dims: 0, 1, 2, 8 at $n = 1, 2, 3, 4$.

(v) **Inverse vacuum character.** $c_H = [q^H] \prod_{n \geq 1}(1 - q^n)^8$: $[1, -8, 20, 0, -70, 64, 56, 0, -125, \ldots]$. Convolution $\sum_{j=0}^{H} c_j V_{H-j} = \delta_{H,0}$ verified through weight 8.

Where the computation is stuck: the derivative-residue operator $R^{(1)}_{ij}$ maps outside the Orlik–Solomon algebra. No finite-dimensional extension closes under $R^{(1)}$. The $R^{(1)}$ barrier is structural: the OPE is not a binary operation on a fixed vector space but a family of operations parametrised by the formal disc. The correct framework is configuration-space integrals on FM compactifications.

The number 84 appears three ways: $204 - 120$ (chiral minus quadratic); $140 - 56$ (Koszul dual excess); $\dim(\mathrm{ad}^{\otimes 3})^{\mathfrak{g}}_{\text{beyond-antisym}}$. It measures the departure of a chiral algebra from a universal enveloping algebra.

15 $\mathcal{W}_\infty$ scalar saturation

Theorem 15.1. $\dim H^2_{\mathrm{cyc}}(\mathcal{W}_\infty, \mathcal{W}_\infty) = 1$.

Proof. Step 1. For each N, BRST pullback along $\mathcal{W}^k_N \leftarrow \widehat{\mathfrak{sl}}_{N,k}$ sends cyclic 2-cocycles to $H^2_{\mathrm{cyc}}(\mathfrak{sl}_N) \cong H^3(\mathfrak{sl}_N) \cong \mathbb{C}$. The ghost contribution vanishes: the exterior algebra $\bigwedge^* \mathfrak{n}_+$ is Koszul self-dual with no nontrivial cyclic deformations. The level direction provides a nonzero cocycle. $\dim H^2_{\mathrm{cyc}}(\mathcal{W}_N) = 1$.

Step 2. Truncation $\mathcal{W}_{N+1} \twoheadrightarrow \mathcal{W}_N$ preserves the unique central-charge direction. Both sides one-dimensional: isomorphism.

Step 3. Sectorwise stabilisation ($N \geq w$ suffices for weight w) gives Mittag–Leffler, killing $\varprojlim^1$: $H^2(\mathcal{W}_\infty) = \varprojlim \mathbb{C} = \mathbb{C}$.

Cross-check. $H^*(\mathfrak{sl}_\infty) = \bigwedge(x_3, x_5, x_7, \ldots)$ by Borel stability; $\dim H^3(\mathfrak{sl}_\infty) = 1$. □

16 The tower, computed

The shadow metric $Q_{\mathrm{Vir}}(t) = (c + 6t)^2 + 80t^2/(5c + 22)$ determines the entire infinite tower. The generating function $H(t) = t^2\sqrt{Q_{\mathrm{Vir}}(t)}$ gives $\mathrm{Sh}_r = a_{r-2}/r$ where a_n satisfies the convolution recursion

$$a_0 = c, \quad a_1 = 6, \quad a_2 = \frac{40}{c(5c+22)}, \quad a_n = -\frac{1}{2c}\sum_{j=1}^{n-1} a_j\, a_{n-j} \qquad (n \geq 3). \tag{1}$$

This is a finite computation at each arity.

$\alpha = 2$

The cubic shadow coefficient α for the Virasoro algebra is 2, independent of the central charge. This is not assumed. It is *determined*.

The shadow growth rate is $\rho = \sqrt{9\alpha^2 + 2\Delta}/(2|\kappa|)$, and $\rho = 1$ gives

$$9\alpha^2 + \frac{80}{5c + 22} = c^2.$$

Clearing denominators: $5c^3 + 22c^2 - 45\alpha^2 c - (198\alpha^2 + 80) = 0$. The known critical cubic is $5c^3 + 22c^2 - 180c - 872 = 0$. Matching: $45\alpha^2 = 180$ and $198\alpha^2 + 80 = 872$. Both give $\alpha^2 = 4$.

Three independent confirmations: (i) the critical cubic; (ii) back-derivation from S_5 via $\alpha = -5\kappa S_5/(6S_4) = 2$; (iii) the Sugawara self-coupling $C_{TT}^{T} = 2$ in the Virasoro OPE.

The first ten shadows

$$
\begin{aligned}
S_2 &= \frac{c}{2} \\
S_3 &= 2 \\
S_4 &= \frac{10}{c(5c+22)} \\
S_5 &= \frac{-48}{c^2(5c+22)} \\
S_6 &= \frac{80(45c+193)}{3\,c^3(5c+22)^2} \\
S_7 &= \frac{-2880(15c+61)}{7\,c^4(5c+22)^2} \\
S_8 &= \frac{80(2025c^2+16470c+33314)}{c^5(5c+22)^3} \\
S_9 &= \frac{-1280(2025c^2+15570c+29554)}{3\,c^6(5c+22)^3} \\
S_{10} &= \frac{256(91125c^3+1050975c^2+3989790c+4969967)}{c^7(5c+22)^4}
\end{aligned}
$$

Verified by three independent implementations: the convolution recursion (16), the master equation recursion $S_r = -(1/(rc))\sum_{3\le j\le k,\, j+k=r+2} \varepsilon(j,k)\, jk\, S_j S_k$, and the existing `ds_shadow_higher_arity` module. 67 tests, all passing.

The denominator theorem

Theorem 16.1 *(Denominator divisibility and exact finite-range saturation).* For all $r \ge 4$: $\mathrm{denom}(S_r)$ divides $(\mathrm{const}) \cdot c^{r-3}(5c+22)^{\lfloor (r-2)/2 \rfloor}$. Equality of the displayed saturation pattern holds for every computed coefficient through $r = 11$. Beyond this range the theorem identifies the only remaining obstruction to equality: cancellation inside the explicit quadratic recursion for the Taylor coefficients of $\sqrt{Q_L}$.

Proof. The Taylor coefficient a_n of $\sqrt{Q_L}$ satisfies $a_n = -(1/(2c))\sum a_j a_{n-j}$. By induction: $\mathrm{denom}(a_n) = c^{n-1}(5c+22)^{\lfloor n/2 \rfloor}$ for $n \ge 2$. Base: $a_2 = 40/[c(5c+22)]$. Step: the product $a_j a_{n-j}$ contributes c^{n-2} in the denominator and $(5c+22)^{\lfloor j/2 \rfloor + \lfloor (n-j)/2 \rfloor}$; the inequality $\lfloor j/2 \rfloor + \lfloor (n-j)/2 \rfloor \le \lfloor n/2 \rfloor$ gives the displayed divisibility. Saturation in the computed range follows by direct evaluation of the same recursion through $r = 11$. The displayed recursion is therefore the exact no-cancellation test for every higher arity. Division by $2c$ raises the c-power to $n-1$. □

No Kac determinant zeros beyond weight 4 appear. The shadow tower sees only the first Gram determinant.

The phase slip

The branch points of Q_{Vir} have argument $\theta(c) \in (\pi/2, \pi)$ for all $c > 0$, since $\operatorname{disc}(Q_{\mathrm{Vir}}) = -320c^2/(5c+22) < 0$ forces complex roots with Re < 0, Im > 0. The Darboux transfer theorem gives approximate sign alternation in S_r, with beat period $\pi/(\pi - \theta)$. As $c \to \infty$, $\theta \to \pi$ and the alternation becomes permanent. At finite c, the pattern has a finite lifetime:

c	θ	first sign break
1	164°	$r = 17$
10	170°	$r \gg 30$
13	171°	$r \gg 30$
26	173°	$r = 35$

The sign breaks at $c = 1$ ($r = 17$) and $c = 26$ ($r = 35$) are verified by exact rational computation.

*

The shadow metric contains the entire tower. What the computation teaches: the denominators know only one Gram determinant, the numerators know the shadow radius, and the signs know the argument of the nearest branch point. The tower is a Taylor expansion of one quadratic form.

The $\mathcal{W}_3$ tower: two variables

For $\mathcal{W}_3$ with generators T (weight 2) and W (weight 3), the shadow lives on a 2-dimensional deformation space (x_T, x_W) with propagator $P = \operatorname{diag}(2/c, 3/c)$. The tower is computed from the same master equation, now as a 2d H-Poisson bracket: $\{f, g\}_H = (\partial_T f)(2/c)(\partial_T g) + (\partial_W f)(3/c)(\partial_W g)$.

Initial data: $\mathrm{Sh}_2 = (c/2)\, x_T^2 + (c/3)\, x_W^2$, $\mathrm{Sh}_3 = 2x_T^3 + 3x_T x_W^2$, $\mathrm{Sh}_4 = \frac{10}{c(5c+22)}\, x_T^4 + \frac{960}{c(5c+22)^2}\, x_T^2 x_W^2 + \frac{2560}{c(5c+22)^3}\, x_W^4$.

Computed through arity 7 by the multi-variable master equation. Two structural results:

(i) *Zero Coxeter anomaly on the T-line.* $\mathrm{Sh}_r(\mathcal{W}_3)|_{x_W=0} = \mathrm{Sh}_r(\mathrm{Vir}_c)$ exactly, through arity 7. The W-generator does not backreact on the T-sector.

(ii) $(5c+22)$*-power increases with W-degree.* At arity r, the coefficient of $x_T^{r-2k} x_W^{2k}$ has $(5c+22)$ to power $\geq \lfloor (r-2)/2 \rfloor + k$ in its denominator (with equality at arities 4 and 5). Each pair of W-insertions costs at least one extra factor.

The T-line is the Virasoro shadow metric. The W-line carries the non-abelian correction: the $\mathcal{W}_3$ shadow "lives over" the Virasoro shadow, with the additional $(5c+22)$ factors encoding the coupling between the two generators through the weight-4 Gram matrix.

17 The vast openness

Everything in §1–§16 can be said in two sentences.

A 1-form with a simple pole on $\mathrm{Conf}_2(X)$ satisfies Arnold's relation, induces a nilpotent differential on dressed configuration spaces, and, when deformed from genus 0 to genus g, acquires a curvature $\kappa(\mathcal{A}) \cdot \omega_g$ controlled by the leading OPE coefficient. The genus-completed differential $D_{\mathcal{A}}$ satisfies

$D^2_{\mathcal{A}} = 0$, so $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is a universal Maurer–Cartan element whose finite-arity projections are the shadow Postnikov tower.

That is what the 1-form told us. What follows is what it does when you push it harder: into deeper strata of moduli space, along ordered paths, through the prime factorisation of its sewing amplitudes, past the death of the derived category at genus 1, into the bulk on the other side of the boundary. Each direction is a computation that did not quite finish.

18 The second boundary

At genus 0, $D^2 = 0$ is Arnold's relation: three simple poles, opposite orientations at the codimension-1 faces of $\overline{C}_3(\mathbb{P}^1)$.

At genus 1, $D^2 \neq 0$. The failure is $\kappa \cdot \omega_1$: one scalar, one Kähler form, from the monodromy of the theta function around its period.

At genus 2, D^2 is again not zero on the nose. The correction that restores nilpotence requires the *planted forest*: a new kind of boundary stratum in the moduli space, invisible at genera 0 and 1.

*

Here is the geometry. The Fulton–MacPherson compactification $\mathrm{FM}_n(\mathbb{P}^1)$ resolves collisions of points on the projective line. At genus $g \geq 1$ the curve has handles, and the collision geometry changes: a bubble can grow from a node where the curve degenerates. These bubbles are the *planted trees*–-rooted trees whose vertices carry genus labels $g(v) \geq 0$, growing out of nodes of Σ_g.

The correct compactification is Mok's logarithmic Fulton–MacPherson space $\mathrm{FM}_n(X|D)$ for simple normal crossings pairs (X, D). Its tropicalisation $\mathrm{Trop}(\mathrm{FM}_n(C|D))$ produces planted forests. The total differential $D_{\mathcal{A}}$ acquires three components:

$$\begin{aligned} d_{\mathrm{sew}} &: \quad \text{self-sewing (identify two punctures, gain a handle)} \\ d_{\mathrm{pf}} &: \quad \text{planted-forest degeneration (bubble tree grows)} \\ \hbar\Delta &: \quad \text{non-separating degeneration (BV Laplacian)} \end{aligned}$$

Each is a residue correspondence on the boundary divisors of $\mathrm{FM}_n(X|D)$. The identity $D^2 = 0$ at the ambient level reduces to **codimension-2 cancellation**: every codimension-2 stratum appears as the intersection of exactly two codimension-1 faces with opposite orientations. This is Mok's normal-crossings theorem.

Every stable graph Γ in the modular bar complex carries a tridegree (g, n, d): loop genus, arity, log boundary depth. The depth d counts internal edges of the planted-forest part–-how far into the boundary of $\overline{\mathcal{M}}_{g,n}$ the stratum lies. The **genus spectral sequence** has $E_1^{p,*}$ isolating data at loop genus p, with differentials $d_r \colon E_r^{p,q} \to E_r^{p+r,q-r+1}$ that are obstruction maps raising genus. This filtration is by loop number, not bar degree–-it is distinct from the PBW spectral sequence.

Two explicit results:

PROPOSITION 18.1 *(Genus-2 planted-forest correction).*

$$\delta_{\mathrm{pf}}^{(2,0)} = \frac{S_3(10S_3 - \kappa)}{48}.$$

For Virasoro: $(40 - c)/48$. For Heisenberg: 0.

Proposition 18.2 *(Self-loop parity vanishing).* A single-vertex graph of type $(0, 2k)$ with k self-loops has amplitude $I_\Gamma = 0$ for all $k \geq 2$, by an odd-parity involution on $\overline{\mathcal{M}}_{0,2k}$.

The **shadow visibility genus** $g_{\min}(S_r) = \lfloor r/2 \rfloor + 1$ is a selection rule: S_4 first contributes at genus 3; S_6 at genus 4. Each new shadow arity requires a new handle. The combinatorics of non-separating self-sewings on Σ_g permit at most $2g - 2$ such operations, and arity r requires $\lfloor r/2 \rfloor$ of them.

19 One quadratic form

Look at the ten shadows of §16 again. The denominators are $c^{r-3}(5c + 22)^{\lfloor (r-2)/2 \rfloor}$. The numerators are polynomials in c with integer coefficients. The signs oscillate with a period that depends on the argument of one complex number. Ten invariants, each a rational function of c, each determined by one convolution recursion. Why should one recursion control everything?

Because the recursion coefficients are the Taylor coefficients of one quadratic form.

The **shadow metric** on a 1-dimensional primary line L of the cyclic deformation complex is

$$Q_L(t) \;=\; (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2, \qquad \Delta \;=\; 8\kappa S_4.$$

The Maurer–Cartan equation projected to arity r, on L, is equivalent to

$$H^2 \;=\; t^4\, Q_L.$$

The generating function $H(t) = t^2\sqrt{Q_L(t)}$ satisfies this identically. Its Taylor coefficients *are* the shadow obstruction tower.

This has a geometric meaning. Define the logarithmic connection

$$\nabla^{\mathrm{sh}} \;=\; d - \frac{Q_L'(t)}{2\, Q_L(t)}\, dt.$$

It has simple poles at the zeros of Q_L, residue $1/2$ at each, and monodromy -1: the Koszul sign. The function $\Phi(t) = \sqrt{Q_L(t)/Q_L(0)}$ is a *flat section* of ∇^{sh}, and the shadow generating function $H(t) = 2\kappa\, t^2\, \Phi(t)$ is its parallel transport weighted by t^2. The shadow obstruction tower is not a sequence of independent invariants. It is the Taylor expansion of a single algebraic function of degree 2.

*

On the Virasoro line, Koszul duality acts on the discriminant:

$$\Delta_{\mathrm{Vir}}(c) + \Delta_{\mathrm{Vir}}(26 - c) \;=\; \frac{6960}{(5c + 22)(152 - 5c)}.$$

At $c = 13$ both factors equal 87, and $\Delta_{\mathrm{Vir}}(13) = 40/87$. The numerator $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$ is the Virasoro conductor numerator.

The shadow growth rate $\rho = \sqrt{9\alpha^2 + 2\Delta}\,/\,(2|\kappa|)$ governs asymptotics: $S_r \sim A\, \rho^r\, r^{-5/2} \cos(r\theta + \varphi)$. Setting $\rho = 1$ gives the critical cubic $5c^3 + 22c^2 - 180c - 872 = 0$ with positive root $c^* \approx 6.125$.

The depth classification is forced by Q_L:

- Q_L constant ($\alpha = \Delta = 0$): class G, $r_{\max} = 2$. Heisenberg.
- Q_L a perfect square ($\Delta = 0$): class L, $r_{\max} = 3$. Affine Kac–Moody.
- Q_L irreducible ($\Delta \neq 0$): class M, $r_{\max} = \infty$. Virasoro, $\mathcal{W}_N$.
- Stratum separation (rank-one escape): class C, $r_{\max} = 4$. $\beta\gamma$.

One quadratic form, four classes, an infinite tower or a finite death. The shadow metric is the complete invariant of shadow depth on each primary line.

20 The monodromy

Move z_1 around z_2 on the curve.

If the OPE is commutative–-$\phi_1(z)\phi_2(w) = \phi_2(w)\phi_1(z)$–-nothing happens. The propagator $\eta_{12} = d\log(z_1 - z_2)$ is symmetric. The bar complex does not see the ordering. This is the E_∞-chiral world: vertex algebras.

If the OPE carries a braiding–-

$$J_a(z)\, J_b(w) - R_{ab}^{cd}(z-w)\, J_d(w)\, J_c(z) \;=\; 0$$

–-the propagator acquires a monodromy. The bar complex on ordered configuration spaces uses R-twisted Arnold relations. This is the E_1-chiral world: Yangians, quantum lattice algebras, the KZ equations.

The R-matrix is the genus-0 binary shadow of $\Theta_{\mathcal{A}}$:

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}).$$

For Kac–Moody: $r(z) = k\,\Omega_{\mathfrak{g}}/z$, where $\Omega_{\mathfrak{g}} = \sum_a J^a \otimes J_a$ is the Casimir. For Virasoro: $r(z) = (c/2)/z^3 + 2T/z$, a quasi-trigonometric solution of the CYBE. The full MC element $\Theta_{\mathcal{A}}$ lives on the modular operad; its restriction to genus 0, arity 2, with ordering, is $r(z)$.

*

Without ordering: Feigin–Frenkel, $k \mapsto -k - 2h^\vee$. With ordering: $R \mapsto R^{-1}$. In a single identity:

$$Y^{\mathrm{dg}}_{\hbar}(\mathfrak{sl}_N)^{!,\mathrm{ch}} \;\cong\; Y^{\mathrm{dg}}_{R^{-1},\hbar}(\mathfrak{sl}_N)^{\mathrm{ch}}.$$

Thick generation of the Yangian derived category is proved for the evaluation-generated core via multiplicity-free ℓ-weights (Chari–Moura): $V_n \otimes L^- \;\cong\; \bigoplus_{j=0}^{n} L^-(n-2j)$ at the character level, verified to depth 60 in the local computation. Extension beyond the natural evaluation core, and the complete DK4/DK5 comparison, remain conjectural.

The **jet principle**: the reduced-weight-q bar window $K_q(\mathcal{A})$ determines $r(z)$ through order z^{-q}. This is finite-dimensional linear algebra at each jet order–-the ordered analogue of the shadow visibility genus.

The $\widehat{\mathfrak{sl}}_3$ barrier of §14 has a Yangian reading. The derivative-residue operator $R^{(1)}$ that maps outside the Orlik–Solomon algebra is the obstruction to closing the first jet in a finite-dimensional quotient. The jet principle says: do not close $R^{(1)}$ in finite dimensions. Compute K_q windowwise. The $\mathfrak{sl}_3$ window at $q = 4$ has $\dim K_4 = 4096$: a computation in a *fixed* vector space. The barrier is not the computation. It is the expectation that the answer should have finitely many terms. The answer has infinitely many terms. They stabilise windowwise.

21 Prime factorisation

Sew the Heisenberg bar complex into a torus. The partition function $Z_{\mathcal{H}}(\tau) = q^{-\kappa/24}\prod_{n\geq 1}(1-q^n)^{-1}$ has Fourier coefficients $a(N)$.

Are they multiplicative?

No. The partition-number coefficients of

$$\prod_{n\geq 1}(1-q^n)^{-1}$$

are not multiplicative: for example $p(6) = 11$ while $p(2)p(3) = 6$. The proved Euler product here is the *q-product* of oscillator modes, not a Dirichlet Euler product for its Fourier coefficients. Any arithmetic Euler product must come from a separately defined scalar zeta or moment lane, not from the partition coefficients themselves.

Now ask about Virasoro.

$$S_{\mathrm{Vir}}(u) \quad \text{is a separately defined Dirichlet-sewing lift, not the Dirichlet series of Virasoro partition coefficients.}$$

No Virasoro Euler-product formula for partition coefficients is proved here. The arithmetic packet claims therefore concern the explicitly constructed sewing or moment transforms only.

*

The **arithmetic packet connection** $\nabla^{\mathrm{arith}}_{\mathcal{A}}$ is a flat meromorphic connection on the module $M_{\mathcal{A}} = \bigoplus_\chi M_\chi$ over the spectral s-line, decomposing into an arithmetic skeleton $\mathrm{Ask}(\mathcal{A}) = M^{\mathrm{ss}}_{\mathcal{A}}$ (semisimple) and an algebraic defect $\mathrm{Def}_{\mathrm{alg}}(\mathcal{A}) = \bigoplus N_\chi M_\chi$ (unipotent monodromy).

The principle: *arithmetic is semisimple; homotopy defect is unipotent*.

The total depth decomposes: $d(\mathcal{A}) = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$. Algebraic depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$. At $d_{\mathrm{arith}} \geq 4$: cusp forms of weight ≥ 12 contribute; the Ramanujan $\tau(n)$ enters the shadow obstruction tower. No standard family has $d_{\mathrm{arith}} \geq 4$. Whether such families exist is open.

22 Two categories

The ordinary derived category is the wrong invariant for the curved higher-genus bar lane.

Positselski's acyclicity phenomena show why: once $m_1^2(a) = [m_0, a]$ is genuinely curved, ordinary exactness and ordinary quasi-isomorphism erase precisely the curvature data that the modular problem needs. The local statement is about the curved bar object under the hypotheses of the curved theorem surface, not a slogan about every curved DGA in every context.

At genus 0, κ does not appear. At genus $g \geq 1$, $m_0 = \kappa \cdot \omega_g$ is the curvature. The bar complex is curved, and ordinary derived methods no longer see the intended invariant.

What replaces it: the **coderived category** $D^{\mathrm{co}}(\mathcal{A}\text{-mod})$, which replaces cones by totalised products. The shadow invariants $Q_g(\mathcal{A})$ live there, not in D^b.

The formal series $\Theta_{\mathcal{A}} = \sum \Theta^{(g)}\hbar^g$ converges *2-dimensionally*: in both genus and arity. Genus convergence: Bernoulli decay $1/(2\pi)^{2g}$. Arity convergence: the shadow growth rate ρ from §19. The partition function of the 1-form is the value of a convergent double series. The partition function of the *string* (genus-factorial growth $(2g)!$) diverges: it requires resummation.

For Heisenberg, the sewing reduces to a Fredholm determinant:

$$Z_g(\mathcal{H}_\kappa;\, \tau) \;=\; \det\nolimits'_{\mathrm{Fred}}\bigl(1 - K_\tau^{(g)}\bigr)^{-\kappa}$$

where $K_\tau^{(g)}$ is the genus-g Bergman kernel operator, computed from the prime form on Σ_g.

*

Three objects remain conjectural:

(i) When does a formal MC element $\Theta_{\mathcal{A}}$ lift to the completed sewing envelope? Sufficient conditions are proved (polynomial OPE + subexponential growth); necessity is open.

(ii) The analytic bar coalgebra $\bar{B}^{\mathrm{an}}(\mathcal{A})$ should form an analytic Koszul pair with $\bar{B}^{\mathrm{an}}(\mathcal{A}^!)$. The algebraic pair is proved; the analytic closure is not.

(iii) Moriwaki's IndHilb-valued factorisation homology works for conformally flat surfaces. Extension to genus $g \geq 2$ requires a curved IndHilb framework that does not yet exist.

23 The bulk

The bar differential extracts residues when points collide along the curve–-the $\mathbb{C}$-direction. The bar coproduct extracts residues when points collide along the boundary–-the $\mathbb{R}$-direction. Together: four kinds of collision.

$$\begin{aligned} d_{\mathrm{res}} &: \text{ bulk points collide (OPE residue)} \\ d_{\mathsf{A}_\infty} &: \text{ boundary points collide (Stasheff)} \\ d_{\mathrm{mix}} &: \text{ bulk point hits boundary (bubbling)} \\ d_{\mathrm{clutch}} &: \text{ curve degenerates (nodal clutching)} \end{aligned}$$

Total: $d_{\mathrm{mod}} = d_{\mathrm{res}} + d_{\mathsf{A}_\infty} + d_{\mathrm{mix}} + d_{\mathrm{clutch}}$, with $d^2_{\mathrm{mod}} = 0$ at genus 0 and $d^2_{\mathrm{mod}} = m_0$ at genus $g \geq 1$.

This is the **Swiss-cheese operation space** on $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_m(\mathbb{R})$. The ordered bar complex extracted from it remains a single-coloured E_1-chiral coalgebra; the genuine $\mathsf{SC}^{\mathrm{ch,top}}$ datum lives on the derived-center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}), \mathcal{A})$.

On the other side of the boundary lives the **chiral derived center** $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = H^*(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}), \delta)$: the universal bulk algebra. It classifies bulk operators acting on the boundary–-*not* twisting morphisms between $\mathcal{A}$ and $\mathcal{A}^!$ (which is what bar-cobar classifies). This is a critical distinction. Bar-cobar is about couplings. The derived center is about the operators that the boundary algebra does not know it has.

The **open/closed MC element**

$$\Theta^{\mathrm{oc}}_{\mathcal{A}} \;:=\; \Theta_{\mathcal{A}} + \sum_{j=1}^{r} \mu^{\mathcal{M}_j}$$

packages both: $\Theta_{\mathcal{A}}$ (bar-intrinsic) and $\mu^{\mathcal{M}_j}$ (A_∞-module structures at punctures). Its MC equation decomposes into: pure closed, pure open, mixed bulk-boundary, and clutching.

The **completed modular Koszul datum**

$$\Pi_X^{\mathrm{oc}}(\mathcal{A}) \;=\; \big(\bar{B}_X,\ \Omega_X,\ \Theta_{\mathcal{A}},\ \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}},\ \mathrm{Tr}_{\mathcal{A}},\ \mathrm{HH}_*,\ \mathfrak{R}^{\mathrm{oc}}_{\bullet},\ \nabla^{\mathrm{hol}}\big)$$

assembles eight components–-bar coalgebra, cobar algebra, universal MC class, derived center, annulus trace, Hochschild homology, nonlinear resonance tower, holographic connection–-from one λ-bracket.

This is the constructive endpoint: given a chiral algebra in the proved lanes, compute Π_X^{oc}. The theorem packages A, B, C, D, and H record structural properties of this datum with their stated scope restrictions. The non-perturbative content is its analytic completion. That is the frontier.

24 Window by window

At weight 4, the Gram matrix of §13 determines everything: the quasi-primary Λ, the structure constant c^2_{334}, the quartic contact invariant $Q^{\mathrm{ct}} = 10/[c(5c+22)]$. Two basis vectors, one 2×2 matrix, one determinant $c^2(5c+22)/2$.

At weight 6: one new quasi-primary enters. Its norm involves $c(5c+22)$ and an S_4-corrected composite. A 3×3 Gram matrix.

At weight 8: two new quasi-primaries. A 5×5 matrix.

Each weight is a finite linear algebra problem. The Koszul dual of Virasoro is the inverse limit of these windows. The question of convergence is the deepest question in the theory.

*

Vir_{26-c} is the depth-zero approximation, not the genuine dual. Self-duality at $c = 13$, **not** $c = 26$. The critical dimension $c = 26$ is where the *dual* curvature vanishes: $\kappa(\mathrm{Vir}_{26-c}) = 0$. The genuine dual is the completed tower of quasi-primary composites, stabilised one weight window at a time. Whether the inverse limit converges to a recognisable vertex algebra depends on whether the Gram spectrum is uniformly bounded away from zero–-which is exactly Kac–Shapovalov nondegeneracy in the bar-relevant range, one of the eight genuinely independent bidirectional equivalences in the Koszulness meta-theorem.

The **resonance rank** $\rho(\mathcal{A})$ measures the dimension of $V_0 = B(\mathcal{A})_0 \cap \ker(d_{\bar{B}^{\mathrm{ch}}})$ in the completed bar complex.

Theorem 24.1 *(Resonance completion).* Every positive-energy chiral algebra has $\rho(\mathcal{A}) < \infty$.

Positive energy implies finite-dimensional weight spaces; weight-by-weight SDR (the chiral propagator preserves weight) bounds ρ by $\dim V_0 < \infty$. For Virasoro: $\rho = 1$. For $\mathcal{W}_3$: $\rho = 1$. For affine KM: $\rho = 0$.

The MC4 completion splits: MC4$^+$ (positive towers: $\mathcal{W}_{1+\infty}$, affine Yangians–-solved by weight stabilisation) versus MC4^0 (resonant towers: Virasoro, non-quadratic $\mathcal{W}_N$–-reduced to a finite resonance problem).

The non-principal corridor. Drinfeld–Sokolov reduction for arbitrary nilpotent $f \in \mathfrak{g}$ produces $\mathcal{W}_k(\mathfrak{g}, f)$. The algebra exists. The obstruction is proving that bar-cobar and Koszul duality *commute* with non-principal reduction. Hook-type nilpotents in type A are proved. The Coxeter anomaly: DS can increase shadow depth (affine at $3 \to \mathcal{W}_3$ at ∞). The type A transpose conjecture:

$$\mathcal{W}_k(\mathfrak{sl}_N, f_\lambda)^! \;\simeq\; \mathcal{W}_{k^\vee_\lambda}(\mathfrak{sl}_N, f_{\lambda^t}).$$

The factorisation envelope. Nishinaka's construction produces $U_X(L)$ from a Lie conformal algebra L: the enveloping vertex algebra on X. The **modular Koszul datum** $\Pi_X(L) = (\mathcal{F}_X, \overline{B}_X, \Theta_L, \mathcal{L}_L, (V^{\mathrm{br}}, T^{\mathrm{br}}), R_4^{\mathrm{mod}})$ assembles factorisation algebra, bar coalgebra, MC class, obstruction Lie algebra, branch data, modular quartic class–-from one Lie conformal algebra.

The modular envelope $U_X^{\mathrm{mod}}(L)$ should be left adjoint of $\mathrm{Prim}^{\mathrm{mod}}$. Its existence is conjectural.

The grand cumulant transform.

$$M(\mathcal{A}) = \big(\mathrm{Cum}_c(\mathcal{A}),\ D_{\mathcal{A}},\ \tau_{\mathcal{A}},\ r_{\mathcal{A}}(z),\ \Theta_{\mathcal{A}}\big).$$

Components 1–3 proved. Component 4 proved windowwise. Component 5 proved. The question: does $M(\mathcal{A})$ determine $\mathcal{A}$ up to quasi-isomorphism?

*

What remains is not a single problem but a boundary: where algebra meets analysis, where formal series meet convergent functions, where the 1-form's predictions become the theory's theorems. The modular Koszul datum $\Pi_X(L)$ is the algebraic side. The sewing envelope $\mathcal{A}^{\mathrm{sew}}$ is the analytic side. The bridge between them is the completion problem. It is finite at each stage and infinite in the limit.

25 The sixth movement

The first five movements of §1–§17 take a single input

$$\{T_\lambda T\} = \partial T + 2T\lambda + \frac{c}{12}\lambda^3$$

and extract the perturbative algebraic shadow of the three-dimensional gravity dictionary: the infinite A_∞ tower, the Koszul dual Vir_{26-c}, the gravitational R-matrix $r(z) = (c/2)/z^3 + 2T/z$, the genus tower $\sum_{g\geq 1} F_g\,\hbar^{2g} = \frac{c}{2}\big(\hat{A}(i\hbar) - 1\big)$, and the Cardy asymptotic $S = 2\pi\sqrt{ch/6}$ under the Brown–Henneaux hypotheses. Movements I–V are perturbative. Movement VI is not.

*

The standard approach to the non-perturbative partition function of 3d gravity is: sum over bulk topologies. Pick a boundary (say the torus T^2). Sum over all 3-manifolds M with $\partial M = T^2$: handlebodies, their $SL(2,\mathbb{Z})$ images, and whatever non-handlebody geometries one can classify. Weight each by $e^{-S_{\mathrm{EH}}[M]}$. This is the Maloney–Witten programme.

The sum fails. For generic c, the Poincaré series $Z_{\mathrm{MW}} = \sum_{\gamma\in\Gamma_\infty\backslash SL(2,\mathbb{Z})} Z_0(\gamma\cdot\tau)$ produces negative Fourier coefficients. No physical Hilbert space has a negative number of states.

Every proposed repair introduces something from outside: ensemble averaging (Cotler–Jensen), non-handlebody corrections (Maxfield–Turiaci), specific Poincaré seeds (Benjamin–Collier–Maloney). They ask: *which topologies should we include?*

The MC element asks a different question.

*

The replacement. The universal Maurer–Cartan element Θ_{Vir_c} of the modular convolution algebra satisfies

$$D_{\mathrm{Vir}_c}(\Theta) + \tfrac{1}{2}[\Theta, \Theta] = 0.$$

This is not a sum over anything. It is a single algebraic identity whose genus-g component reads

$$d^{\mathrm{tree}}\Theta^{(g)} \;=\; -\sum_{\substack{g_1+g_2=g\\ g_i<g}} [\Theta^{(g_1)}, \Theta^{(g_2)}] \;-\; \Theta^{(g-1)}_{\mathrm{clutch}}.$$

The right side involves only genera strictly below g. The left side is the tree-level differential. On the Koszul locus (which is where Virasoro lives: one-loop exactness implies Koszulness), the tree-level complex is acyclic and carries a contracting homotopy h. So $\Theta^{(g)} = -h(\text{source})$ is determined uniquely.

No free parameters at any genus. No choice of which topologies to include. The MC equation determines the partition function at genus g from the genus-0 λ-bracket, exactly as $d_{\overline{B}}^2 = 0$ is the boundary-of-boundary identity on $\overline{\mathcal{M}}_{g,n}$. The sum over topologies is not repaired. It is *replaced* by algebraic induction from one quartic pole.

*

The anomaly datum. The Koszul dual is $B = \mathrm{Vir}_{26-c}$, with curvature $\lambda = \kappa(B)\cdot\omega_1 = \frac{26-c}{2}\,\omega_1$. The **transgression algebra**

$$B_\lambda \;=\; B\langle\eta\rangle \,/\, (\eta^2 - \lambda), \qquad |\eta| = 1, \quad d_{B_\lambda}(\eta) = 0,$$

adjoins a degree-1 generator η with $\eta^2 = \lambda$ (Clifford, not exterior) and differential $d_{B_\lambda}(b) = d_B(b) + [\eta, b]$ for $b \in B$. The secondary anomaly

$$u \;=\; \eta^2 \;=\; \lambda \;=\; \tfrac{26-c}{2}\,\omega_1$$

is closed, central, and acts as the scalar $(26-c)/2$ on standard modules. The genus-g Clifford completion

$$\mathfrak{G}_g^{\mathrm{grav}}(c) \;=\; \mathfrak{G}_g(B_\Theta)$$

adjoins $2g$ handle operators $\alpha_1, \beta_1, \dots, \alpha_g, \beta_g$ with $\alpha_i\beta_j + \beta_j\alpha_i = \delta_{ij}\,u$.

*

The dichotomy. The Clifford anticommutator controlled by u splits quantum gravity into two regimes.

When $u \neq 0$ (i.e. $c \neq 26$): the Clifford algebra is a matrix algebra after inverting u. Genus-g is Morita equivalent to genus-0. All genus corrections are resummable. The $\hat{A}$-genus captures everything. This is the perturbative regime: different genera see the same module category.

When $u = 0$ (i.e. $c = 26$): the Clifford anticommutator vanishes. The handle operators become exterior generators on $H_1(\Sigma_g)$. The theory sees genuine surface topology. The genus tower collapses ($\kappa_{\text{eff}} = 0$), and gravity becomes topological: $Z_{\text{grav}} = 1$.

Different values of c are different theories, not different phases of one theory. The transition at $c = 26$ is a parametric bifurcation in the algebraic structure: as the parameter c crosses 26, the Clifford algebra degenerates from 2^g-dimensional matrix factors to 2^{2g}-dimensional exterior algebras. The secondary anomaly u is the bifurcation parameter.

Observation 25.1. The two critical values of c play distinct roles:
- $c = 13$: Koszul self-duality ($\mathrm{Vir}_{13}^! = \mathrm{Vir}_{13}$). The dichotomy parameter is $u = 13/2 \neq 0$: self-duality lives in the perturbative regime.
- $c = 26$: anomaly cancellation ($u = 0$). The Clifford/exterior bifurcation. This is the bosonic string.

Self-duality is a symmetry enhancement. Anomaly cancellation is a structural bifurcation. They are separated by 13 units of central charge.

*

Complementarity as entanglement. Theorem C of the main work gives, at every genus g,

$$H^*(\overline{\mathcal{M}}_g, \mathcal{Z}(\mathrm{Vir}_c)) \;=\; Q_g(\mathrm{Vir}_c) \;\oplus\; Q_g(\mathrm{Vir}_{26-c}).$$

The $+1$-eigenspace of the Verdier involution is the boundary sector. The -1-eigenspace is the bulk sector. The $(-(3g-3))$-shifted symplectic structure makes both Lagrangian.

In gravity, this is holographic entanglement:
- The boundary sector $Q_g(\mathrm{Vir}_c)$ is the CFT contribution.
- The bulk sector $Q_g(\mathrm{Vir}_{26-c})$ is the gravitational (Koszul dual) contribution.
- The symplectic pairing between the Lagrangian subspaces is the entanglement between boundary and bulk at genus g.

At genus 1: $Q_1(\mathrm{Vir}_c) = k \cdot \kappa$ (with $\kappa = c/2$) and $Q_1(\mathrm{Vir}_{26-c}) = k \cdot \lambda_1$, with $\langle \kappa, \lambda_1 \rangle = 1$. In the genus-1 algebraic model the boundary/bulk pairing has unit pairing. The Bekenstein–Hawking entropy $S = 2\pi\sqrt{ch/6}$ is the saddle-point asymptotic of the modular S-transform applied to this factorisation.

Whether the Ryu–Takayanagi formula is a consequence of the Lagrangian structure is an open question. At genus 1, the structures match: $\langle \kappa_1, \lambda_1 \rangle = 1$ on $\overline{\mathcal{M}}_{1,1}$ reproduces the BTZ factorisation. At higher genus the identification is conjectural.

*

Unitarity as a quasi-isomorphism. The bar-cobar equivalence

$$\Omega(\overline{B}(\mathrm{Vir}_c)) \;\simeq\; \mathrm{Vir}_c$$

is Theorem B. It says: the bar coalgebra $\overline{B}(\mathrm{Vir}_c)$ classifies twisting morphisms (couplings between Vir_c and its Koszul dual Vir_{26-c}), and the cobar $\Omega(\overline{B}(\mathrm{Vir}_c)) \simeq \mathrm{Vir}_c$ recovers the boundary algebra itself. The bulk is not $\overline{B}(\mathrm{Vir}_c)$ but the derived centre $C^\bullet_{\mathrm{ch}}(\mathrm{Vir}_c, \mathrm{Vir}_c)$ (AP34).

No cohomological information is lost in this algebraic model. A state that enters a BTZ black hole (encoded as an MC deformation $\delta\alpha$ at weight h) can be recovered from the boundary via the cobar functor. This is not a statement about evaporation dynamics; it is a statement about the algebraic invertibility of the holographic map.

The information paradox, at this level of structure, takes a precise form: the bar-cobar equivalence guarantees that no *cohomological* information is lost in the holographic encoding. Whether this implies that no *dynamical* information is lost (as in black hole evaporation) depends on the relationship between the algebraic encoding and the physical Hamiltonian; the algebraic framework does not address this relationship. The quasi-isomorphism is a necessary condition for no-information-loss, not a sufficient one.

*

The architecture, in one paragraph. Start from $\{T_\lambda T\} = \partial T + 2T\lambda + (c/12)\lambda^3$. The quartic pole forces an infinite A_∞ tower. The bar complex encodes it as a dg coalgebra; Koszul duality produces Vir_{26-c}. The Koszul dual curvature $\kappa(B) = (26-c)/2$ controls the transgression. The anomaly completion of the Koszul dual by this curvature produces the transgression algebra $(\mathrm{Vir}_{26-c})_\Theta$ with secondary anomaly $u = \lambda = \kappa(B)\cdot\omega_1 = \frac{26-c}{2}\,\omega_1$. The proposed genus-$g$ Clifford completion $\mathfrak{G}_g^{\mathrm{grav}}(c)$ is a model for the genus-g line-operator algebra; any Morita-triviality statement belongs to this finite Clifford model and must be checked against the full open-sector category. The MC recursion determines the partition function $Z_g = \mathrm{str}_{\mathfrak{G}_g/B_\Theta}(\Theta^{(g)})$ (the super-trace of the MC element through the Clifford factor) at every genus from genus-0 data alone. This is the bridge: the partition function (a section on $\overline{\mathcal{M}}_g$) and the line-operator algebra (a Clifford extension of B_Θ) are connected by the super-trace. Complementarity gives a Lagrangian decomposition of the genus-g cohomology into boundary and bulk sectors. Bar-cobar equivalence gives algebraic invertibility of the holographic encoding. One quartic pole, six movements: the proved algebra, and an honest accounting of the interpretive identifications it suggests.

*

What the Maloney–Witten programme treats as a problem of *enumeration* (which topologies to sum over), the MC element treats as a problem of *algebra* (which Maurer–Cartan equation to solve). The sum over topologies is a shadow of the genus expansion of Θ_{Vir_c}. But Θ exists before the expansion. The partition function is the character of $\mathfrak{G}_g^{\mathrm{grav}}$, not a sum over fillings.

Question 25.2. Does the graded character $\hat{Z}_1^{\mathrm{grav}}(c;\tau) := \chi_{\mathrm{gr}}(\mathfrak{G}_1^{\mathrm{grav}}(c);\, q)$ have non-negative Fourier coefficients for all $c > 1$? If so, the Maloney–Witten negativity is resolved by the algebraic construction. If not, the violation would identify the precise obstruction to a unitary 3d gravity at that value of c.

Question 25.3. What is the explicit genus-2 gravitational partition function $Z_2(\mathrm{Vir}_c)$ computed by the MC recursion? The stable graph combinatorics of $\overline{\mathcal{M}}_2$ (seven stable graphs at (g=2, n=0), including the banana graph with $|\mathrm{Aut}| = 8$) should yield a computable answer. Does it agree with the $\hat{A}$-genus prediction $F_2 = (c/2)\cdot 7/5760 = 7c/11520$?

Question 25.4. The complementarity decomposition at genus $g \geq 2$ has not been computed for Virasoro. What is $\dim Q_2(\mathrm{Vir}_c)$? At genus 2, $\dim H^*(\overline{\mathcal{M}}_2) = 8$ (Poincaré polynomial $1 + 3t^2 + 3t^4 + t^6$), so the Verdier involution splits this 8-dimensional space into two 4-dimensional Lagrangian halves. The boundary/bulk partition is determined by $\kappa = c/2$: which Mumford–Morita–Miller classes land in which eigenspace?

26 The inner music

The 1-form does not describe bar-cobar duality from outside, as a functor between categories. It constructs it from inside, as an extraction of residues at collisions. The functor is a consequence of the construction, not the other way around.

*

The following are all the same object:

- the curvature $m_0 = \kappa \cdot \omega_g$ of the curved A_∞ structure on the genus-g bar complex;
- the conformal anomaly of the bosonic string;
- the failure of the classical BRST operator to square to zero;
- the obstruction class in $R^*(\overline{\mathcal{M}}_g)$;
- the leading coefficient of the Taylor expansion of the categorical logarithm;
- the residue at the branch point of the multi-valued propagator on Σ_g;
- the modular characteristic $\kappa(\mathcal{A})$ in Theorem D;
- the genus-1 Eisenstein coefficient;
- the $\hat{A}$-genus generating function evaluated at the leading OPE coefficient;
- the scalar projection of the universal MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$;
- the curvature of the Koszul dual: $\kappa(B) = (26-c)/2$, vanishing at $c = 26$;
- the bifurcation parameter for the Clifford/exterior dichotomy: $u = \eta^2 = \lambda$, acting by $\kappa(B)$ on standard modules;
- the Clifford anticommutator of handle operators: $\alpha_i\beta_j + \beta_j\alpha_i = \delta_{ij}\, u$;
- the cosmological constant, measured in units of the Lagrangian splitting between the ± 1-eigenspaces of the Verdier involution;
- the source term in the MC recursion that determines $\Theta^{(g)}$ from $\Theta^{(g-1)}$;
- the obstruction to Morita triviality of the genus-g Clifford completion.

Most items (1, 3–10, 15) are the algebraic modular characteristic $\kappa(\mathcal{A})$, an invariant of any chiral algebra. Items 11–14 and 16 involve the Koszul dual curvature $\kappa(B) = (26-c)/2$, which is the curvature of the dual algebra $B = \mathrm{Vir}_{26-c}$. For the bosonic string, the effective curvature $\kappa_{\text{eff}} = \kappa(\text{matter}) + \kappa(\text{ghost}) = c/2 + (-13) = (c-26)/2$ vanishes at $c = 26$; note the opposite sign: $\kappa_{\text{eff}} = -\kappa(B)$. Item 2 (the conformal anomaly *of the bosonic string*) is likewise a gravitational quantity: the worldsheet anomaly that determines the critical dimension is κ_{eff}, not the bare $\kappa(\mathcal{A})$. The two scalars coincide only when the algebra is the full bosonic string. The list records not a tautology but a convergence: the same scalar organises both the algebraic engine and the gravitational application.

*

A 1-form with a simple pole.

A relation among three.

The rest follows.

27 Three roads to a shadow

On a primary line L of the cyclic deformation complex, the shadow coefficients S_r can be computed by three independent methods. Each method is exact. Each method is finite at every arity. Their agreement is a theorem.

The algebraic road. The shadow metric $Q_L(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 2\Delta)\, t^2$ is a quadratic polynomial. The generating function $H(t) = t^2\sqrt{Q_L(t)}$ gives

$$S_r \;=\; \frac{1}{r}\,[t^{r-2}]\,\sqrt{Q_L(t)}.$$

One square root, infinitely many coefficients. The quadratic has two roots; the square root is algebraic of degree 2; the entire tower is determined by three numbers (κ, α, Δ).

The recursive road. The MC equation projected to arity r expresses the next coefficient as a polynomial in earlier coefficients after the initial data S_2, S_3, S_4 have been fixed. Equivalently one expands

$$H(t)^2 = t^4 Q_L(t) \tag{2}$$

coefficient by coefficient. The closed form supplies the same recursion without committing to a misleading universal denominator formula: for the Virasoro T-line the all-arity denominator statement is a divisibility theorem with saturation verified in the computed range.

The analytic road. The ordinary generating function for the shadow coefficients has branch points at the zeros of $Q_L(t)$. The growth is geometric (Gevrey-0), so the factorially divided Borel transform $\sum S_r s^r/\Gamma(r+1)$ is entire. Analytic continuation is needed for the original algebraic series, not Borel resummation of a factorially divergent one. For class M lanes, the nearest branch points produce oscillatory Darboux asymptotics. The subexponential power depends on the lane: the Virasoro T-line has a linear leading recurrence, while the $\mathcal{W}_3$ W-line has the Catalan-type algebraic recurrence.

*

Three methods, one answer on the verified primary line. The agreement is not accidental: the algebraic road solves the Riccati equation $H^2 = t^4 Q_L$; the recursive road is its power-series expansion; the analytic road is the branch-geometry of the same algebraic function. The three roads meet at Q_L.

	Shadow obstruction tower	**Bessel function**
Closed form	$t^2\sqrt{Q_L(t)}$	$\sqrt{2/(\pi z)}\cos(z - \nu\pi/2 - \pi/4)$
Power series	$\sum S_r\, t^r$ via MC recursion	$\sum(-1)^k (z/2)^{\nu+2k}/(k!\,\Gamma(\nu+k+1))$
Analytic continuation	branch cut of the ordinary algebraic series	$\int_0^\pi \cos(\nu\theta - z\sin\theta)\, d\theta/\pi$

The analogy with classical special functions is exact at the structural level: ODE, power series, integral representation. What distinguishes the shadow obstruction tower is that the ODE is *algebraic* (degree 2, not transcendental), and the three invariants (κ, α, Δ) that parametrise Q_L are themselves computable from the OPE.

28 A recurring discriminant

The discriminant $(1 - 3x)(1 + x)$ recurs in three rank-one computations.

$\widehat{\mathfrak{sl}}_2$ bar cohomology. The Riordan numbers $R(n)$, with generating function

$$R(x) = \frac{1 + x - \sqrt{(1 - 3x)(1 + x)}}{2x(1 + x)}.$$

Virasoro bar cohomology. The Motzkin differences $M(n{+}1) - M(n)$, with generating function satisfying the cubic $x^3 P^2 - (1{-}2x)(1{-}x^2)P + (1{-}x)(1{-}x^2) = 0$.

$\beta\gamma$ bar cohomology. Generating function $\sqrt{(1 + x)/(1 - 3x)}$.

Growth rate 3 for all three. Convergence radius 1/3 for all three. The discriminant $(1 - 3x)(1 + x) = 1 - 2x - 3x^2$ has roots at $x = 1/3$ and $x = -1$.

*

Why 3? The bar differential extracts residues at collisions using the propagator $\eta_{ij} = d\log(z_i - z_j)$. At each collision, a two-pole OPE (z^{-2} and z^{-1}) produces three independent residue extractions: the level term, the bracket term, and their cross-term. The growth rate 3 = number of independent residue extractions per step.

Why −1? The bar complex uses desuspension: the Koszul sign. The $\mathbb{Z}/2$-symmetry of the desuspension contributes a root at $x = -1$. The two factors of the discriminant have separate origins: $(1 - 3x)$ is the OPE, $(1 + x)$ is the grading.

In the computed rank-one/contact lanes, Drinfeld–Sokolov reduction carries the branch locus compatibly: $\widehat{\mathfrak{sl}}_2$ and Virasoro share $(1 - 3x)(1 + x)$ after the bar generating functions are compared. This is a computed pattern in the rank-one sector, not a universal Koszul invariant for arbitrary chiral algebras.

Algebra	Discriminant	Growth	Radius
$\widehat{\mathfrak{sl}}_2$	$(1 - 3x)(1 + x)$	$3^n/n^{3/2}$	$1/3$
Virasoro	$(1 - 3x)(1 + x)$	$3^n/n^{3/2}$	$1/3$
$\beta\gamma$	$(1 - 3x)(1 + x)$	$3^n/\sqrt{n}$	$1/3$
$\widehat{\mathfrak{sl}}_3$	$1 - 3x - x^2$	$\approx 3.3^n$	$(\sqrt{13} - 3)/2$

The $\mathfrak{sl}_3$ discriminant breaks the pattern: $1 - 3x - x^2$ has roots involving $\sqrt{13}$, and $13 = \dim(\mathfrak{sl}_3)$. The recurrence depth of the generating function appears to equal $\operatorname{rank}(\mathfrak{g}) + 1$ in the computed cases; no all-rank theorem is asserted here.

29 The depth creation mechanism

Free bosons have $S_4 = 0$. The shadow obstruction tower terminates at arity 2: class G. No quartic shadow, no infinite tower, no contact invariant.

The Sugawara tensor $T = \sum_a {:}\partial\phi_a \partial\phi_a{:}/(2k)$ is quadratic in the free fields. Its shadow obstruction tower inherits S_4 from same-index double contractions and from the DS/Sugawara nonlinear projection:

$$S_4(T) \;=\; \frac{1}{\kappa(T)} \sum_a \langle \partial\phi_a\, \partial\phi_a\, \partial\phi_a\, \partial\phi_a \rangle_{\text{conn}} \quad \text{after the interacting projection.}$$

This is nonzero because the Sugawara construction mixes the bosons. The quadratic nonlinearity creates the quartic from nothing.

Three views of the same phenomenon.

(i) The HPL view. The Heisenberg–plus–Lie decomposition of the affine OPE separates into a Heisenberg part $(k\kappa^{ab}/(z-w)^2)$ and a Lie part $(f^{ab}{}_c J^c/(z-w))$. After DS reduction, the ghost-number obstruction kills the coproduct of the Heisenberg part; the r-matrix of the Lie part survives. The quartic shadow is the first nontrivial interaction between Heisenberg and Lie data in the reduced algebra.

(ii) The spectral sequence view. The PBW spectral sequence at E_1 contains the affine generators. The d_1 differential kills them: at E_2, only Virasoro and higher-spin composites survive. These composites carry quartic poles by double Wick contraction. The spectral sequence is the mechanism by which a 2-pole algebra becomes a 4-pole algebra.

(iii) The Miura view. $T = \sum(\partial\phi)^2$ creates quartic terms from Gaussian double contractions: $\langle(\partial\phi_a)^2(\partial\phi_b)^2\rangle_{\text{conn}} = 2\delta_{ab}$. Independent bosons have zero connected mixed four-point function for $a \neq b$; the quartic invariant appears after summing same-index contractions and passing through the interacting Sugawara/DS projection.

*

The depth increases from class L to class M. The mechanism is the nonlinearity of the Sugawara construction. Free becomes interacting, finite tower becomes infinite tower. The transition is not continuous: $S_4 = 0$ for free fields, $S_4 = 10/[c(5c+22)]$ for Virasoro. No intermediate value. The quartic invariant is either absent or nonzero; the tower either terminates or runs forever.

30 The decisive test

$F_2(V_{E_8}) = 7/720$.

The bar complex says: $\kappa(V_L) = \text{rank}(L)$. For E_8, $\kappa = 8$, hence $F_2 = 8 \cdot 7/5760 = 7/720$.

The Siegel–Weil formula says: $\Theta^{(2)}_{E_8} = E^{(2)}_4$, the genus-2 Siegel Eisenstein series of weight 4. The Fourier coefficient $a(\text{diag}(1,1)) = 30240 = 240 \cdot 126$. Each of the 240 roots of E_8 has exactly 126 orthogonal roots.

These agree. The machine works.

*

The agreement is a theorem: for any unimodular lattice L of rank d,

$$F_2(V_L) \;=\; d \cdot \frac{7}{5760},$$

and this equals the second coefficient of $\log \Theta_L^{(2)}$ at the cusp. The proof: the shadow tower of a lattice vertex algebra terminates (class G, all $S_r = 0$ for $r \geq 3$), so $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ at all genera, where λ_g^{FP} is the Faber–Pandharipande coefficient of the $\hat{A}$-genus. The lattice theta function $\Theta_L^{(g)} = E_{d/2}^{(g)}$ by Siegel–Weil, and the logarithmic derivative at the cusp extracts the same Bernoulli coefficient.

Lattice	d	$\kappa = d$	$F_2 = \kappa \cdot 7/5760$
$\mathbb{Z}$ (Heisenberg)	1	1	7/5760
D_4	4	4	7/1440
E_8	8	8	7/720
$E_8 \oplus E_8$	16	16	7/360
Leech	24	24	7/240

Every entry is independently verifiable: κ from the OPE ($\kappa = d$ for a rank-d lattice VOA), F_2 from the $\hat{A}$-genus (7/5760), and $\Theta_L^{(2)}$ from lattice theta functions in the Siegel modular forms literature. The three computations use different mathematics. They give the same number.

31 The $\mathcal{N} = 2$ surprise

The $\mathcal{N} = 2$ superconformal algebra is the first algebra outside the standard Lie-theoretic landscape where the shadow obstruction tower was computed. Three results were unexpected.

(a) The anomaly ratio (CORRECTED). The naive channel sum $\kappa = c/2 + c/3 + c/3 = 7c/6$ is **wrong**: it uses Zamolodchikov metric eigenvalues, which are not anomaly ratios. The correct kappa comes from the Kazama–Suzuki coset decomposition of the $\mathcal{N} = 2$ SCA from $\mathfrak{sl}(2)_k$:

$$\kappa(\mathcal{N}{=}2) \;=\; \kappa(\mathfrak{sl}(2)_k) + \kappa(\text{fermion pair}) - \kappa(U(1)_{\text{denom}}) \;=\; \frac{3(k+2)}{4} + \frac{1}{2} - \left(\frac{k}{2} + 1\right) \;=\; \frac{k+4}{4}.$$

In terms of $c = 3k/(k+2)$: $\kappa = (6-c)/(2(3-c))$.

(b) Additive complementarity (CORRECTED). $c' = 6 - c$, so $c + c' = 6$ (additive, like Virasoro's $c + c' = 26$). The Feigin–Frenkel involution for $\mathfrak{sl}(2)$ (dual Coxeter number $h^\vee = 2$) acts as $k \mapsto -k-4$, giving

$$c' \;=\; \frac{3(k+4)}{k+2}, \qquad c + c' \;=\; \frac{3k}{k+2} + \frac{3(k+4)}{k+2} \;=\; \frac{3(2k+4)}{k+2} \;=\; 6.$$

The complementarity sum $\kappa + \kappa' = 1$ is constant. The Koszul involution $c \mapsto 6 - c$ has its fixed point at $c = 3$; however, $\kappa(3) = (6-3)/(2(3-3))$ has a pole there (the coset denominator vanishes at the free-field limit $k \to \infty$). Self-duality at $c = 3$ holds at the level of the central charge involution, not at the level of κ.

(c) The chiral ring obstruction. The chiral ring $R = \mathbb{C}[x]/(x^{K-1})$ of an $\mathcal{N} = 2$ theory at level K is a truncated polynomial ring. It is *not* Koszul: $\mathrm{Ext}_R^n(\mathbb{C}, \mathbb{C}) = \mathbb{C}$ for all $n \geq 0$ (a 2-periodic resolution). Koszulness of the full $\mathcal{N} = 2$ superconformal algebra does not descend to the chiral ring.

Warning 31.1. The direct OPE channel sum ($\kappa = 7c/6$) used Zamolodchikov metric eigenvalues in place of anomaly ratios, and is wrong (AP1). The Kazama–Suzuki coset computation ($\kappa = (6-c)/(2(3-c))$, $c' = 6 - c$, $\kappa + \kappa' = 1$) is the correct one. The naive formula $c' = 9/c$ (from setting $\kappa' = c'/2$ and solving $7c/6 + 7c'/6 = \text{const}$) is a consequence of the wrong kappa and is likewise dismissed.

32 What the tables contain

580 shadow coefficients. 15 algebras. 30 arities each. Every entry an exact rational number.

$$S_{10}(\mathrm{Vir},\ c = 1/2) = \frac{3795318931456}{5764801}.$$

$$S_{15}(\mathrm{Vir},\ c = 13) = -\frac{83470593873707008}{259042567755018935661111}.$$

$$F_3(\mathrm{Vir},\ c = 1/2) = \frac{31}{3870720}.$$

These numbers encode how deeply the bar complex of a chiral algebra sees moduli-space geometry. They are Taylor coefficients of a square root: algebraic of degree 2, controlled by three invariants (κ, α, S_4), branching over two points in the t-plane.

The denominator of S_r at arity $r \geq 4$ is (const) $\cdot\, c^{r-3}(5c + 22)^{\lfloor (r-2)/2 \rfloor}$ (Theorem 16.1). No Kac determinant beyond weight 4 appears. The numerator is a polynomial in c with integer coefficients whose degree grows linearly in r. The signs oscillate with period governed by $\arg\big(\text{root of } Q_L\big)$: a beat frequency between the OPE poles and the Koszul sign.

Every number is computable. None existed before this work.

33 The arithmetic of the shadow obstruction tower

The shadow metric $Q_L(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 2\Delta)\, t^2$ is a quadratic polynomial. Its square root $\sqrt{Q_L}$ generates the shadow obstruction tower. Everything in §19–§32 follows from this single algebraic function of degree 2.

The structure of this function is the structure of a quadratic extension over the function field. The literal invariants are the branch discriminant, the Galois involution, and the branch-point growth rate. The language of rings of integers, regulators, splitting of primes, and Dedekind zeta functions is only a dictionary unless a separate arithmetic specialisation is proved.

*

The extension. Fix a primary line L with data (κ, α, S_4) and function field $F = \mathbb{Q}(c)(t)$. The shadow metric $Q_L \in F$ is a quadratic polynomial. Define

$$K_L \ :=\ F(\sqrt{Q_L})\ =\ \mathbb{Q}(c)(t)(\sqrt{Q_L}).$$

This is a quadratic extension of F. Its Galois group is $\mathrm{Gal}(K_L/F) \cong \mathbb{Z}/2$, generated by the involution $\sigma\colon \sqrt{Q_L} \mapsto -\sqrt{Q_L}$.

The shadow generating function $H(t) = t^2\sqrt{Q_L(t)} \in K_L$ is an element of the extension that does not lie in the base field. The shadow obstruction tower $\{S_r\}_{r\geq 2}$ consists of the Taylor coefficients of $\sqrt{Q_L}$. After localising the coefficient ring at the displayed denominators these coefficients form a natural coefficient order, not the full integral closure of the quadratic function-field extension.

The Galois involution σ sends $\sqrt{Q_L}$ to $-\sqrt{Q_L}$ and therefore $H(t)$ to $-H(t)$. This sign is the Koszul sign: the monodromy -1 of the shadow connection ∇^{sh} around a branch point is the nontrivial element of $\mathrm{Gal}(K_L/F)$.

Proposition 33.1 *(Branch discriminant of the shadow metric).* The discriminant of the quadratic polynomial $Q_L(t)$ is

$$\mathrm{disc}_t(Q_L) \;=\; -32\,\kappa^2\,\Delta,$$

where $\Delta = 8\kappa S_4$ is the critical discriminant. For Virasoro: $\mathrm{disc}_t(Q_{\mathrm{Vir}}) = -320\,c^2/(5c{+}22)$.

Proof. The discriminant of the quadratic $At^2 + Bt + C$ with $A = 9\alpha^2 + 2\Delta$, $B = 12\kappa\alpha$, $C = 4\kappa^2$ is $B^2 - 4AC = 144\kappa^2\alpha^2 - 16\kappa^2(9\alpha^2 + 2\Delta) = -32\kappa^2\Delta$. □

*

Proposition 33.2 *(Square-class behaviour of the shadow metric).* Let L be a primary line with critical discriminant $\Delta_L = 8\kappa_L S_{4,L}$. The branch behaviour of the shadow metric is constrained by Δ_L and by the square class of Q_L:

(i) *Split* ($\Delta_L = 0$, $\kappa_L \neq 0$). The shadow metric Q_L is a perfect square: $Q_L = (2\kappa + 3\alpha\, t)^2$. The extension is trivial on L; $\sqrt{Q_L} \in F$. The shadow obstruction tower terminates after finitely many terms. **Classes G and L.** Heisenberg ($\alpha = 0$: terminates at arity 2) and affine Kac–Moody ($\alpha \neq 0$: terminates at arity 3).

(ii) *Nonsquare separable* ($\Delta_L \neq 0$ and Q_L is not a square in the coefficient field). The extension K_L/F is genuine; $\sqrt{Q_L} \notin F$. The shadow obstruction tower is infinite and the Taylor coefficients S_r grow as $A\,\rho^r\, r^{-5/2}\cos(r\theta + \varphi)$. **Class M.** Virasoro and $\mathcal{W}_N$.

(iii) *Ramified* ($\kappa_L = 0$). The shadow metric degenerates: $Q_L(t) = (9\alpha^2 + 2\Delta)\,t^2$, with a double zero at $t = 0$. The curvature vanishes, the connection has an irregular singularity at the origin, and the shadow obstruction tower reduces to a monomial. This occurs on charged strata where the weight-changing line has $\kappa = 0$. **Class C** (stratum separation: the quartic invariant lives on a stratum whose self-bracket exits the complex).

Proof. (i) If $\Delta = 0$ then $S_4 = 0$ (since $\kappa \neq 0$), so the coefficient of t^2 in Q_L is $9\alpha^2$, and $Q_L = (2\kappa + 3\alpha\, t)^2$. The recursion $S_r = -(1/(2r\kappa)) \sum \varepsilon(j,k)\; jk\, S_j S_k$ yields $S_r = 0$ for all $r \geq 4$ when $S_4 = 0$ (each subsequent term vanishes by induction). If also $\alpha = 0$: $S_3 = 0$ and the tower terminates at $r = 2$.

(ii) If $\Delta \neq 0$ then $\mathrm{disc}(Q_L) = -32\kappa^2\Delta \neq 0$, so Q_L has two distinct roots. If, in addition, Q_L is not a square in the coefficient field, then $\sqrt{Q_L}$ is algebraic of degree 2 over F. The Taylor coefficients of an algebraic function with branch points at $t_\pm$ grow as $|t_\pm|^{-r}\, r^{-3/2+\epsilon}$ by the Darboux transfer theorem; the precise asymptotics $A\,\rho^r\, r^{-5/2}\cos(r\theta + \varphi)$ follow from the exponent $-1/2$ of the branch-point singularity $(t - t_\pm)^{1/2}$, giving $r^{-1-1/2} = r^{-3/2}$, combined with the additional r^{-1} from the Taylor extraction $S_r = [t^{r-2}]\sqrt{Q_L}/(r)$.

(iii) If $\kappa = 0$ then the constant and linear terms of Q_L vanish, and $Q_L(t) = (9\alpha^2 + 2\Delta)\, t^2$ has a double zero at $t = 0$. The flat section $\Phi(t) = \sqrt{Q_L(t)/Q_L(0)}$ is undefined (the connection has an irregular point at the origin). $\square$

*

The four-class partition G/L/C/M is a square-class table for the shadow metric on primary lines. The analogy with genus theory is useful but not literal: a primary line is not a prime ideal, and Δ_L is not a Legendre symbol. What is proved is the vanishing/nonsquare/degenerate pattern of Q_L and the resulting termination or nontermination of the shadow tower.

Class	**Splitting**	Δ_L	K_L/F	**Number field analogue**
G	totally split	$0\ (\alpha = 0)$	trivial	$\mathbb{Q}(\sqrt{1}) = \mathbb{Q}$
L	split	$0\ (\alpha \neq 0)$	trivial	$\mathbb{Q}(\sqrt{d^2}) = \mathbb{Q}$
C	ramified	$\kappa = 0$	degenerate	ramified prime
M	inert	$\neq 0$	genuine degree-2	$\mathbb{Q}(\sqrt{d})$, $d < 0$ squarefree

*

The coefficient order. In $\mathbb{Q}(\sqrt{d})$, the ring of integers is $\mathbb{Z}[\omega]$ where $\omega = \sqrt{d}$ or $(1 + \sqrt{d})/2$ depending on d mod 4. In the shadow extension K_L, the analogous object used here is the localised coefficient order generated by the shadow coefficients:

$$\mathcal{O}_{K_L} \;:=\; \mathbb{Z}[\kappa, \alpha, S_4]\big[\{S_r\}_{r\geq 5}\big] \;\subset\; K_L.$$

The shadow coefficients S_r for $r \geq 5$ are elements of the localised coefficient field generated recursively by (27). They are not algebraic integers over $\mathbb{Z}[\kappa, \alpha, S_4]$ in the ordinary number-theoretic sense; the displayed denominators are part of the structure.

For Virasoro: $S_r \in \mathbb{Q}(c)$ for all $r \geq 2$, and the denominator pattern (Theorem 16.1) gives

$$\operatorname{denom}(S_r) \;=\; (\text{const}) \cdot c^{r-3}(5c{+}22)^{\lfloor (r-2)/2 \rfloor}.$$

Two primes appear: $c = 0$ and $c = -22/5$. These are the *ramified primes* of the Virasoro shadow obstruction tower: the places where the shadow coefficient has a pole. They are the zeros of the weight-4 Gram determinant $\det G_4 = c^2(5c{+}22)/2$. The shadow obstruction tower's poles are the representation theory's zeros.

No Kac determinant beyond weight 4 appears. The quadratic extension sees only the first nontrivial Gram matrix.

*

The growth rate analogue. In a real quadratic field $\mathbb{Q}(\sqrt{d})$, the fundamental unit $\varepsilon = x_1 + y_1\sqrt{d}$ generates all units via ε^n. The regulator $R = \log|\varepsilon|$ measures the size of the unit group. (The specialised shadow fields $K(c_0) = \mathbb{Q}(\sqrt{d(c_0)})$ are *imaginary* quadratic, so their unit group is $\{\pm 1\}$; the analogy below concerns the branch-point growth rate, not literal units.)

In the shadow obstruction tower: the growth rate

$$\rho = \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|} = |t_+|^{-1},$$

where t_+ is the zero of Q_L nearest the origin, plays the role of the fundamental unit. The shadow "regulator" is $\log\rho$, measuring the exponential growth of the tower.

The convergence condition $\rho < 1$ means $|t_+| > 1$: the branch points of $\sqrt{Q_L}$ lie outside the unit disk. The Taylor series $\sqrt{Q_L(t)} = \sum a_n t^n$ converges for $|t| < |t_+|$, and the shadow obstruction tower converges as a formal power series in the arity parameter. Setting $\rho = 1$ yields the critical cubic

$$5c^3 + 22c^2 - 180c - 872 = 0, \qquad c^* \approx 6.125,$$

the boundary between convergent and divergent shadow obstruction towers: the analogue of the Minkowski bound that controls the class number of a number field.

For Virasoro at $c = 13$ (the self-dual point): $\rho(13) \approx 0.467 < 1$. The shadow obstruction tower converges at the self-dual point.

*

The Galois involution and Koszul duality. The Galois involution of K_L/F is $\sigma\colon \sqrt{Q_L} \mapsto -\sqrt{Q_L}$. Koszul duality acts on the shadow extension by $\mathcal{A} \mapsto \mathcal{A}^!$: for Virasoro, $c \mapsto 26 - c$. These are different involutions that commute.

The Galois involution σ is the monodromy of ∇^{sh}: transport around a branch point of $\sqrt{Q_L}$ returns $-\sqrt{Q_L}$. This is the Koszul sign of bar-cobar duality: the desuspension s^{-1} contributes a sign at each bar element.

The Koszul involution $\mathcal{A} \mapsto \mathcal{A}^!$ acts on the coefficients of Q_L, not on $\sqrt{Q_L}$ itself. For Virasoro it sends $c \to 26 - c$, so $\kappa \to (26 - c)/2$, $\alpha \to \alpha'$, $S_4 \to S_4'$. The complementarity identity

$$\Delta(\mathcal{A}) + \Delta(\mathcal{A}^!) = \frac{6960}{(5c+22)(152-5c)} \tag{3}$$

is the constraint that relates the discriminant of the extension K_L to the discriminant of the Koszul dual extension $K_{L^!}$. The numerator $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$ is universal.

At $c = 13$: both factors in the denominator equal 87, $\Delta(13) = 40/87$, and $\Delta + \Delta' = 80/87 = 6960/87^2$. The self-dual point is where the two extensions K_L and $K_{L^!}$ coincide.

*

The complementarity sum as class invariant. The complementarity sum $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$ measures how far the Koszul pair $(\mathcal{A}, \mathcal{A}^!)$ deviates from perfect anti-symmetry.

Family	$\kappa + \kappa'$	Status
Kac–Moody	0	anti-symmetric (FF involution around 0)
Free fields	0	anti-symmetric
Virasoro	13	anti-symmetric around $c = 13$
$\mathcal{N} = 2$	1	anti-symmetric around $c = 3$

In number theory, the class number $h(d)$ of $\mathbb{Q}(\sqrt{d})$ measures the failure of unique factorisation. The complementarity sum plays the analogous role for the Koszul pair: it measures the failure of perfect duality. When $\kappa + \kappa' = 0$ (as for Kac–Moody), the pair is "principal" (the analogue of class number 1). When $\kappa + \kappa' \neq 0$ (as for Virasoro), the pair has a nontrivial "class invariant."

*

The Dirichlet series. For a number field K, the Dedekind zeta function $\zeta_K(s) = \sum_{\mathfrak{a}} N(\mathfrak{a})^{-s}$ encodes the arithmetic. For a quadratic field: $\zeta_{\mathbb{Q}(\sqrt{d})}(s) = \zeta(s) \cdot L(s, \chi_d)$ where χ_d is the Kronecker character.

In the shadow programme, a separately defined Dirichlet-sewing lift $S_{\mathcal{A}}(u)$ may play an analogous role when it has actually been constructed. It is not the Dirichlet series of partition coefficients, and the following formulas are templates rather than Dedekind zeta identities:

- Heisenberg: the oscillator q-product factors mode-by-mode. Any scalar zeta moment attached to it is a separate analytic transform, not the Dedekind zeta of a shadow quadratic field.
- Virasoro: no Euler-product formula for Virasoro partition coefficients is proved here. Defects in a Dirichlet-sewing lift must be checked in that transform, not imported from the q-series.

The Heisenberg sewing has no shadow obstruction tower (class G, $\Delta = 0$): its number field is $\mathbb{Q}$ itself, and the Dirichlet series factors completely. The Virasoro sewing has an infinite shadow obstruction tower (class M, $\Delta \neq 0$): its number field is a genuine quadratic extension, and the Euler product fails.

*

The functional equation. The Dedekind zeta satisfies the functional equation $\zeta_K(s) = |d_K|^{1/2-s}\, \gamma(s)\, \zeta_K(1-s)$, where d_K is the discriminant. The shadow obstruction tower satisfies a complementarity identity (33) that plays the same structural role: it relates the discriminant of $\mathcal{A}$ to the discriminant of $\mathcal{A}^!$, with a universal numerator independent of the family.

The complementarity identity is not a functional equation in the classical sense (the shadow obstruction tower does not satisfy a self-referential relation under $s \to 1 - s$). It is the Koszul-dual constraint: the quadratic extension determined by $\mathcal{A}$ and the quadratic extension determined by $\mathcal{A}^!$ are related by a universal formula. The constant 6960 is the analogue of the conductor.

*

What this is. The shadow obstruction tower is not a sequence of independent invariants. It is not an analogy with number theory. It is the Taylor expansion of $\sqrt{Q_L}$ at the origin, and $\sqrt{Q_L}$ defines a quadratic extension $K_L = \mathbb{Q}(c)(t)(\sqrt{Q_L})$ of the function field $F = \mathbb{Q}(c)(t)$. The entire tower is the arithmetic of this extension: the discriminant is $-32\kappa^2\Delta$; the splitting table is the G/L/C/M classification; the growth rate is the reciprocal of the nearest branch point; the Koszul sign is the Galois involution; the complementarity identity is the duality constraint on discriminants.

Number-field dictionary	Shadow obstruction tower
$\mathbb{Q}(\sqrt{d})$	$\mathbb{Q}(c)(t)(\sqrt{Q_L})$
Discriminant d	branch discriminant $-32\kappa^2\Delta$
Ring of integers $\mathbb{Z}[\omega]$	localised coefficient order
Splitting of primes	square-class depth classification
Fundamental unit ε	branch growth rate $\rho = \lvert t_+\rvert^{-1}$
Regulator $\log\lvert\varepsilon\rvert$	$\log\rho$ as growth analogue
Minkowski bound	Critical cubic c^*
Galois involution	Koszul sign (monodromy -1)
Conductor	6960
Dedekind zeta ζ_K	separately defined sewing transform
Class number	Complementarity sum $\kappa + \kappa'$
Genus theory	Four depth classes

The left column is realised over $\mathbb{Q}$. The right column is realised over $\mathbb{Q}(c)(t)$ and through Taylor coefficients of an algebraic function. Only the rows supported by the function-field quadratic extension are literal; the number-field language is a disciplined dictionary, not a transfer of every Dedekind theorem.

34 The specialisation functor

The shadow extension $K_L = \mathbb{Q}(c)(t)(\sqrt{Q_L})$ of §33 lives over the function field $\mathbb{Q}(c)$. It is a generic quadratic extension: the generic fibre of a family parametrised by the central charge. The closed fibres are obtained by specialising c to rational values.

For each $c_0 \in \mathbb{Q}$ with $c_0 \neq 0$ and $5c_0 + 22 \neq 0$, the *specialised shadow obstruction tower* is the sequence $S_r(c_0) = (1/r)[t^{r-2}]\sqrt{Q_L(t)}|_{c=c_0}$, which consists of specific rational numbers. The critical discriminant specialises to

$$\Delta(c_0) \;=\; \frac{40}{5c_0 + 22} \;\in\; \mathbb{Q},$$

The discriminant of the shadow metric $Q_L(t) = (9\alpha^2 + 2\Delta)\, t^2 + 12\kappa\alpha\, t + 4\kappa^2$ as a polynomial in t is $\operatorname{disc}(Q_L) = -32\kappa^2\Delta < 0$ (for $\kappa, \Delta > 0$), so the zeros of Q_L are complex and $\mathbb{Q}(\sqrt{\operatorname{disc}(Q_L)})$ is an *imaginary* quadratic number field.

Definition 34.1 *(Specialised shadow field).* For $c_0 \in \mathbb{Q}$ with $c_0(5c_0 + 22) \neq 0$, the *specialised shadow field* is

$$K(c_0) \;:=\; \mathbb{Q}\big(\sqrt{-2\Delta(c_0)}\big) \;=\; \mathbb{Q}\Big(\sqrt{d(c_0)}\Big)$$

where $d(c_0) < 0$ is the squarefree part of $-2\Delta(c_0)$, computed as follows. Write $5c_0 + 22 = a/b$ in lowest terms. Then $-2\Delta(c_0) = -80b/a$, and $K(c_0) = \mathbb{Q}(\sqrt{-80ab}) = \mathbb{Q}(\sqrt{\operatorname{sqfree}(-80ab)})$. Since $-80 = -2^4 \cdot 5$, the squarefree part is $-5 \cdot \operatorname{sqfree}(ab)$ when $\gcd(\operatorname{sqfree}(ab), 5) = 1$, with appropriate adjustment otherwise.

*

Computation. The minimal model central charges are $c_{p,q} = 1 - 6(p-q)^2/(pq)$ for coprime integers $2 \le p < q$. Each gives a specific quadratic field.

$c = 1/2$ (Ising model, $M(3,4)$). $5c + 22 = 49/2$, so $a = 49$, $b = 2$. $-80ab = -80 \cdot 49 \cdot 2 = -7840 = -2^5 \cdot 5 \cdot 7^2$. $\mathrm{sqfree}(-7840) = -10$. $K(1/2) = \mathbb{Q}(\sqrt{-10})$. Discriminant -40 (since $-10 \equiv 2 \pmod 4$, disc $= 4 \cdot (-10)$). Class number $h = 2$.

The shadow obstruction tower of the Ising model lives over an imaginary quadratic field.

$c = 7/10$ (tricritical Ising, $M(4,5)$). $5c + 22 = 51/2$, so $a = 51$, $b = 2$. $-80ab = -80 \cdot 51 \cdot 2 = -8160 = -2^5 \cdot 3 \cdot 5 \cdot 17$. $\mathrm{sqfree}(-8160) = -2 \cdot 3 \cdot 5 \cdot 17 = -510$. $K(7/10) = \mathbb{Q}(\sqrt{-510})$. Discriminant $4 \cdot (-510) = -2040$. Class number $h = 16$.

$c = 4/5$ (three-state Potts, $M(5,6)$). $5c + 22 = 26$, so $a = 26$, $b = 1$. $-80ab = -80 \cdot 26 = -2080 = -2^5 \cdot 5 \cdot 13$. $\mathrm{sqfree}(-2080) = -2 \cdot 5 \cdot 13 = -130$. $K(4/5) = \mathbb{Q}(\sqrt{-130})$. Discriminant $4 \cdot (-130) = -520$. Class number $h = 4$.

$c = 1$ (free boson compactified). $5c + 22 = 27 = 3^3$, so $a = 27$, $b = 1$. $-80ab = -80 \cdot 27 = -2160 = -2^4 \cdot 3^3 \cdot 5$. $\mathrm{sqfree}(-2160) = -3 \cdot 5 = -15$. $K(1) = \mathbb{Q}(\sqrt{-15})$. Discriminant -15 (since $-15 \equiv 1 \pmod 4$). Class number $h = 2$.

$c = 13$ (self-dual point). $5c + 22 = 87 = 3 \cdot 29$, so $a = 87$, $b = 1$. $-80ab = -80 \cdot 87 = -6960 = -2^4 \cdot 3 \cdot 5 \cdot 29$. $\mathrm{sqfree}(-6960) = -3 \cdot 5 \cdot 29 = -435$. $K(13) = \mathbb{Q}(\sqrt{-435})$. Discriminant -435 (since $-435 \equiv 1 \pmod 4$). Class number $h = 4$.

The complementarity constant $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$ appears as $-80 \cdot 87 = -6960$, so the self-dual shadow field is $\mathbb{Q}(\sqrt{-435})$ where $435 = 6960/16$.

$c = 25$ (Koszul dual of $c = 1$). $5c + 22 = 147 = 3 \cdot 7^2$, so $a = 147$, $b = 1$. $-80ab = -80 \cdot 147 = -11760 = -2^4 \cdot 3 \cdot 5 \cdot 7^2$. $\mathrm{sqfree}(-11760) = -3 \cdot 5 = -15$. $K(25) = \mathbb{Q}(\sqrt{-15})$.

The same field as $c = 1$.

$c = 26$ (critical string). $5c + 22 = 152 = 2^3 \cdot 19$, so $a = 152$, $b = 1$. $-80ab = -80 \cdot 152 = -12160 = -2^7 \cdot 5 \cdot 19$. $\mathrm{sqfree}(-12160) = -2 \cdot 5 \cdot 19 = -190$. $K(26) = \mathbb{Q}(\sqrt{-190})$. Discriminant $4 \cdot (-190) = -760$. Class number $h = 4$.

c	$\Delta(c)$	$d(c)$	disc	h	$\rho(c)$
$1/2$	$80/49$	-10	-40	2	12.53
$7/10$	$80/51$	-510	-2040	16	8.94
$4/5$	$20/13$	-130	-520	4	7.81
1	$40/27$	-15	-15	2	6.24
13	$40/87$	-435	-435	4	0.47
25	$40/147$	-15	-15	2	0.24
26	$5/19$	-190	-760	4	0.23

The fields are *imaginary* quadratic: the zeros of $Q_L(t)$ are complex ($\mathrm{disc}(Q_L) < 0$). The unit group is $\{\pm 1\}$ (no fundamental unit).

*

Proposition 34.2 *(Koszul field-preservation criterion).* The Koszul pair $(\mathrm{Vir}_c, \mathrm{Vir}_{26-c})$ gives the same specialised shadow field $K(c) = K(26-c)$ if and only if the ratio

$$\frac{5c+22}{152-5c} \quad \in \quad \mathbb{Q}^{\times 2} \tag{4}$$

is a perfect square in $\mathbb{Q}^{\times}$.

Proof. $K(c) = \mathbb{Q}(\sqrt{-80(5c+22)})$ and $K(26-c) = \mathbb{Q}(\sqrt{-80(152-5c)})$ (up to squarefree extraction from the denominator, absorbed by clearing denominators). Two quadratic fields $\mathbb{Q}(\sqrt{a})$ and $\mathbb{Q}(\sqrt{b})$ coincide iff a/b is a perfect square. The factor -80 cancels in the ratio. □

Example 34.3 *(The pair $c = 1$, $c = 25$).* $(5+22)/(152-5) = 27/147 = 9/49 = (3/7)^2$. Both give $K = \mathbb{Q}(\sqrt{-15})$.

Example 34.4 *(The self-dual point $c = 13$).* $(65+22)/(152-65) = 87/87 = 1 = 1^2$. Trivially the same field $K = \mathbb{Q}(\sqrt{-435})$.

Remark 34.5 *(Failure for minimal models).* For the Ising model $c = 1/2$, the Koszul dual is $c' = 51/2$. The ratio $(49/2)/(299/2) = 49/299$, and $299 = 13 \cdot 23$ is squarefree, so $49/299$ is not a perfect square. The fields are different: $K(1/2) = \mathbb{Q}(\sqrt{-10})$ while $K(51/2) = \mathbb{Q}(\sqrt{-1495})$ where $1495 = 5 \cdot 13 \cdot 23$.

More generally: for every minimal model central charge $c_{p,q}$ with $p < 10$, the Koszul pair gives different imaginary quadratic fields. The criterion (34.2) requires the specific arithmetic coincidence $\mathrm{sqfree}(5c+22) = \mathrm{sqfree}(152-5c)$, which fails generically.

Among integers $c \in [0, 26]$, it holds at $c = 1$, $c = 13$, and $c = 25$.

*

Proposition 34.6 *(Complementarity constrains the field pair).* For any $c_0 \in \mathbb{Q}$ with $c_0(5c_0+22)(152-5c_0) \neq 0$, the discriminants of the Koszul pair satisfy

$$\Delta(c_0) \cdot \Delta(26-c_0) = \frac{1600}{(5c_0+22)(152-5c_0)},$$

and the complementarity identity

$$\Delta(c_0) + \Delta(26-c_0) = \frac{6960}{(5c_0+22)(152-5c_0)}.$$

The numerators $1600 = 40^2$ and $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$ are independent of c_0.

Proof. Direct computation: $\Delta(c)\,\Delta(26-c) = 40/(5c+22) \cdot 40/(152-5c)$. The sum is $40(152-5c+5c+22)/((5c+22)(152-5c)) = 40 \cdot 174/((5c+22)(152-5c)) = 6960/((5c+22)(152-5c))$. □

The pair $(K(c), K(26-c))$ is not determined by the complementarity identity alone, but the identity constrains it: the two discriminants are roots of the quadratic $x^2 - (6960/P)\,x + 1600/P = 0$ where $P = (5c+22)(152-5c)$.

*

What fails. The shadow growth rate $\rho(c) = \sqrt{(180c + 872)/(5c + 22)}\,/\,c$ is a smooth function of c. The class number $h(d(c))$ of the specialised shadow field $K(c)$ is a function of the squarefree part $d(c) < 0$, which depends on c through the prime factorisation of $(5c + 22)_{\mathrm{num}} \cdot (5c + 22)_{\mathrm{den}}$. This is a wildly discontinuous function of c: a small change in c can change the squarefree part arbitrarily.

Observation 34.7 *(No relation between ρ and h).* The growth rate ρ and the class number h arise from different constructions applied to the same input. ρ comes from the branch point of $\sqrt{Q_L}$ (OPE data). h comes from the ideal class group of $\mathbb{Q}(\sqrt{d(c)})$ (prime factorisation data).

At $c = 1/2$: $\rho \approx 12.53$, $h = 2$. At $c = 13$: $\rho \approx 0.467$, $h = 7$.

No algebraic relation connects them. The proof: $\rho(c)$ is a smooth algebraic function of c, while $h(d(c))$ is a step function with discontinuities at every c where the squarefree part of $-80(5c + 22)$ changes. A smooth function cannot equal a step function.

*

What the specialisation sees. The specialisation functor $c \mapsto K(c)$ sends the generic shadow extension $K_L/\mathbb{Q}(c)(t)$ to a specific quadratic number field for each rational c. This functor is:

(i) Not injective: $c = 1$ and $c = 25$ give the same field $\mathbb{Q}(\sqrt{-15})$.
(ii) Not continuous: nearby rational values of c can give completely different fields (different squarefree parts, different class numbers).
(iii) Koszul-equivariant only at special points: the Koszul involution $c \mapsto 26 - c$ preserves the field iff $(5c + 22)/(152 - 5c) \in \mathbb{Q}^{\times 2}$, which holds for $c = 1, 13$ and fails generically.

The family of quadratic fields $\{K(c_{p,q})\}$ attached to the minimal model tower is a new arithmetic invariant of 2d CFT. Its structure is controlled by the prime factorisation of $5c_{p,q} + 22$.

Conjecture 34.8 *(Class number growth along the minimal model tower).* The class number $h(d(c_{p,q}))$ grows without bound as $p + q \to \infty$ along the minimal model tower. Equivalently: the squarefree parts $d(c_{p,q})$ produce quadratic fields of increasing arithmetic complexity.

This is a number-theoretic conjecture about the factorisation of the linear forms $5c_{p,q} + 22$, analogous to the classical conjecture on class numbers of imaginary quadratic fields $\mathbb{Q}(\sqrt{-p})$ for primes p. The shadow obstruction tower reduces the question to the behaviour of the denominator polynomial $5c + 22$ under prime factorisation as c ranges over the minimal model spectrum.

35 The descent chain

The preceding sections (§5–§34) construct, one level at a time, a chain of identifications between structures in operadic algebra and structures in analytic number theory. This section subjects the chain to the Beilinson test: which links are theorems, which are structural analogies, and which are false.

35.1 The five levels

Level	Object	Involution	Status
0	Operadic Koszul duality	$\mathcal{A} \mapsto \mathcal{A}^!$	Theorem (A)
1	Bar construction as Fourier kernel	$\bar{B}_X(\mathcal{A}) \xrightarrow{\mathbb{D}_{\mathrm{Ran}}} \bar{B}_X(\mathcal{A}^!)$	Theorem (A)
2	Shadow obstruction tower as quadratic extension	$\sigma\colon \sqrt{Q_L} \mapsto -\sqrt{Q_L}$	Proved (§33)
3	Dirichlet-sewing lift $S_{\mathcal{A}}(u)$	Koszul $\mathcal{A} \mapsto \mathcal{A}^!$ on weights	Partially defined (§21)
4	Zeros of ε^c_s on critical lines	$s \mapsto c/2 + 1 - s$	Proved for lattice VOAs

The claim under examination: these five levels are connected by a single descent, and the involution at each level is the shadow of a single Fourier duality.

35.2 Level 0 to Level 1: the bar construction as Fourier transform

Proposition 35.1 *(Bar = categorical Fourier).* The bar construction $\bar{B}_X(\mathcal{A})$ satisfies:

(a) $\mathbb{D}_{\mathrm{Ran}}(\bar{B}_X(\mathcal{A})) \simeq \bar{B}_X(\mathcal{A}^!)$ (Theorem A: convolution to pointwise).
(b) $\Omega_X(\bar{B}_X(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (Theorem B: inversion on the Koszul locus).
(c) $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) \cong H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ (Theorem C: Plancherel).

These three properties characterise the classical Fourier transform: intertwining convolution with multiplication, self-inversion, and Parseval's identity.

This is a theorem, not an analogy. The bar construction is literally a Fourier transform in the sense of a monoidal involution on factorisation (co)algebras on $\mathrm{Ran}(X)$, with Verdier duality playing the role of complex conjugation.

Warning 35.2 *(Scope of the identification).* The identification "bar = Fourier" is precise at the categorical level: $\bar{B}_X$ is a functor between ∞-categories of factorisation algebras and coalgebras on $\mathrm{Ran}(X)$, with properties (a)–(c) paralleling $\mathcal{F}\colon L^2 \to L^2$. It is *not* a Fourier transform in the L^2 sense: there is no Hilbert space, no Plancherel measure, and no pointwise evaluation. The kernel $\eta_{ij} = d\log(z_i - z_j)$ is a section of $\omega_{\overline{C}_n(X)}(\log D)$, not a character of an abelian group.

The one case where the identification is literally classical: Heisenberg on an elliptic curve, where $\bar{B}_X(\mathcal{H}_k)$ is the Fourier–Mukai kernel on $\mathrm{Jac}(E) \times \mathrm{Jac}(E)^\vee$. For non-abelian algebras, the bar construction is the non-abelian Fourier transform, as the Grothendieck simultaneous resolution is the non-abelian Springer resolution.

Verdict on Level 0→1: PROVED. The bar construction satisfies the three defining properties of a Fourier transform (intertwining, inversion, Plancherel) as theorems of the monograph (Theorems A, B, C). The kernel $d\log(z_i - z_j)$ is the chiral analogue of $e^{-2\pi i x\xi}$.

35.3 Level 1 to Level 2: from bar curvature to quadratic extension

The bar curvature $\kappa(\mathcal{A}) = \mathrm{Res}_{s=1} F_1(\mathcal{A})$ seeds the shadow metric $Q_L(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 2\Delta)\, t^2$. The shadow obstruction tower $\{S_r\}_{r\geq 2}$ consists of the Taylor coefficients of $\sqrt{Q_L(t)}$. This defines a quadratic extension $K_L = \mathbb{Q}(c)(t)(\sqrt{Q_L})$ of the function field $F = \mathbb{Q}(c)(t)$, with:

- Galois group $\mathbb{Z}/2$, generated by $\sigma : \sqrt{Q_L} \mapsto -\sqrt{Q_L}$,
- discriminant $-32\kappa^2\Delta$,
- splitting table = G/L/C/M classification.

Proposition 35.3 *(The Galois involution is the Koszul sign).* The monodromy of the shadow connection $\nabla^{\mathrm{sh}} = d - Q'_L/(2Q_L)\,dt$ around each branch point of $\sqrt{Q_L}$ is -1. This is the Koszul sign: the desuspension s^{-1} in the bar construction contributes $(-1)^n$ at bar degree n. The Galois involution of K_L/F is the action of s^{-1} on the shadow generating function.

This is proved (the monodromy computation is direct). But the identification of the Galois involution with the Koszul sign is a *consequence* of the algebraic structure, not a descent of the Fourier involution. More precisely:

Warning 35.4 *(Two involutions, not one).* The bar-cobar Verdier duality (Level 1) sends $\mathcal{A}$ to $\mathcal{A}^!$ by acting on $\mathrm{Ran}(X)$. The Galois involution of the shadow extension (Level 2) sends $\sqrt{Q_L}$ to $-\sqrt{Q_L}$ by acting on the arity variable t. These are different involutions acting on different spaces. The Koszul involution $\mathcal{A} \mapsto \mathcal{A}^!$ acts on the *coefficients* of Q_L (sending $\kappa \to \kappa'$, $\alpha \to \alpha'$, $S_4 \to S'_4$); it does *not* negate $\sqrt{Q_L}$.

The complementarity identity $\Delta(\mathcal{A}) + \Delta(\mathcal{A}^!) = 6960/[(5c+22)(152-5c)]$ relates the discriminants of the two extensions K_L and $K_{L^!}$. It is a constraint between K_L and $K_{L^!}$, not a symmetry of K_L alone. At $c = 13$: $K_L = K_{L^!}$, and the Koszul involution preserves the extension; at generic c: $K_L \neq K_{L^!}$ (Remark 34.5).

To summarise: the Galois involution (σ) is intrinsic to a single algebra $\mathcal{A}$ and its shadow obstruction tower. The Koszul involution ($\mathcal{A} \mapsto \mathcal{A}^!$) relates two algebras and their two shadow obstruction towers. They coincide only at the self-dual point.

Verdict on Level 1→2: PROVED that the shadow tower is a quadratic extension with Galois involution = Koszul sign. FALSE that this Galois involution is the "descent" of the Verdier duality from Level 1. The Galois involution and the Koszul involution are two distinct $\mathbb{Z}/2$-actions that commute but do not coincide.

35.4 Level 2 to Level 3: the Dirichlet-sewing lift

The Dirichlet-sewing lift $S_{\mathcal{A}}(u) = \zeta(u+1)\sum_i(\zeta(u) - H_{w_i-1}(u))$ is defined from the bosonic weight multiset $W(\mathcal{A}) = \{w_i\}$. For Heisenberg ($W = \{1\}$): $S_{\mathcal{H}}(u) = \zeta(u)\zeta(u+1)$, with full Euler product. For Virasoro ($W = \{2\}$): $S_{\mathrm{Vir}}(u) = \zeta(u+1)(\zeta(u) - 1)$, with broken Euler product.

Warning 35.5 *(The Dirichlet-sewing lift is not derived from the shadow obstruction tower).* The function $S_{\mathcal{A}}(u)$ is defined from the weight multiset $W(\mathcal{A})$, which is genus-0 data (the conformal weights of the generating fields). The shadow obstruction tower $\{S_r\}$ is genus-1 and higher data (the MC element $\Theta_{\mathcal{A}}$, projected to arities $r \geq 2$).

These are logically independent. On recurrence lanes, the shadow obstruction tower is determined by its full initial and recursive OPE data; the Dirichlet-sewing lift is determined by $W(\mathcal{A})$. For the four-class landscape:
- Heisenberg: $\kappa(\mathcal{H}_1) = 1$, $\alpha = 0$, $S_4 = 0$, $W = \{1\}$, $S_{\mathcal{H}}(u) = \zeta(u)\zeta(u+1)$.
- Virasoro: $\kappa = c/2$, $\alpha = 2$, $S_4 = 10/[c(5c+22)]$, $W = \{2\}$, $S_{\mathrm{Vir}}(u) = \zeta(u+1)(\zeta(u) - 1)$.

The Dirichlet-sewing lift sees only the weight multiset; the shadow obstruction tower sees the OPE structure constants. They are related only insofar as both derive from the same chiral algebra, but neither determines the other.

In particular: two algebras with the same weight multiset but different OPE coefficients have the same $S_{\mathcal{A}}(u)$ but different shadow obstruction towers. Two algebras with the same recurrence data but different weight multiplicities can have the same shadow obstruction tower but different $S_{\mathcal{A}}(u)$.

Verdict on Level 2→3: the link is BROKEN. The Dirichlet-sewing lift and the shadow obstruction tower are independent invariants of the chiral algebra. Neither descends from the other. The Koszul involution acts on $S_{\mathcal{A}}(u)$ by replacing $W(\mathcal{A})$ with $W(\mathcal{A}^!)$: this is a well-defined operation, but it is not the "descent" of the Galois involution of the shadow extension, which acts on $\sqrt{Q_L}$ and has no knowledge of $W(\mathcal{A})$.

35.5 Level 3 to Level 4: zeros and functional equations

For lattice vertex algebras, the constrained Epstein zeta ε^c_s factors into L-functions via the Hecke decomposition of the theta function (Theorem **??**). The functional equation of ε^c_s follows from the modular invariance of Z, with the symmetry $s \mapsto c/2 + 1 - s$ determined by the weight.

Warning 35.6 *(Structural obstruction to a "shadow Riemann hypothesis").* The scattering matrix $\varphi(s) = \Lambda(1-s)/\Lambda(s)$ has poles at complex spectral parameters $s = \rho/2$, where ρ ranges over the nontrivial zeros of ζ. The MC equation and the shadow obstruction tower constrain the spectral coefficients on the *real* spectral axis $s = 1/2 + it$, $t \in \mathbb{R}$. The zeta zeros live at *complex t* (Remark **??** in `arithmetic_shadows.tex`).

This separation is structural. The constructions in this tree currently connect the MC equation to real-axis spectral coefficients and, in lattice cases, to Hecke L-functions. They do not produce a functor from the shadow obstruction tower to the nontrivial zero set of ζ. The "shadow Riemann hypothesis" (all zeros of ε^c_s on critical lines) is a consequence of GRH for the constituent L-functions when such a factorisation is present, not a consequence of the MC equation.

The Li coefficient test (`compute/lib/dirichlet_sewing.py`): the modified Li coefficients $\tilde{\lambda}_n(\mathcal{H})$ for the Heisenberg sewing lift $S_{\mathcal{H}}(u) = \zeta(u)\zeta(u+1)$ are positive for $n = 1, \ldots, 6$ and **negative at** $n = 7$. The shifted product and the chosen regularisation do not give the standard Li criterion for ζ itself. The sign change therefore does not bear on RH; it shows only that this sewing-lift regularisation lacks the positivity properties needed for a "shadow RH."

Verdict on Level 3→4: PROVED on the Hecke arithmetic lanes where the relevant theta or partition function admits a Hecke eigen-decomposition. For the connected Dirichlet-sewing lift, Euler products survive only on the all-weight-1 lane; higher conformal weights introduce harmonic subtraction terms and break multiplicativity. No mechanism in the MC equation forces zeros onto critical lines.

35.6 The descent chain: what is proved and what fails

Link	Status	Obstruction
$0 \to 1$	**Proved**	None. Bar = categorical Fourier.
$1 \to 2$	**Proved***	Two involutions, not one. *The shadow obstruction tower is a quadratic extension; the Galois involution is the Koszul sign. But the Galois involution is *not* the descent of Verdier duality.
$2 \to 3$	**Broken**	The Dirichlet-sewing lift and the shadow obstruction tower are independent invariants. Neither descends from the other.
$3 \to 4$	**Weight-1 / Hecke lanes only**	Higher-weight sewing defects break Euler products. Structural obstruction: OPE data $\not\Rightarrow$ zeta zeros.

35.7 The four Dirichlet series: a disambiguation that matters

The preceding analysis of the descent chain reveals that the arithmetic programme involves not one but *four* distinct Dirichlet series, each measuring a different invariant of the chiral algebra. The failure to distinguish them has been the source of every false claim about "shadow L-functions" and "shadow Riemann hypotheses" in the development of this chapter.

	Object	Input data	Axis	Arithmetic depth
(a)	Shadow-metric Epstein $\varepsilon_{Q_L}(s)$	(κ, α, Δ)	Arity (genus 0)	Quadratic field
(b)	Constrained Epstein ε_s^c	Primary spectrum $\{(\Delta_i, d_i)\}$	Genus (genus 1)	L-functions (lattice)
(c)	Shadow Dirichlet $\zeta_{\mathcal{A}}(s)$	Shadow tower $\{S_r\}$	Arity (genus 0)	Entire, no Euler product
(d)	Categorical zeta $\zeta_{\mathfrak{g}}^{\mathrm{DK}}(s)$	$\{\dim V_\lambda\}$	Lie-theoretic	$\zeta(s)$ for $\mathfrak{sl}_2$

Plus the Dirichlet-sewing lift $S_{\mathcal{A}}(u) = \zeta(u{+}1)\sum_i(\zeta(u) - H_{w_i-1}(u))$, built from the weight multiset $W(\mathcal{A})$–-a fifth object independent of both (a)–(c) and (d).

The resolution of the Benjamin–Chang tension. Benjamin–Chang [?BenjaminChang22] proved that the nontrivial Riemann zeta zeros ρ_n appear as poles at $s = (1 + \rho_n)/2$ in the spectral decomposition of the constrained Epstein (b) on $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$, via the scattering factor $\zeta(2s)/\zeta(2s{-}1)$ in the Eisenstein series E_s. This is a theorem about the spectral theory of the *moduli space* $\mathcal{M}_{1,1}$: the Riemann zeta zeros enter because the Eisenstein series' functional equation is governed by the Euler product of $\zeta(s)$.

The "negative results" (shadow zeta $\notin$ Selberg class, no Euler product, c-duality $\neq$ s-reflection) concern object (c), the shadow Dirichlet series. There is no tension: (b) and (c) are different Dirichlet series measuring different invariants. The shadow zeta (c) lives on the arity axis at genus 0; the constrained Epstein (b) lives on the genus axis at genus 1. The Riemann zeta enters through the spectral theory of $\mathcal{M}_{1,1}$ (the genus axis) and through the Dirichlet-sewing lift (the weight multiset), not through the shadow obstruction tower (the arity axis).

In the bigraded shadow algebra $\mathcal{A}^{\mathrm{sh}}_{r,g}$: the shadow obstruction tower is the $(r, 0)$ slice, the constrained Epstein is extracted from the $(0, 1)$ slice, and the genus tower $F_g = \kappa\lambda_g^{\mathrm{FP}}$ is the $(0, g)$ slice. The descent chain breaks at Level $2 \to 3$ because it attempts to connect orthogonal axes of the bigraded structure. The correct picture is the *fan* (§?? below): three independent projections from a single MC element, with the Riemann zeta entering at the genus projection (via $\mathcal{M}_{1,1}$) but absent from the arity projection (the shadow obstruction tower).

35.8 The positivity frontier: from genus 1 to genus 2

The integrability–positivity dichotomy is the root obstruction to connecting the bar-cobar programme to the Riemann Hypothesis. The MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ is *integrability* (flatness); every spectral conclusion (Ramanujan bounds, L-value non-vanishing, zero locations) requires *positivity* (sign-definiteness). These are logically independent.

At genus 1, three independent obstructions are proved: (i) unfolding erasure (the Rankin–Selberg integral is holomorphic at all scattering poles), (ii) real–complex separation (MC constrains the real spectral axis; zeta zeros live at complex parameters), (iii) dimensional underdetermination (three numbers κ, α, S_4 cannot determine infinitely many zeros). All three are **genus-1 theorems** (Theorem **??**).

At genus ≥ 2, the structural obstruction is bypassed: the Böcherer bridge (Theorem **??**) provides algebraic access to central L-values $L(\frac{1}{2}, \pi_f \times \chi_D)$, and the off-diagonal sewing blocks R_{12}, R_{21} break the Fock-diagonality that enables genus-1 erasure.

The surviving research programme: the genus-2 bar Laplacian and $\mathrm{Sp}(4)$ **Arthur packets.**

The genus-2 bar Laplacian $\Delta_B^{(2)} = d_B^{(2)}(d_B^{(2)})^* + (d_B^{(2)})^* d_B^{(2)}$ inherits a 2×2 block structure from the genus-2 sewing kernel: diagonal = two genus-1 Laplacians, off-diagonal = inter-handle coupling through the Schur complement $S = K_{q_2} + R_{21}(1 - K_{q_1})^{-1} R_{12}$. The programme proceeds in four steps.

Step 1: The $\mathrm{Sp}(4)$ *Arthur packet discrimination.* Automorphic representations of $\mathrm{Sp}(4)$ have a richer structure than $\mathrm{GL}(2)$ via Arthur's classification:

- *Stable (tempered) packets*: genuine Siegel modular forms, satisfying the Ramanujan conjecture (Weissauer, for holomorphic cusp forms of sufficiently high weight).
- *Endoscopic (non-tempered) packets*: Saito–Kurokawa lifts from $\mathrm{GL}(2)$, which do *not* satisfy the full Ramanujan bound for $\mathrm{Sp}(4)$.

The key observation: the off-diagonal sewing coupling R_{12}, R_{21} detects whether a Siegel modular form is a Saito–Kurokawa lift. An SK lift factors through the genus-1 sewing (it lifts from $\mathrm{GL}(2)$), so its off-diagonal coupling is "reducible." A genuine Siegel eigenform has "irreducible" off-diagonal coupling. The spectral gap of $\Delta_B^{(2)}$ could force the representations to lie in the stable (tempered) class, which *is* the Ramanujan condition.

Step 2: The Böcherer bridge converts to L-values. The proved Böcherer factorisation (Theorem **??**, via Furusawa–Morimoto/DPSS) gives

$$\frac{|B_f(D)|^2}{\langle \mathrm{SK}(f), \mathrm{SK}(f) \rangle} = \alpha(r, k)\, L(\tfrac{1}{2}, \pi_f)\, L(\tfrac{1}{2}, \pi_f \times \chi_D).$$

The genus-2 MC data determines the Fourier coefficients $a_\Lambda(T)$ of the genus-2 theta series. These feed into the Böcherer coefficient $B_f(D)$. If the spectral gap forces the relevant amplitudes to be nonzero (through an anti-cancellation mechanism), then $L(\frac{1}{2}, \pi_f \times \chi_D) \neq 0$.

Step 3: Three sub-gaps (honest assessment).

Gap I: *Spectral gap* $\rightarrow$ *amplitude lower bound.* A spectral gap provides exponential decay of off-diagonal matrix elements, not a lower bound. The gap controls coherence length–-it prevents arbitrary cancellation among Fourier modes–-but converting this to a pointwise lower bound on $B_f(D) = \sum_{[T]} a_{\mathrm{cusp}}(T)/\varepsilon(T)$ requires showing the Schur complement structure prevents destructive interference.

Gap II: *Individual vs. aggregate positivity.* The existing bracket positivity (Proposition ??) gives positivity of a *weighted sum* of L-values (the genus-2 Beurling kernel), not individual L-values. The spectral gap would need to interact with the Petersson inner product to isolate individual eigenform contributions.

Gap III: *Planted-forest anti-cancellation.* The planted-forest corrections from FM_3 (the cubic obstruction) provide the cusp-projection channel. The Schur complement (positive operator + positive correction) ensures the total determinant is bounded below. But converting this to non-vanishing of *individual*Böcherer coefficients is the precise open problem.

Step 4: Why genus 2 might work where genus 1 provably fails.

(1) Genus-1 structural separation is a *theorem*: the MC equation is algebraically exhausted, the RS unfolding erases scattering data, the arithmetic content is pluriharmonic.
(2) Genus-2 non-diagonality is *proved*: the off-diagonal blocks R_{12}, R_{21} break the Fock-diagonal structure.
(3) The Böcherer channel bypasses RS unfolding *entirely*: it extracts L-values from Fourier coefficients, not from Eisenstein spectral decomposition.
(4) $\mathrm{Sp}(4)$ Arthur packets provide the discrimination mechanism that $\mathrm{GL}(2)$ lacks: the distinction between tempered and non-tempered is precisely what the off-diagonal coupling measures.
(5) Weissauer's theorem already proves Ramanujan for genus-2 holomorphic cusp forms using exactly the mechanism of detecting SK lifts via their factorisation structure–-which is what the off-diagonal sewing coupling computes.

The relationship to RH. The genus escalation principle extends to all genera as an automorphic programme: at genus g, conjectural Böcherer-type formulas access $L(\frac{1}{2}, \mathrm{Sym}^{g-1} f)$. Such formulae can control automorphic local parameters and central L-values. They do not imply the Riemann Hypothesis for ζ; zeta-zero locations enter through the Eisenstein scattering determinant and require separate RH/GRH input. The gaps at higher genus: Böcherer-type formulas are conjectural at $g \geq 3$, and Langlands functoriality $\mathrm{GL}(2) \to \mathrm{GL}(g)$ is open for $g \geq 6$ (Newton–Thorne resolves it for holomorphic eigenforms of regular algebraic weight, but not for Maass forms).

35.9 The algebra–geometry–arithmetic trichotomy

The partition function $Z(\tau)$ is the meeting point of three mathematical worlds. The bar complex $\overline{B}(\mathcal{A})$ encodes the *algebra*: the OPE data that determines Z as a q-series. The hyperbolic Laplacian Δ_H on $\mathcal{M}_{1,1} = \mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$ encodes the *geometry*: the spectral theory of the modular surface whose continuous spectrum is governed by the scattering matrix $\varphi(s) = \Lambda(1-s)/\Lambda(s)$, with poles at $s = \rho/2$ (the zeta zeros). The Riemann zeta function encodes the *arithmetic*: the distribution of primes, which controls the lattice $\mathrm{SL}(2,\mathbb{Z})$ acting on $\mathfrak{H}$.

The bar complex does not encode the zeta zeros. It encodes the algebra $\mathcal{A}$, which produces $Z(\tau)$. The zeros are a property of how $Z(\tau)$ *sits on the modular surface*–-they come from the geometry of $\mathcal{M}_{1,1}$, not from the algebra of the chiral product. The chain is:

$$\overline{B}(\mathcal{A}) \;\to\; Z(\tau) \text{ as } q\text{-series} \;\to\; Z(\tau) \text{ on } \mathcal{M}_{1,1} \xrightarrow{\Delta_H} \text{spectral decomposition } \ni \zeta \text{ zeros.}$$

The bar complex controls the first arrow; the zeta zeros enter at the last.

What the CFT "knows." The zeta zeros correspond to the *equidistribution rate* of $Z(\tau)$ on the modular surface. The Eisenstein spectrum controls how fast a function on $\mathcal{M}_{1,1}$ approaches its average. For $Z(\tau)$ with $d(n) \geq 0$ (unitarity), the equidistribution rate is controlled by the spectral gap λ_1 of Δ_H–-and this gap *is* the Selberg eigenvalue conjecture, which *is* the Ramanujan conjecture for GL(2) Maass forms.

The CFT does not "know" where the zeros are through its algebraic structure. It "knows" through its *equidistribution rate on the moduli space*–-a property requiring the full analytic structure of $Z(\tau)$ as a function on a hyperbolic surface, not just its q-expansion coefficients. The bar complex gives you the q-series; what you need is the hyperbolic geometry of the function that q-series defines.

See Volume II, §**??**, for the full development.

35.10 The Koszul fixed point and the zeta fixed point

Proposition 35.7 *(The two fixed points are unrelated).* The Koszul fixed point $c = 13$ and the critical line $\mathrm{Re}(s) = 1/2$ arise from different structures:

(i) $c = 13$ is the fixed point of $c \mapsto 26 - c$ (the Koszul involution on the Virasoro family). It is determined by $c + c' = 26$ (Freudenthal–de Vries), which depends on the Virasoro central charge formula $c = 1 - 6(p-q)^2/(pq)$ and the Sugawara construction. This is CFT data.

(ii) $\mathrm{Re}(s) = 1/2$ is the fixed line of $s \mapsto 1 - s$ (the functional equation of ζ). It is determined by the Euler product and the gamma factors. This is number-theoretic data.

The coincidence $c/2 = 13/2$ and $s = 1/2$ share the symbol "1/2" but arise from unrelated arithmetic. The critical line of ζ is a consequence of the functional equation $\xi(s) = \xi(1-s)$; the self-duality of Vir_{13} is a consequence of $26 - 13 = 13$. No mechanism connects them.

The temptation to identify $c = 13$ with $s = 1/2$ is a special case of the meta-principle (AP9): the same name ("fixed point of an involution") applied to different objects does not make them the same.

35.11 What would close the chain?

To close the descent chain, one would need a theorem of the following form:

Conjecture 35.8 *(Closed descent, if true).* There exists a natural transformation $\Phi\colon \mathrm{ShTow}(\mathcal{A}) \to S_{\mathcal{A}}$ from the shadow obstruction tower functor to the Dirichlet-sewing lift, such that:

(i) Φ intertwines the Galois involution σ of K_L with the operation $\mathcal{A} \mapsto \mathcal{A}^!$ on $S_{\mathcal{A}}$.

(ii) The functional equation of $S_{\mathcal{A}}(u)$ descends from the complementarity identity $\Delta(\mathcal{A}) + \Delta(\mathcal{A}^!) = 6960/[(5c+22)(152-5c)]$.

(iii) The Euler product of $S_{\mathcal{A}}(u)$ (when it exists) factors through the shadow spectral decomposition.

No such natural transformation is known. The obstruction is at step (i): the shadow obstruction tower lives over the function field $\mathbb{Q}(c)(t)$ (a quadratic extension in the arity variable t), while the Dirichlet-sewing lift lives over the complex u-plane (the spectral variable). The two variables parametrise different structures: t is the arity, u is the Mellin conjugate of the eigenvalue q^n of the sewing operator. To connect them would require identifying the arity expansion of $\sqrt{Q_L}$ with the Mellin transform of the sewing determinant, and no such identification exists.

35.12 Is the claim "they are the same involution" correct?

No.

The descent chain contains three distinct involutions:

(i) **Verdier duality** $\mathbb{D}_{\mathrm{Ran}}$ on $\mathrm{Ran}(X)$: exchanges $\bar{B}(\mathcal{A})$ and $\bar{B}(\mathcal{A}^!)$. Acts on factorisation (co)algebras. This is the Fourier involution.

(ii) **Galois involution** σ of K_L/F: sends $\sqrt{Q_L} \mapsto -\sqrt{Q_L}$. Acts on the shadow extension. This is the monodromy of ∇^{sh}, intrinsic to a single algebra.

(iii) **Koszul involution** $\mathcal{A} \mapsto \mathcal{A}^!$: sends $c \mapsto 26 - c$ (for Virasoro), $k \mapsto -k - 2h^\vee$ (for Kac–Moody). Acts on the space of algebras.

These three involutions are all of order 2. They are all related to the bar construction. But they act on different objects, in different categories, with different fixed points:

Involution	Acts on	Fixed point
Verdier	$\mathrm{FactCoalg}(\mathrm{Ran}(X))$	self-dual coalgebras
Galois	K_L	F (the base field)
Koszul	families of chiral algebras	self-dual algebras ($c = 13$, etc.)

The claim "they are the SAME involution (Fourier) acting at different levels" conflates these three. The correct statement:

Observation 35.9 *(The three involutions).* (i) The Verdier duality is a theorem (Theorem A).

(ii) The Galois involution of the shadow extension is a consequence of $\sqrt{Q_L}$ being algebraic of degree 2 (the MC equation forces a quadratic, not a higher-degree, algebraic function).

(iii) The Koszul involution is a structure on the moduli of chiral algebras, not on the bar construction itself.

(iv) Items (i) and (iii) are related: Verdier duality intertwines $\bar{B}(\mathcal{A})$ with $\bar{B}(\mathcal{A}^!)$, so the Koszul involution is the "shadow on moduli" of Verdier duality.

(v) Items (ii) and (iii) are related: the complementarity identity constrains $\Delta(\mathcal{A})$ and $\Delta(\mathcal{A}^!)$.

(vi) Items (i) and (ii) are *not directly related*: the Galois involution is intrinsic to a single algebra and its shadow obstruction tower, while Verdier duality requires a pair $(\mathcal{A}, \mathcal{A}^!)$.

The three involutions form a commutative diagram at the self-dual point, but they are three distinct morphisms in three distinct categories.

35.13 The two-critical-line phenomenon

The shadow–spectral correspondence (Theorem **??**) establishes that for lattice VOAs, the number of critical lines of ε_s^c equals the shadow depth minus 1. The two-critical-line phenomenon (E_8: lines at $\mathrm{Re}(s) = 1/2$ and $\mathrm{Re}(s) = 7/2$) is a feature, not an obstruction. It reflects the Eisenstein-only Hecke decomposition $\Theta_{E_8} = E_4$: no cusp form, so no third critical line.

The passage from one critical line (rank ≤ 2, Eisenstein only) to multiple critical lines (rank ≥ 8, Eisenstein plus cuspidal) is the arithmetic face of the transition from class G/L to class M: the shadow obstruction tower acquires arithmetic depth when the theta function has cuspidal components.

Whether the two-critical-line structure of E_8 is an "obstruction" depends on what one is trying to prove. For the shadow–spectral correspondence, it is confirming evidence. For a hypothetical "shadow RH" that would force all zeros onto a single line, it is a definitive refutation: E_8 has zeros on two distinct lines, so no single-critical-line hypothesis can hold for all modular Koszul algebras.

35.14 Summary

The descent chain is a sequence of independently interesting structures that share a common origin in the bar construction. The first two links (bar = Fourier, shadow obstruction tower = quadratic extension) are theorems. The claim that "they are the same involution at different levels" is false: there are three distinct involutions. The link from the shadow obstruction tower to the Dirichlet-sewing lift is broken: these are independent invariants. The link from the Dirichlet-sewing lift to the zeros of L-functions is proved for lattice algebras and structurally obstructed for non-lattice algebras.

What the bar construction does provide: a single algebraic object ($\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$) whose finite-order projections simultaneously produce the shadow obstruction tower (an arithmetic invariant), the genus tower (a topological invariant), and the Koszul dual (an algebraic invariant). These three outputs are facets of $\Theta_{\mathcal{A}}$, not levels of a single descent. The correct picture is a *fan*, not a *chain*: three projections from one object, not five levels of one reduction.

$$\begin{tikzcd} & \Theta_{\mathcal{A}} \arrow[dl] \arrow[d] \arrow[dr] & \\ \text{Shadow obstruction tower} & \text{Genus tower} & \text{Koszul dual} \\ \text{(arithmetic)} & \text{(topological)} & \text{(algebraic)} \end{tikzcd}$$

36 What the machine computes and what it does not

The preceding thirty-five sections build a single machine. The input is a chiral algebra $\mathcal{A}$: its OPE coefficients, its conformal weights, its central charge. The output is a cascade of invariants: five theorems, a shadow obstruction tower, a depth classification, a genus expansion, a shadow metric, a quadratic extension, an Epstein zeta function.

This section says what the machine computes, what it does not, and where the boundary lies.

36.1 What the machine computes

Seven outputs. Each is scoped to its stated domain; primary-line and scalar-shadow formulas are not asserted for arbitrary multi-generator data.

(a) The primary-line shadow obstruction tower. For a fixed scalar primary line with OPE data (κ, α, S_4), the corresponding shadow tower $\{S_r\}_{r\geq 2}$ is the sequence of Taylor coefficients

$$S_r \;=\; \frac{1}{r}\,[t^{r-2}]\,\sqrt{Q_L(t)}, \qquad Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2, \quad \Delta = 8\kappa S_4.$$

Each S_r is a rational function of the structure constants. The generating function $H(t) = t^2\sqrt{Q_L(t)}$ is algebraic of degree 2. Proved: the Riccati-type identity $H^2 = t^4 Q_L$ is the MC equation projected to each arity (§19). Computed: 580 exact rational shadow coefficients across 15 algebras to arity 30 (§32), verified by three independent implementations (§27).

(b) The depth classification. Every chirally Koszul algebra falls into one of four classes, determined by the critical discriminant $\Delta = 8\kappa S_4$:

Class	Criterion	$r_{\max}$	Examples
G	$\alpha = \Delta = 0$	2	Heisenberg
L	$\alpha \neq 0, \Delta = 0$	3	affine KM
C	$\kappa = 0$ (stratum separation)	4	$\beta\gamma$
M	$\Delta \neq 0$	∞	Virasoro, $\mathcal{W}_N$

Proved: the splitting behaviour of Q_L forces $r_{\max} \in \{2, 3, 4, \infty\}$ with no other values (Proposition 33.2). Verified computationally for all standard families through arity 20.

(c) The shadow connection. The logarithmic connection $\nabla^{\mathrm{sh}} = d - Q_L'/(2Q_L)\,dt$ has simple poles at the zeros of Q_L, residue $1/2$ at each zero, and monodromy -1: the Koszul sign. The flat sections are $\sqrt{Q_L(t)/Q_L(0)}$. The Galois group of the shadow extension $K_L = F(\sqrt{Q_L})$ is $\mathbb{Z}/2$. Proved by direct computation (§19, §33). The complementarity identity $\Delta(\mathcal{A}) + \Delta(\mathcal{A}^!) = 6960/[(5c+22)(152-5c)]$ constrains the discriminant pair with a universal numerator $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$.

(d) The genus expansion. For uniform-weight modular Koszul algebras:

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}, \qquad \sum_{g \geq 1} \lambda_g^{\mathrm{FP}}\, \hbar^{2g} = \hat{A}(i\hbar) - 1.$$

The $\hat{A}$-genus generates the tower from a single scalar. At all genera. The bar propagator $d \log E(z, w)$ has weight 1 regardless of the conformal weight of the field (§12); all edge-level Hodge data is standard. Genus-1 universality $F_1 = \kappa/24$ is proved on the stated theorem lanes. The decisive test: $F_2(V_{E_8}) = 8 \cdot 7/5760 = 7/720$ matches the Siegel–Weil formula (§30).

(e) The bar-cobar machine. Five main theorems.

- (i) **Theorem A** (adjunction): $\mathbb{D}_{\mathrm{Ran}}(\bar{B}(\mathcal{A})) \simeq \bar{B}(\mathcal{A}^!)$. Bar intertwines convolution with pointwise.
- (ii) **Theorem B** (inversion): $\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ on the Koszul locus.
- (iii) **Theorem C** (complementarity): $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) \cong H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$.
- (iv) **Theorem D** (modular characteristic): $\mathrm{obs}_g = \kappa(\mathcal{A}) \cdot \lambda_g$ for uniform-weight algebras.
- (v) **Theorem H** (Hochschild): $\mathrm{ChirHoch}^*(\mathcal{A})$ polynomial, Koszul-functorial.

All proved. Together with the Koszulness programme (eight genuinely independent bidirectional equivalences plus one one-way chiral Hochschild consequence, one conditional Lagrangian criterion, one one-directional D-module purity), the MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$, and the all-genera sewing theorem.

(f) The Epstein zeta. For rational c, the binary shadow metric Q_L defines a shadow-metric Epstein zeta with the standard binary reflection $s \mapsto 1 - s$ (up to the dual form and determinant factors). This is distinct from the constrained lattice Epstein zeta attached to a rank-c lattice, whose completed functional equation reflects the ambient lattice dimension. The specialised shadow field $K(c) = \mathbb{Q}(\sqrt{d(c)})$ is a specific quadratic number field at each rational central charge. Proved (§34). Computed: the Ising model lives over $\mathbb{Q}(\sqrt{-10})$; the self-dual point $c = 13$ over $\mathbb{Q}(\sqrt{-435})$; the Koszul pair $c = 1$ and $c = 25$ over the same field $\mathbb{Q}(\sqrt{-15})$.

(g) The gravitational coproduct. For any principal DS reduction, the ghost-number obstruction kills the bar coproduct: $\varepsilon \circ \Delta^{\mathrm{grav}} = 0$ at all arities. The genus expansion of the reduced algebra inherits the DS cascade: shadow depth increases from class L to class M (§29). Verified for $N = 2, \ldots, 5$ to arity 20.

36.2 What the machine does not compute

Six things were attempted or hoped for. Each fails, and the failure is structural.

(h) Universal Langlands. The chain $\mathcal{A} \to S_{\mathcal{A}}(u) \to$ functional equation $\to$ automorphic representation does not close universally. Three obstructions:

(i) The Dirichlet-sewing lift $S_{\mathcal{A}}(u)$ is defined from the weight multiset $W(\mathcal{A}) = \{w_i\}$, which is genus-0 data (§21). The Koszul involution $c \mapsto 26 - c$ acts on the shadow extension, not on $S_{\mathcal{A}}(u)$: the lift forgets the central charge.

(ii) For Virasoro ($W = \{2\}$): $S_{\mathrm{Vir}}(u) = \zeta(u{+}1)(\zeta(u) - 1)$. The subtracted harmonic polynomial breaks the Euler product. At weight ≥ 2, no prime factorisation.

(iii) Route C (automorphic counting) requires finitely many spectral atoms. Class M has $r_{\max} = \infty$.

The shadow obstruction tower and the Dirichlet-sewing lift are logically independent: the tower is determined by the relevant OPE recurrence data, while the lift is determined by $W(\mathcal{A})$. Two algebras with the same weight multiset but different OPE coefficients have the same $S_{\mathcal{A}}(u)$ but different shadow obstruction towers (Warning 35.5). The link at Level 2 $\to$ 3 of the descent chain is broken (§35).

(i) Weight-defect obstruction to Euler products. The Epstein zeta function ε_s^c gives an L-function for the shadow quadratic field. The automorphic L-function (the object connected to the Ramanujan conjecture) requires additional structure: an Euler product, $\mathrm{GL}(n)$ twists, the CPS converse theorem. The connected Dirichlet-sewing lift has an Euler product only on the all-weight-1 lane; higher-weight generators introduce the harmonic subtraction $H_{w_i-1}(u)$ and break multiplicativity.

For lattice vertex algebras, the Hecke decomposition $\Theta_L = E_{d/2} + \sum_f c_f \cdot f$ factors ε_s^c into products of classical L-functions (§33). Ramanujan bounds then follow only for the appropriate automorphic constituents: Deligne for integral-weight holomorphic cusp eigenforms, Weissauer for genus-2 Siegel stable packets, and separate arguments for half-integral or vector-valued cases. The mechanism depends on having an actual Hecke decomposition; it is not available from the weight-defective sewing lift alone.

The MC equation constrains the spectral coefficients on the real axis $s = 1/2 + it$. The nontrivial zeros of ζ live at complex t. The constructions here do not give a map from OPE recurrence data to that zero set (Warning 35.6).

(j) A single descent. The three involutions (Verdier duality on $\mathrm{Ran}(X)$, Galois involution of K_L/F, Koszul involution $\mathcal{A} \mapsto \mathcal{A}^!$) are three distinct $\mathbb{Z}/2$-actions on three distinct objects (Observation 35.9). The claim "they are levels of a single Fourier duality" is false. The Verdier involution acts on factorisation coalgebras. The Galois involution acts on the shadow extension. The Koszul involution acts on the moduli of chiral algebras. They commute at the self-dual point $c = 13$ but do not coincide.

The correct picture is a fan, not a chain:

$$\begin{tikzcd} & \Theta_{\mathcal{A}} \arrow[dl] \arrow[d] \arrow[dr] & \\ \text{Shadow obstruction tower} & \text{Genus tower} & \text{Koszul dual} \end{tikzcd}$$

Three independent projections from one MC element. Not five levels of one reduction.

(k) Multi-generator universality at $g \geq 2$. The formula $\mathrm{obs}_g = \kappa \cdot \lambda_g$ is proved for uniform-weight algebras (all generators have the same conformal weight) at all genera. For multi-weight algebras at $g \geq 2$, the correct form is $F_g = \kappa\lambda_g + \delta F_g^{\mathrm{cross}}$. The $\mathcal{W}_3$ genus-2 computation gives a nonzero cross-channel correction, so scalar purity is not universal on the multi-weight lane; it fails for interacting multi-weight algebras such as $\mathcal{W}_3$.

At genus 1: $\mathrm{obs}_1 = \kappa \cdot \lambda_1$ is proved on the stated theorem lanes. At genus 2 and beyond, for algebras like $\mathcal{W}_3$ with generators of different weights, the edge-level Hodge bundle is $\mathcal{E}_1$ but vertex-level multi-generator data produces extra tautological contributions.

(l) The shadow Riemann hypothesis. No well-posed "shadow RH" exists.

The modified Li coefficients $\tilde{\lambda}_n(\mathcal{H})$ for the Heisenberg sewing lift $S_{\mathcal{H}}(u) = \zeta(u)\zeta(u+1)$ are positive for $n = 1, \dots, 6$ and negative at $n = 7$. This is a failure of the sewing-lift regularisation to supply Li positivity, not a statement about RH for ζ itself (Warning 35.6). The E_8 Epstein zeta has zeros on two critical lines ($\mathrm{Re}(s) = 1/2$ and $\mathrm{Re}(s) = 7/2$), so no single-critical-line hypothesis holds for all modular Koszul algebras (§35).

The interacting Gram positivity–-unconditional for $c > 0$–-is the correct substitute. It is a positivity statement about Dirichlet coefficients, not a zero-location statement.

(m) The smooth–arithmetic bridge. The shadow growth rate $\rho(\mathcal{A}) = \sqrt{9\alpha^2 + 2\Delta}/(2|\kappa|)$ is a smooth algebraic function of the central charge. The class number $h(d(c))$ of the specialised shadow field is a function of the squarefree part $d(c)$, which depends on the prime factorisation of $5c + 22$: a wildly discontinuous function of c. A smooth function cannot equal a step function. No relation between ρ and h exists (Observation 34.7).

At $c = 1/2$: $\rho \approx 12.53$, $h = 2$. At $c = 13$: $\rho \approx 0.467$, $h = 7$. The two quantities inhabit different worlds: ρ lives in the analytic world of branch points; h lives in the arithmetic world of ideal class groups. (The specialised shadow fields are *imaginary* quadratic, so their unit group is $\{\pm 1\}$; the analogue of the fundamental unit does not exist.)

36.3 What survives

The programme is not "Langlands for chiral algebras." The Dirichlet-sewing lift does not close into automorphic representations. The shadow RH does not exist. The descent chain is a fan, not a chain. Say this clearly, because the temptation to overstate is the most persistent enemy of the work.

The programme is a machine with five moving parts.

Part 1: the shadow obstruction tower machine. Input: OPE data (κ, α, S_4). Output: an infinite sequence of rational invariants S_r, the Taylor coefficients of an algebraic function of degree 2. This is exact, finite at each arity, and verified across 580 coefficients. The shadow metric Q_L determines everything: the tower, the depth, the growth rate, the Galois involution, the connection, the monodromy, the denominators. One quadratic form.

Part 2: the depth classifier. Input: the shadow metric Q_L. Output: an assignment of every chirally Koszul algebra to one of four classes (G/L/C/M), with structural consequences: finite vs. infinite tower, convergent vs. divergent series, Gaussian vs. mixed shadow, terminated vs. persistent obstruction. The classification is complete: no algebra escapes the four classes. Shadow depth does not characterise Koszulness (all four classes are Koszul); it classifies complexity within the Koszul world.

Part 3: the genus expansion machine. Input: the modular characteristic $\kappa(\mathcal{A})$. Output: exact genus-g free energies $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ for uniform-weight algebras, generated by $\hat{A}(i\hbar) - 1$. Verified against Siegel–Weil at genus 2 for E_8, against the Fredholm determinant for Heisenberg at all genera, against Bernoulli asymptotics $(2\pi)^{-2g}$ for the decay rate.

Part 4: the Epstein zeta machine. Input: Q_L at rational c. Output: an Epstein zeta function ε_s^c with functional equation, a specialised quadratic number field $K(c) = \mathbb{Q}(\sqrt{d(c)})$, and a splitting table of shadow primes. For lattice VOAs, the Hecke decomposition factors ε_s^c into classical L-functions, and unconditional Ramanujan follows by Deligne. For non-lattice algebras, the Epstein zeta exists but does not factor, and the L-function programme remains open.

Part 5: the bar-cobar machine. Input: a chiral algebra $\mathcal{A}$. Output: the theorem packages A, B, C, D, and H, the MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$, the Koszulness meta-theorem (eight genuinely independent bidirectional equivalences plus Barr–Beck–Lurie, chiral Hochschild, Lagrangian, and $\mathcal{D}$-module purity clauses), the completed modular Koszul datum $\Pi_X^{\mathrm{oc}}(\mathcal{A})$, the sewing envelope, the coderived shadows $Q_g(\mathcal{A})$. This is the algebraic engine. Everything else (the shadow obstruction tower, the genus expansion, the Epstein zeta, the depth classification) is a projection of $\Theta_{\mathcal{A}}$.

36.4 The boundary

The boundary between what is proved and what is open runs through three places.

(I) Multi-weight universality at $g \geq 2$. Uniform-weight: proved at all genera. Multi-weight: proved at genus 1, open at $g \geq 2$. The gap is a single tautological identity: does the bar construction produce λ_g at higher genus for multi-channel algebras? The cyclic direction is fixed by rigidity; the Mumford direction is not.

(II) Non-lattice L-functions. The Epstein zeta exists for every rational c. Its factorisation into automorphic L-functions requires the Hecke algebra of the theta function, which exists only for lattice VOAs. Extending the L-function programme to non-lattice algebras requires new mathematics at the intersection of graph-sum operads and quasi-modular Hecke algebras. The Dirichlet-sewing lift is the wrong bridge: it forgets the central charge and breaks the Euler product. The right bridge, if it exists, must pass through the shadow tower itself–-through Q_L, not through $W(\mathcal{A})$.

(III) The analytic frontier. The algebraic bar-cobar machine is proved. The analytic completion–-sewing envelopes, coderived categories, IndHilb factorisation homology–-is proved at the level of sufficient conditions (polynomial OPE growth + subexponential sector growth implies convergence) and for the Heisenberg algebra (Fredholm determinant, Bergman reduction). The general analytic bar-cobar duality (an analytic Koszul pair $\bar{B}^{\mathrm{an}}(\mathcal{A}) \dashv \bar{B}^{\mathrm{an}}(\mathcal{A}^!)$) is conjectural. Moriwaki's IndHilb framework works for conformally flat surfaces; extension to genus $g \geq 2$ requires a curved IndHilb theory that does not yet exist.

36.5 The computations that anchor the claims

The preceding subsections appeal to specific computations. Here they are, in one place.

(1) 580 shadow coefficients across 15 algebras to arity 30, verified by three independent implementations: the convolution recursion, the MC recursion, and the `ds_shadow_higher_arity` module (§16, §32).
(2) The E_8 decisive test: $F_2(V_{E_8}) = \kappa \cdot 7/5760 = 4 \cdot 7/5760 = 7/1440$, matching the Siegel–Weil genus-2 theta coefficient (§30).
(3) The denominator theorem: $\mathrm{denom}(S_r) = c^{r-3}(5c+22)^{\lfloor (r-2)/2 \rfloor}$ for all $r \geq 4$ (Theorem 16.1). Two primes, both from the weight-4 Gram determinant.
(4) The Li coefficient sign change: $\tilde{\lambda}_7(\mathcal{H}) < 0$ for the Heisenberg sewing lift, verified in four independent test suites.
(5) The DS-shadow intertwining: $\mathrm{Sh}_r(\mathcal{W}_N) = \mathrm{DS}_k(\mathrm{Sh}_r(\widehat{\mathfrak{sl}}_N))$ to arity 20 at levels $N = 2, \ldots, 5$.
(6) The $\mathcal{W}_3$ T-line independence: the W-generator does not backreact on the T-sector to arity 7 (§16).
(7) The $\mathcal{N} = 2$ Kazama–Suzuki kappa: $\kappa(\mathcal{N}{=}2) = (k+4)/4$, with $\kappa + \kappa' = 1$ (§31).
(8) The shadow growth rate at the self-dual point: $\rho(13) \approx 0.467 < 1$. The tower converges.
(9) The Koszul field-preservation test: $c = 1$ and $c = 25$ give $K = \mathbb{Q}(\sqrt{-15})$; the Ising model $c = 1/2$ gives $K = \mathbb{Q}(\sqrt{-10})$ (Proposition 34.2).

(10) The critical cubic $5c^3 + 22c^2 - 180c - 872 = 0$, determining $\alpha^2 = 4$ from two independent equations (§16).

These ten computations use different mathematics (algebraic, analytic, number-theoretic, combinatorial, representation-theoretic). Each is independently verifiable. They agree because they are projections of the same object.

*

This is what the 1-form computes. An algebraic function of degree 2, controlled by three OPE invariants, with a genus expansion generated by the $\hat{A}$-genus, a depth classification into four classes, a quadratic number field at each rational central charge, and an Epstein zeta function with a functional equation.

It does not compute automorphic L-functions for non-lattice algebras. It does not prove a shadow Riemann hypothesis. It does not close the chain from Koszul duality to analytic number theory. It does not determine λ_g at $g \geq 2$ for multi-weight algebras.

A smaller true theorem beats a larger false one. These five machines are the true theorem.

37 The single object

$\Theta_{\mathcal{A}}$.

Two characters. One object. Everything else is a projection.

The five theorems are five faces of $\Theta_{\mathcal{A}}$:

(A) $\Theta_{\mathcal{A}}$ exists: the bar-cobar adjunction produces the MC element, because $D^2_{\mathcal{A}} = 0$ on $\overline{\mathcal{M}}_{g,n}$.
(B) $\Theta_{\mathcal{A}}$ determines $\mathcal{A}$: bar-cobar inversion recovers the algebra from its categorical logarithm.
(C) $\Theta_{\mathcal{A}}$ splits complementarily: the genus-g obstruction decomposes into Lagrangian halves under the Verdier involution.
(D) $\Theta_{\mathcal{A}}$ has scalar shadow κ: on the uniform-weight lane, the leading arity-2 projection controls the scalar genus tower via the $\hat{A}$-genus. Multi-weight algebras also carry cross-channel terms.
(E) $\Theta_{\mathcal{A}}$ has polynomial coefficient ring: the chiral Hochschild ring is Koszul-functorial, concentrated in three degrees.

The shadow obstruction tower is a primary-line projection of $\Theta_{\mathcal{A}}$. The r-matrix is its ordered binary genus-zero projection. The scalar genus expansion is its uniform-weight genus projection, with $F_g = \kappa\lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$ in the multi-weight case. The chiral Hochschild complex is the tangent/coefficient complex on the stated generic/PBW locus. The depth classification is the square-class behaviour of the quadratic shadow metric on a primary line.

*

On a verified primary line: $\Theta_{\mathcal{A}} = t^2\sqrt{Q_L}$.

A square root of a quadratic. Three numbers determine that primary-line tower: κ (the leading OPE coefficient), α (the cubic shadow), S_4 (the quartic contact invariant). The quadratic $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2$ defines a Galois extension $K_L = F(\sqrt{Q_L})$ with monodromy -1: the Koszul sign. A separate quadratic-form specialisation can give an Epstein zeta function at rational c, but this is not automatic for every chiral algebra. The branch discriminant $\Delta = 8\kappa S_4$ participates in the depth classification; the growth rate $\rho = |t_+|^{-1}$ controls the ordinary series radius.

*

In three dimensions: $\Theta_{\mathcal{A}}$ is the closed projection of $\Theta^{\mathrm{oc}}_{\mathcal{A}}$. The open sector $\mathcal{C}_{\mathrm{op}}$ is primitive. The bulk is the derived center. Modularity is trace plus clutching. The gravitational coproduct is primitive because the graviton is composite and the ghost sector kills the splitting channel. The bar differential extracts residues. The bar coproduct separates words. The bar curvature bends the theory. These are one object seen from three directions.

*

The descent from $\Theta_{\mathcal{A}}$ to arithmetic is a fan, not a chain. Three projections share a source but do not compose:

Projection	**Output**	**Structure**
Categorical	$\mathcal{A} \mapsto \mathcal{A}^!$ (Koszul dual)	Verdier duality on $\mathrm{Ran}(X)$
Spectral	$\{S_r\} = [t^{r-2}]\sqrt{Q_L}$ (shadow obstruction tower)	quadratic extension K_L/F
Modular	$\{F_g\} = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (genus tower)	Mumford classes on $\overline{\mathcal{M}}_g$

The Koszul involution, the Galois involution, and any functional equation involution are three different symmetries. For Virasoro the Koszul self-dual point is $c = 13$; this does not identify the three involutions as one symmetry.

*

The machine converts a chirally Koszul algebra, on the relevant proved lane, into:
(i) a quadratic extension with Galois theory,
(ii) an Epstein zeta function with functional equation when a separate quadratic-form specialisation is present,
(iii) a scalar genus expansion controlled by the $\hat{A}$-genus,
(iv) a depth classification (G/L/C/M),
(v) a gravitational coproduct (primitive for DS reductions).

It does not convert arbitrary chiral algebras into automorphic L-functions. That requires lattice or rationality structure. The universal Langlands statement is false. The Riccati algebraicity, the ghost-number primitivity, and the Epstein zeta construction are true.

*

What	Where it lives	What determines it
The algebra $\mathcal{A}$	vertex algebras on X	the OPE
The MC element $\Theta_{\mathcal{A}}$	$\mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$	$D^2_{\mathcal{A}} = 0$
The primary-line shadow obstruction tower $\{S_r\}$	Taylor coefficients of $\sqrt{Q_L}$	three numbers (κ, α, S_4) on that line
The genus expansion $\{F_g\}$	Hodge classes on $\overline{\mathcal{M}}_g$	κ, the $\hat{A}$-genus, and cross terms when present
The r-matrix $r(z)$	collision residue on $\mathrm{Conf}_2(X)$	binary ordered projection of $\Theta_{\mathcal{A}}$
The quadratic field K_L	$\mathbb{Q}(c)(t)(\sqrt{Q_L})$	the discriminant $\Delta = 8\kappa S_4$
The L-function $L(s, \chi_d)$, when present	an external arithmetic specialisation	a proved quadratic-form or theta constructio
The gravitational coproduct	HPL transfer through DS	ghost-number obstruction on the proved DS/
The depth class	$\{G, L, C, M\}$	square class and termination pattern of Q_L

One ordered MC element, several scoped projections. In lattice and other genuinely arithmetic cases some projections compose with Hecke or theta-function structures. In the general chiral case the shadow tower, Dirichlet-sewing transforms, and representation-theoretic spectra are different functors and must not be identified without a theorem.

38 Attack-heal status ledger

The following entries record mathematically substantive status changes: proved statements, conditional statements, and claims that were destroyed or narrowed.

38.1 The gravitational coproduct is primitive

Let Δ_z^{Vir} denote the transferred coproduct on the Virasoro algebra obtained by Drinfeld–Sokolov reduction from the affine coproduct on $\widehat{\mathfrak{sl}}_2$.

Theorem 38.1. Δ_z^{Vir} is strictly primitive at all arities:

$$(p \otimes p) \circ \Delta_z^{\mathrm{Vir}}(T) \;=\; T \otimes 1 + 1 \otimes T,$$

where p is the projection to ghost number 0.

Status: proved on the Virasoro/principal-DS transfer lane. The physics no-hair interpretation below is heuristic.

The mechanism is ghost-number obstruction. The BRST charge $h = c + \phi b$ carries ghost degree -1. Every HPL tree correction to the coproduct at arity $r \geq 2$ passes through h-decorated internal edges, producing elements of ghost number $\neq 0$ on at least one tensor factor. Source trees (with root on the left) contribute ghost $+1$; target trees (with root on the right) contribute ghost -1 on one factor. In both cases, $p \otimes p$ kills the correction.

*

This is not an accident of low arity. The ghost-number filtration is an exact functor on the category of transferred operations, and DS reduction preserves the filtration strictly on the proved lane.

The consequence is structural. The transferred product m_z^{Vir} is infinitely complicated: the full A_∞ tower is nonzero at all arities (class M, shadow depth $r_{\max} = \infty$). The transferred coproduct is trivial. In the (m, Δ)-complexity plane, gravity sits at $(\infty, 0)$: infinite product complexity, zero coproduct complexity. Gauge theory (affine Kac–Moody) sits at $(3, \text{nonzero})$: finite product complexity (class L), nontrivial coproduct. The Sugawara embedding confirms this: the affine coproduct applied to the Sugawara stress tensor $T = \kappa^{ab}{:}J_a J_b{:}/(2(k + h^\vee))$ produces a nontrivial cross term

$$\Delta_z^{\mathrm{KM}}(T) - T \otimes 1 - 1 \otimes T \;=\; \frac{\Omega}{(k + h^\vee)\, z}, \qquad \Omega = \kappa^{ab}\, J_a \otimes J_b.$$

Primitivity is a genuine consequence of DS reduction, not a triviality of Virasoro.

*

The physical interpretation: gravity does not produce particles. It braids them, it curves them, it dresses their propagators with an infinite tower of higher products–-but the coproduct, which governs particle creation from the vacuum, is trivial. The bar coproduct on $\bar{B}(\mathcal{A})$ is a coalgebra structure on the factorisation coalgebra; DS reduction kills its transferred image on the quotient. This is the coalgebraic shadow of the no-hair theorem: the gravitational sector has no nontrivial coalgebra structure at any arity.

38.2 The shadow depth escalator

The standard landscape divides into four depth classes. The following pattern emerged from the 95-agent computation and was verified across all families.

Current-algebra systems (Chern–Simons, M2-brane holomorphic-topological theories) are class L: their shadow towers terminate at arity 3, and the complementarity sum vanishes: $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$.

W-algebra systems (gravity, $\mathcal{W}_3$ minimal models, M5-brane theories) are class M: their shadow obstruction towers are infinite, and $\kappa + \kappa' \neq 0$.

Drinfeld–Sokolov reduction is the functor that sends L to M. Three structural mechanisms operate simultaneously:

(i) The cubic shadow S_3 doubles: affine $\mathfrak{sl}_N$ has S_3^{KM}; the reduced $\mathcal{W}_N$ has $S_3^{\mathrm{W}} = 2S_3^{\mathrm{KM}}$ (universal for principal reduction at all N).

(ii) The quartic shadow S_4 is created from zero: the Jacobi identity kills the quartic invariant for affine algebras ($S_4^{\mathrm{KM}} = 0$, hence $\Delta = 0$, hence class L), while the BRST differential creates a nonzero quartic on the quotient ($S_4^{\mathrm{W}} \neq 0$, $\Delta \neq 0$, hence class M). DS does not merely modify the quartic; it creates it from nothing.

(iii) The shadow growth rate ρ decreases with N: the convergence radius of the shadow obstruction tower improves under reduction. At $N = 2$: $\rho(13) \approx 0.467$. At $N = 3$: ρ is smaller. The tower becomes better-behaved as the algebra becomes more complicated.

The shadow depth classification is not a property of the algebra alone. It is a property of the algebra relative to its pre-reduction ancestor. DS is the escalator that lifts the tower from finite to infinite.

38.3 The first multi-channel genus-2 amplitude

The $\mathcal{W}_3$ algebra at central charge c has two generators: T (weight 2) and W (weight 3). At genus 2, the scalar uniform-weight term is no longer the whole amplitude. The multi-weight cross-channel correction is computed from the mixed stable-graph channels.

PROPOSITION 38.2 *(computed W_3 genus-2 correction).*

$$F_2(\mathcal{W}_3) \;=\; \frac{7c}{6912} + \frac{c+204}{16c}.$$

Equivalently,

$$\delta F_2^{\text{cross}}(\mathcal{W}_3) = \frac{c+204}{16c}.$$

The cross-channel decomposition by graph topology is

$$F_2^{\text{banana}} = \frac{3}{c}, \qquad F_2^{\text{theta}} = \frac{9}{2c},$$
$$F_2^{\text{lollipop}} = \frac{1}{16}, \qquad F_2^{\text{mixed}} = \frac{21}{4c}.$$

The lollipop contribution is c-independent. This is an exact cancellation, verified by direct computation: the c-dependent factors from the propagator ($\sim 1/c$), the vertex ($\sim c$), and the self-loop normalisation ($\sim c/48$) combine to produce a pure rational number. Explicitly:

$$F_2^{\text{lollipop}} = \frac{3}{c} \cdot \frac{2}{c} \cdot c \cdot \frac{c/48}{2} = \frac{1}{16}.$$

The cross-channel term vanishes at $c = -204$, outside the positive-unitary range. Dropping the W-channel does not recover the Virasoro formula; Virasoro has its own scalar computation $F_2(\text{Vir}) = (c/2) \cdot 7/5760 = 7c/11520$. The W_3 formula is a counterexample to universal scalar genus-2 control for multi-weight algebras.

38.4 The bc reclassification

The bc ghost system (conformal weights λ and $1-\lambda$, central charge $c = -2(6\lambda^2 - 6\lambda + 1)$) is class C in the standard depth table.

PROPOSITION 38.3. The bc ghost system has shadow depth $r_{\max} = 4$: class C (contact/quartic), the same class as $\beta\gamma$.

The bc system has two generators (b of weight λ, c of weight $1-\lambda$), and the shadow class is determined by the two-generator structure, not by the statistics of either generator alone. The quartic shadow $S_4 \neq 0$ for bc at generic λ, but the stratum separation mechanism (the same mechanism that terminates $\beta\gamma$ at arity 4) applies: the quartic contact invariant lives on a charged stratum whose self-bracket exits the complex by rank-one rigidity.

Thus bc belongs with the contact/quartic class, not the Gaussian class.

38.5 The four-test boundary

The modular Koszul machine has a precise interface: four independent tests that exhaust its proved output. Beyond these four, nothing further is known.

Test	Statement	Status
1	$D^2_{\mathcal{A}} = 0$	proved (all families, all genera)
2	$\mathrm{obs}_g = \kappa \cdot \lambda_g$	proved (uniform-weight)
3	$Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) \cong H^*(\overline{\mathcal{M}}_g, \mathcal{Z})$	proved
4	HS-sewing convergence	proved (polynomial growth)

The four tests are independent. Multi-weight algebras at $g \geq 2$ pass tests 1, 3, and 4 but the status of test 2 is open. This is the precise location of the boundary: a single test, for a single class of algebras, at genera above 1.

What lies beyond the boundary: multi-weight universality at $g \geq 2$, the identity BV/BRST = bar at higher genus, and non-perturbative completion. These are three distinct problems with no known logical dependencies between them.

38.6 Two orthogonal axes

The MC element $\Theta_{\mathcal{A}}$ lives in a bigraded space with two independent filtrations.

The genus axis ($g = 0, 1, 2, \ldots$), controlled by the genus spectral sequence and the sewing construction. The genus-g component $\Theta^{(g)}_{\mathcal{A}} \in H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ is determined inductively by lower genera via the MC equation. At the scalar uniform-weight level: $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$; for multi-weight algebras this is only the scalar summand. The genus axis is vertical: higher genus means more topology.

The arity axis ($r = 2, 3, 4, \ldots$), controlled by the arity filtration and the shadow Postnikov tower. The arity-r component $\Theta^{(\leq r)}_{\mathcal{A}}$ is determined by the MC equation at arity $\leq r$, with obstruction class $o_{r+1}(\mathcal{A}) \in H^2(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}})$. The arity axis is horizontal: higher arity means more nonlinearity.

*

These two axes are orthogonal at the scalar projection. The scalar shadow κ determines the uniform-weight scalar summand at all genera, while multi-weight algebras can see cross-channel data. Higher-arity shadows are invisible at genus 1 and become visible at

$$g_{\min}(S_r) = \lfloor r/2 \rfloor + 1.$$

The cubic shadow S_3 first appears at genus 2. The quartic S_4 at genus 3. The quintic S_5 also first appears at genus 3. The arity axis unfolds as genus increases, but the genus axis does not unfold as arity increases.

The MC equation couples the two axes: the genus-g, arity-r component of $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ receives contributions from all (g', r') with $g' \leq g$ and $r' \leq r$. The coupling is triangular in genus (genus g depends on genera $< g$) and triangular in arity (arity r depends on arities $< r$). Both filtrations are exhaustive, and their intersection is the origin: $g = 0$, $r = 2$, the λ-bracket. The entire machine is an algebraic induction from a quartic pole.

38.7 The free fermion at Heisenberg depth

Both the Heisenberg algebra and the free fermion are class G: shadow depth $r_{\max} = 2$. Both terminate at arity 2: no cubic, no quartic, no infinite tower. The mechanism is different.

For Heisenberg: the OPE $\alpha(z)\alpha(w) = k/(z-w)^2$ has no simple pole. The bar differential extracts only the level (the double pole), producing the scalar $\kappa = k$. There is no Lie bracket to generate higher-arity corrections.

For the free fermion: the OPE $\psi(z)\psi(w) = 1/(z-w)$ has a simple pole, but the cyclic forms $\langle\psi,\dots,\psi\rangle_{\mathrm{cyc}}$ vanish at arities ≥ 3 by antisymmetry: cyclic symmetry forces $\langle\psi^{\otimes n}\rangle_{\mathrm{cyc}} = 0$ for odd n and sign cancellation kills even n.

The two class G algebras differ in every other respect:

	Heisenberg	**Free fermion**
Sewing	determinant: $\det(1-qA)^{-1}$	Pfaffian
Partition function	$\eta(\tau)^{-2k}$	$\theta_\sigma(\tau)/\eta(\tau)$
Spin structure	none	2^{2g} choices at genus g
Bosonisation	$\mathcal{F}\otimes\mathcal{F}\cong\mathcal{H}_1$	$\mathcal{H}_1\cong\mathcal{F}\otimes\mathcal{F}$

The bosonisation isomorphism $\mathcal{F}\otimes\mathcal{F}\cong\mathcal{H}_1$ is *not* Koszul duality. It is a tensor product decomposition in the category of vertex algebras, not a bar-cobar adjunction. The free fermion is self-dual in the familiar quadratic/free-field sense; a full chiral Koszul-duality identification should be cited separately before being used as a theorem.

38.8 The modular Koszul triple

The binary genus-zero seed of the machine is the triple

$$(\mathcal{A},\ \mathcal{A}^!,\ r(z)),$$

where $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is the collision r-matrix: the binary genus-0 projection of the universal MC element. Equivalently, $r(z)$ is the image of $\Theta_{\mathcal{A}}$ under the forgetful map from the modular convolution algebra to the genus-0, arity-2 component.

On the proved principal DS-shadow lane, DS reduction is compatible with the binary triples. For the principal reduction $\widehat{\mathfrak{sl}}_N \to \mathcal{W}_N$:

$$\big(V_k(\mathfrak{sl}_N),\ V_{k'}(\mathfrak{sl}_N),\ k\,\Omega/z\big) \longmapsto \big(\mathcal{W}_N^k,\ \mathcal{W}_N^{k'},\ r^{\mathcal{W}}(z)\big).$$

The ghost sector supplies the BRST correction implementing this compatibility on the principal lane. It should not be described as the kernel of a global functor on triples, nor as a quotient map on the full modular datum. Non-principal and global triple functoriality remain frontier problems.

The triple embeds into larger structures by successive enrichment:

(1) The **modular Koszul datum** $\Pi_X(\mathcal{A})$: the triple plus the factorisation coalgebra $\bar{B}_X(\mathcal{A})$, the shadow tower $\{S_r\}$, and the quartic resonance class R_4^{mod}. Six-fold datum.

(2) The **completed modular Koszul datum** $\Pi_X^{\mathrm{oc}}(\mathcal{A})$: the package plus the chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ (universal bulk) and the annulus trace $\mathrm{Tr}_{\mathcal{A}} \simeq HH_*(\mathcal{A})$. Eight-fold datum.

The triple is the seed. The full MC element also needs the factorisation coalgebra, determinant/trace line, branch data, and quartic resonance information. Those data are forgotten by the binary triple.

38.9 $\Theta_{\mathcal{A}}$ as protagonist

The five main theorem packages can be organised as scoped projections controlled by the ordered MC element. This is a useful architecture, not an identity theorem: Theorem C has a conditional BV upgrade, Theorem H is generic/PBW chiral Hochschild, and Theorem D has scalar and cross-channel lanes.

$$\begin{aligned}
\text{Theorem A (Verdier/Koszul):}\quad & \mathbb{D}_{\mathrm{Ran}}(\bar{B}(\mathcal{A}^{i})) \simeq \bar{B}(\mathcal{A}^{!}) \quad \Longleftarrow \quad \Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}) \\
\text{Theorem B (inversion):}\quad & \Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A} \quad \Longleftarrow \quad \Theta_{\mathcal{A}} \text{ bar-intrinsic} \\
\text{Theorem C (complementarity):}\quad & Q_g \oplus Q^!_g \quad \Longleftarrow \quad \text{derived-centre/complementarity projection of } \Theta_{\mathcal{A}} \\
\text{Theorem D (modular char.):}\quad & F_g = \kappa \cdot \lambda^{\mathrm{FP}}_g + \delta F^{\mathrm{cross}}_g \quad \Longleftarrow \quad \text{scalar shadow of } \Theta_{\mathcal{A}} \\
\text{Theorem H (Hochschild):}\quad & \mathrm{ChirHoch}^*(\mathcal{A}) \quad \Longleftarrow \quad \text{generic/PBW tangent complex of } \Theta_{\mathcal{A}}
\end{aligned}$$

The MC element is defined as $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$, where $D_{\mathcal{A}}$ is the full bar differential and d_0 is the genus-0 collision differential. It is MC because $D^2_{\mathcal{A}} = 0$. The shadow Postnikov obstruction tower consists of finite-order projections $\Theta^{(\leq r)}_{\mathcal{A}}$. The full element lives in the completed modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \mathrm{Conv}_{\mathrm{str}}(C^!_{\mathrm{ch}}, \mathcal{A})$.

In the open/closed framework, $\Theta_{\mathcal{A}}$ is the *closed projection* of the full open/closed MC element $\Theta^{\mathrm{oc}} = \Theta_{\mathcal{A}} + \sum \mu^{M_j}$, where the μ^{M_j} are open-sector operations. The primitive object is the open-sector factorisation dg-category $\mathcal{C}_{\mathrm{op}}$; the boundary algebra $\mathcal{A}_b = \mathrm{End}(b)$ is a Morita-dependent chart; the bulk is the chiral derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$. The bar complex classifies twisting morphisms $\mathrm{Tw}(C, \mathcal{A})$ for the coalgebra input C; it is not literally a space of generic twisting morphisms between $\mathcal{A}$ and $\mathcal{A}^!$. The derived centre classifies bulk operators acting on the boundary. These are different functors applied to related bar data.

38.10 The corrected Picard–Fuchs equation

The shadow connection $\nabla^{\mathrm{sh}} = d - Q'_L/(2Q_L)\,dt$ has flat sections $f(t) = C\sqrt{Q_L(t)}$. The flatness condition $\nabla^{\mathrm{sh}} f = 0$ is

$$f' = \frac{Q'_L}{2Q_L} f.$$

Differentiating and eliminating f via the first-order equation gives the second-order ODE satisfied by the flat sections:

$$2Q_L f'' + Q'_L f' - Q''_L f = 0.$$

The derivation is direct: from $2Q_L f' = Q'_L f$, differentiate to get $2Q_L f'' + 2Q'_L f' = Q''_L f + Q'_L f'$, hence $2Q_L f'' + Q'_L f' - Q''_L f = 0$.

The equation has regular singular points at the zeros of Q_L (the branch points of the shadow extension). The local exponents at each zero are 0 and 1/2: the flat section $\sqrt{Q_L}$ has a square-root branch point, and the second solution is regular. The monodromy around each singular point is -1: the Koszul sign.

39 The Koszul–Epstein frontier

The shadow obstruction tower of a modular Koszul algebra $\mathcal{A}$ produces numerical invariants $\{S_r(\mathcal{A})\}_{r\geq 2}$. In the lattice corner these invariants interact with theta functions and standard L-functions. Outside that corner, any graph-sum or Rankin–Selberg bridge is a separate construction with separate hypotheses. The question is whether the algebraic structure of the Maurer–Cartan element $\Theta_{\mathcal{A}}$ constrains analytic zero-location or positivity properties beyond automorphic spectral theory. No such theorem is known here.

39.1 What is proved

(i) *MC recursion and Carleman determinacy.* The MC equation at arity $r+1$ imposes a polynomial relation on the shadow coefficients $\{S_2, \ldots, S_{r+1}\}$ (Theorem ??). For $r \geq 2m+2$ (where m is the number of spectral atoms), the system is overdetermined (Proposition ??). The Carleman condition holds (Proposition ??): $\sum |S_r|^{-1/(2r)} = \infty$. Therefore the Hamburger moment problem has at most one solution for the full moment sequence satisfying the stated hypotheses. Carleman determinacy is uniqueness from all moments, not a proof that a finite parameter list determines zeta zeros or automorphic spectra.

(ii) *Route C coefficient-level prime-locality.* If the low-arity shadow data (κ, α) are determined by Hecke-equivariant character data, then the MC recursion propagates Hecke equivariance to all arities (Theorem ??). This is a coefficient-level propagation theorem under the stated Hecke/recursion hypotheses. It is not an amplitude-level Euler product or a zero-location theorem.

(iii) *Shadow–spectral correspondence for lattice VOAs.* Shadow depth equals the number of critical lines minus one (Theorem ??).

(iv) *Verdier–Hecke commutation.* At self-dual points in the proved arithmetic lane, Verdier duality commutes with Hecke operators and can force the constrained Epstein zeta to factor into standard L-functions (Theorem ??).

39.2 What is false and dismissed

Seven false ideas, each identified during the Beilinson pass on the research programme (raeeznotes105), are now permanently dismissed.

(i) *"Bar–cobar forces Epstein zeros onto the critical line."* False. Davenport–Heilbronn counterexamples exist for generic Epstein zeta functions of non-unimodular lattices ($h(D) \geq 2$). Bar–cobar provides homotopy equivalence, not arithmetic positivity.

(ii) *"The MC equation implies Li positivity."* False. The MC equation is a flatness condition (quadratic consistency). Li positivity is a sign-definiteness condition. The Rankin–Selberg transform connecting them is not sign-preserving: the Mellin transform of a positive function can be negative.

(iii) *"The shadow obstruction tower descends to a functional equation of $S_{\mathcal{A}}(u)$."* False. The shadow obstruction tower acts on the arity variable t; the sewing lift acts on the spectral variable u. The Virasoro complementarity identity $\kappa + \kappa' = 13$ is an algebraic constraint in c, orthogonal to the u-variable analytic structure. The function $S_{\mathcal{A}}(u)$ depends only on the conformal weight multiset $W(\mathcal{A}) = \{w_i\}$ and is independent of the central charge c.

(iv) *"Narain radius variation gives zero-location leverage."* False. Varying the radius changes the coefficient factor, not the zero locations.

(v) *"The ζ-proxy is a valid test."* False. The proxy does not preserve the safe point $\sigma = 1/2$.

(vi) *"Bare spectral measures close the quartic gap."* False. The mismatch for Liouville and A_2-Toda is four to five orders of magnitude. The crossing-weighted bilinear residue kernel (involving the actual DOZZ/Toda structure constants and the modular transform kernel) is required.

(vii) *"The Li coefficients for the Heisenberg sewing lift are all positive."* False. The completed Li coefficients turn negative at $n = 974$, driven by the exponential growth of contributions from zeros on $\Re = -1/2$. The uncompleted Li coefficients turn negative at $n = 7$. Both failures are structural consequences of the two-critical-line architecture of $\zeta(u)\zeta(u+1)$, not evidence against the Riemann hypothesis.

39.3 The structural obstruction

The shadow spectral measure ρ (atoms at effective couplings λ_i of the MC element $\Theta_{\mathcal{A}}$) and the automorphic spectral measure (Maass eigenvalues, scattering poles of the Laplacian on $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$) are different objects (Remark ??). The map from ρ to the automorphic spectrum goes through the Rankin–Selberg integral. This map is injective for lattice VOAs but ill-conditioned for non-lattice theories. The structural obstruction (Remark ??) is that this map is not surjective: the shadow spectral measure does not determine the automorphic spectral measure on the moduli space. In particular, the MC equation constrains data on the real spectral axis, while the zeta zeros live at complex spectral parameters.

39.4 Honest status

The Koszul–Epstein programme produced genuine arithmetic results about the shadow obstruction tower: Carleman-type determinacy, Route C prime-locality, the shadow–spectral correspondence. These results stand on their own mathematical merit. They illuminate the arithmetic structure of modular Koszul algebras.

The programme did not produce, and cannot produce in its current form, a critical-line theorem or a positivity result. The gap between the algebraic content (MC flatness, moment determination, Hecke propagation) and the analytic conclusion (zero location, sign definiteness) remains unbridged. No mechanism for positivity has been identified. No construction of the conjectured modular surface with the required polarisation exists.

The correct statement of the surviving programme is: *the MC element $\Theta_{\mathcal{A}}$, through the graph sum and Rankin–Selberg integral, determines a regularised two-variable L-object $L_{\mathcal{A}}(s,u)$ whose quartic resonance section is the first nonlinear place where Koszul duality constrains the automorphic spectral decomposition beyond the character level. Whether this constraint carries critical-line or positivity information is open, with no positive evidence and no definitive obstruction.*

40 The spectral question

The constrained Epstein series

$$\varepsilon_s^c(\mathcal{A}) = \sum_{\Delta \in \mathrm{Spec}_{\mathrm{sc}}(\mathcal{A})} (2\Delta)^{-s},$$

where the sum runs over scalar primary dimensions Δ with multiplicity, is the genus-one spectral invariant that couples to the scattering resonances of $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$. It is the Eisenstein spectral coefficient of the primary-counting function $\hat{Z}^c(\tau) = y^{c/2}\,|\eta(\tau)|^{2c}\,Z(\tau,\bar{\tau})$. The question of this section: does the bar construction see this object? The answer is structured. Five partial constructions each capture a projection; none captures the full spectrum; the gap is structural; and the deepest proposed identification is false.

40.1 Five partial captures

40.1.0.1 (i) Zhu's algebra. Zhu's algebra $A(V)$ captures the weight-0 structure of the representation category. For $V = V_c$ (universal Virasoro): $A(V_c) \cong \mathbb{C}[x]$, polynomial in $[\omega]$. The simple modules of $A(V_c)$ correspond to highest-weight representations, and the augmentation ideal encodes the ground-state structure. But $A(V)$ sees only the weight-0 part of the category: the lowest weights Δ of simple modules, not the full primary spectrum at arbitrary conformal weight. Higher Zhu algebras $A_n(V)$ extend the reach to weight n, but no single A_n captures the entire tower. The Zhu filtration gives a spectral sequence $E_2 = \mathrm{Ext}^*_{A(V)}(\mathbb{C}, \mathbb{C}) \Rightarrow H^*(B(V))$ at weight 0, but it does not extend to a full spectral sequence in weight. *Partial: captures the set of lowest weights* $\{\Delta_0\}$*, not the multiplicities or the higher spectrum.*

40.1.0.2 (ii) Associated variety. The associated variety $X_V = \mathrm{Specm}(R_V)$ of the Zhu C_2-algebra $R_V = V/C_2(V)$ captures the singular support of V. For affine Kac–Moody at non-critical level: $X_{V_k(\mathfrak{g})} = \overline{\mathbb{O}}$, the closure of a nilpotent orbit (Arakawa). For Virasoro: $X_{V_c} = \mathbb{C}$. The associated variety controls the *asymptotic* growth of the character (Arakawa–Kawasetsu): $\chi_V(q) \sim q^{-c/24}(1-q)^{-\dim X_V}$ as $q \to 1$. This gives the leading Weyl-type asymptotics of the primary-counting function, hence the leading pole of ε^c_s at $s = c/2$. But it captures only the leading asymptotics, not the full Dirichlet series. *Partial: captures the growth rate* $\dim X_V$ *and the leading pole.*

40.1.0.3 (iii) Bar cohomology. For a chirally Koszul algebra $\mathcal{A}$, the bar cohomology $H^*(B(\mathcal{A}))$ is concentrated in bar degree 1 (Theorem **??**, item (i)). The graded dimension $\dim H^1(B(\mathcal{A}))_h$ at conformal weight h counts the generators of the Koszul dual algebra $\mathcal{A}^!$ at weight h:

$$\dim H^1(B(\mathcal{A}))_h \;=\; \dim(\mathcal{A}^!)_h.$$

For the Virasoro algebra: $\dim H^1(B(\mathrm{Vir}))_2 = 1$ (the generator T), $\dim H^1(B(\mathrm{Vir}))_3 = 0$, $\dim H^1(B(\mathrm{Vir}))_4 = 1$, and in general the dimensions are given by the Motzkin difference formula $M(h+1) - M(h)$. These dimensions are *independent of the central charge c* (the Lie bracket on $\mathrm{Vir}_- = \bigoplus_{n\geq 2} \mathbb{C}\, L_{-n}$ has no central extension: $[L_{-m}, L_{-n}] = (n-m)\, L_{-m-n}$ with $\delta_{m+n,0} = 0$ for $m, n \geq 2$).

The quasi-primary count at weight h is a *different* sequence. For the universal Virasoro vacuum module:

$$d_{\mathrm{qp}}(h) \;=\; p_{\geq 2}(h) - p_{\geq 2}(h-1),$$

the number of states at weight h minus the number of L_{-1}-descendants (partitions of h into parts ≥ 2 minus partitions of $h-1$ into parts ≥ 2). This sequence is 0, 1, 0, 1, 1, 1, 2, 2, 3, 4, 5, 6, 9, . . . The bar cohomology dimensions $\dim H^1(B(\mathrm{Vir}))_h$ are 0, 1, 0, 1, 0, 2, 0, 5, 0, 12, 0, 30, . . . (Motzkin differences), which are supported on even weights and grow as $3^h/(h\sqrt{\pi h})$. These two sequences do not agree.

Bar cohomology captures the Koszul dual, not the primary spectrum.

40.1.0.4 (iv) Conformal blocks.

The space of conformal blocks $\mathcal{V}(\Sigma_{g,n}; V_1, \ldots, V_n)$ for a vertex algebra V on a genus-g surface with n insertions is the coinvariant space $V_1 \otimes \cdots \otimes V_n/(\text{holomorphic sections of } V \otimes \omega_\Sigma)$. At genus 1 with no insertions, the conformal block is the character $\chi_V(\tau) = \mathrm{tr}_V\, q^{L_0 - c/24}$. The primary-counting function $\hat{Z}^c$ is constructed from this character by stripping the Casimir factor. So the genus-1 conformal block *does* capture the full spectrum (it is the generating function of the spectrum). But conformal blocks are not invariants of the bar complex: they depend on the surface, the insertions, and the full representation theory of V, which the bar construction does not control. *Full capture at genus 1, but external to the bar machine.*

40.1.0.5 (v) Shadow obstruction tower.

The primary-line shadow tower $\{S_r\}_{r\geq 2}$ is an arity-varying, genus-0 invariant determined by the three OPE numbers (κ, α, S_4) on that line. The constrained Epstein ε_s^c is a weight-varying, genus-1 invariant determined by the full scalar primary spectrum (an infinite sequence). The shadow obstruction tower determines a shadow-metric Epstein zeta $\varepsilon_{Q_L}(s)$ (a binary quadratic form zeta), which is a *different* object from the constrained Epstein ε_s^c. The relationship is a chain of projections:

$$\text{full CFT data} \xrightarrow{\text{primary spectrum}} \varepsilon_s^c \xrightarrow[\text{(information loss)}]{\text{low moments}} \varepsilon_{Q_L} \xrightarrow{\text{Taylor coeff.}} \{S_r\}.$$

The shadow obstruction tower sees the quadratic approximation to the spectral data. For lattice VOAs, the shadow–spectral correspondence recovers the full L-function package. For non-lattice algebras, the projection loses infinitely many degrees of freedom. *Partial: captures three moments of the spectral data.*

40.2 The gap: shadow versus spectrum

The shadow obstruction tower lives on the genus-0 side of the bar complex (the MC element $\Theta_{\mathcal{A}}$ is a graph sum over planted trees on the disc, with arity r counting the number of boundary insertions). The constrained Epstein lives on the genus-1 side (the partition function $Z(\tau, \bar{\tau})$ is a trace over the vacuum module, which is a genus-1 datum).

	Shadow obstruction tower	**Constrained Epstein**
Genus	0	1
Variable	Arity r	Weight h
Parameters	$3\ (\kappa, \alpha, S_4)$	$\infty\ (\{\Delta_i, m_i\})$
Functional equation	Complementarity ($c \to 26 - c$)	Modular ($s \to c/2 + 1 - s$)
c-dependence	Rational in c	Transcendental in c

The shadow obstruction tower and the constrained Epstein are related by the same chiral algebra, but they extract different projections. Neither determines the other. The shadow obstruction tower is determined by three numbers; the constrained Epstein encodes infinitely many independent primary dimensions.

40.3 The deepest question: is $\varepsilon_s^c = \sum H^1(B(V))_h \cdot (2h)^{-s}$?

Proposition 40.1 *(Bar cohomology $\neq$ quasi-primary count).* The Dirichlet series $\sum_{h\geq 2} \dim H^1(B(\mathcal{A}))_h \cdot (2h)^{-s}$ is *not* the constrained Epstein zeta $\varepsilon_s^c(\mathcal{A})$, for any chirally Koszul algebra $\mathcal{A}$ with more than one generator.

Proof. Two independent obstructions.

(a) Different sequences. For the Virasoro algebra: $\dim H^1(B(\mathrm{Vir}))_h$ is the Motzkin difference $M(h+1) - M(h)$, which vanishes at all odd weights and is independent of c. The quasi-primary count $d_{\mathrm{qp}}(h) = p_{\geq 2}(h) - p_{\geq 2}(h-1)$ is nonzero at odd weights ($d_{\mathrm{qp}}(5) = 1$, $d_{\mathrm{qp}}(7) = 2$) and also independent of c for the universal Virasoro. These are different sequences.

At weight 5: $\dim H^1(B(\mathrm{Vir}))_5 = 0$, $d_{\mathrm{qp}}(5) = 1$. The bar cohomology misses the quasi-primary at weight 5 because the Koszul dual $\mathrm{Vir}^!$ has no generator at weight 5 (the dual algebra has generators only at even weights, by the parity of the Virasoro bracket).

(b) The bar construction sees the Koszul dual, not the original. By the Koszul property, $H^1(B(\mathcal{A}))_h \cong (\mathcal{A}^!)^{\mathrm{gen}}_h$: bar cohomology in degree 1 counts generators of the *dual* algebra, not quasi-primaries of the *original*. For Kac–Moody: $\dim H^1(B(\hat{\mathfrak{g}}))_1 = \dim \mathfrak{g}$ (the dual has $\dim \mathfrak{g}$ generators at weight 1), while the quasi-primary count at weight 1 is also $\dim \mathfrak{g}$ (the currents J^a). The coincidence at weight 1 is a consequence of the KM algebra being generated in weight 1, so the generators of $\mathcal{A}$ and the generators of $\mathcal{A}^!$ both live at weight 1. For Virasoro, with a weight-2 generator, the sequences diverge starting at weight 4.

The constrained Epstein ε^c_s sums over primary dimensions of the *original* algebra $\mathcal{A}$ (with multiplicity, over all representations appearing in the partition function). Bar cohomology counts states of the *Koszul dual* $\mathcal{A}^!$. These are different algebras (unless $\mathcal{A}$ is self-dual at $c = 13$), and even at $c = 13$ the quasi-primary count is not the bar cohomology dimension. $\square$

Remark 40.2 *(Why the identification is tempting).* For free-field algebras (Heisenberg, free fermion), the quasi-primary count at weight h equals the bar cohomology dimension at weight h, because the algebra is *generated* by a weight-1 field and the Koszul dual is the symmetric/exterior algebra on the same space. The coincidence is a special-case phenomenon (AP3): it holds when the algebra is both free and generated in weight 1, which forces the Koszul dual to have the same Hilbert series as the quasi-primary tower. For algebras with generators of weight ≥ 2, the Koszul dual's weight distribution diverges from the quasi-primary distribution.

40.4 What does capture the full spectrum?

No single object in the bar-cobar machine captures the full primary spectrum. The spectrum is an invariant of the *representation category* $\mathcal{A}$-mod, not of the algebra $\mathcal{A}$ alone. The bar complex $B(\mathcal{A})$ is built from the algebra (the vacuum module), not from its modules. The partition function $Z(\tau, \bar{\tau})$, which encodes the full spectrum, requires the *category of modules* together with a modular-invariant pairing on characters.

Observation 40.3 *(The categorical gap).* The bar construction is a functor from chiral algebras to factorisation coalgebras. The primary spectrum is a function from the *representation category* to Dirichlet series. These have different domains. The bar complex sees the algebra; the spectrum sees the category.

For rational VOAs, Zhu's correspondence provides a bridge: the simple modules of $A(V)$ biject with simple modules of V, so the set of primary dimensions $\{\Delta_i\}$ is determined by $A(V)$. But the multiplicities in the partition function are determined by the modular S-matrix, which is categorical data (the braiding of the modular tensor category V-mod), not algebra data.

The genus-1 factorisation homology $\int_{E_\tau} \mathcal{A}$ is a single object that sees both the algebra and the genus-1 categorical data: it computes the character (via the trace over the vacuum representation) and thereby encodes the full primary spectrum. But it is a genus-1 object, not a genus-0 object, and it is not a projection of the MC element $\Theta_{\mathcal{A}}$ (which lives at genus 0). The genus-1 amplitude $F_1(\mathcal{A}) = \kappa(\mathcal{A})/24$ is the *integrated* form of this datum: one number extracted from the full partition function. The rest of the partition function (the full Fourier expansion in q) is inaccessible to the shadow obstruction tower.

40.5 The single object that sees both

The full modular Koszul datum

$$\big(\mathcal{A},\ B(\mathcal{A}),\ \Theta_{\mathcal{A}},\ \mathcal{A}\text{-mod}^{\mathrm{ss}},\ Z(\tau,\bar{\tau})\big)$$

sees both the shadow data (from $\Theta_{\mathcal{A}}$) and the spectral data (from Z). But this is the *entire theory*, not a single mathematical object. The question was whether a single object captures both. The answer: the partition function $Z(\tau,\bar{\tau})$ captures the spectral data by definition, and it determines the shadow data (by extracting OPE coefficients from the q-expansion). The reverse is false: the shadow data does not determine the partition function.

This asymmetry is structural. The partition function encodes the full genus-1 trace, which involves all states of the algebra at all weights. The shadow obstruction tower encodes three OPE coefficients extracted from genus-0 collision residues. The genus-1 object is strictly richer.

For the Koszul–Epstein programme, the implication is decisive: the constrained Epstein ε^c_s cannot be recovered from the bar complex alone. It requires the partition function, which is external to the bar construction. The bar complex provides the *algebraic* constraints on ε^c_s (the MC equation, the shadow obstruction tower, the depth classification), but not the *spectral* content (the primary dimensions and multiplicities).

40.6 Summary

Construction	**Captures spectrum?**	**Bar-internal?**	**What it sees**
Zhu's algebra $A(V)$	Partially (weight 0)	No (Zhu $\neq$ bar)	Lowest weights
Associated variety X_V	Partially (asymptotics)	No	Growth rate
Bar cohomology $H^*(B)$	No (Koszul dual)	Yes	Generators of $\mathcal{A}^!$
Conformal blocks	Fully (genus 1)	No	Full primary spectrum
Shadow obstruction tower	Partially (3 moments)	Yes (from $\Theta_{\mathcal{A}}$)	κ, α, S_4
Factorisation homology	Fully (genus 1)	No (requires trace)	Full partition function

The bar complex is the algebraic engine. The primary spectrum is the analytic output. The two are connected by the representation category, which the bar complex constrains but does not determine. A smaller true theorem: the bar complex determines the Koszul dual and the shadow obstruction tower, which together give three constraints on the spectrum. A larger false theorem: the bar complex determines the full spectrum.

41 Algebraic integration and the spectral question

The naive claim that "integration over moduli space is analytic, not algebraic" is false. The bar complex defines a sheaf on $\overline{\mathcal{M}}_g$ whose algebraic invariants are nontrivial and proved. But the deeper question, whether these algebraic invariants capture the primary spectrum, has a structured answer that is neither trivially positive nor trivially negative.

41.1 What is algebraically available

The manuscript already constructs three algebraic objects from the bar complex over moduli space:

41.1.0.1 (1) The virtual bar K-class. $V_{\mathcal{A}} := [R\pi_{g*}\overline{B}_X^{(g)}(\mathcal{A})] \in K^0(\overline{\mathcal{M}}_g)$ (Definition ??). This is the derived pushforward of the bar complex along the universal curve $\pi_g : \overline{\mathcal{C}}_g \to \overline{\mathcal{M}}_g$. By the perfectness criterion (Lemma ??: Koszul acyclicity + finite-dimensional fiber cohomology $\Rightarrow$ perfect complex), this is well-defined in the bounded derived category $D^b(\overline{\mathcal{M}}_g)$.

41.1.0.2 (2) The center sheaf. $\mathcal{Z}_{\mathcal{A}} := R^*\pi_{g*}(\overline{B}_X^{(g)}(\mathcal{A}), d_{\mathrm{fib}}) \in \mathrm{Loc}(\overline{\mathcal{M}}_g)$ (Theorem ??(ii)). Its fiber over $[\Sigma_g]$ is the chiral homology $H_*^{\mathrm{ch}}(\Sigma_g, \mathcal{A})$. This is a local system carrying a flat Gauss–Manin connection.

41.1.0.3 (3) The ambient complementarity complex. $C_g(\mathcal{A}) = R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ (Theorem ??(iii)). This is the derived global sections of the center sheaf: a purely algebraic operation on a coherent/constructible sheaf on a proper Deligne–Mumford stack.

41.1.0.4 (4) The GRR computation. By Grothendieck–Riemann–Roch on the universal curve (Proposition ??):

$$c_1(\det V_{\mathcal{A}}) = \kappa(\mathcal{A}) \cdot \lambda, \qquad \mathrm{ch}(V_{\mathcal{A}}) \leftrightarrow \Delta_{\mathcal{A}}(x), \qquad \mathrm{Hol}(V_{\mathcal{A}}) = \Pi_{\mathcal{A}}.$$

These are the scalar level (the modular characteristic κ), the spectral level (the discriminant Δ), and the holonomy level (the periodicity profile Π). All three are computed algebraically from the K-class $V_{\mathcal{A}}$.

So the bar complex *does* define a sheaf on $\overline{\mathcal{M}}_g$ whose $R\Gamma$ is computable. The derived global sections, the Chern character, the monodromy representation–-all of these are algebraic operations on algebraic objects. The statement "integration over moduli space is analytic, not algebraic" is a false idea in the Beilinson sense. Derived global sections $R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ *is* algebraic integration, and it produces the ambient complementarity complex, which is a proved algebraic invariant.

41.2 What $R\Gamma$ does not capture

The question now becomes precise: does $R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ capture the primary spectrum?

The answer is **no**, and the obstruction is not "analysis vs. algebra" but something deeper: the bar complex is built from the *vacuum module's OPE data*, not from the representation category.

(i) *The sheaf $\mathcal{Z}_{\mathcal{A}}$ is an algebra invariant.* The center sheaf $\mathcal{Z}_{\mathcal{A}}$ is functorial in morphisms of chiral algebras. Its fiber $H_*^{\mathrm{ch}}(\Sigma_g, \mathcal{A})$ depends only on the algebra $\mathcal{A}$ and the curve Σ_g, not on the category $\mathcal{A}$-mod or the modular invariant.

(ii) *The primary spectrum is a categorical invariant.* Two non-isomorphic modular-invariant partition functions for the same chiral algebra produce different constrained Epstein series but the same center sheaf $\mathcal{Z}_{\mathcal{A}}$. Example: the D-type and E-type modular invariants for $\widehat{\mathfrak{su}(2)}_k$ at $k \geq 10$ (Cappelli–Itzykson–Zuber) have the same bar complex but different primary spectra.

(iii) *The partition function is a trace, not a global section.* The character $\chi_V(\tau) = \mathrm{tr}_V\, q^{L_0 - c/24}$ is a *trace* over a module, not a derived global section of a sheaf on $\overline{\mathcal{M}}_g$. The genus-1 factorization homology $\int_{E_\tau} \mathcal{A}$ computes this trace, but it depends on the genus-1 curve E_τ (hence on τ) and produces a function on $\mathcal{M}_{1,1}$, not a single cohomology class. The *integrated* invariant $F_1(\mathcal{A}) = \kappa(\mathcal{A})/24$ is the single number obtained by extracting the leading Hodge class from $R\Gamma$; the full partition function (the q-expansion) is not captured by $R\Gamma$ of the center sheaf.

(iv) *The GRR hierarchy is strictly coarser than the spectrum.* The three-level hierarchy $\kappa \to \Delta \to \Pi$ captures: (L1) one number (κ), (L2) the bar cohomology dimensions at all weights (the Koszul dual's Hilbert series), and (L3) the monodromy of the bar cohomology local system. None of these is the primary spectrum. The bar cohomology dimensions count generators of $\mathcal{A}^!$, not primaries of $\mathcal{A}$ (Proposition 40.1). The monodromy of $\mathcal{Z}_{\mathcal{A}}$ encodes the action of $\mathrm{MCG}(\Sigma_g)$ on chiral homology, which is the Hitchin/KZ connection–-an invariant of the algebra, not of its module category.

41.3 The precise obstruction

The obstruction to recovering the spectrum from $R\Gamma$ is not "analysis vs. algebra." It is the distinction between two algebraic categories:

	Algebra invariants	**Categorical invariants**
Input	$\mathcal{A}$ (the chiral algebra)	($\mathcal{A}$-mod, M_{ij})
Construction	$B(\mathcal{A})$ (bar complex)	partition function $Z(\tau, \bar{\tau})$
Sheaf on $\overline{\mathcal{M}}_g$	$\mathcal{Z}_{\mathcal{A}}$ (center sheaf)	conformal blocks $\mathcal{V}_{g,n}$
$R\Gamma$	$C_g(\mathcal{A})$ (complementarity)	$\chi_V(\tau)$ (character)
GRR output	$\kappa \cdot \lambda$	$\sum m_i \chi_i(\tau)\overline{\chi_j(\tau)}$
Information	Finitely many invariants	Infinitely many primary dims

The right column requires the module category and a modular invariant as additional input. No algebraic operation on the *left column* can produce the *right column*, because the domain is strictly smaller. This is a theorem (Proposition **??**), not a limitation of current technology.

41.4 What would bridge the gap

Three constructions could, in principle, extend the bar complex's reach toward the spectrum:

41.4.0.1 (a) The module bar construction with all modules. The module bar complex $\overline{B}_{\mathcal{A}}(\mathcal{M})$ for a chiral module $\mathcal{M}$ produces a comodule over $\overline{B}(\mathcal{A})$ (Chapter **??**). If one applies the bar construction to *every* simple module L_i of $\mathcal{A}$, one obtains a family of comodules $\{\overline{B}_{\mathcal{A}}(L_i)\}_{i \in I}$ whose K-classes $[R\pi_{g*}\, \overline{B}_{\mathcal{A}}(L_i)] \in K^0(\overline{\mathcal{M}}_g)$ carry the conformal dimension h_i in their Chern character (the fiber Euler characteristic is $\chi(L_i) = q^{h_i - c/24} \prod(1 - q^n)^{-1}$). Summing with the modular invariant multiplicities:

$$\sum_{i,j} M_{ij}\, \mathrm{ch}\big(R\pi_{1*}\, \overline{B}_{\mathcal{A}}(L_i)\big) \cdot \overline{\mathrm{ch}(R\pi_{1*}\, \overline{B}_{\mathcal{A}}(L_j))}$$

would recover the partition function at genus 1, and hence the primary spectrum. But this construction takes $\mathcal{A}$-mod and M_{ij} as *input*, not output; it does not extract the spectrum from the bar complex alone.

41.4.0.2 (b) The factorization homology sheaf as a D-module. The factorization homology $\int_X \mathcal{A}$, viewed as a function of $X \in \mathcal{M}_g$, defines a D-module on $\mathcal{M}_g$ (the conformal blocks local system with its flat connection). The *spectral data of this D-module*–-its characteristic variety, monodromy, and irregular singular structure–-is an algebraic invariant that encodes more than $R\Gamma$ of the center sheaf. For integrable affine Kac–Moody algebras, this D-module is the Hitchin connection on the Verlinde bundle, and its monodromy recovers the modular S-matrix (hence the fusion rules, hence the primary dimensions via the Verlinde formula). Whether a similar construction works for non-rational algebras (Virasoro at generic c) is open: the conformal blocks local system has irregular singularities along the boundary of $\overline{\mathcal{M}}_g$, and the spectral theory of such irregular D-modules is a frontier topic (Boalch, Sabbah, Mochizuki).

41.4.0.3 (c) A motivic spectral zeta. A construction $\zeta_V^{\mathrm{mot}}(s) := \sum_i [L_i] \cdot (L_0|_{L_i})^{-s} \in K_0(\mathcal{A}\text{-mod}) \otimes \mathbb{C}$ would categorify the constrained Epstein series. For rational VOAs this reduces to a finite sum; for irrational VOAs it requires a continuous K-theory with weight topology. The functional equation would come from the modular tensor category structure (the S-matrix as categorical Frobenius). No such construction exists. The structural obstruction: the spectral decomposition on $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$ uses the hyperbolic Laplacian, which is a real-analytic operator without algebraic meaning. A categorical/motivic version would require categorifying the Roelcke–Selberg decomposition–-a problem at the frontier of geometric Langlands.

41.5 Beilinson verdict

The claim "the bar complex cannot see the spectrum because integration over moduli space is analytic" was a false idea. The corrected diagnosis:

(a) The bar complex *does* define a sheaf $\mathcal{Z}_{\mathcal{A}}$ on $\overline{\mathcal{M}}_g$ with well-defined algebraic invariants ($R\Gamma$, ch, Hol, GRR pushforward). These invariants are proved and nontrivial.

(b) The algebraic invariants of $\mathcal{Z}_{\mathcal{A}}$ capture the *algebra's* modular structure (κ, the bar cohomology dimensions, the KZ/Hitchin monodromy), not the *module category's* spectral content (the primary dimensions and multiplicities).

(c) The obstruction is not "analysis vs. algebra" but *algebra vs. category*: the bar complex is functorial in $\mathcal{A}$, while the spectrum is functorial in $(\mathcal{A}\text{-mod}, M_{ij})$. These are different functors with different domains.

(d) The "analytic" step that is genuinely unavoidable is the spectral decomposition on the modular curve $\mathrm{SL}(2,\mathbb{Z})\backslash\mathfrak{H}$ (Roelcke–Selberg), which maps the partition function (a categorical invariant) to the constrained Epstein series (an analytic invariant). This step involves the hyperbolic Laplacian and has no known algebraic/categorical substitute. But the partition function *itself* is an algebraic invariant of the module category (it is a finite sum of characters for rational VOAs, determined by the modular tensor category structure). The "analytic" content resides entirely in the last step: from characters to spectral coefficients.

(e) Status: **OPEN** (the algebraic construction exists but does not reach the spectrum; the gap is algebra-vs-category, not algebra-vs-analysis). The precise frontier question is whether the D-module structure of the conformal blocks local system on $\mathcal{M}_g$ (whose construction is algebraic via Beilinson–Drinfeld) has spectral invariants that recover the primary spectrum at genus 1 without invoking the Roelcke–Selberg decomposition. For rational VOAs with the Hitchin connection, the monodromy of this D-module *does* recover the S-matrix, hence the primary dimensions. For non-rational VOAs, this is a geometric Langlands frontier.

42 The real-to-complex frontier

The bar complex produces spectral data that is inherently *real*: the shadow coefficients $S_r(\mathcal{A})$ are rational functions of the central charge c, the MC equation constrains these coefficients along the real spectral axis, and the Carleman condition ensures that the resulting moment problem has a unique solution on the real line. The zeros of the Riemann zeta function live at complex spectral parameters. The central question of the arithmetic frontier is whether any mathematical mechanism bridges this gap.

After a complete investigation across eight candidate approaches, the definitive assessment is as follows.

42.1 The eight candidates

42.1.0.1 1. Deligne categories. The Deligne category $\underline{\mathrm{Rep}}(S_t)$ for $t \notin \mathbf{Z}_{\geq 0}$ gives a semisimple rigid tensor category interpolating between representation categories of symmetric groups. The idea: the shadow obstruction tower at non-integer c lives in a Deligne-type interpolation, and the analytic continuation in c might produce complex spectral data.

Status: dead end. The Deligne category is a purely algebraic interpolation in a *categorical* variable (the rank of the symmetric group). It does not produce analytic continuation in a *spectral* variable (the Mellin parameter s of an L-function). The shadow obstruction tower at non-integer c is already well-defined by rational interpolation of the OPE coefficients; no Deligne category is needed. The continuation in c gives a continuation of the shadow *coefficients* $S_r(c)$, not a continuation of the sewing Dirichlet series $S_{\mathcal{A}}(u)$ off the real spectral axis. These are different variables parametrising different structures.

42.1.0.2 2. Wick rotation / analytic continuation in c. Rotate c into the complex plane and study the shadow obstruction tower at complex c.

Status: dead end. For one conformal weight w, the sewing Dirichlet series is $S_{\mathcal{A}}(u) = \zeta(u+1)\,(\zeta(u) - H_{w-1}(u))$; for a weight multiset $W(\mathcal{A}) = \{w_i\}$ it is

$$S_{\mathcal{A}}(u) = \zeta(u+1) \sum_i (\zeta(u) - H_{w_i-1}(u)).$$

In either case it is independent of c. The central charge enters the shadow obstruction tower through the OPE structure constants, not through the weight multiset. Continuing c to complex values changes the shadow coefficients $S_r(c)$ but does *not* change $S_{\mathcal{A}}(u)$, which depends only on the conformal weights of the generators. The two variables are orthogonal: c acts on the arity axis; u acts on the prime axis. No rotation in one can reach the other.

42.1.0.3 3. Analytic continuation of the shadow obstruction tower. Continue the arity variable t in the shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ to complex values, obtaining a Riemann surface.

Status: genuine mathematics, no bridge. The shadow generating function is algebraic of degree 2 over $\mathbf{Q}(c)(t)$. Its analytic continuation produces a branched double cover of the t-line, ramified at the zeros of the shadow metric Q_L. The Galois group $\mathrm{Gal}(K_L/F) = \mathbf{Z}/2$ is the Koszul sign. The branch points determine a quadratic number field at each rational c (for instance $\mathbf{Q}(\sqrt{-10})$ for the Ising model).

This is a proved result about the *algebraic* structure of the shadow obstruction tower. But the resulting number field lives on the arity axis, not on the spectral axis. The discriminant of Q_L at a rational specialisation of c determines a quadratic Dirichlet L-function $L(s, \chi_d)$, which *is* an object on the spectral axis. However, this L-function is a classical object from algebraic number theory (determined by the Legendre symbol); it is not a new L-function produced by the bar construction. The bridge from the arity axis to the spectral axis goes through the graph sum, which is an irreducible intermediary, not an integral transform.

42.1.0.4 4. Resonances and the completed bar complex. The MC4 resonance-filtered bar-cobar machine (Theorem **??**) produces a completed bar complex $\hat{B}(\mathcal{A})$ with a finite resonance rank $\rho(\mathcal{A})$. The resonance data might carry complex spectral information.

Status: dead end for zeta zeros. The resonance rank classifies the *algebraic* completion difficulty of the bar-cobar tower. For Virasoro, $\rho = 1$ (the curvature m_0 is the single resonance). The resonance data controls the convergence of the shadow partition function (the growth rate $\rho(\mathcal{A})$ determines the radius of convergence), which is a statement about the arity expansion. It does not constrain the analytic properties of L-functions. The completed bar complex solves the MC4 problem (bar-cobar for non-quadratic infinite-generator algebras), which is a homological algebra problem, not an analytic number theory problem.

42.1.0.5 5. Borel resummation and resurgence. The shadow partition function $Z^{\mathrm{sh}}(\hbar) = \sum_{g\geq 1} F_g\, \hbar^{2g}$ is convergent for the scalar (leading κ) sector. The string partition function $\sum F_g (2g)!\, \hbar^{2g}$ is factorially divergent. Borel resummation might connect the perturbative (convergent) shadow series to the non-perturbative (resurgent) string series.

Status: genuine mathematics, wrong target. Borel resummation applies when a formal power series has known factorial growth. The shadow series *converges absolutely* for $|\hbar| < 2\pi$ (the Bernoulli decay gives coefficients of order $(2\pi)^{-2g}$, not $(2g)!$). There is nothing to resummate. The Borel transform of the string series $\sum (2g)!\, \hbar^{2g}$ has a non-trivial Stokes structure (action singularities on the Borel plane), and the resurgent trans-series involves non-perturbative effects (D-branes, ZZ instantons) that are *external* to the bar construction. The bar complex is the perturbative algebraic engine; it does not access non-perturbative data.

More precisely: the shadow series captures the *tautological* genus expansion $F_g = \kappa\, \lambda_g^{\mathrm{FP}}$, which is a polynomial in κ at each genus. The string partition function involves the *full* genus expansion including non-tautological classes on $\overline{\mathcal{M}}_g$. These are different objects. The shadow obstruction tower is the Hodge-tautological projection of the full amplitude; the non-tautological part is invisible to the bar complex.

42.1.0.6 6. Spectral curves and topological recursion. The Eynard–Orantin topological recursion produces a spectral curve from genus-0 data, and the higher-genus amplitudes $\omega_{g,n}$ are residues on this curve. The spectral curve has complex branch points, which might serve as a bridge to complex spectral data.

Status: genuine mathematics, not a bridge to zeta zeros. The shadow CohFT (Theorem **??**) satisfies equivariance and splitting. The flat identity is conditional (the vacuum $|0\rangle$ must lie in the generating space V). Without the flat unit, Teleman reconstruction does not apply, and the shadow CohFT does not arise from a Frobenius manifold or a spectral curve in the standard sense.

Even with a spectral curve, the relationship to zeta zeros would require identifying the branch points of the curve with the zeros of an L-function. No such identification exists. The spectral curve encodes the genus-0 OPE data (the same three parameters κ, α, S_4 that determine the shadow obstruction tower), not the analytic number theory of the partition function. The "spectral" in "spectral curve" refers to the spectrum of a matrix model, not to the automorphic spectrum.

42.1.0.7 7. Non-commutative geometry (Connes–Marcolli). The Bost–Connes system produces a quantum statistical mechanical system whose partition function is $\zeta(\beta)$ and whose KMS phase structure encodes arithmetic symmetry breaking. The zeta zeros are not KMS states. Connes-style trace formulae approach RH through a Weil-type positivity on a non-commutative space.

Status: the most structurally compatible approach, but no bridge has been constructed. The Bost–Connes system has a Hecke algebra of symmetries that is compatible with the Verdier–Hecke commutation (Theorem **??**). The non-commutative geometry approach replaces the function field with a non-commutative adele class space, and the Weil positivity criterion becomes a trace-class condition on a specific operator. In principle, the bar complex at the self-dual point $c = 13$ (where Verdier self-duality gives an involution on the factorization coalgebra) could feed into the non-commutative trace formula through the sewing determinant.

In practice, no such construction exists. The Bost–Connes system is defined over $\mathbf{Q}$ (the rational numbers as a number field); the bar complex is defined over $\mathbf{C}$ (complex algebraic curves). The arithmetic descent from function fields to number fields (Gap B in the Langlands analysis) is the fundamental obstruction, and it applies with full force here. The non-commutative geometry approach provides a framework in which the question could in principle be asked, but the bar construction does not supply the input data that the framework requires.

42.1.0.8 8. Analytic Langlands. The analytic Langlands programme (Etingof–Frenkel–Kazhdan) studies the spectral decomposition of the Hecke operators on the space of half-densities on $\mathrm{Bun}_G(X)$ for a curve X over $\mathbf{C}$. The opers (which the bar complex at critical level computes, Theorem **??**) appear as the spectral side.

Status: the most mathematically natural connection, but pertains to the geometric (function-field) Langlands, not the arithmetic (number-field) Langlands. The bar complex at critical level $k = -h^\vee$ provides a chain-level model for the oper side of the geometric Langlands correspondence. At generic level, the shadow obstruction tower deforms the oper data. This interpolation across levels is genuinely new and is not visible in the original Langlands framework.

However, the geometric Langlands programme operates over function fields (algebraic curves over $\mathbf{C}$), where the analogue of the Riemann Hypothesis is Weil's theorem (proved by Deligne). The number-field Riemann Hypothesis requires arithmetic descent, which is the deepest unsolved problem in the Langlands programme and is not addressed by the bar construction. The analytic Langlands provides the correct categorical framework for the oper/shadow interpolation, but it does not bridge the real-to-complex spectral gap for number-field L-functions.

42.2 The structural verdict

Three of the eight approaches produce genuine mathematics (analytic continuation of the shadow obstruction tower, spectral curves, analytic Langlands). The remaining five are dead ends for the real-to-complex question. None of the eight bridges the gap from the bar complex to zeta zeros.

The structural reason is a scope theorem for the constructions currently present here, not a theorem forbidding every possible finite definition of an entire function.

(i) *The primary-line shadow tower is determined by three numbers.* The MC equation on a single primary line determines that infinite shadow obstruction tower from (κ, α, S_4). These three OPE coefficients are a finite-dimensional algebraic datum.

(ii) *The zeros of an L-function encode infinitely many independent quantities.* The zero set $\{\rho_n\}$ of $\zeta(s)$ is a countably infinite collection of complex numbers, each carrying independent transcendental data (the imaginary parts γ_n are believed to be algebraically independent).

(iii) *No functor constructed here reaches the zero set.* The present bar/Koszul/shadow functors do not produce the analytic spectral parameter or the Hadamard product data of the relevant L-function. Dimension counting is heuristic evidence for the gap, not an independent impossibility theorem.

The real/complex divide is therefore a **hard boundary for the current single-algebra construction**: the bar complex of a fixed chiral algebra $\mathcal{A}$ produces algebraic shadow data, while zero locations require analytic spectral data not constructed by the bar functor.

42.3 The bootstrap escape and why it fails

The single-algebra obstruction suggests a multi-algebra bootstrap: use the MC constraints from *all* algebras simultaneously and ask whether the intersection of constraint loci forces zeros onto the critical line. This is the content of the "simultaneous-family closure" mechanism.

The bootstrap fails for three independent reasons.

(i) *The relevant sewing transforms, when defined, live in the spectral variable rather than the central-charge variable.* For the Virasoro family, varying c changes the shadow coefficients but does not by itself supply new spectral Dirichlet data. Varying c changes the shadow coefficients but not the spectral variable u in any constructed transform. The bootstrap therefore has no proved leverage on zero locations.

(ii) *The Rankin–Selberg unfolding erases the MC structure.* The unfolded integral depends only on the Fourier coefficients of the partition function, not on the convolution-algebraic structure that the MC equation controls (Proposition ??). The MC equation constrains the coefficients c_j in the spectral decomposition $\hat{Z}^c = c_E\, E_k + \sum c_j\, f_j$, but the L-functions $L(s, f_j)$ are properties of the eigenforms f_j, which are independent of the algebra $\mathcal{A}$.

(iii) *The MC equation is a flatness condition, not a positivity condition.* The Li criterion requires positive definiteness of a specific bilinear form. The MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ is a quadratic integrability condition. The chain MC $\longrightarrow$ Rankin-Selberg $\longrightarrow$ Li positivity does not close because the Rankin–Selberg transform is not sign-preserving.

42.4 The Beilinson verdict

The following false ideas are dismissed:

(i) "Bar-cobar homotopy equivalence forces Epstein zeros onto the critical line." *False.* Davenport–Heilbronn counterexamples exist for generic Epstein. Bar-cobar gives homotopy equivalence, not arithmetic positivity.

(ii) "The MC equation implies Li positivity." *False.* MC is flatness; Li is positivity.

(iii) "The shadow obstruction tower descends to a functional equation of $S_{\mathcal{A}}(u)$." *False.* The shadow obstruction tower acts on the arity variable t; the sewing lift acts on the spectral variable u. Complementarity does not produce a functional equation because $S_{\mathcal{A}}(u)$ is c-independent.

(iv) "The fixed point $c = 13$ of the Koszul involution is related to the critical line $\mathrm{Re}(s) = \frac{1}{2}$ of ζ." *False.* These are fixed loci of different involutions on different spaces (AP9: same name, different object). The Koszul involution depends on the Virasoro central charge formula; the zeta functional equation depends on the Euler product. These have no common input.

(v) "A surface positivity mechanism (Hodge index on a modular surface) converts quartic compatibility into Li-type positivity." *False.* No such surface has been constructed; no polarisation has been identified; and the Li tests for the Heisenberg sewing lift fail in the recorded computations: the modified derivative/uncompleted coefficients first fail at $n = 7$, while the completed spectral zero-sum normalisation first fails at $n = 974$.

(vi) "The three involutions (Verdier, Galois, Koszul) are the same involution acting at different levels." *False.* They are three distinct order-2 transformations acting on different objects in different categories with different fixed points. Their coincidence at the self-dual locus $c = 13$ is an accident of the fixed-point condition, not a structural identification.

42.5 What survives

The investigation produced the following genuine mathematics, all of which stands independently of any connection to the Riemann Hypothesis.

(i) *The MC recursion and Carleman-type determinacy.* On the lanes where the recurrence is proved, the MC equation at arity $r + 1$ determines S_{r+1} from the preceding shadow data. Carleman-type growth/determinacy applies to the resulting full sequence when a positive Hamburger/Stieltjes moment problem has been constructed. For generic class M branch-cut data the associated measure can be complex, so this is not yet ordinary Carleman uniqueness. It is not a theorem that three parameters (κ, α, S_4) determine an arbitrary spectral measure outside those recurrence hypotheses.

(ii) *Route C prime-locality.* The MC recursion can propagate Hecke equivariance from low-arity character data to higher arities on the arithmetic lanes where the Hecke action and recurrence hypotheses have both been constructed.

(iii) *The shadow-spectral correspondence for lattice VOAs.* Shadow depth equals the number of critical lines minus 1.

(iv) *Verdier–Hecke commutation on the arithmetic lane.* At self-dual points, assuming $Z_{\mathcal{A}}$ is a modular form in $M_k(\Gamma_0(N))$ and admits a Hecke eigen-decomposition, Verdier–Hecke commutation forces the corresponding L-function factorisation.

(v) *The Galois structure of the shadow obstruction tower.* The shadow generating function generates a quadratic extension K_L/F with Galois group $\mathbf{Z}/2$ (the Koszul sign). The discriminant at each rational c determines a quadratic number field. The splitting of primes in this field governs the mod-p behaviour of the shadow obstruction tower.

(vi) *The bar construction as non-abelian Fourier transform.* For abelian chiral algebras, the bar construction literally specialises to the Fourier–Mukai transform. For non-abelian algebras, it is a genuine generalisation that shares four structural properties with the Fourier transform (intertwining, inversion, Plancherel, characteristic function) but involves irreducibly n-body interactions with no factored kernel.

(vii) *The critical-level bar complex computes opers.* $H^n(B(\hat{\mathfrak{g}}_{-h^\vee})) = \Omega^n(\mathrm{Op}_{\mathfrak{g}^\vee}(D))$. This is a genuine piece of local geometric Langlands.

42.6 The deepest structural question

Is the real/complex divide *fundamental* (no bridge exists) or *contingent* (a bridge exists but has not been found)?

The answer depends on the scope.

(i) *For a single algebra in the present construction:* no bridge is constructed. The primary-line shadow data do not include the analytic spectral variable or the zero set.

(ii) *For the standard landscape simultaneously:* the obstruction is that $S_{\mathcal{A}}(u)$ is c-independent for each family, so varying c gives no spectral information. The bootstrap has nothing to bootstrap.

(iii) *For a hypothetical extension of the bar complex that accesses the full partition function:* the question becomes whether the representation category (which determines the primary spectrum and hence ε_s^c) can be reconstructed from the bar complex. The answer is no: the bar complex determines the Koszul dual $\mathcal{A}^!$ and the shadow tower, but not the representation category. The primary spectrum is a representation-theoretic invariant; the bar complex is an algebra-theoretic invariant. They live in different categories.

The bar complex is an algebraic object (a factorisation coalgebra on $\mathrm{Ran}(X)$ with coefficients in rational functions of c). Zeta zeros are analytic objects (the zero set of an entire function constructed from a Dirichlet series with multiplicative coefficients). The current construction supplies no functor from the former data to the latter.

The three genuine connections between the bar complex and number theory are all at the *algebraic* level: the Feigin–Frenkel centre (opers = critical-level bar cohomology), the Hecke–Verdier commutation on its proved lane, and the shadow quadratic function fields ($\mathbb{Q}(c)(t)(\sqrt{Q_L})$). None of these accesses the *analytic* properties (zero locations, growth rates, functional equations) of the associated L-functions. The algebraic structure constrains the input to the Langlands machine; it does not constrain the output.

43 The modular direction

The bar construction of a chiral algebra on a curve computes two things at once. At genus 0, it is a factorisation coalgebra on $\mathrm{Ran}(X)$: the five theorems of Volume I. At all genera, it is an algebra over the Feynman transform of the commutative modular operad: the Getzler–Kapranov structure. These are two different algebraic structures living in two different geometric directions. Neither reduces to the other.

The factorisation structure lives on the curve X. Points collide; the local propagator $\eta_{ij} = d\log(z_i - z_j)$ extracts residues; Arnold's relation and Borcherds/Jacobi kill the genus-zero square. At higher genus the prime/theta propagator introduces curvature. The modular structure lives on the collection $\{\overline{\mathcal{M}}_{g,n}\}$ of moduli spaces of stable curves. Curves degenerate; the clutching maps ξ_{sep} and ξ_{ns} sew them together; the boundary-of-boundary relation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ forces $D^2 = 0$. Point-collision and curve-degeneration are categorically different operations: the first is factorisation (keeping apart), the second is composition (bringing together).

Observation 43.1 *(Two codimension-1 operations).* A chiral algebra $\mathcal{A}$ on a curve X admits two independent "codimension-1" enhancements:

(i) **The Swiss-cheese centre** (the chiral Deligne–Tamarkin theorem). The chiral Hochschild cochain complex $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ is the terminal local $\mathsf{SC}^{\mathrm{ch,top}}$-algebra. This adds one *topological* direction $\mathbb{R}$, producing a 3d holomorphic-topological theory on $X \times \mathbb{R}$. The mechanism is the operadic centre construction.

(ii) **The modular convolution** (the Getzler–Kapranov Feynman transform). The modular convolution dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \prod_{g,n} \mathrm{Hom}_{\Sigma_n}(C_*(\overline{\mathcal{M}}_{g,n}), \mathrm{End}_{\mathcal{A}}(n))$ extends genus-0 data to all genera. The Maurer–Cartan element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ packages the full genus expansion. The mechanism is the Feynman transform: a convolution of the modular cooperad of stable curves with the endomorphism operad of $\mathcal{A}$.

The first adds a spatial dimension. The second adds the genus tower. They are compatible (the open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}} = \Theta_{\mathcal{A}} + \sum_j \mu^{\mathcal{M}_j}$ packages both), but they are not the same construction.

The analogy with Tamarkin's theorem is this. Tamarkin proves: for an E_d-algebra A, the Hochschild cochain complex $C^*(A, A)$ is an E_{d+1}-algebra. The mechanism is the Swiss-cheese centre, and the extra direction is topological. The chiral Deligne–Tamarkin theorem is the $d = 2$ (holomorphic) incarnation: the extra direction is $\mathbb{R}$. No known instance of the Hochschild construction adds a holomorphic direction.

The modular convolution is a *different* functor. Both are convolutions against a cooperad: Tamarkin uses the little-discs cooperad; Getzler–Kapranov uses the modular cooperad of stable curves. Both produce Maurer–Cartan-controlling Lie-type algebras. The distinction is that Tamarkin adds a spatial dimension, while Getzler–Kapranov adds the genus.

43.1 The quantum L_∞ structure

The genus-completed bar differential $D_{\mathcal{A}} = \sum_{g \geq 0} \hbar^g d^{(g)}_{\mathcal{A}}$ decomposes into five components:

$$D = d_{\mathrm{int}} + [\tau, -] + d_{\mathrm{sew}} + d_{\mathrm{pf}} + \hbar\Delta.$$

The first two are curve-side (the internal differential and the bar–cobar twisting cochain). The last three are moduli-side: d_{sew} inserts a separating node, d_{pf} corrects for codimension-≥ 2 boundary strata of the log-FM compactification, and $\hbar\Delta$ is the non-separating clutching $\overline{\mathcal{M}}_{g,n+2} \to \overline{\mathcal{M}}_{g+1,n}$ that raises genus by 1.

The resulting algebraic structure is a completed modular convolution L_∞-type algebra: operations $\{\ell^{(g)}_n\}$ indexed by arity $n \geq 1$ and genus $g \geq 0$, satisfying the quantum master equation

$$\sum_{n \geq 1} \sum_{g \geq 0} \frac{\hbar^g}{n!} \ell^{(g)}_n (\Theta^{\otimes n}_{\mathcal{A}}) = 0. \tag{5}$$

At genus 0 this is the classical L_∞ MC equation. The genus-1 term is the one-loop anomaly: $\kappa(\mathcal{A}) \cdot \omega_1$. Higher genera are nonlinear interference terms determined inductively by the genus-g MC component.

43.2 The BCOV structural analogy

The genus-by-genus decomposition of (43.1) is:

$$\begin{aligned} \text{genus } 0: &\quad d_{\Theta^{(0)}}(\Theta^{(0)}) + \tfrac{1}{2}[\Theta^{(0)}, \Theta^{(0)}] = 0, \\ \text{genus } g: &\quad d_{\Theta^{(0)}}(\Theta^{(g)}) = -\tfrac{1}{2} \sum_{g_1+g_2=g} [\Theta^{(g_1)}, \Theta^{(g_2)}] - \Delta(\Theta^{(g-1)}) - d^{(g)}_{\mathrm{sew}} - d^{(g)}_{\mathrm{pf}}. \end{aligned} \tag{6}$$

Compare with the BCOV holomorphic anomaly equation for the genus-g free energy F_g on the moduli space of complex structures of a Calabi–Yau threefold:

$$\bar{\partial}_{\bar{i}} F_g = \tfrac{1}{2}\, \bar{C}_{\bar{i}}^{\,jk}\Big(D_j D_k F_{g-1} + \sum_{r=1}^{g-1} D_j F_r \cdot D_k F_{g-r} \Big).$$

Observation 43.2 *(BCOV–shadow structural parallel).* The two equation systems share the same combinatorial backbone:

(1) *Inductive genus determination:* genus-g data is determined by genera $< g$.
(2) *Bilinear term:* a sum $\sum_{g_1+g_2=g}$ (genus-g_1) $\cdot$ (genus-g_2) from separating degenerations.
(3) *Genus-raising operator:* $\Delta(\Theta^{(g-1)})$ in the MC hierarchy, $\bar{C}^{jk} D_j D_k F_{g-1}$ in BCOV. Both produce genus-g data from genus-$(g-1)$ data via non-separating degeneration.

Both are instances of a single universal pattern: a genus recursion governed by a modular operad, with the bilinear term from separating clutching and the genus-raising term from non-separating clutching.

The decisive asymmetries are:

- The BCOV left-hand side is $\bar{\partial}_{\bar{i}} F_g$ (an anti-holomorphic derivative on CY moduli). The MC left-hand side is $d_{\Theta^{(0)}}(\Theta^{(g)})$ (the tree-twisted differential in the deformation complex).
- The shadow obstruction tower has no holomorphic anomaly: $\Theta^{(g)} \in H^*(\overline{\mathcal{M}}_g)$ is algebraic, so $\bar{\partial} = 0$ on this algebraic projection. The BCOV free energies F_g are analytic functions on CY moduli with genuine holomorphic anomaly ($\bar{\partial} F_g \neq 0$).
- The MC hierarchy has two additional terms ($d^{(g)}_{\mathrm{sew}}$ and $d^{(g)}_{\mathrm{pf}}$) from the log-FM boundary geometry that have no BCOV counterpart.
- The BCOV genus expansion diverges ($(2g)!$ growth). The shadow partition function converges (Bernoulli decay $1/(2\pi)^{2g}$).

The shadow tower is algebraic and narrower: it lives on the curve-side modular bar construction, not on a CY3 moduli space. The BCOV Kodaira–Spencer chiral algebra is one specific example: the boundary chiral algebra of 6d holomorphic Chern–Simons on $\mathbb{C}^3$.

43.3 The shadow connection as BCOV propagator

Warning 43.3 *(The BCOV analogy is combinatorial, not structural).* The MC hierarchy (43.2) and the BCOV holomorphic anomaly equation share a combinatorial backbone (genus recursion governed by a modular operad), but they are *not* structurally isomorphic. The decisive obstructions:

(1) The shadow obstruction tower has *no holomorphic anomaly*: $\Theta^{(g)} \in H^*(\overline{\mathcal{M}}_g)$ is algebraic, so $\bar{\partial} = 0$. The BCOV equation computes $\bar{\partial} F_g \neq 0$. The chiral left-hand side is identically zero.
(2) The BCOV propagator S^{ij} is a symmetric 2-tensor on CY moduli (finite-dimensional, special Kähler). The shadow connection ∇^{sh} is a scalar connection on a 1D arity line. These are different tensor types on different spaces.
(3) The BCOV propagator is non-canonical (it depends on a Kähler metric). The bar propagator $d \log E(z, w)$ is canonical. No metric choice, no anomaly.
(4) The MC hierarchy has two additional terms ($d^{(g)}_{\mathrm{sew}}$, $d^{(g)}_{\mathrm{pf}}$) from the log-FM boundary geometry with no BCOV counterpart.
(5) The shadow partition function converges ($1/(2\pi)^{2g}$ Bernoulli decay); the BCOV expansion diverges ($(2g)!$ growth).

The correct physics analogue of the MC hierarchy is the DVV/Virasoro recursion for CohFTs, not the BCOV holomorphic anomaly. The shadow CohFT satisfies Virasoro constraints via Teleman reconstruction; the Eynard–Orantin recursion is identified as the scalar MC shadow.

Observation 43.4 *(What does survive: the Gauss–Manin variation).* For a 1D family $\{\mathcal{A}_s\}_{s\in B}$ of chirally Koszul algebras (e.g., $V_k(\mathfrak{g})$ as k varies), the shadow generating function $H(t;s) = t^2\sqrt{Q_L(t;s)}$ satisfies the first-order ODE

$$\frac{\partial}{\partial s}H(t;s) = \frac{t^2}{2H(t;s)}\cdot\frac{\partial}{\partial s}Q_L(t;s),$$

which is a Gauss–Manin-type variation on the 1D theory space. The coefficients are $\kappa'(s)$, $\alpha'(s)$, $S_4'(s)$–-the derivatives of three OPE invariants along the family. This is an immediate corollary of the Riccati algebraicity theorem. The flat modular connection $D_\Theta = d + [\Theta_{\mathcal{A}}, -]$ on bar cohomology over $\mathcal{M}_g$ is the algebraic flatness condition. Calling its genus expansion BCOV-type is a combinatorial analogy, not a structural identification with the BCOV anomaly equation.

44 The modular cooperad conjecture

The chiral Deligne–Tamarkin theorem (Observation 43.1(i)) adds one topological direction at genus 0. The modular convolution (Observation 43.1(ii)) adds the genus tower. The natural completion of this picture is to extend the genus-0 centre theorem to all genera:

Conjecture 44.1 *(The modular open/closed cooperad (CT-2)).* The open-sector factorisation dg-category $\mathcal{C}_{\mathrm{op}}$ admits a modular cooperad structure $\delta^{\mathrm{op}}_{g,n} : \mathcal{C}_{\mathrm{op}} \to \mathcal{C}_{\mathrm{op}}^{\otimes n}$ whose cocomposition maps are compatible with clutching on bordered Fulton–MacPherson compactifications, whose trace recovers the annulus identification $\mathrm{HH}_*(\mathcal{A}_b) \simeq \mathrm{Tr}_{\mathcal{A}}$ at genus 1, and whose full structure determines and is determined by the open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$.

The three theorems in the Tamarkin lineage then form a single hierarchy:

Tamarkin:	A is E_d	$\Longrightarrow$ $C^*(A,A)$ is E_{d+1}
Chiral Deligne–Tamarkin:	$\mathcal{A}$ is chiral	$\Longrightarrow$ $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\mathcal{A})$ is Swiss-cheese
CT-2:	$\mathcal{A}$ is chirally Koszul	$\Longrightarrow$ conjecturally Swiss-cheese extends to modular Swiss-cheese

Each row adds one layer of structure: Tamarkin adds one topological direction; the chiral centre adds the $\mathbb{R}$-direction at genus 0; CT-2, if constructed, would extend this to all genera via the cooperad on $\mathcal{C}_{\mathrm{op}}$.

Remark 44.2 *(Revised status of CT-2).* The status of CT-2 decomposes as follows.

What is now known to exist.

(a) The bordered FM compactification $\overline{C}^{\mathrm{oc}}_{k,\vec{m}}(X,D,\tau)$ is *fully constructed* at all genera with all five blowup steps proved (Construction constr:bordered-fm: interior, boundary, mixed, nodal). The four-type boundary decomposition and normal-crossings property are proved (Theorem thm:bordered-fm-properties).

(b) The codimension-2 identities ($d^2_{\mathrm{bdy}} = 0$, $d^2_{\mathrm{mix}} = 0$, $[d_{\mathrm{bdy}}, d_{\mathrm{mix}}] = 0$, $[d_{\mathrm{bdy}}, d_{\mathrm{sew}}] + [d_{\mathrm{mix}}, d_{\mathrm{pf}}] = 0$) follow from $\partial^2 = 0$ on the manifold with corners. These are *proved*.

(c) The doubling strategy (form the double $d\Sigma = \Sigma \cup_{\partial\Sigma} \bar{\Sigma}$, FM-compactify the double, restrict to the $\mathbb{Z}/2$-fixed locus) *works* for the interval-bordered case and is already proved in Vol II (Theorem thm:doubling-rwi).

(d) The annular bordered FM (cyclic ordering on circles) is proved in Vol II (Theorem thm:bordered-annular) but marked conjectural in Vol I (concordance line 3327). This is a cross-volume status inconsistency to be reconciled.

(e) The chain-level MC equation $(D^{\mathrm{oc}})^2 = 0$ and the open-closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ are proved at all genera.

The gap is categorical, not geometric. The manifolds with corners exist. The chain-level identities are proved. What remains is the passage from *chains on bordered FM* to *functors between dg-categories*. The revised sub-problem list:

(P1′) *The clutching correspondence.* For each stable bordered graph Γ, the clutching map $\xi_\Gamma\colon \prod_v \overline{C}^{\mathrm{oc}}_{k_v,\vec{m}_v} \to \overline{C}^{\mathrm{oc}}_{k,\vec{m}}$ is a proper map of manifolds with corners. *Constructed* (Step 5 of Construction constr:bordered-fm). The fiber product decomposition at nodal strata is equation (**??**).

(P2′) *The chain-level cooperad.* A proper clutching map gives pushforward on chains. A cooperadic map in the opposite direction requires choosing a cochain, Borel–Moore/Gysin, or correspondence model with the needed orientations and transversality. Coassociativity should follow from $\partial^2 = 0$ on corner strata once such a chain model is fixed.

Status: the geometric ingredients (properness + corner cancellation) are proved. The actual chain/cochain cooperad is a construction target, not a packaging formality.

(P3′) *The categorical lift.* The open-closed convolution algebra $\mathfrak{g}^{\mathrm{oc}}$ is defined as $\mathrm{Hom}(C_*(\overline{C}^{\mathrm{oc}}), \mathrm{End}^{\mathrm{oc}})$. An MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ in this convolution produces, at each stable bordered graph Γ, a multilinear operation $\Theta^{\mathrm{oc}}_\Gamma\colon \mathcal{A}^{\otimes k} \otimes \bigotimes \mathcal{M}_j^{\otimes m_j} \to \mathcal{A} \otimes \bigotimes \mathcal{M}_j$. These operations are the *structure maps* of the cooperad evaluated on the MC element.

The categorical lift asks: do these structure maps assemble into *functors* $\delta^{\mathrm{op}}_{g,n}\colon \mathcal{C}_{\mathrm{op}} \to \mathcal{C}_{\mathrm{op}}^{\otimes n}$?

The answer depends on the *chart-independence* of the construction. The MC element $\Theta^{\mathrm{oc}}_{\mathcal{A}}$ is computed in a chart $\mathcal{A}_b = \mathrm{End}(b)$. Different charts give gauge-equivalent MC elements (by Morita invariance of the cyclic deformation complex). The cooperad on $\mathcal{C}_{\mathrm{op}}$ must be chart-independent.

Strategy: the Brochier–Woike machine [`?BrochierWoike2022`] constructs modular functors from E_2-algebras in ribbon categories. The chiral Deligne–Tamarkin theorem provides the E_2 input ($C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A})$ is E_2). The Calabi–Yau structure on $\mathcal{C}_{\mathrm{op}}$ provides the ribbon condition. The resulting modular functor is Morita-invariant by construction. Its dual is the cooperad on $\mathcal{C}_{\mathrm{op}}$.

Status: the E_2 input and CY structure are proved. The verification that they satisfy the Brochier–Woike hypotheses is open.

(P4′) *Decomposition into existence, compatibility, uniqueness.* CT-2 decomposes into three layers:

  (i) *Existence*: $\mathcal{C}_{\mathrm{op}}$ admits a modular cooperad structure. This is the categorical lift (P3′ above).

  (ii) *Compatibility*: the closed projection of the cooperad recovers $\Theta_{\mathcal{A}}$. This follows from existence plus the Getzler–Kapranov correspondence (which is proved at the chain level).

  (iii) *Uniqueness*: the cooperad is determined by $\Theta^{\mathrm{oc}}_{\mathcal{A}}$. This requires a categorical Getzler–Kapranov correspondence (an $(\infty,2)$-categorical integration theorem), which does not yet exist.

The depth-dependent road map. For class G (Heisenberg), the cooperad has no higher coherences beyond binary clutching. The categorical lift is simplest here: the modular functor is determined by the single scalar κ, and the Brochier–Woike machine should apply directly. For class L (affine KM), one ternary cocomposition. For class M (Virasoro), the cooperad is expected to require a completed/pro-object treatment. This is a scalar-shadow roadmap only: finite arity depth of the closed scalar tower does not by itself bound dg-categorical modular-cooperad coherences.

44.1 The CT-2 proof programme: a definitive road map

The conjecture decomposes into a five-phase proof strategy.

The core architecture. The conjecture CT-2 (Conjecture 44.1) decomposes into three layers (P4′):

(i) *Existence*: C_{op} admits a modular cooperad structure;

(ii) *Compatibility*: its closed projection recovers $\Theta_{\mathcal{A}}$;

(iii) *Uniqueness*: the cooperad is determined by $\Theta^{\mathrm{oc}}_{\mathcal{A}}$.

Layer (ii) follows from (i) by the Getzler–Kapranov correspondence (proved at the chain level). Layer (iii) requires a categorical Getzler–Kapranov theorem (an $(\infty, 2)$-categorical integration theorem) that does not yet exist. The proof programme targets (i), with (ii) automatic and (iii) deferred to the $(\infty, 2)$-frontier.

The gap identified by both swarms is *categorical, not geometric*: the bordered FM compactification exists at all genera (Construction constr:bordered-fm), the geometric ingredients are proved (properness and corner cancellation), and $D^2_{\mathrm{oc}} = 0$ holds at all genera. The obstacle is the passage from chains on bordered FM to functors between dg-categories.

There are two complementary strategies. Strategy A (Phases 1–4) is a class-by-class construction that exploits the shadow depth filtration. Strategy B (Phase 5) uses the Brochier–Woike machine to bypass the class-by-class approach. Both are viable; Strategy B is more efficient if its hypotheses can be verified.

Phase	Target	Shadow depth	Key step	Difficulty
1	Heisenberg (class G)	$r_{\max} = 2$	One-particle Fredholm	Medium
2	All class G	$r_{\max} = 2$	Wick theorem	Medium
3	Classes L and C	$r_{\max} \in \{3, 4\}$	Finitely many cocompositions	Medium–High
4	Class M (Vir, W_N)	$r_{\max} = \infty$	Pro-cooperad + Mittag-Leffler	High
5	All Koszul (Brochier–Woike)	any	Verify BW hypotheses	Medium–High

Phase 1: CT-2 for Heisenberg

Proposition 44.3 *(Heisenberg determinant-valued clutching theorem).* Let $\mathcal{H}_\kappa$ be the rank-one Heisenberg chiral algebra and let Γ be a stable bordered graph with sewing parameters $|q_e| < 1$. The one-particle Bergman sewing operators attached to the edges of Γ define, after second quantisation and Fock trace, scalar amplitudes

$$A_\Gamma(\mathcal{H}_\kappa) = \mathrm{Tr}_{\mathrm{Sym}\,A^2(D)}\,\Gamma(K_\Gamma)^\kappa = \det(1 - K_\Gamma)^{-\kappa}.$$

These amplitudes are invariant under $\mathrm{Aut}(\Gamma)$, are associative under graph substitution at the one-particle operator level, and send the non-separating self-sewing graph to the Fock trace. At genus one this gives

$$A_{\Gamma_{\mathrm{ns}}} = \prod_{n\geq 1}(1 - q^n)^{-\kappa} = (q^{-1/24}\eta(q))^{-\kappa}.$$

Thus the Heisenberg CT-2 data are proved at the scalar/Fock amplitude level. This proposition does not construct a modular cooperad on C_{op} in dgCat; that requires the finite-model functor-packaging lemma stated below.

Theorem 44.4 *(Heisenberg determinant-line modular cooperad package).* Let $\mathcal{H}_\kappa$ be the rank-1 Heisenberg chiral algebra at level $\kappa \neq 0$. Let $\mathsf{Fock}_\kappa^{\det}(g, n)$ be the one-object Picard dg-category whose endomorphism line over a bordered stable curve is the completed determinant line of the Heisenberg one-particle sewing operator,

$$\det(1 - K_\Gamma)^{-\kappa}.$$

Then $\mathsf{Fock}_\kappa^{\det}$ is a modular cooperad valued in Picard dg-categories:

$$\partial_\Gamma^{\det} : \mathsf{Fock}_\kappa^{\det}(g, n) \longrightarrow \bigotimes_{v \in V(\Gamma)} \mathsf{Fock}_\kappa^{\det}(g_v, n_v).$$

Its scalar fibre functor is exactly the determinant amplitude of Proposition 44.3. Consequently:

(a) the cocomposition maps are compatible with clutching on the bordered FM compactification $\overline{C}^{\mathrm{oc}}_{k,\vec{m}}$;
(b) the trace recovers the annulus identification $\mathrm{HH}_*(\mathcal{H}_{\kappa,b}) \simeq \mathrm{Tr}_{\mathcal{H}_\kappa}$ at genus 1;
(c) the genus-g closed projection recovers $F_g(\mathcal{H}_\kappa) = \kappa \cdot \lambda_g^{\mathrm{FP}}$;
(d) the full structure is determined by the open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{H}_\kappa}$.

This is the canonical categorical package beyond the scalar Fock determinant. A stronger dgCat-valued modular cooperad on the full open sector $C_{\mathrm{op}}(\mathcal{H}_\kappa)$ is equivalent to choosing finite functorial Fock models over all bordered boundary strata whose determinant line is the Picard envelope above.

Proof. For a stable graph Γ, Heisenberg sewing is Gaussian: all higher cumulants vanish and every sewing edge contributes the Fredholm determinant line of the one-particle operator K_e. For two disjoint sewings the one-particle operators act on orthogonal summands, hence

$$\det(1 - (K_{e_1} \oplus K_{e_2})) = \det(1 - K_{e_1}) \det(1 - K_{e_2}).$$

For nested sewings, the Schur-complement identity for Fredholm determinants gives the same line isomorphism as sewing first inside the component and then along the outer edge. These two identities are precisely the modular cooperad associativity relations; automorphism equivariance follows because the determinant line is functorial under permutation of equal sewing blocks. Applying the fibre functor from the Picard dg-category to complex lines recovers $A_\Gamma(\mathcal{H}_\kappa) = \det(1 - K_\Gamma)^{-\kappa}$, and the non-separating one-loop graph gives the eta product. The final assertion is just the universal property of the Picard envelope: any finite dg-category model with the same sewing functors maps to its determinant line, and lifting back requires exactly such a choice of finite functorial models. □

Theorem 44.5 *(Finite Fock obstruction and the nuclear replacement).* For $\kappa \neq 0$ there is no finite dg-category $\mathcal{D}_{\mathrm{fin}}$ with finite-dimensional total Hochschild trace whose sewing trace recovers the full Heisenberg determinant

$$\prod_{n \geq 1} (1 - q^n)^{-\kappa}.$$

The strongest functorial lift of Theorem 44.4 is instead the pro-nuclear Fock cooperad

$$\mathsf{Fock}_\kappa^{\mathrm{nuc}} = \varprojlim_N \mathsf{Fock}_\kappa^{\leq N},$$

where $\mathsf{Fock}_\kappa^{\le N}$ is the finite-energy truncation generated by oscillator modes of energy at most N. Its categorical trace is the convergent limit of finite traces and its determinant-line shadow is $\mathsf{Fock}_\kappa^{\det}$.

Proof. The trace of a finite dg-category with finite-dimensional total Hochschild homology is a finite Laurent polynomial in the sewing parameter after diagonalising the energy operator on its finite trace space. The Heisenberg Fock trace is the infinite product $\prod_{n\ge 1}(1-q^n)^{-\kappa}$, with nonzero coefficients in arbitrarily high energy. Hence no finite trace object can realise the full determinant. Truncating oscillator modes at energy N gives a finite trace

$$\prod_{1\le n\le N}(1-q^n)^{-\kappa},$$

and the transition maps are the evident deletion of the top oscillator mode. The inverse limit in nuclear complete dg-categories has trace the analytic limit of these finite traces. Fredholm multiplicativity of determinants gives compatibility with all clutching maps. Passing to determinant lines is exactly the Picard shadow constructed above. □

Main steps.

Step 1. *The Heisenberg open sector requires a finite dualizable model.* The Weyl/Heisenberg module category is not Morita-trivial to Vect at this generality. The finite-energy truncations form the compact approximating system; the full determinant is their pro-nuclear limit. The proved input is the Heisenberg sewing theorem, not categorical Morita triviality.

Step 2. *One-particle Bergman reduction.* On a genus-g surface, the sewing amplitude is a Fredholm determinant: $Z_g(\mathcal{H}_\kappa;\tau) = \det'_{\mathrm{Fred}}(1-K_\tau^{(g)})^{-\kappa}$ where $K_\tau^{(g)}$ is the genus-g Bergman kernel operator. The clutching correspondence ξ_Γ at a stable bordered graph Γ factors through tensor products of one-particle propagators. *Existing result*: Theorem thm:heisenberg-one-particle-sewing.

Step 3. *From amplitudes to functors.* The clutching amplitudes assemble into dg-functors on the pro-nuclear Fock cooperad; determinant identities give their Picard shadow. A finite dg-category recovers only a finite-energy truncation. This is the categorical packaging problem, not linear algebra already solved.

Step 4. *Genus-1 verification.* The genus-1 trace must recover the annulus identification. For Heisenberg, $\mathrm{HH}_*(\mathrm{dgVect}^\kappa)\simeq\mathbb{C}$, and the cooperad trace maps to $\kappa/24 = F_1(\mathcal{H}_\kappa)$. *Existing result*: annulus trace theorem (Theorem thm:thqg-annulus-trace).

Step 5. *Compatibility with $\Theta_{\mathcal{H}}$.* The closed projection of the cooperad is the scalar κ at each genus. Since $\mathcal{H}_\kappa$ is class G (shadow depth 2), the modular cooperad is entirely determined by its binary cocomposition (separating clutching) and the genus-raising map (non-separating clutching). Both are one-parameter families controlled by κ. Compatibility with the MC element $\Theta_{\mathcal{H}} = \kappa\cdot\eta\otimes\Lambda$ is then a scalar identity at each (g,n).

Invoked results.

- Heisenberg sewing theorem (Theorem thm:heisenberg-sewing): PROVED.
- One-particle Bergman reduction (Theorem thm:heisenberg-one-particle-sewing): PROVED.
- Bordered FM compactification (Construction constr:bordered-fm): PROVED.
- Chain-level coassociativity ($\partial^2=0$): PROVED.
- Annulus trace theorem (Theorem thm:thqg-annulus-trace): PROVED.

What the theorem closes. The Picard-valued categorical package is now closed by Theorem 44.4: the determinant lines, their clutching isomorphisms, and their scalar Fock fibres form a modular cooperad. The only stronger statement not claimed there is a dgCat-valued lift on a chosen finite dualizable open-sector model; that lift requires the extra functor packaging datum named in the theorem.

Phase 1 proof: CT-2 for Heisenberg (class G)

The Picard-valued programme above is carried out by assembling the existing sewing and bordered-FM results with the determinant line functor. The stronger finite-model programme below is not a Morita-triviality assertion for the whole Heisenberg module category; it asks for a dgCat lift of the already constructed Picard modular cooperad after choosing a finite dualizable open-sector model.

Conjecture 44.6 *(CT-2 finite-model Heisenberg cooperad).* Let $\mathcal{H}_\kappa$ be the rank-1 Heisenberg chiral algebra at level $\kappa \neq 0$, and let $\mathcal{C}_{\mathrm{op}}$ be a finite dualizable open-sector model whose categorical trace recovers the Heisenberg sewing determinant. Then, conjecturally, $\mathcal{C}_{\mathrm{op}}$ admits a modular cooperad structure

$$\{\partial_\Gamma\}_{\Gamma \in \mathcal{G}^{\mathrm{bord}}}$$

indexed by stable bordered graphs, satisfying:

(i) *Well-definedness and equivariance.* Each cocomposition ∂_Γ is a well-defined bounded linear map, equivariant under $\mathrm{Aut}(\Gamma)$.

(ii) *Coassociativity.* For composable stable bordered graphs $\Gamma_1 \circ_e \Gamma_2$: $(\partial_{\Gamma_1} \otimes \mathrm{id}) \circ \partial_{\Gamma_2} = \partial_{\Gamma_1 \circ_e \Gamma_2}$.

(iii) *Self-sewing = trace.* The non-separating degeneration cocomposition is the categorical trace: $\partial_{\Gamma_{\mathrm{ns}}} = \mathrm{Tr}_{\mathcal{C}_{\mathrm{op}}}$. At genus 1, this recovers the annulus identification $\mathrm{HH}_*(\mathcal{H}_{\kappa,b}) \simeq \mathrm{Tr}_{\mathcal{H}_\kappa}$.

(iv) *Closed-sector compatibility.* The closed projection of the cooperad recovers the genus tower: $\pi_{\mathrm{cl}}(\partial_\Gamma(Z_g)) = \kappa \cdot \lambda_g^{\mathrm{FP}}$ for all g.

(v) *Scalar Quillen-norm amplitude.* The genus-g scalar analytic amplitude, after Quillen-norm trivialisation of the determinant line, is

$$Z_g(\mathcal{H}_\kappa;\, \Omega) = (\det \operatorname{Im} \Omega)^{-\kappa/2} \cdot (\det{}'_\zeta \Delta_{\Sigma_g})^{-\kappa/2}.$$

(vi) *Determination by* Θ^{oc}. The cooperad is determined by the open/closed MC element $\Theta^{\mathrm{oc}}_{\mathcal{H}_\kappa}$: the Getzler–Kapranov correspondence identifies cooperad cocompositions with evaluations of the MC element on stable bordered graphs.

Proof programme and proved reductions. The reductions below establish the analytic amplitude identities. The missing step is the promotion of these identities to dg-functorial cocomposition maps on the chosen finite open-sector model.

Setup: the Heisenberg data. The Heisenberg algebra $\mathcal{H}_\kappa$ has OPE $J(z)\,J(w) \sim \kappa/(z-w)^2$ (purely quadratic: no simple pole, hence no Lie bracket). Its shadow Postnikov tower terminates at arity 2: $\Theta_{\mathcal{H}_\kappa} = \kappa \cdot \eta \otimes \Lambda$, with all higher obstructions vanishing ($o_r = 0$ for $r \geq 3$). The modular characteristic is $\kappa(\mathcal{H}_\kappa) = \kappa$. The Koszul dual is $\mathcal{H}_\kappa^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ with $\kappa(\mathcal{H}_\kappa^!) = -\kappa$.

The Heisenberg sewing theorem (Theorem thm:heisenberg-sewing) and its one-particle refinement (Theorem thm:heisenberg-one-particle-sewing) establish:

(S1) The sewing envelope of $\mathcal{H}_\kappa$ is $\mathrm{Sym}\, A^2(D)$ (Moriwaki).

(S2) The completed bar differential is the closure of the Gaussian collision operator on Bergman tensors.

(S3) Pair-of-pants sewing is the second quantisation $\Gamma(R)$ of the one-particle Bergman restriction $R\colon A^2(D_{\mathrm{out}}) \to A^2(D_{\mathrm{in},1}) \otimes A^2(D_{\mathrm{in},2})$, with $|R_{n,m}| \le Cq^{(n+m)/2}$.

(S4) The sewing operator T_q on the reduced Bergman space $A^2_0(\partial D)$ has eigenvalues q^n $(n \ge 1)$ and is trace class: $\|T_q\|_1 = q/(1-q)$.

(S5) The genus-g scalar Quillen-norm amplitude is the Fredholm determinant $\det(1-K_g)^{-\kappa}$ on the one-particle Bergman space.

Step 1: the open-sector category $\mathcal{C}_{\mathrm{op}}$ is semisimple.

The semisimple Fock sector has objects $\mathcal{F}_\lambda$ $(\lambda \in \mathbb{C})$, generated by highest-weight vectors $|\lambda\rangle$ with $\alpha_0\,|\lambda\rangle = \lambda\,|\lambda\rangle$ and $\alpha_n\,|\lambda\rangle = 0$ for $n > 0$. No claim is made here that the full category $\mathrm{Perf}(\mathcal{H}_\kappa)$ is semisimple or Morita-trivial. The finite objects are the finite-energy truncations retained by the open boundary condition; the full Fock sector is their pro-nuclear completion. On these sectors the fusion rule is $\mathcal{F}_\lambda \boxtimes \mathcal{F}_\mu = \mathcal{F}_{\lambda+\mu}$ (abelian group law on charges).

Each finite-energy truncation of $\mathcal{F}_\lambda$ is dualizable in dgCat: $\mathcal{F}_\lambda^\vee \simeq \mathcal{F}_{-\lambda}$ under the contragredient (Shapovalov) pairing. The full determinant is not the trace of a single finite model; it is the trace of the pro-nuclear inverse system of finite-energy models.

Step 2: the cooperad cocomposition via one-particle Bergman reduction.

Fix a stable bordered graph Γ of type (g,n) with vertex set $V(\Gamma)$, edge set $E(\Gamma)$, and leg set $L(\Gamma)$ with $|L(\Gamma)| = n$. Each vertex v carries a genus $g_v \ge 0$ with $2g_v - 2 + n_v > 0$ (stability) and $g = \sum_v g_v + |E(\Gamma)| - |V(\Gamma)| + 1$ (genus identity).

Define the cocomposition

$$\partial_\Gamma\colon \mathrm{End}_{\mathcal{C}_{\mathrm{op}}}(\mathcal{F}_\lambda) \longrightarrow \bigotimes_{v\in V(\Gamma)} \mathrm{End}_{\mathcal{C}_{\mathrm{op}}}(\mathcal{F}_\lambda)$$

by the sewing formula: decompose a genus-g amplitude into vertex amplitudes connected by sewing operators. By (S3), each internal edge $e \in E(\Gamma)$ contributes a sewing operator T_{q_e} on the Bergman space $A^2(D)$, and each vertex v contributes the second quantisation $\Gamma(R_v)$ of its vertex restriction operator. Concretely:

$$\partial_\Gamma(Z) = \bigotimes_{v\in V(\Gamma)} Z_v, \qquad Z_v = \mathrm{Tr}_{\{e\,:\,e\text{ at }v\}}\Big(\Gamma\Big(\prod_{e\text{ at }v} T_{q_e}\Big)^{\kappa}\Big).$$

At the one-particle level:

$$\partial_\Gamma^{(1)}\colon K_g \longmapsto \bigotimes_{v\in V(\Gamma)} K_{g_v,n_v},$$

where K_{g_v,n_v} is the one-particle sewing kernel of the surface Σ_{g_v,n_v} at vertex v.

Well-definedness. Each T_{q_e} is trace class on $A^2(D)$ with $\|T_{q_e}\|_1 = q_e/(1-q_e)$ and operator norm $\|T_{q_e}\| = q_e < 1$ (by (S4)). The second quantisation $\Gamma(T)$ is trace class on $\mathrm{Sym}\,A^2(D)$ whenever T is trace class with $\|T\| < 1$. Hence ∂_Γ is a well-defined bounded linear map.

Equivariance. An automorphism $\sigma \in \mathrm{Aut}(\Gamma)$ permutes edges and vertices. The trace over Fock states is invariant under permutation of sewing channels (the trace is cyclically invariant and the Bergman operators on disjoint edges commute). Hence ∂_Γ descends to the quotient by $\mathrm{Aut}(\Gamma)$.

This verifies the amplitude-level part of (i); the dg-functorial lift remains part of the conjecture.

Step 3: coassociativity = multiplicativity of Fredholm determinants.

We establish (ii) in three sub-steps.

Sub-step 3a: reduction to one-particle operators. By (S3), every multi-particle sewing amplitude on $\mathrm{Sym}\, A^2(D)$ is the second quantisation of a one-particle operator: $A_P = \Gamma(R)$. By (S5), the trace of any composed sewing operator on the Fock space factors through the one-particle trace:

$$\mathrm{Tr}_{\mathrm{Sym}}\big(\Gamma(T_1 \circ T_2)\big) = \det(1 - T_1 \circ T_2)^{-\kappa}.$$

It suffices to establish coassociativity at the one-particle level.

Sub-step 3b: one-particle operators compose associatively. On $A^2(D)$, the sewing operators $T_{q_{e_1}}, T_{q_{e_2}}$ associated to distinct internal edges e_1, e_2 act on disjoint tensor factors of $A^2(D)^{\otimes |E(\Gamma)|}$ (each edge connects the Bergman spaces of two disks at the node). Operators on disjoint tensor factors commute:

$$T_{q_{e_1}} \circ T_{q_{e_2}} = T_{q_{e_2}} \circ T_{q_{e_1}} \quad \text{on } A^2(D_{e_1}) \otimes A^2(D_{e_2}).$$

For composable degenerations $\Gamma_1 \circ_e \Gamma_2$: degenerating first along Γ_2 and then along Γ_1 produces the one-particle operator

$$K_{\Gamma_1 \circ_e \Gamma_2} = \Big(\prod_{e' \in E(\Gamma_1)} T_{q_{e'}}\Big) \circ \Big(\prod_{f \in E(\Gamma_2)} T_{q_f}\Big).$$

By commutativity of operators on disjoint edges, the reverse composition gives the same operator. This is the associativity of operator composition on $A^2(D)$.

Sub-step 3c: Fredholm determinant multiplicativity. For disjoint sewing kernels K_1, K_2 on disjoint one-particle Hilbert spaces:

$$\det(1 - K_1 \oplus K_2) = \det(1 - K_1) \cdot \det(1 - K_2).$$

This is the standard multiplicativity of Fredholm determinants under direct sum. For composable degenerations where the sewing kernels share a Bergman space, the key identity is that operator composition is associative: $(K_1 \circ K_2) \circ K_3 = K_1 \circ (K_2 \circ K_3)$. The Fredholm determinant, as a spectral invariant of the composite operator, inherits associativity. The second quantisation functor $\Gamma(-)$ is monoidal (preserves composition), so

$$\mathrm{Tr}_{\mathrm{Sym}}\big(\partial_{\Gamma_1} \circ \partial_{\Gamma_2}(\cdot)\big) = \mathrm{Tr}_{\mathrm{Sym}}\big(\partial_{\Gamma_1 \circ_e \Gamma_2}(\cdot)\big),$$

because both sides reduce to the same one-particle operator $\prod_{e \in E(\Gamma_1 \circ_e \Gamma_2)} T_{q_e}$ on $A^2(D)$.

Step 4: self-sewing = trace, and the annulus at genus 1.

The non-separating degeneration Γ_{ns} (one vertex of genus $g-1$ with one self-loop) identifies two half-edges and traces over the corresponding Fock space:

$$\partial_{\Gamma_{\mathrm{ns}}}(Z) = \sum_{\mu} \langle \mu | \, Z \, | \mu \rangle = \mathrm{Tr}_{\mathcal{F}}(Z).$$

At the one-particle level, the self-loop sewing operator is T_q on a single copy of $A^2(D)$, and the trace is $\mathrm{Tr}_{\mathrm{Sym}}(\Gamma(T_q)^{\kappa}) = \det(1 - T_q)^{-\kappa}$. At genus 1 (one pair of pants with a self-loop):

$$Z_1 = \det(1 - T_q)^{-\kappa} = \prod_{n=1}^{\infty} (1 - q^n)^{-\kappa} = (q^{-1/24}\, \eta(q))^{-\kappa},$$

reproducing the scalar Heisenberg Quillen-norm amplitude (Theorem thm:heisenberg-one-particle-sewing).

This is the categorical trace $\mathrm{Tr}_{\mathcal{C}_{\mathrm{op}}}$ on the dualizable category $\mathcal{C}_{\mathrm{op}}(\mathcal{H}_\kappa)$. Agreement with the annulus trace theorem (Theorem thm:thqg-annulus-trace) follows: the Hochschild homology of the semisimple category $\mathcal{C}_{\mathrm{op}}$ is $\mathrm{HH}_*(\mathcal{C}_{\mathrm{op}}) \simeq \bigoplus_\lambda \mathbb{C}$, concentrated in degree 0 (semisimple E_1-algebras have $\mathrm{HH}_n = 0$ for $n \geq 1$).

This verifies the trace identity at amplitude level.

Step 5: compatibility with $\Theta_{\mathcal{A}}$ and the genus-g scalar amplitude.

Closed-sector compatibility. The closed projection π_{cl} of the open/closed MC element (Theorem thm:thqg-oc-projection(i)) sends $\Theta^{\mathrm{oc}}_{\mathcal{H}_\kappa} \mapsto \Theta_{\mathcal{H}_\kappa} = \kappa \cdot \eta \otimes \Lambda$. The genus-$g$ component is $\Theta_{\mathcal{H}_\kappa}|_g = \kappa \cdot \lambda_g^{\mathrm{FP}} = F_g(\mathcal{H}_\kappa)$, the genus tower of Theorem D. The cooperad cocomposition in the closed sector is the clutching identity $\xi_\Gamma^*(F_g) = \sum F_{g_1} \cdot F_{g_2} \cdot \prod_e(\text{propagator at } e)$, where the propagator at each edge is $d \log E(z, w)$ (the weight-1 Bergman kernel) and the scalar projection extracts κ from each propagator factor. Compatibility holds because π_{cl} commutes with sewing operations (Theorem thm:thqg-mc-forced-consistency(a,c)).

This verifies the closed scalar projection, conditional on the open/closed packaging.

Scalar Quillen-norm amplitude. After determinant-line trivialisation, the genus-g scalar analytic amplitude is the total trace:

$$\begin{aligned} Z_g(\mathcal{H}_\kappa;\, \Omega) &= \sum_{\Gamma \in \mathcal{G}_{g,0}} \frac{1}{|\mathrm{Aut}(\Gamma)|} \mathrm{Tr}\big(\partial_\Gamma(\mathrm{id})\big) \\ &= \sum_{\text{pants decompositions}} \frac{1}{|\mathrm{Aut}|} \prod_{\alpha=1}^{3g-3} \det(1 - T_{q_\alpha})^{-\kappa} \\ &= (\det \mathrm{Im}\, \Omega)^{-\kappa/2} \cdot ({\det}'_\zeta \Delta_{\Sigma_g})^{-\kappa/2}. \end{aligned}$$

The first equality is the definition. The second uses the one-particle reduction: each graph Γ specifies a pants decomposition, and the cocomposition at each vertex is the second-quantised Bergman operator, whose trace is the Fredholm determinant. The third is the Polyakov–Alvarez formula, which converts the product of one-particle determinants to the period-matrix determinant times the zeta-regularised Laplacian determinant. This is a Quillen-norm scalar amplitude, not the untrivialised holomorphic conformal block. The summation over pants decompositions is redundant: all decompositions give the same Fredholm determinant (the scalar amplitude is independent of the decomposition).

This gives the determinant formula for the analytic amplitudes; the equality with a categorical partition function is exactly the finite-model functoriality obligation.

Determination by Θ^{oc}. Since $\mathcal{H}_\kappa$ is class G (shadow depth $r_{\max} = 2$), the modular cooperad is entirely determined by binary cocomposition (separating clutching) and the genus-raising map (non-separating clutching). Both are one-parameter families controlled by the single scalar κ. The Getzler–Kapranov correspondence identifies the cooperad structure maps with evaluations of $\Theta^{\mathrm{oc}}_{\mathcal{H}_\kappa}$ on stable bordered graphs: the MC equation $D^{\mathrm{oc}}\Theta^{\mathrm{oc}} + \frac{1}{2}[\Theta^{\mathrm{oc}}, \Theta^{\mathrm{oc}}] + \hbar\Delta_{\mathrm{clutch}}(\Theta^{\mathrm{oc}}) = 0$ is the chain-level identity encoding all cooperad relations simultaneously. For the Heisenberg, this reduces to a scalar identity at each (g, n) because $\Theta_{\mathcal{H}} = \kappa \cdot \eta \otimes \Lambda$ is purely scalar.

This is the expected mechanism for (vi), once the finite-model cooperad has been constructed. $\square$

Remark 44.7 *(Proved inputs and open composition).* The proof above depends on five existing results:

(1) The Heisenberg sewing theorem (Theorem thm:heisenberg-sewing): the sewing envelope is $\mathrm{Sym}\, A^2(D)$ (clause (i) depends on Moriwaki; clauses (ii)–(iv) are proved in this work).

(2) The one-particle Bergman reduction (Theorem thm:heisenberg-one-particle-sewing): pair-of-pants sewing is $\Gamma(R)$, every closed amplitude is $\det(1 - K)^{-\kappa}$.
(3) The bordered FM compactification (Construction constr:bordered-fm): existence of the manifold with corners at all genera, with codimension-2 corner cancellation giving $\partial^2 = 0$.
(4) The annulus trace theorem (Theorem thm:thqg-annulus-trace): $\int_{S^1_p} \mathcal{C}_{\mathrm{op}} \simeq \mathrm{HH}_*(\mathcal{C}_{\mathrm{op}})$.
(5) The open/closed MC equation (Theorem thm:thqg-oc-mc-equation): $(D^{\mathrm{oc}})^2 = 0$ and $\Theta^{\mathrm{oc}}_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{oc}})$ at all genera.

These are the proved or previously invoked inputs. The new content specific to CT-2 is the *composition* problem: verify that the Fredholm-determinant amplitudes on a finite dualizable open-sector model assemble into a cooperad. The operator-algebraic identities on $A^2(D)$ are necessary; by themselves they do not prove the dg-categorical packaging.

Remark 44.8 *(Heisenberg features used by the test case).* The test-case proof programme exploits three features of the chosen finite Heisenberg model:
(a) *One-particle reduction.* All multi-particle amplitudes factor through one-particle Bergman operators (Wick's theorem). This reduces the cooperad from an infinite-dimensional categorical structure to finite-dimensional linear algebra on $A^2(D)$.
(b) *Semisimplicity in the retained Fock sector.* The finite model is chosen so that the retained Fock sector is semisimple. This is a model assumption to be checked against the sewing trace, not a claim about the whole Heisenberg module category.
(c) *Gaussian termination.* The shadow obstruction tower terminates at arity 2, so the only MC data entering the cooperad structure is the scalar κ. There are no higher cocompositions beyond binary clutching and genus-raising.

For non-Gaussian algebras (classes L, C, M), all three properties fail, and the proof must use the full open/closed MC element with higher-arity shadow data.

Phase 2: CT-2 for all class G algebras

Class G consists of all algebras with shadow depth $r_{\max} = 2$: the shadow obstruction tower terminates at the quadratic level, so $\Theta_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda$ and the cooperad is controlled by a single scalar. The class includes:
(i) Rank-r Heisenberg $\mathcal{H}_\kappa^{\oplus r}$ (direct sum of rank-1 systems);
(ii) Free fermions Ff_c (symplectic bosons, bc-systems);
(iii) Lattice VOAs V_Λ for even unimodular lattices;
(iv) Any tensor product of class G algebras.

What generalises from Heisenberg. The key structural fact is that the scalar shadow tower of a class G algebra terminates at arity 2. This is not the same as saying that every OPE pole is at most quadratic: lattice vertex operators can have higher poles, but their projected shadow obstruction tower still terminates. Thus the expected finite cooperad is controlled, after projection and choice of finite open-sector model, by binary cocomposition and genus-raising clutching. The extension from rank 1 to rank r replaces the scalar Fredholm determinant by a matrix-valued one: $\det'_{\mathrm{Fred}}(1 - K^{(g)})^{-Q}$ where Q is the charge matrix. The Wick theorem for multi-component free fields provides the factorisation.

What is specific to rank-1 Heisenberg. The one-particle reduction (Step 2 of Phase 1) requires a finite dualizable open-sector model compatible with the sewing trace. For rank-r Heisenberg one expects a finite model with r generators and a matrix-valued Fredholm determinant. For lattice VOAs one must keep the finite coset sectors. For free fermions, the Clifford algebra is Morita-trivial only in the finite even-dimensional nondegenerate case; odd and completed sectors must be handled separately.

Lattice VOAs. V_Λ for an even lattice decomposes as $V_\Lambda = \bigoplus_{\lambda \in \Lambda^*/\Lambda} \mathcal{H}_\Lambda \otimes e^\lambda$. The open-sector category has $|\Lambda^*/\Lambda|$ simple objects (lattice cosets). The cooperad decomposes as a direct sum over these cosets. Each summand is a one-particle cooperad (Heisenberg with charge shifted by λ). Compatibility is the lattice theta function identity $\Theta_\Lambda(\tau) = \sum_\lambda q^{|\lambda|^2/2}$. The lattice sewing theorem (Theorem thm:lattice-sewing) provides the analytic input.

Free fermions. For finite even-dimensional nondegenerate V, the Clifford algebra $\mathrm{Cl}(V, q)$ is a matrix algebra. For odd dim V, the module category has two simple objects. Completed or field-theoretic fermionic sectors require the same finite-model trace check as the Heisenberg case. The expected amplitudes are controlled by a Pfaffian analogue of the Fredholm determinant.

The main work is verifying the functor packaging lemma of Phase 1 in the multi-generator setting, using the Wick theorem.

Phase 3: CT-2 for classes L and C

Class L has shadow depth $r_{\max} = 3$ (affine Kac–Moody): the cubic shadow $C(\mathcal{A})$ is nonzero but the quartic shadow vanishes. Class C has shadow depth $r_{\max} = 4$ ($\beta\gamma$-systems): the quartic contact shadow Q^{contact} is nonzero but all higher shadows vanish.

The finite cooperad target. For class L, the scalar shadow data provide binary and ternary chain-level cocomposition candidates, controlled by κ and $C(\mathcal{A})$ respectively. For class C, the scalar data also provide a quaternary candidate controlled by $Q^{\mathrm{contact}}(\mathcal{A})$. The truncations $\Theta_{\mathcal{A}}^{\leq 3}$ and $\Theta_{\mathcal{A}}^{\leq 4}$ do not by themselves prove a finite cooperad in dgCat: scalar shadow depth bounds the displayed chain-level operations, not all categorical coherences. The categorical lift remains a construction target.

The construction. At each truncation level r, the shadow obstruction tower $\Theta_{\mathcal{A}}^{\leq r}$ is an MC element in the truncated convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}, \leq r}$. The Getzler–Kapranov correspondence (at the chain level) converts this MC element to a cooperad structure with operations of arity $\leq r$. The categorical lift at finite arity r requires lifting the chain-level cocomposition maps to functors and proving the natural coherences. Its finiteness is a target theorem, not a formal consequence of scalar truncation.

Affine Kac–Moody: the main challenge. For $V_k(\mathfrak{g})$, the open-sector category $\mathcal{C}_{\mathrm{op}}(V_k(\mathfrak{g}))$ is *not* Morita-trivial in general. At integrable level k, it is semisimple with $|P_+^k(\mathfrak{g})|$ simple objects (integrable highest-weight modules). The cooperad acts on this finite semisimple category. The binary cocomposition is the standard sewing (fusion); the ternary cocomposition is the cubic shadow map, which can be constructed from the three-point functions on $\overline{\mathcal{M}}_{0,3}$ and the cubic shadow $C(\mathcal{A})$.

The one-particle reduction does NOT apply here: the module category is not generated by a single object for $\mathfrak{g} \neq \mathfrak{sl}_2$ at generic level. Instead, the cooperad must be constructed on the full category, using the finiteness of the set of simples at integrable level.

At non-integrable (generic) level, the module category is not semisimple, and any cooperad must be constructed on a derived or completed category. Shadow truncation supplies finitely many displayed chain-level operations; it does not by itself prove that the categorical coherence problem is finite.

$\beta\gamma$: the contact class. The $\beta\gamma$-system has shadow depth 4, with $Q_{\beta\gamma}^{\mathrm{contact}}$ controlling the quartic chain-level cocomposition candidate. The open-sector category still requires a finite dualizable model before the one-particle reduction can be used categorically. The main new ingredient is the quaternary cocomposition functor to be constructed from the quartic clutching law via Mok's degeneration formula (Construction constr:arity4-degeneration).

The main new mathematics is the functor packaging at arity 3 (for class L) and arity 4 (for class C), requiring explicit construction of higher cocomposition functors from the bordered FM clutching correspondence. For affine KM at integrable level, this connects to factorisation homology (Moriwaki 2026, Brochier–Woike); for generic level, new techniques are needed.

Phase 4: CT-2 for class M via pro-cooperad

Class M has shadow depth $r_{\max} = \infty$ (Virasoro, W_N): the shadow obstruction tower does not terminate, and the cooperad has cocompositions at every arity.
The pro-cooperad construction. Conjecturally, the shadow obstruction tower $\Theta_{\mathcal{A}} = \varprojlim \Theta_{\mathcal{A}}^{\leq r}$ (Theorem thm:recursive-existence) presents $\Theta_{\mathcal{A}}$ as an inverse limit of finite-arity MC elements. Once the finite categorical lifts in Phase 3 are constructed, each truncation $\Theta_{\mathcal{A}}^{\leq r}$ should determine a finite cooperad $O_r(\mathcal{A})$ on $C_{\mathrm{op}}(\mathcal{A})$. The restriction maps $O_{r+1} \to O_r$ (forgetting the $(r+1)$-ary cocomposition) make $\{O_r(\mathcal{A})\}_{r\geq 2}$ a projective system.

Definition (Pro-cooperad). The *modular pro-cooperad* on $C_{\mathrm{op}}(\mathcal{A})$ is

$$O(\mathcal{A}) := \varprojlim_{r} O_r(\mathcal{A})$$

in the category of modular cooperads in dgCat, provided the finite lifts and transition coherences exist.

Convergence. The expected projective system $\{O_r\}$ should satisfy the Mittag-Leffler condition: the transition maps $O_{r+1} \to O_r$ should be surjective on morphism complexes (the truncation forgets but does not modify lower-arity operations). If this is proved, the Mittag-Leffler theorem gives a well-defined inverse limit and

$$R^1 \varprojlim O_r = 0.$$

The convergence of the algebraic inverse limit is a direct consequence only after the categorical transition maps have been constructed; the algebraic convergence of the shadow obstruction tower (Theorem thm:recursive-existence) supplies the chain-level input. Analytic convergence of the sewing amplitudes is a separate estimate, not automatic from scalar shadow depth.
The Virasoro case. For Vir_c with $c \neq 0, 26$, the open-sector category $C_{\mathrm{op}}(\mathrm{Vir}_c)$ is the category of Virasoro modules. The cooperad at truncation level r has cocompositions controlled by the shadow obstruction tower $\{\kappa, C, Q, S_5, S_6, \ldots, S_r\}$ where $S_j = S_j(\mathrm{Vir}_c)$ is the weight-j shadow. The shadow growth rate $\rho(\mathrm{Vir}_c) \approx 0.234$ at $c = 26$ ensures that the higher shadows decay, so the pro-cooperad converges at a definite rate.

The genus-g closed projection is $F_g(\mathrm{Vir}_c) = (c/2) \cdot \lambda_g^{\mathrm{FP}}$, recovering the modular characteristic.

Three challenges: (i) constructing the arity-r cocomposition functor for arbitrary r (the chain-level version exists; the categorical lift is needed); (ii) verifying Mittag-Leffler at the categorical level; (iii) controlling convergence in the non-semisimple setting. Phase 4 depends on Phases 1–3.

Phase 5 (Alternative): The Brochier–Woike path

The Brochier–Woike theorem [`?BrochierWoike2022`] classifies modular functors via factorisation homology. The input is an E_2-algebra in a ribbon category; the output is a modular functor. Dualising the modular functor yields the modular cooperad.
The strategy.

(i) Verify that the chiral derived center $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ is an E_2-algebra in the appropriate sense. *Status*: the chiral Deligne–Tamarkin theorem (Observation 43.1(i)) provides the E_2 structure at genus 0. The Swiss-cheese theorem (Theorem thm:thqg-swiss-cheese) proves that the pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}), \mathcal{A})$ is the terminal $\mathsf{SC}^{\mathrm{ch,top}}$-algebra. This is the E_2 input.

(ii) Verify the ribbon/Calabi–Yau condition on the open-sector category $\mathcal{C}_{\mathrm{op}}$. *Status*: for chirally Koszul algebras, the cyclic pairing on the deformation complex provides a non-degenerate trace. This is the perfectness condition of Proposition prop:lagrangian-perfectness, which is proved for all standard families. Whether this suffices for the BW ribbon condition requires verification.

(iii) Verify the finiteness conditions. BW requires the E_2-algebra to be dualizable in the Morita $(\infty, 2)$-category. For chirally Koszul algebras with finite-dimensional weight spaces (positive-energy axiom), this is expected to reduce to finite-generation of the module category. That reduction and finite-generation verification are case-by-case hypotheses, not consequences of shadow depth alone.

(iv) Apply BW to obtain the modular functor, then dualise to obtain the cooperad.

(v) Verify compatibility with the MC element $\Theta_{\mathcal{A}}$ and the shadow obstruction tower.

Advantages. The BW path bypasses the class-by-class analysis (Phases 1–4) entirely: if the hypotheses are verified for a given chirally Koszul algebra, the modular cooperad would follow for that algebra. Uniform verification across the standard landscape remains part of the programme.

Obstacles.

(a) BW works in the setting of E_2-algebras in ribbon categories, which is a topological (not holomorphic) framework. The passage from the chiral E_2 structure (defined via FM compactification of $\mathbb{C}$, not of $\mathbb{R}^2$) to the BW framework requires a comparison theorem between holomorphic and topological configuration spaces. This comparison exists at genus 0 (Kontsevich formality provides it) but must be extended to all genera.

(b) BW requires 5 simultaneous conditions:
   (a) an E_2-algebra input,
   (b) a ribbon/CY structure on the target category,
   (c) dualizability,
   (d) compatibility with the genus-g FM compactification,
   (e) the resulting modular functor must be equipped with a flat connection (for varying the curve).

   No published reference verifies all five simultaneously for any chiral algebra. The manuscript provides (a), (b) is available from the cyclic pairing, (c) is a case-by-case Morita-dualizability hypothesis for which positive energy and finite-dimensional weight spaces are evidence, (d) is the main new content, and (e) is the shadow connection ∇^{sh}.

(c) The BW modular functor is a *topological* modular functor (valued in chain complexes). Extracting the *holomorphic* content (the modular forms, the shadow tower, the $\hat{A}$-generating function) from the topological modular functor requires an additional step: the Hodge-to-de Rham comparison. This is the content of Theorem D (the modular characteristic), which provides the bridge.

Summary

(I) *Immediate target* (Phase 1): prove CT-2 for Heisenberg. All geometric ingredients exist; the new content is the categorical packaging.

(II) *First extension* (Phase 2): extend to all class G. The Wick theorem provides the multi-generator generalisation.

(III) *Second extension* (Phase 3): extend to classes L and C. Requires explicit construction of ternary and quaternary cocomposition functors. The affine KM case at integrable level connects to the existing modular functor literature.

(IV) *Full result* (Phase 4): extend to class M via pro-cooperad. Requires the pro-cooperad framework and Mittag-Leffler convergence.

(V) *Alternative path* (Phase 5): verify Brochier–Woike hypotheses for all chirally Koszul algebras. If successful, bypasses Phases 1–4. The Heisenberg case serves as a test.

(VI) *Uniqueness* (deferred): the categorical Getzler–Kapranov correspondence requires $(\infty, 2)$-categorical technology that does not yet exist. This is a long-term frontier problem, independent of CT-2 existence.

45 Chromatic speculations

The scalar shadow obstruction tower lives at chromatic height 0. The generating function $F(\mathcal{A}; \hbar) = \kappa(\mathcal{A})/\hbar^2 \cdot [\hat{A}(i\hbar) - 1]$ is the $\hat{A}$-genus, associated to the additive formal group law $F(x, y) = x + y$. The Bernoulli numbers appear because they are the Taylor coefficients of the Todd pushforward on the universal curve, not because of a formal group coincidence.

The bar complex on curves of different type, however, uses propagators of different chromatic height:

Curve	*Propagator*	*Formal group*	*Height*	*Bar–cobar theory*
$\mathbb{P}^1$	$1/(z - w)$	$\hat{\mathbf{G}}_a$	∞ (rational)	classical
$\mathbb{C}^*$	$\pi \cot(\pi z)$	$\hat{\mathbf{G}}_m$	1 (K-theory)	trigonometric
E_τ	ϑ_1'/ϑ_1	$\hat{E}_\tau$	2 (elliptic)	elliptic

The rational/trigonometric/elliptic hierarchy of quantum groups is therefore a height-filtered analytic shadow: the rational propagator is additive, the trigonometric propagator is multiplicative, and the elliptic propagator is attached to the formal group of the fixed elliptic curve. The Fay identity on E_τ is the genus-1 MC equation with elliptic propagator. The ordinary dg bar–cobar construction over $\mathbb{C}$, however, is not yet a positive-chromatic object.

Theorem 45.1 *(Height-zero truncation and the spectral obstruction).* For a fixed elliptic curve E_τ, the period-corrected elliptic bar differential D_τ is a flat coderivation; the unmodified fiberwise collision differential has curvature

$$d_{\text{fib}}^2 = \kappa(\mathcal{A})\, \omega_1.$$

Thus the fixed-fiber elliptic bar coalgebra is theorem-level after Fay correction.

The ordinary dg bar–cobar construction over $\mathbb{C}$ is the rational height-zero shadow of any chromatic refinement. After Eilenberg–Mac Lane realization it is an $H\mathbb{C}$-module, and hence cannot carry positive Morava-local information. A TMF-valued chromatic bar–cobar duality would require additional spectrum-level data: an E_2 sheaf over $\mathcal{M}_{1,1}$, an orientation or map to tmf, identification of the associated formal group with the universal elliptic formal group, and compatibility of chiral Koszul duality with the tmf-dualizing structure.

Proof. Flatness at fixed E_τ is exactly Fay's trisecant identity for the period-corrected elliptic propagator; without the period correction the collision differential squares to the genus-one Hodge curvature $\kappa(\mathcal{A})\omega_1$. This proves the fixed-fiber elliptic bar statement.

For the chromatic assertion, the bar complex used here is a chain complex of complex vector spaces. Its spectral realization is therefore an $H\mathbb{C}$-module. Rational Eilenberg–Mac Lane modules have no nontrivial positive Morava-local layers: the $K(n)$-local data for $n > 0$ are invisible after passage to $H\mathbb{C}$. Consequently any genuine TMF refinement cannot be extracted from the classical dg bar complex alone; it must be supplied by a spectrum-level sheaf/orientation carrying the universal elliptic formal group. $\square$

Theorem 45.2 *(TMF lift criterion for chromatic bar–cobar).* The height-zero elliptic bar complex attached to a fixed E_τ admits a genuine tmf-valued chromatic lift if and only if there is a sheaf

$$\mathscr{B}_{\mathrm{tmf}}(\mathcal{A}) \in \mathrm{QCoh}_{\mathsf{E}_2}(\mathcal{M}_{\mathrm{ell}};\mathrm{tmf})$$

whose completed stalk at the point E_τ is the period-corrected elliptic bar coalgebra $B_{E_\tau}(\mathcal{A})$ after extension of scalars to $\mathbb{C}$, whose cusp restriction is the Tate trigonometric bar complex, and whose dualising object transports Koszul duality under the Witten orientation. Under these data, chromatic bar–cobar duality is the base-change identity

$$\Omega_{\mathrm{tmf}}\mathscr{B}_{\mathrm{tmf}}(\mathcal{A}) \simeq \mathscr{A}_{\mathrm{tmf}}$$

in the tmf-linear category over $\mathcal{M}_{\mathrm{ell}}$.

Thus the positive-chromatic problem is not a hidden consequence of the complex dg bar construction. It is exactly the existence and descent of the tmf-linear sheaf above.

Proof. If $\mathscr{B}_{\mathrm{tmf}}(\mathcal{A})$ exists, completion at E_τ gives the complex elliptic formal group and recovers the Fay-corrected elliptic differential; completion at the Tate cusp gives the multiplicative formal group and the trigonometric bar complex. The Witten orientation supplies the dualising comparison needed for Koszul duality in tmf-modules, so the displayed base-change identity is the chromatic lift.

Conversely, a genuine tmf-valued chromatic lift must vary over $\mathcal{M}_{\mathrm{ell}}$, because the universal elliptic formal group is classified there; its complex fibre and Tate fibre are exactly the two stated completions. The duality statement forces compatibility with the tmf dualising object. These necessary data are also sufficient by descent for quasi-coherent tmf-modules on the elliptic moduli stack. $\square$

Theorem 45.3 *(String-oriented tmf realisation).* The criterion of Theorem 45.2 is realised on the full subcategory of chiral bar complexes whose determinant anomaly is represented by a string-oriented virtual bundle ξ over $\mathcal{M}_{\mathrm{ell}}$. In this case

$$\mathscr{B}_{\mathrm{tmf}}(\mathcal{A}) := \pi_!\big(B^{\mathrm{ch}}(\mathcal{A}) \otimes \mathrm{Th}_{\mathrm{tmf}}(\xi)\big) \in \mathrm{QCoh}_{\mathsf{E}_2}(\mathcal{M}_{\mathrm{ell}};\mathrm{tmf})$$

is the required tmf-valued bar sheaf, where $\mathrm{Th}_{\mathrm{tmf}}(\xi)$ is the Thom object defined by the Ando–Hopkins–Strickland Witten orientation. Its complex fibre is the elliptic bar coalgebra, its Tate fibre is the trigonometric bar complex, and its dualising object is the Witten-oriented determinant line.

The obstruction to this realisation is exactly the failure of the string orientation of ξ: equivalently the anomaly class $p_1(\xi)/2$ does not vanish in the orientation theory used by tmf. Thus a tmf lift is constructed on the string-oriented locus and is not asserted outside it.

Proof. The Hopkins–Miller sheaf of E_∞ ring spectra over $\mathcal{M}_{\mathrm{ell}}$ carries the universal elliptic formal group. The Witten orientation supplies a Thom class in tmf for string-oriented virtual bundles. Tensoring the chiral bar coalgebra with this Thom object and applying the tmf pushforward $\pi_!$ therefore produces a quasi-coherent tmf-module over the elliptic moduli stack. Completion at an elliptic curve recovers the corresponding complex elliptic formal group, hence the elliptic bar coalgebra; completion at the

Tate cusp recovers the multiplicative formal group and hence the trigonometric bar complex. The same Thom class is the dualising datum for Koszul duality. If $p_1(\xi)/2$ is nonzero, the Thom class does not exist in tmf, so the displayed construction cannot be made. □

Remark 45.4 *(Non-scalar shadow data versus chromatic data).* The non-scalar shadow obstruction tower (arity ≥ 3) may organise deformation complexity in a way that resembles a height filtration, but this is not positive-chromatic information until the construction is spectralized. The G/L/C/M depth classification is a theorem of the dg chiral bar complex; a TMF-module refinement is a separate spectrum-level problem.

46 The string field theory perspective

The bar construction is a string field theory functor. This is not an analogy. The A_∞ operations m_n extracted from the chiral Hochschild complex are Zwiebach's closed string field theory vertices: $m_1 = Q_{\mathrm{BRST}}$, m_2 is the string product, m_3 is the four-string vertex, and higher m_n are the contact terms. The action is

$$S[\Upsilon] = \tfrac{1}{2}\langle \Upsilon, Q\Upsilon\rangle + \sum_{n\geq 3} \frac{1}{n!}\langle \Upsilon, m_n(\Upsilon, \ldots, \Upsilon)\rangle.$$

This is proved in the manuscript as a consequence of Zwiebach's theorem (1993), restated as Theorem **??**.

The modular MC hierarchy $D\Theta + \frac{1}{2}[\Theta, \Theta] + \hbar\Delta\Theta = 0$, read genus by genus, is the genus expansion of this string field theory. The string field is the MC element $\Theta_{\mathcal{A}}$. The vertices are $\ell_n^{(g)}$. The BV operator Δ (non-separating clutching) is the loop contraction.

Warning 46.1 *(Heuristic boundary).* The genus-0 identification (BV complex ≃ bar complex) is *proved*: the BRST–bar quasi-isomorphism at $c = 26$ and for Kac–Moody at all levels $k \neq -h^\vee$. The full chain-level identification at genus $g \geq 1$ is *heuristic* (Remark rem:modular-qme-bv in the BV chapter, and Conjecture conj:master-bv-brst in the concordance). The MC equation is proved at all genera; what is conjectural is the identification of the algebraic bar differential with the BV path-integral quantisation. Do not confuse the proved algebraic identity $D^2_{\mathcal{A}} = 0$ with the conjectural physical identification BV = bar.

46.1 Four objects and their string-theoretic roles

From a single chiral algebra $\mathcal{A}$, the constructions of the monograph produce four objects. Their string-theoretic roles are:

Object	*String-theoretic role*
$\mathcal{A}$ (the chiral algebra)	Open-string boundary algebra. Lives on the boundary $\{0\} \times \mathbb{R}$ of $\mathbb{C}_{\geq 0} \times \mathbb{R}$.
$\overline{B}(\mathcal{A})$ (bar coalgebra)	Classifies *twisting morphisms*: universal couplings between $\mathcal{A}$ and $\mathcal{A}^!$. This is the open-string *configuration space*, not the closed-string state space.
$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$	The universal *bulk* algebra: closed-string observables. Computed by chiral Hochschild cochains, not by the bar complex.
$\Theta^{\mathrm{oc}}_{\mathcal{A}} = \Theta_{\mathcal{A}} + \sum_j \mu^{\mathcal{M}_j}$	The full open/closed string field. Its closed projection $\Theta_{\mathcal{A}}$ is the genus expansion. Its open projection gives the A_∞ module equations. Its mixed projection is the bulk-to-boundary coupling.

The critical distinction (AP-OC, AP25): the bar complex classifies *twisting morphisms* (couplings), while the derived centre classifies *bulk operators* (observables). Open/closed string duality is not "bar = bulk"; it is the *interplay* between the bar complex and the derived centre, mediated by the Swiss-cheese structure.

46.2 Kodaira–Spencer gravity

The BCOV Kodaira–Spencer chiral algebra appears in the monograph's Vol II examples as the boundary chiral algebra of 6d holomorphic Chern–Simons on $\mathbb{C}^3$. The passage from open-string (the gauged $\beta\gamma$ algebra) to closed-string (gravitational) degrees of freedom is controlled by *Koszul duality* ($\mathcal{A} \mapsto \mathcal{A}^!$ via Verdier duality on bar cohomology) and the *derived centre* ($\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ as the universal bulk), not by bar-cobar inversion ($\Omega(\overline{B}(\mathcal{A})) \simeq \mathcal{A}$, which recovers the original open-string algebra).

The structural parallel between BCOV and the modular MC hierarchy arises because *both are string field theories*: both are governed by the modular operad of stable curves, with bilinear genus-splitting and BV genus-raising. The BCOV system is the *analytic* string field theory (metric-dependent propagator, holomorphic anomaly, divergent genus expansion). The shadow obstruction tower is the *algebraic* string field theory (canonical propagator, no anomaly, convergent).

For the Virasoro algebra at $c = 26$ coupled to bc ghosts, $\kappa_{\mathrm{eff}} = \kappa(\mathrm{matter}) + \kappa(\mathrm{ghost}) = c/2 + (-13) = 0$: anomaly cancellation. This kills the *scalar* genus tower: $F_g(\mathcal{A}_{\mathrm{tot}}) = 0$ for all $g \geq 1$. The modular characteristic $\kappa(\mathcal{A})$ is the gravitational coupling constant; $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ is the genus-g gravitational free energy. For the combined matter + ghost system, the scalar free energies vanish. By AP31 ($\kappa = 0 \not\Rightarrow \Theta = 0$), the higher-arity shadow obstruction tower may survive at $\kappa_{\mathrm{eff}} = 0$, but this is not computed in the manuscript.

Observation 46.2 *(Three functors from the bar complex).* From $\overline{B}(\mathcal{A})$, three distinct functors produce three distinct objects:

(i) *Reconstruction*: $R_X(\mathcal{A}) := \Omega(\overline{B}(\mathcal{A})) \simeq \mathcal{A}$ (Theorem B). Recovers the original algebra.
(ii) *Koszul duality*: $K_X(\mathcal{A}) := \Omega(D_{\mathrm{Ran}}(\overline{B}(\mathcal{A}))) \simeq \mathcal{A}^!$ (Verdier intertwining). Produces the dual boundary condition.
(iii) *Derived centre*: $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = \mathrm{RHom}(\Omega(\overline{B}(\mathcal{A})), \mathcal{A}) \simeq C^{\bullet}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ (Theorem H + bar resolution). Computes the universal bulk.

Open/closed duality is functor (iii): the passage from the bar coalgebra (the open-string resolution) to the derived centre (the closed-string observables) via RHom.

Observation 46.3 *(The genus tower restores directionality).* At genus 0, the Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ enforces one-way information flow: closed → open (bulk acts on boundary), with no open-to-closed maps. At genus 1, the nonseparating clutching Δ_{ns} maps the annulus trace $\mathrm{Tr}_{\mathcal{A}} \in \mathrm{HH}_*(\mathcal{A}_b)$ (open-sector data) to $\kappa(\mathcal{A}) \cdot \lambda_1$ (closed-sector class): the first open-to-closed map. At every subsequent genus, iterated clutching provides further open-to-closed maps. The full modular MC equation $D^{\mathrm{oc}}\Theta^{\mathrm{oc}} + \frac{1}{2}[\Theta^{\mathrm{oc}}, \Theta^{\mathrm{oc}}] + \hbar\Delta_{\mathrm{clutch}}(\Theta^{\mathrm{oc}}) = 0$ is fully bidirectional.

The genus tower is the mechanism by which the genus-0 directionality constraint is progressively lifted.

47 Cross-volume cascade from the Vol II Polyakov programme

Session notes, 2 April 2026. Five findings from Vol II that propagate back to Vol I.

47.1 DS ≠ Koszul confirmed (AP33)

The Drinfeld–Sokolov transport formulas and the Koszul conductor formulas are genuinely different invariants. For the principal $\mathcal{W}$-algebra of $\widehat{\mathfrak{sl}}_N$, the canonical central charge is

$$c(\mathcal{W}_N^k) = (N-1) - N(N^2-1)\frac{(k+N-1)^2}{k+N},$$

and the Koszul conductor is

$$K_N = 2(N-1) + 4N(N^2-1).$$

Thus one must not replace the Koszul conductor by a putative DS sum under $k \mapsto -k-2N$.

- DS transport acts on the level and depends on the chosen convention for the shifted level.
- Koszul duality gives the trace-form conductor K_N above; for $N = 3$ this is 100.

The three operations–-Koszul duality, FF involution, and negative-level substitution–-share surface similarities but differ once the canonical central-charge formula is used.

47.2 Shapovalov obstruction for closed-form m_k

A closed form for the Virasoro scalar polynomials P_k (where $m_k\big|_{\mathrm{scalar}} = (c/12)\,P_k$) is equivalent to inverting the Shapovalov form at all levels–-the same problem as constructing all Virasoro singular vectors. The generating function is the graviton self-energy resolvent

$$G(t;\lambda) = \mathrm{Tr}_{\mathfrak{h}}\left(\frac{1}{1-t\,K(\lambda)}\right),$$

where $K(\lambda)$ is the collision kernel at conformal weight λ. At the symmetric point $\lambda = (c-1)/24$ (where $\Delta = \bar{\Delta}$), a universal factor $(x+2)$ appears in P_k for all $k \leq 5$–-this is the conformal weight Ward identity forcing a zero at $x = -2$. Whether the factor persists at all arities is tied to the question of whether the Shapovalov determinant has a universal factorisation beyond the Kac determinant formula.

47.3 AP-OC Steinberg refinement

The three-functor picture from Observation 46.2 acquires a slogan: *the bar complex is the E_1 coalgebraic engine for the Swiss-cheese programme, as the Steinberg variety presents the Hecke algebra.* More precisely, $\overline{B}(\mathcal{A})$ is the presenting coalgebra from which the boundary algebra, the dual boundary, and the bulk are extracted by three distinct functors:

(i) $\Omega(\overline{B}(\mathcal{A})) \simeq \mathcal{A}$ (cobar = boundary reconstruction).

(ii) $\Omega(D_{\mathrm{Ran}}(\overline{B}(\mathcal{A}))) \simeq \mathcal{A}^!$ (Verdier dual cobar = Koszul dual = dual boundary).

(iii) $\mathrm{RHom}(\Omega(\overline{B}(\mathcal{A})), \mathcal{A}) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ (chiral Hochschild cochains = bulk).

The Steinberg variety $\widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}}$ presents the Hecke algebra $H(W)$ via convolution on the resolution; the bar complex $\overline{B}(\mathcal{A})$ is the ordered E_1 presenting coalgebra whose bar-cobar, Verdier-dual, and Hochschild functors recover the boundary algebra, the dual boundary, and the bulk. The Swiss-cheese datum itself lives on the derived-center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}), \mathcal{A})$ computed from $\overline{B}(\mathcal{A})$ via the bar resolution. In both cases, the presenting object is a *correspondence*, not the algebra it produces.

47.4 BP conductor conventions

For the Bershadsky–Polyakov algebra the central-charge sum is convention-sensitive. In the Arakawa convention used by the current local ledger,

$$c(BP_k) = 2 - 24\frac{(k+1)^2}{k+3}, \qquad c(k) + c(-k-6) = 196.$$

In the Fateev–Lukyanov/free-field convention the corresponding sum is 50. The κ-complementarity constant is $98/3$.

47.5 Ghost formula fails for non-principal reductions

The BV ghost formula $\alpha = \sum_s 2(6s^2 - 6s + 1)$ (summed over spins s of the $\mathcal{W}$-algebra generators) gives the correct Koszul self-dual central-charge conductor for principal $\mathcal{W}_N$-algebras: $\alpha_2 = 26$ (Virasoro), $\alpha_3 = 100$ (W_3, spins 2, 3), and so forth. But for non-principal reductions it fails. For the Bershadsky–Polyakov algebra B^k (spins $1, \frac{3}{2}, \frac{3}{2}, 2$):

- Ghost formula: $\alpha = 2(6\cdot 1 - 6 + 1) + 2\cdot 2(6\cdot \frac{9}{4} - 6\cdot\frac{3}{2} + 1) + 2(6\cdot 4 - 12 + 1) = 2 + 22 + 26 = 50$.
- FF/DS transport is convention-sensitive. In the Arakawa normalisation $c(BP_k) = 2 - 24(k+1)^2/(k+3)$ and $c(k) + c(-k-6) = 196$; in the Fateev–Lukyanov/free-field convention the conductor is 50. The corresponding κ-complementarity constant is $98/3$, not a spin-only invariant.
- Direct Koszul computation: $c^* = \alpha_{\mathrm{Koszul}}/2$ requires the full A_∞ analysis, not just the spin content.

The discrepancy arises because the Koszul involution for a non-principal $\mathcal{W}$-algebra is *not determined by the spin content alone*: the non-linear OPE structure (which depends on the choice of nilpotent orbit, not just the conformal weights) contributes to the Koszul dual central charge. The ghost formula $\alpha = \sum 2(6s^2 - 6s + 1)$ is a *principal-reduction formula*, valid precisely when the generators are in bijection with the exponents of $\mathfrak{g}$ and the OPE is controlled by the Miura transform.

Observation 47.1 *(The non-principal frontier).* For principal $\mathcal{W}_N$: the ghost formula, the FF transport, and the Koszul computation all agree. For non-principal reductions: the ghost formula can disagree with both FF transport and Koszul duality. The correct general statement is that the *Koszul self-dual point* is the unique c^* satisfying $c^* = c^{*\prime}$ (where $c^{*\prime}$ is the Koszul dual central charge), and this requires knowing the full bar complex of $\mathcal{A}$, not just its spin content. For W-algebras, $\kappa + \kappa'$ is a nonzero constant (e.g., 13 for Virasoro), so the self-dual point is not characterised by $\kappa + \kappa' = 0$ (AP24).

48 The formal group of the bar complex

The spectral parameter λ in the λ-bracket $\{a_\lambda b\}$ is not a decoration. It carries a formal group law, and the formal group determines which endomorphisms–-in particular, which rotations–-the theory admits.

Observation 48.1 *(The sesquilinearity axiom forces* $\hat{\mathbf{G}}_a$*).* The sesquilinearity axiom

$$\{\partial a_\lambda b\} = -\lambda\{a_\lambda b\}$$

forces the composition of spectral parameters to be additive: $F(\lambda, \mu) = \lambda + \mu$. Indeed, ∂ is the infinitesimal translation $a(z) \mapsto a(z + \varepsilon)$, and the eigenvalue equation $\partial \mapsto -\lambda$ identifies λ as the Fourier dual of translation. The Fourier dual of the additive group is the additive group. No other formal group law is compatible with sesquilinearity: a multiplicative law $F(\lambda, \mu) = \lambda + \mu + \lambda\mu$ would require $\{\partial a_\lambda b\} = -(1 + \lambda)\{a_\lambda b\} \cdot \log(1 + \lambda)/\lambda$, which is not polynomial in λ.

Observation 48.2 *(The endomorphism ring of* $\hat{\mathbf{G}}_a$*).* Over any $\mathbb{C}$-algebra R, the endomorphism ring of the additive formal group is

$$\mathrm{End}(\hat{\mathbf{G}}_{a/R}) = R.$$

Every element $\alpha \in R$ defines an endomorphism $[\alpha] : x \mapsto \alpha x$, and these exhaust all endomorphisms since $\hat{\mathbf{G}}_a$ has no non-linear endomorphisms in characteristic zero. In particular, $[i] \in \mathrm{End}(\hat{\mathbf{G}}_{a/\mathbb{C}})$ is an endomorphism: the *Wick rotation* $\lambda \mapsto i\lambda$ is a legitimate automorphism of the spectral-parameter formal group. This is why the Wick rotation $x \mapsto ix$ (equivalently $t \mapsto it$ in the topological direction of $\mathbb{C}_z \times \mathbb{R}_t$) is well-defined in the classical theory: it lives inside $\mathrm{End}(\hat{\mathbf{G}}_a) = \mathbb{C}$.

Remark 48.3 *(Height of* $\hat{\mathbf{G}}_a$ *over* $\mathbf{F}_p$*).* Over $\mathbf{F}_p$, the additive formal group has height ∞, not height 1. The p-series is $[p](x) = px = 0$, which vanishes identically–-the defining condition of infinite height. The formal groups of finite height relevant to the chromatic tower (§45) are $\hat{\mathbf{G}}_m$ (height 1) and $\hat{E}_\tau$ (height ≤ 2), not $\hat{\mathbf{G}}_a$.

Observation 48.4 *(Spectralization contracts the endomorphism ring).* Upon spectralization (passing from chain complexes to spectra, i.e. from $\hat{\mathbf{G}}_a$ to formal groups of finite height), the endomorphism ring contracts:

$$\begin{array}{lll} \hat{\mathbf{G}}_a & : \quad \mathrm{End} = \mathbb{C} & \text{(all constants; height } \infty \text{ over } \mathbf{F}_p) \\ \hat{\mathbf{G}}_m & : \quad \mathrm{End} = \mathbb{Z}_p & \text{(height 1)} \\ \hat{E}_\tau & : \quad \mathrm{End} = \text{order in } \mathbb{Q}(\sqrt{-d}) & \text{(height} \leq 2) \end{array}$$

The ring $\mathbb{Z}_p$ does not contain i for $p \neq 2$ unless $p \equiv 1 \pmod 4$, and even then $i \in \mathbb{Z}_p$ is a p-adic integer, not the complex Wick rotation. For a generic elliptic curve, $\mathrm{End}(\hat{E}_\tau) = \mathbb{Z}$, which excludes $[i]$ entirely.

The sole exception is the CM curve E_i with j-invariant $j = 1728$: here $\mathrm{End}(\hat{E}_i) \supseteq \mathbb{Z}[i]$, so $[i]$ is an endomorphism. This is the unique elliptic curve over $\mathbb{C}$ (up to isomorphism) whose formal group admits a Wick rotation. It is an arithmetic accident, not a structural feature.

Observation 48.5 *(The Wick rotation is intrinsically rational).* The classical bar-cobar theory of §§1–46 is a theory of chain complexes over $\mathbb{C}$. Every chain complex is an $H\mathbb{C}$-module, hence rational, hence chromatic height 0. The endomorphism ring $\mathrm{End}(\hat{\mathbf{G}}_a) = \mathbb{C}$ reflects this: the theory admits *all* constant endomorphisms, including $[i]$.

Positive chromatic height enters only after spectralization–-the passage from chain complexes to genuine spectra, replacing $\hat{\mathbf{G}}_a$ by a formal group of finite height. This passage contracts the endomorphism ring from $\mathbb{C}$ to an arithmetic object ($\mathbb{Z}_p$ or an order in a quadratic field). The Wick rotation, which requires $[i] \in \mathrm{End}$, is generically excluded after this contraction. The classical bar-cobar theory is height 0 *because* its endomorphism ring is large enough to contain all rotations; the height-0 diagnosis of AP-CHR is a consequence of $\mathrm{End}(\hat{\mathbf{G}}_a) = \mathbb{C}$, not an independent axiom.

49 Borel structure of the shadow obstruction tower

The scalar genus tower converges. A class M shadow tower has a finite ordinary arity radius; evaluating the arity series at $t = 1$ diverges only in the parameter range $\rho > 1$. These are different analytic phenomena, and their Borel structures are different. The distinction resolves a paradox about Wick rotation.

The scalar tower is entire. The Faber–Pandharipande coefficients satisfy $\lambda_g^{\mathrm{FP}} \sim 2/(2\pi)^{2g}$ as $g \to \infty$. The genus tower

$$F_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} \;\sim\; \frac{2\kappa}{(2\pi)^{2g}}$$

decays exponentially in g. The mechanism: the Bernoulli numbers $|B_{2g}| \sim 2(2g)!/(2\pi)^{2g}$ carry factorial growth, but λ_g^{FP} divides by exactly the $(2g)!$ from the $\hat{A}$-genus denominator. The cancellation is exact–-$|B_{2g}|/(2g)! \sim 2/(2\pi)^{2g}$–-and the Borel transform $\mathcal{B}F(\xi) = \sum_{g\geq 1} F_g\, \xi^{2g}/\Gamma(2g+1)$ is entire. There are no singularities on the Borel plane: no Stokes lines, no resurgent trans-series, no ambiguity.

This is *not* Gevrey-1 growth (factorial) tamed by Borel resummation. It is honest convergence: the series $\sum F_g\, \hbar^{2g}$ converges absolutely for $|\hbar| < 2\pi$. The convergence reflects two facts: compactness of $\overline{\mathcal{M}}_g$ (which bounds the integration domain) and polynomial growth of λ_g^{FP} in the tautological ring (which is *sub*-factorial). The $\hat{A}$-genus is a convergent power series; the genus tower inherits this convergence.

The shadow obstruction tower has finite ordinary radius. For class M algebras (Virasoro, $\mathcal{W}_N$), the shadow coefficients S_r grow as

$$S_r \;\sim\; C \cdot \rho^r \cdot r^{-5/2} \cos(r\theta + \varphi), \qquad \rho \;=\; \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}.$$

The ordinary generating series therefore has radius $1/\rho$. Specialising it at $t = 1$ is parameter-dependent. For Virasoro the positive critical threshold is the real root $c_* \approx 6.1254$ of $5c^3 + 22c^2 - 180c - 872 = 0$; on the corresponding positive interval $0 < c < c_*$, $\rho > 1$ and the arity series diverges at $t = 1$. For $\rho < 1$, the same series converges at $t = 1$ but still has finite ordinary radius. The growth is Gevrey-0 (geometric), not Gevrey-1 (factorial): $|S_r| \lesssim C\rho^r$, not $|S_r| \lesssim C^r\, r!$. The Borel transform $\sum S_r\, \xi^r/r!$ converges everywhere (entire), but the *ordinary* series has finite radius and requires analytic continuation past $|t| = 1/\rho$, not Borel resummation.

*

Ordinary analytic continuation. The ordinary generating function of the shadow tower has branch singularities at the nearest zeros of $Q_L(t)$, with radius $1/\rho$. Since the growth is geometric rather than factorial, the factorially divided Borel transform is entire; there are no Borel-plane singularities at $1/\rho$ in this transform. Wick rotation can rotate the ordinary branch directions, but the following Stokes story is a conjectural analytic-continuation programme, not a proved Borel-resummation theorem:

- *Euclidean frame.* ordinary branch points at $t = \pm 1/\rho$ on the *real* axis.
- *Lorentzian frame.* ordinary branch points at $t = \pm i/\rho$ on the *imaginary* axis.

This is an ordinary analytic-continuation distinction, not a proved Euclidean/Borel paradox.

Resolution. This issue concerns only the shadow obstruction tower (S_r, the arity expansion), not for the scalar tower (F_g, the genus expansion). The scalar tower converges outright: its Borel transform is entire, so neither frame encounters singularities.

*

The S-transform as a possible continuation. The shadow series $\sum S_r t^r$ has branch singularities at radius $1/\rho$ in the algebraic variable. A natural boundary is not proved. In modular examples, the modular S-transform may provide an analytic continuation between cusp expansions, but this requires the full partition function and its modular representation. The modular S-transform $\tau \mapsto -1/\tau$ acts on the modular variable q and on the full modular-representation data. It does not act on the arity variable t of the bar shadow series. Thus a cusp expansion near $q \to 1$ can only be used as an analogy or an extra modular input; it is not literally the shadow series and does not by itself resum the bar obstruction tower.

This is only an analogy with resummation. It is not the Borel transform of the shadow series, and the S-matrix belongs to the module-category/modular-representation data, not to the bar complex alone.

Observation 49.1 *(Two towers, two Borel types).* The genus tower $\{F_g\}$ and the shadow obstruction tower $\{S_r\}$ have different Borel structures:

	Genus tower F_g	**Shadow obstruction tower S_r**
Growth	$O((2\pi)^{-2g})$, exponential decay	$O(\rho^r)$, geometric growth
Gevrey class	Convergent (no Gevrey class)	Gevrey-0 (geometric)
Borel transform	Entire	Entire; ordinary series has finite radius
Analytic continuation	Not needed	ordinary branch continuation; S-transform only with extra modular data
Stokes phenomena	None	not proved for the factorial Borel transform
Source	Compactness of $\overline{\mathcal{M}}_g$	Irreducibility of Q_L (class M)

The convergence of the genus tower is a theorem of intersection theory on $\overline{\mathcal{M}}_g$. The finite-radius branch structure of the shadow obstruction tower is a theorem of the quadratic extension K_L/F; divergence at $t = 1$ is parameter-dependent. The two analytic behaviours coexist because they probe different directions of the double series $\Theta_{\mathcal{A}} = \sum_{g,r} \Theta_r^{(g)} \hbar^{2g} t^r$.

50 The seven-face programme — Vol I drafts (April 2026)

50.1 Overview: one object, seven faces

The April 2026 restructuring drafts a comparison between seven independent strands of the chiral literature. The collision residue

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \;\in\; \mathrm{Hom}(\mathcal{A} \otimes \mathcal{A},\, \mathcal{A})((z))$$

is the arity-2, genus-0 projection of the bar$-$intrinsic Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$ proved by Theorem thm:mc2-bar-intrinsic. The programme is to prove that each construction presents the same $r(z)$ after fixing the trace form, spectral parameter, prime-form/local-coordinate convention, and associated-graded classical normalisation. This is verified only on the scoped theorem lanes, not as a blanket theorem about all seven presentations in the literature. The seven faces are: (1) the bar–cobar twisting morphism of Theorem A; (2) the Dimofte–Niu–Py A_∞ Yang–Baxter MC element on line operator categories; (3) the Khan–Zeng 2025 Poisson vertex $r-$matrix arising from the classical limit of the chiral OPE; (4) the Gaiotto–Zeng 2026 leading commuting Hamiltonian on W_N vacuum modules; (5) the Drinfeld Yangian $r-$matrix in the original 1985 form; (6) the Sklyanin Poisson bracket on classical $r-$matrices; and (7) the Gaudin Hamiltonian on tensor products of evaluation modules.

50.2 Sketches of the seven-face master theorem proof

The proof proceeds in three layers.

Layer 1: Construction. Define $r(z) := \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ as the binary genus-0 collision residue of the universal MC element. This construction is intrinsic after the above normalisations are fixed; the comparison with external r-matrices is not independent of trace-form and spectral-parameter conventions.

Layer 2: Identification, face by face. For each of the seven external constructions, exhibit an explicit isomorphism between the external $r(z)$ and the internal collision residue. The seven identifications are themselves theorems of the rectified Part III: bar–cobar identification (DNP), commuting differentials (GZ26), KZ classical/quantum bridge (KZ25), Gaudin/Yangian identification, Yangian/Sklyanin quantization, shadow depth/operator order, and the Drinfeld–Kohno spectral identification on prefundamental modules. Each identification is one–directional in construction but bidirectional in conclusion: the external object is recovered from the bar–intrinsic $r(z)$ by collision residue, and the bar–intrinsic $r(z)$ is recovered from the external object by uniqueness of the genus-0 tangent class.

Layer 3: Coherence. The seven identifications are mutually compatible: the diagram of pairwise comparisons commutes. This is not automatic. The DNP identification and the Gaudin identification, for instance, refer to different module categories (line operators vs. evaluation tensor products), and the agreement requires the categorical CG closure of MC3 plus the multiplicity–free $\ell-$weight argument of Theorem thm:categorical-cg-all-types. Similarly, the KZ25/GZ26 agreement requires the Khan–Zeng comparison of the classical PVA $r-$matrix with the leading Hamiltonian flow.

50.3 The c-independent identity $S_3(\mathrm{Vir}) = 2$

A finite identity that survives the entire central charge family. The arity-3 shadow projection of the Virasoro MC element evaluated on the stress tensor has a remarkable presentation:

$$S_3(\mathrm{Vir}) = 2.$$

Here S_3 is the normalized arity-3 shadow coefficient after projecting the OPE to the T-channel; it is not a literal quotient of the operator $T_{(1)}T$ by the scalar channel $T_{(3)}T$. The factor 2 is independent of c. This is the witness of class–M non–formality through a finite algebraic identity: while the full shadow tower of Virasoro is infinite (class M, $r_{\max} = \infty$), the cubic shadow on the stress–tensor line collapses to a c–independent constant. The same identity holds in DNP25's commuting–differentials picture (Theorem thm:gz26-commuting-differentials) and in the Gaiotto–Zeng 2026 GZ26 framework, providing a third independent verification of the $S_3 = 2$ value.

Open question: is there a similar c–independent identity for higher S_r on the stress–tensor line? Numerical evidence suggests $S_4(\mathrm{Vir})$ depends on c as $S_4 = 10/(c(5c+22))$ with no constant slice. The cubic collapse may be special to arity 3.

50.4 The W_3 holographic datum $H(W_3)$ — open computations

The W_3 holographic Koszul datum $H(W_3) = (W_3, W_3^!, C_{W_3}, r_{W_3}(z), \Theta_{W_3}, \nabla^{\mathrm{hol}}_{W_3})$ has been partially computed. The compute module `compute/lib/theorem_w3_holographic_datum_engine.py` verifies the following entries by at least three independent paths:

- Modular characteristic $\kappa(W_3) = 5c/6$, verified via direct OPE computation, trace-form duality $c \mapsto 100 - c$, and DS reduction from $\widehat{\mathfrak{sl}}_3$ at the principal nilpotent.
- Cubic shadow coefficient $\alpha_3 = 100$, central-charge conductor $K_3 = c + c^! = 100$, complementarity sum $\kappa(W_3) + \kappa(W_3^!) = (5/6) \cdot 100 = 250/3$, and self-dual value $\kappa^* = 125/3$ at $c^* = 50$. This is a third witness (after Virasoro's 13 at $c \mapsto 26 - c$ and Heisenberg's vanishing 0) that the complementarity sum is family–specific: it vanishes for KM and free fields, and obeys $\kappa + \kappa^! = \rho \cdot K$ with $\rho = 5/6$ and $K = 100$ for principal W_3 (Theorem D, AP24 qualified).
- Multi–weight cross–channel correction at genus 2: $\delta F_2(W_3) = (c + 204)/(16c) > 0$ for all $c > 0$. This is the computational refutation of the universal–scalar conjecture on the multi-weight genus-2 lane.
- Action of the leading W_3 commuting Hamiltonian Λ_0 on a primary $|h\rangle$: $\Lambda_0|h, w\rangle = (h^2 - 9h/5)|h, w\rangle$, computed from the GZ26 commuting differentials and verified against the Bouwknegt–Schoutens W_3 Hamiltonian table.

Open: what is the W_4 analogue of Λ_0? Is there a uniform formula for W_N that unifies the eigenvalues across the rank? Preliminary computations suggest $\Lambda_0(W_N)|h\rangle = h^2 - (N+1)h/(N+2) +$ (lower order) but the lower–order terms have not been verified beyond $N = 4$.

50.5 Open questions on the seven-face programme

- Does the seven–face identification extend to non–principal W–algebras beyond Bershadsky–Polyakov? The hook–type cases in type A are partially understood, but the seven faces have only been verified for principal DS reductions.
- Is there an eighth face arising from quantum K–theory of moduli spaces? Maulik–Okounkov's K–theoretic R–matrix on instanton moduli is a natural candidate, but its identification with the bar–intrinsic $r(z)$ requires a chiral lift of the K–theoretic stable envelopes.
- Does the trichotomy $r(z) = \tau$ vs. $r(z) \neq \tau$ (derived from local OPE) vs. $r(z) \neq \tau$ (independent input) survive in the q–deformed setting? The q–deformed bar complex has not yet been constructed, and the spectral parameter behaviour at $|q| < 1$ is unclear.

- What is the seven–face story for affine Yangians beyond $\mathfrak{sl}_N$? The affine super Yangian $Y(\mathfrak{gl}_1)$ provides one face for the simplest toric CY3, but the chiral side has not been worked out.

51 Frontier research programme (April 2026)

Eight open problems at the boundary of the seven-face programme, ordered by mathematical accessibility.

51.1 Spectral proof of Theorem B via Bethe completeness

The Gaudin model on evaluation modules of $Y(\mathfrak{g})$ has Bethe ansatz eigenstates whose completeness (Mukhin-Tarasov-Varchenko, for finite-dimensional $\mathfrak{g}$-modules) is equivalent to the surjectivity of the Miura oper map. The conjecture: this completeness, extended to weight-graded tensor products of Verma modules, gives a spectral proof of bar-cobar inversion.

The physics: the Bethe equations are the saddle-point conditions of the genus-0 MC free energy with spectral parameter insertions. The Bethe roots are the positions where the shadow connection has maximal symmetry. Completeness states that these saddle points account for the full partition function. In the geometric Langlands programme, Bethe completeness is the surjectivity of the map from opers to local systems; its equivalence with bar-cobar inversion would embed Theorem B into the Langlands correspondence.

The obstruction: infinite-dimensional representations. The Bethe equations for Verma modules are transcendental (not polynomial), and spectral simplicity fails at resonant levels. The critical level $k = -h^\vee$ is a wall. The minimal input: weight-by-weight reduction using the weight-space finiteness of chirally Koszul algebras.

51.2 Seven-face categorification

Each face of $r(z)$ is a functor from the category of modular Koszul chiral algebras to a target category (twisting morphisms, MC elements, Poisson brackets, commuting Hamiltonians, R-matrices, Poisson structures, Gaudin systems). The seven-way agreement is a system of natural isomorphisms. The conjecture: these assemble into a 2-functor valued in a 2-category $\mathrm{ChirKos}_2$, with coherent 2-isomorphisms satisfying hexagonal relations.

The physics: line operators in 3d holomorphic-topological QFT form a braided monoidal 2-category (the Kapustin-Rozansky-Witten structure). The seven faces are seven extractions of the braiding data from the bar complex. The 2-categorical coherence encodes the physical consistency of braiding, fusion, and modular transformations.

The obstruction: the seven categories live in different worlds (dg coalgebras, braided monoidal categories, D-modules on configuration spaces). The Lurie-Haugseng framework provides foundations but not the specific construction. The minimal input: coherence for the triangle F1-F5-F7 (bar-cobar, Yangian, Gaudin), all algebraic.

51.3 Shifted-symplectic complementarity via Sklyanin

The complementarity theorem (Theorem C) decomposes $Q_g(\mathcal{A}) + Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$. The conjecture: this decomposition is the Lagrangian intersection of two Lagrangian subvarieties in a (-1)-shifted symplectic stack $T_{\mathrm{comp}}(\mathcal{A})$, and the Sklyanin Poisson bracket is the genus-0 arity-2 shadow of this structure.

The physics: the (-1)-shifted symplectic structure is the BV structure of the 3d HT sigma model. The complementarity decomposition is the BV antibracket pairing between the boundary algebra $\mathcal{A}$ and its Koszul dual $\mathcal{A}^!$. The Lagrangian intersection computes the partition function of the bulk theory as a state in the boundary Hilbert space. At the perturbative level, the Sklyanin bracket controls the tree-level antibracket; the higher-arity corrections (C, Q, and the full $\Theta_{\mathcal{A}}$) are loop corrections.

The obstruction: the PTVV framework applies to derived Artin stacks; the convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is pro-nilpotent. The passage from Sklyanin (genus 0, arity 2) to PTVV (all genera, all arities) requires the Kontsevich-Pridham correspondence at the modular level.

51.4 Higher-genus seven faces beyond genus 1

At genus 1, the seven faces specialize to the KZB connection, the elliptic r-matrix, and the elliptic Gaudin model (proved for affine KM, Chapter ??). At genus ≥ 2, the bar propagator becomes the Szegő kernel on a higher-genus surface, the period matrix $\tau \in \mathfrak{H}_g$ replaces the single parameter, and the theory of automorphic r-matrices is needed.

The physics: the genus-g seven-face identification encodes the partition function of the 3d HT theory on $\Sigma_g \times \mathbb{R}$. The genus-2 case controls the first gravitational correction beyond the torus. The Hodge-bundle curvature $\kappa \cdot \omega_g$ is the modular anomaly measuring the failure of factorization. For class M (Virasoro), the genus-g collision residue produces higher-order automorphic differential operators coupling to the full Torelli moduli.

The obstruction: the theory of automorphic r-matrices (Etingof-Varchenko) exists for elliptic curves but not for higher-genus surfaces. The B-cycle monodromy introduces Hodge-bundle curvature that couples the z-dependence to the τ-dependence. The minimal input for genus 2: Enriquez's flat connections on configuration spaces of algebraic curves.

51.5 Koszulness from Sklyanin for classes C and M

For classes G and L (affine KM), the vanishing $H^2_\pi((\mathfrak{g}^!)^*, \{-,-\}_{\mathrm{STS}}) = 0$ gives a Poisson-geometric proof of Koszulness (Theorem ??). For class M (Virasoro, $\mathcal{W}_N$), the collision residue has higher-order poles producing differential Poisson brackets of order ≥ 2.

The physics: differential Poisson brackets of order ≥ 2 arise from higher-spin currents. The Virasoro bracket involves spin 2 (OPE pole order 4); the $\mathcal{W}_N$ brackets involve spin N (OPE pole order $2N$). The deformation quantization of these brackets is the vertex algebra quantization problem for higher-spin theories: passing from the classical PVA to the quantum vertex algebra through a chain-level deformation controlled by the higher-order Poisson cohomology.

The obstruction: no theory of differential Poisson cohomology for brackets of order ≥ 2 exists. For class C ($\beta\gamma$), the collision residue vanishes ($k_{\max} = 0$), so the Sklyanin bracket is trivial; Koszulness comes from the quartic contact invariant at arity 4.

51.6 Multi-weight genus ≥ 5

The cross-channel correction $\delta F_g^{\mathrm{cross}}$ for multi-weight algebras ($\mathcal{W}_3$, $\mathcal{W}_N$) is computed at genera 2, 3, 4. At genus 2: $\delta F_2(\mathcal{W}_3) = (c + 204)/(16c)$, decomposed into four graph topologies. The growth $\delta F_g / \delta F_{g-1}$ is factorial.

The physics: the factorial growth is the multi-weight analogue of the $(2g)!$ divergence of the bosonic string partition function. The cross-channel correction measures the failure of the scalar truncation

$\Theta^{\min}_{\mathcal{A}} = \kappa \cdot \eta \otimes \Lambda$. The generating function (if it exists) encodes the resurgent structure of the multi-weight theory: the Stokes constants of the Borel-resummed partition function.

The computation at genus ≥ 5 requires enumerating > 2000 stable graphs and is algorithmic.

51.7 Non-principal W-algebras beyond hook type

The hook-type transport duality is conditional on the DS/bar compatibility hypotheses. Beyond hook type, DS–Koszul commutation is not a formal consequence of the Slodowy slice or the generator count. The obstruction is the nonabelian BRST nilradical: $Q_{\mathrm{gh}} \neq 0$, so the Kazhdan spectral sequence and the ordered-bar lift require new control. The sl_5 partition $(3, 2)$ is the minimal non-hook test case. It has 8 strong generators / centralizer dimension 8, but the BRST nilradical has $\dim \mathfrak{n}_+ = 10$, grades $1/2, 1, 3/2, 2$, four-step nilpotence, ten nonzero bracket families, and ghost sectors $-10, \ldots, 10$. Koszul duality for this case is the first decisive non-hook problem, not a theorem deducible from spin data.

The physics: non-hook W-algebras arise from boundary conditions in 4d $\mathcal{N} = 2$ gauge theories with non-regular punctures (Gaiotto's class S programme). Their Koszul duality controls the Coulomb-branch/Higgs-branch duality. The seven-face programme for non-hook W-algebras would give a complete description of line operators for non-regular boundary conditions.

Proposition 51.1 *(The first non-hook obstruction in $\mathfrak{sl}_5$).* For the nilpotent orbit of type $(3, 2)$ in $\mathfrak{sl}_5$, the Drinfeld–Sokolov BRST differential has a nonzero ghost–ghost component

$$Q_{\mathrm{gh}} = \frac{1}{2} \sum_{i,j,k} f^k_{ij}\, c^i c^j b_k \neq 0$$

coming from the nonabelian positive Slodowy nilradical. Hence the hook-type proof of DS/bar commutation cannot apply: the associated Kazhdan spectral sequence has possible differentials generated by the ten bracket families of $\mathfrak{n}_+$, and Koszul duality for this orbit is equivalent to proving their cancellation in the ordered bar complex.

Proof. The hook argument uses abelianity of the BRST positive nilradical to remove Q_{gh} and identify the DS complex with a Koszul complex for linear constraints. For the good grading of the $(3, 2)$ orbit, $\mathfrak{n}_+$ has graded pieces $1/2, 1, 3/2, 2$ and ten nonzero bracket families. Therefore the structure constants f^k_{ij} do not vanish, so the quadratic ghost term $c^i c^j b_k$ is present in the BRST charge. This term changes bar length and ghost number simultaneously, which is exactly the mechanism excluded in the hook proof. The stated cancellation problem is the remaining spectral-sequence condition. □

Theorem 51.2 *(The decisive $(3, 2)$ computation).* For the good grading of the nilpotent orbit $(3, 2) \subset \mathfrak{sl}_5$, the ordinary hook-type quadratic Koszul duality statement is false. The obstruction is already visible on the root string

$$E_{12},\ E_{23},\ E_{13} \subset \mathfrak{n}_+, \qquad [E_{12}, E_{23}] = E_{13}.$$

which contributes the BRST term

$$c_{12} c_{23} b_{13}.$$

Consequently the first non-hook duality problem is not an ordinary quadratic Koszul problem. Its correct target is the inhomogeneous BRST–Koszul complex with the Chevalley–Eilenberg quadratic ghost differential retained. A positive theorem for the $(3, 2)$ W-algebra is equivalent to acyclicity of that inhomogeneous complex away from Kazhdan degree zero.

Proof. In the hook cases used by the earlier proof, the positive Slodowy nilradical is abelian in the relevant associated graded, so the BRST differential is the Koszul differential for linear constraints. For partition $(3, 2)$ the good grading contains the displayed root string. Since $[E_{12}, E_{23}] = E_{13}$, the Chevalley–Eilenberg part of the DS differential contains $c_{12}c_{23}b_{13}$ with nonzero coefficient. This term is quadratic in ghosts and lowers the antighost mode; it is not a linear Koszul constraint and it is not removed by passing to the associated graded. Hence ordinary quadratic Koszul duality cannot be the correct statement. Keeping the CE term gives precisely the inhomogeneous BRST–Koszul complex, and the spectral-sequence criterion for a positive theorem is its acyclicity outside degree zero. □

51.8 BV/BRST = bar at genus ≥ 2

The BV/bar identification is proved in the coderived category D^{co} by the coacyclic-curvature argument. At the ordinary chain level, classes G and L are proved; class C is conditional on harmonic decoupling; class M is false in the ordinary holomorphic chain category but becomes coacyclic in D^{co}.

The physics: BV/BRST = bar at genus g is the statement that perturbative quantization of the 3d HT sigma model on $\Sigma_g \times \mathbb{R}$ is equivalent to the algebraic bar complex. At genus 0 this is established physics (Costello-Gwilliam). At genus ≥ 1 the coderived theorem gives the algebraic replacement for the curved discrepancy, while ordinary chain-level identification has the class-dependent status above.

The decisive test: $F_2^{\mathrm{bar}}(\mathfrak{sl}_2, k{=}1) = 21469/69120$. The BV 2-loop integral on $\overline{\mathcal{M}}_{2,0} \times \mathbb{R}$ for holomorphic Chern-Simons should produce the same value.

52 Twelve frontier research directions

The following directions emerged from the investigation of the relationship between the chiral bar complex and the Swiss-cheese operad. Each represents a genuine open problem at the interface of algebra, geometry, and physics.

52.1 F1. BV/BRST = bar in the coderived category

The BV/BRST complex of a 3d holomorphic-topological field theory on $\Sigma_g \times \mathbb{R}$ encodes the perturbative quantum gauge symmetry: the cohomological mechanism by which unphysical degrees of freedom decouple. The bar complex $B^{(g)}(\mathcal{A})$ encodes the factorization structure: how observables compose when insertion points collide. That these two complexes should be quasi-isomorphic is the statement that quantum gauge symmetry *is* factorization.

The identification is proved at genus 0 for all families, and at genus 1 for shadow classes G, L, and C. For class M (Virasoro, $\mathcal{W}_N$), the identification *fails* at the ordinary chain level: the quartic harmonic discrepancy $\delta_4^{\mathrm{harm}} \sim Q^{\mathrm{contact}} \cdot \kappa/\mathrm{Im}(\tau)$ is non-holomorphic and hence not a coboundary of the holomorphic bar differential. The resolution (Theorem **??**): in Positselski's coderived category $D^{\mathrm{co}}(\mathcal{A})$, the curvature $d^2 = m_0$ absorbs the discrepancy, and the identification holds for *all* shadow classes.

What remains. The universal coderived comparison is proved; the remaining refinements are stronger completed or ordinary chain-level models, harmonic decoupling for class C, and a physical principle explaining why the worldsheet theory selects D^{co} rather than the bounded direct-sum chain category.

52.2 F2. The $(3,2)$ nilpotent in $\mathfrak{sl}_5$: gateway to non-principal DS-KD

Drinfeld–Sokolov reduction extracts $\mathcal{W}$-algebras from affine Kac–Moody algebras by gauging a nilpotent subalgebra. For non-principal nilpotents, the resulting $\mathcal{W}$-algebras describe boundary conditions of 4d $\mathcal{N}=2$ theories at Argyres–Douglas points. The DS-KD intertwining (bar-cobar commutes with DS) is proved when $\mathfrak{n}_+$ is abelian (all hook-type partitions in type A). For the $(3,2)$ partition of 5, the BRST nilradical has $\dim(\mathfrak{n}_+) = 10$, good-grading pieces $\frac{1}{2}, 1, \frac{3}{2}, 2$, four-step nilpotence, and ten nonzero bracket families. The number 8 belongs to the centralizer/strong-generator count, not to $\dim \mathfrak{n}_+$.

The BRST complex has matrix sizes $\le 3000 \times 3000$ (sparse) at the hardest weight. The computation is feasible in sympy, decomposed by ghost number $(-10, \dots, 10$, i.e. 21 sectors).

52.3 F3. Genus-5 cross-channel: the Borel-determining computation

The genus expansion of a multi-weight chiral algebra receives cross-channel corrections $\delta F_g^{\text{cross}}$ from mixed-propagator graphs. The scalar tower $F_g^{\text{scal}} = \kappa \cdot \lambda_g^{\text{FP}}$ converges (Gevrey-0, $\hat{A}$-algebraicity). The cross-channel tower grows factorially: the ratio $\delta F_g^{\text{cross}}/F_g^{\text{scal}}$ reaches $42/31$ at $g = 3$ and ~ 24 at $g = 4$, consistent with but not proving Gevrey-1 behaviour. The interval $A_{\text{cross}}/A_{\text{scal}} \in [1.7,\ 3.1]$ is a three-data-point extrapolation, not a controlled instanton theorem.

The genus-5 computation (~ 4000 stable graphs, 3–8 hours on 1 core) would determine the Gevrey shift parameter and test whether a single A_{cross} governs the data. The denominator structure $D_3 = 24 \cdot 5760 = \text{denom}(\hat{A}_1) \cdot \text{denom}(\hat{A}_2)$ suggests $\hat{A}$-genus arithmetic controls the cross-channel normalizations.

52.4 F4. Admissible $\mathfrak{sl}_3$ Koszulness

For $\mathfrak{sl}_2$, the simple quotient $L_k(\mathfrak{sl}_2)$ is chirally Koszul at all admissible levels (structural argument from single-weight null vector + Kac–Wakimoto character formula). For $\mathfrak{sl}_3$ the null-vector ideal has generators at *multiple* conformal weights, defeating the single-generator argument. The decisive computation: the Li-bar E_2 page for $k = -3/2$ ($p=3$, $q=2$), the first admissible level where nulls enter the bar range. The C_2 algebra is finite-dimensional ($\dim < 100$).

52.5 F5. Restricted DK-4 on the evaluation core

The Drinfeld–Kohno theorem connects the monodromy of the KZ connection to the R-matrix of $U_q(\mathfrak{g})$. DK-4 extends this to the full formal moduli problem. The evaluation-core restriction is a finite-dimensional problem at each evaluation point. The reduction chain for type A reduces DK-4 to a single mixed-tensor coefficient identity, verified on the factorization side. The gap: pointwise data does not automatically assemble into a global algebraic identification.

52.6 F6. DK-5 = categorical E_1 primacy

The conjectural full triple bridge $\text{Fact}_{E_1}(X; \mathcal{A}) \simeq \text{Mod}^{\text{comp}}(Y_{\mathcal{A}}^{\text{dg}}) \simeq \text{Rep}^{\text{spec}}(\text{QG}(R_{\mathcal{A}}))^{\text{op}}$ would make the braided monoidal category of line operators the primitive datum from which modular tensor categories, conformal blocks, and genus-g partition functions are derived. The Bridge Criterion Theorem reduces DK-5 to three sub-targets (B1, B2, B4). The $\mathfrak{sl}_2$ case is proved; the general simple-type thick-generation and global algebraic identification remain conjectural.

52.7 F7. The grand completion

The modular cumulant transform packages the full bar-cobar machine into a single algebraic object equivalent to the chiral algebra up to homotopy. Two sub-conjectures (cumulant recognition, jet principle) plus an equivalence of model categories extending the proved genus-0 Quillen equivalence. The principal open structural problem. Rated VERY HARD.

52.8 F8. Analytic realization: three-layer gap

The algebraic bar-cobar machine must extend to convergent, Hilbert-space-valued factorization theory. Three layers of gap: (1) sewing envelope construction for interacting algebras, (2) metric independence of the conformally flat 2-disk algebra at chain level, (3) higher-genus coderived shadow. Proved for Heisenberg and lattice VOAs.

52.9 F9. E_1 Verdier on ordered configurations

The Verdier intertwining $\mathbb{D}_{\mathrm{Ran}}\overline{B}(\mathcal{A}) \simeq \overline{B}(\mathcal{A}^!)$ is a statement about the symmetric bar on $\mathrm{Ran}(X)$. A naive ordered analogue $\mathbb{D}_{\mathrm{Ran}}(\overline{B}^{\mathrm{ord}}(\mathcal{A})) \simeq \overline{B}^{\mathrm{ord}}(\mathcal{A}^!)$ does *not* exist: the ordered bar lives on $\mathrm{Conf}_n^<(X)$, not on $\mathrm{Ran}(X)$. The correct E_1 analogue is opposite-duality: $\overline{B}^{\mathrm{ord}}(\mathcal{A}^{\mathrm{op}}) = \overline{B}^{\mathrm{ord}}(\mathcal{A})^{\mathrm{cop}}$. Developing $\mathbb{D}_{\mathrm{Conf}^<}$ (Verdier duality on ordered configurations) or a ribbon Ran space is a genuine open direction in higher algebra.

52.10 F10. Resurgence: test A_{cross} at genus 5

The scalar lane has a controlled action $A_{\mathrm{scal}} = (2\pi)^2$. For cross-channel data, three data points (genera 2, 3, 4) suggest an effective interval $A_{\mathrm{cross}}/A_{\mathrm{scal}} \in [1.7,\ 3.1]$, but no single-action theorem is known. A fourth point (genus 5) would test whether a Gevrey ansatz of the form $E_g \sim C \cdot \beta^g \cdot \Gamma(2g + b)$ actually governs the data.

52.11 F11. Cross-channel generating function

For $\mathcal{W}_3$, no one-variable $\hat{A}$-type, geometric, or algebraic ansatz matches $\delta F_g^{\mathrm{cross}}$: the obstructions are inhomogeneous c-scaling, super-linear ratio growth, and irreducible numerators. The broader closed-form problem for $\mathcal{W}_N$ remains open; any generating function is expected to be multi-variable and non-separable.

52.12 F12. Scalar saturation beyond algebraic families

The residual content of the scalar saturation conjecture is $\dim H^2_{\mathrm{cyc}}(\mathcal{A}, \mathcal{A}) = 1$ for non-algebraic-family modular Koszul algebras. Three candidate families need checking: non-GKO cosets, 4d $\mathcal{N}$=2 quiver VOAs, and admissible-level simple quotients at rank ≥ 2. No counterexample is known.

53 Cross-volume notes from the Vol III 180-agent session (2026-04-13)

The following observations from the Vol III 180-agent session (waves 10–12) bear on Vol I structures. Full details are in Vol III working notes, Section ??.

53.1 Derived PBW corrections = shadow tower $S_k/(k-1)!$

The chiral envelope $U^{\mathrm{ch}}(\mathfrak{g})$ has a derived PBW filtration whose k-th correction term has coefficient $S_k/(k-1)!$, where S_k is the k-th shadow invariant from the Vol I MC recursion engine. This gives a new characterisation of the G/L/C/M shadow classes: they classify chiral envelopes by the failure of PBW. Class **G**: PBW holds exactly. Class **M**: infinitely many corrections, PBW is maximally non-split.

Combined with the identification $S_k = \delta^{(k)}$ (shadow = A_∞ coproduct correction, established in the earlier waves), this means the PBW filtration failure of $U^{\mathrm{ch}}(\mathfrak{g})$ is *exactly* the perturbative expansion of the Yangian coproduct.

53.2 Stokes = wall-crossing conjecture

For class **M** chiral algebras (Borel-summable by the discriminant condition $\mathrm{disc}(Q_L) < 0$ from the Vol I shadow tower computation), the Stokes automorphism of the Borel sum is conjecturally equal to the Kontsevich–Soibelman wall-crossing automorphism. Evidence: for the conifold, both are $\exp(\mathrm{Li}_2(X_\gamma))$. If true, this unifies the Vol I resurgence programme (F10, F11) with the KS wall-crossing formalism. The shadow tower controls the perturbative coefficients; the Stokes data controls the non-perturbative completion.

53.3 The 3d $\longrightarrow$ 6d lift: 24% survival rate

A systematic survey (Vol III, `3d_6d_obstruction_catalogue`, 99 tests) found that only 24% of 3d holomorphic CS structures survive the lift to 6d. The obstructions are: ZTE failure (integrability), anomaly cancellation ($c_2 = 0$), and holomorphicity ($\Omega^{6,0}$ exists only on CY_3). The survivors are exactly the structures that factor through CY_3. This constrains the scope of the bar-cobar machine at higher dimensions: the Vol I bar complex at genus ≥ 2 is sensitive to the E_3 content that the 6d theory probes.

54 Session 2026-04-17: Platonic synthesis for Vol I

Session's findings bearing on Vol I structures. Comprehensive programme-wide synthesis is at `../chiral-bar-cobar-vol2/working_notes.tex` Section **??**.

54.1 Universal K_W as Chevalley–Shephard–Todd invariant

For principal $\mathcal{W}(\mathfrak{g})$ at non-critical level:

$$\kappa_W(\mathfrak{g}) \;=\; 2\,\mathrm{rank}(\mathfrak{g}) \;+\; 12 \sum_{i=1}^{\mathrm{rank}(\mathfrak{g})} (d_i - 1)\, d_i \;=\; 12\, p_2(\mathfrak{g}) \;-\; 2\,\mathrm{rank}(\mathfrak{g}) \cdot \big(3\, h(\mathfrak{g}) + 5\big),$$

where $d_i = m_i + 1$ are Chevalley–Shephard–Todd fundamental degrees, $p_2(\mathfrak{g}) = \sum_i d_i^2$ is the second power sum, and $h(\mathfrak{g})$ is the Coxeter number. Universal across all simple $\mathfrak{g}$ (simply-laced and non-simply-laced); verified against Feigin–Frenkel screening-charge central-charge formula at A_{N-1}, B_r, C_r, D_r, $E_{6,7,8}$, F_4, G_2.

Companion affine formula:

$$\kappa_V(\mathfrak{g}) \;=\; 2\,\dim(\mathfrak{g}),$$

uniformly in $k \neq -h^\vee$, derived from Sugawara $c = k\ \dim/(k+h^\vee)$. Super version: $\kappa_V = 2\,\mathrm{sdim}$ when $h^\vee \neq 0$; fails at $h^\vee = 0$ super types ($\mathfrak{psl}(n|n)$, $\mathfrak{osp}(2m|2m)$, $D(2,1;\alpha)$) where Sugawara does not exist.

The exponent-based form is Langlands-invariant: $\kappa_W(B_r) = \kappa_W(C_r)$ for all r. This is the structural signature that Langlands duality $\mathfrak{g} \leftrightarrow \mathfrak{g}^L$ preserves Lie-algebra invariants (Vol I) and permutes root-datum invariants (Vol II, FF-Langlands-naturality).

54.2 Theorem A upgrade: $(\infty, 2)$-properad with R-twisted descent

Theorem A promotes to an $(\infty,2)$-categorical properad-level equivalence on the Francis–Gaitsgory factorisation ambient over $\mathrm{Ran}(X)$ with $\otimes^*$ tensor. Ordered B^{ord} and symmetric B^Σ bar complexes are related by R-matrix-twisted Σ_n-descent along $\mathrm{Ran}^{\mathrm{ord}}(X) \to \mathrm{Ran}(X)$. FORM-A and FORM-B are independent proofs, each unconditional on its locus.

Ambient category triple atlas (resolves "(D-mod, $\otimes^!$)" conflation): (i) (D-mod($\mathrm{Ran}(X)$), $\otimes^*$) for FORM-A; (ii) IndCoh($\mathrm{Ran}(X)$) for Yangian / critical Kac–Moody; (iii) $\mathrm{D}^b_c(\mathrm{Ran}(X))$ with MHM for tensor-Arakelov Theorem D. The BD chiral pseudo-tensor $\otimes^{\mathrm{ch}}$ is never an ambient –- it generates operations inside $\mathrm{Fact}(X)$.

54.3 Periodic-CDG admissible-level programme for simple $\mathfrak{g}$

Conjecture periodic-cdg for $\widehat{\mathfrak{g}}$ at admissible $k = p/q$ is expected to close for simply-connected simple $\mathfrak{g}$ after the following inputs are verified:

- **Simply-laced**: Brylinski–Deligne K_2-reciprocity (IHES 94 (2001) Thm 4.7) plus Arkhipov–Gaitsgory κ-twisted Serre t-exactness on $\mathrm{Pic}(\mathrm{GR}_G) = \mathbb{Z}$ (BFM04, BFM05).
- **Non-simply-laced**: Frenkel–Hernandez lacing automorphism $\Phi_{\mathfrak{g}}(T^{(n)}_\alpha) = (-r_{\mathfrak{g}})^n T^{(n)}_{\alpha^*}$ with lacing $r_{\mathfrak{g}} \in \{2,3\}$, extending simply-laced to B_r, C_r, F_4, G_2.

Residual frontiers: (i) degenerate admissible lane ($X_{L_k} \neq \{0\}$, Creutzig–Kanade–Linshaw logarithmic), (ii) non-admissible generic k (Part VII Massey structure), (iii) critical-level FF regime (a separate adjunction regime, not the fifth G/L/C/M/B archetype and not a limit of admissible levels).

Thus the Kac–Moody/bar-cobar adjunction at admissible negative level is a proved theorem only on the lanes where these reciprocity, Serre-exactness, and lacing-transport hypotheses have been checked.

54.4 Exotic-nilpotent DS-Hochschild chain-level closure

Theorem 54.1 *(DS–Hochschild bridge: internal-hom exactness criterion).* On the generic principal lane the coalgebra-side DS/bar comparison is available: DS reduction commutes with the bar coalgebra comparison on the scoped hypotheses already used in Theorem H. The derived-centre statement is equivalent to the following internal-hom exactness criterion, degree by degree in the bar resolution:

$$H^0_{\mathrm{DS}}\big(C^p_{\mathrm{ch}}(A,A)\big) \longrightarrow C^p_{\mathrm{ch}}(W,W) \quad \text{is a quasi-isomorphism,} \qquad H^q_{\mathrm{DS}}\big(C^p_{\mathrm{ch}}(A,A)\big) = 0 \quad (q>0).$$

When the internal-hom terms admit a bounded-below Kazhdan filtration whose associated graded is a free Koszul complex for the DS constraints, this criterion holds and the DS–Hochschild bridge is a theorem:

$$\mathrm{ChirHoch}^\bullet(W,W) \simeq H^\bullet_{\mathrm{DS}}\big(\mathrm{ChirHoch}^\bullet(A,A)\big).$$

For non-principal or exotic nilpotents the same display remains the exact test; the obstruction is precisely failure of this filtered free-Koszul condition.

Proof. Filter the chiral Hochschild bicomplex by bar degree and Kazhdan degree. The DS differential is vertical and the Hochschild differential is horizontal. If the internal-hom terms satisfy the displayed vanishing, the vertical spectral sequence collapses at E_1 to $C^\bullet_{\mathrm{ch}}(W, W)$, so the total complex computes $\mathrm{ChirHoch}^\bullet(W, W)$. Conversely, if the derived-centre comparison is an isomorphism functorially in each bar degree, the vertical cohomology must be concentrated in degree zero and equal to the internal-hom term for W; otherwise nonzero $q > 0$ classes survive to the total complex or force unavailable differentials across bar degree.

The filtered free-Koszul hypothesis proves the vanishing: the associated graded DS complex is the Koszul resolution of a regular sequence of DS constraints, so its cohomology is degree zero. Bounded-below convergence transfers the statement from the associated graded complex to the original internal-hom complex. □

The DS-Hochschild bridge $\mathrm{ChirHoch}^\bullet(\mathcal{W}_k(\mathfrak{g}, f)) \simeq H^\bullet_{\mathrm{DS}}(\mathrm{ChirHoch}^\bullet(V_k(\mathfrak{g})))$ extends to good-graded and exotic nilpotents f only after proving a three-step Galois descent at branched covers of degree $d_f \in \{1, 2, 3, 6\}$:

(i) Degree-d_f cover $\widetilde{X} \to X$ where fractional Kazhdan weights integralise.
(ii) Integer-weight DS-Hochschild bridge on the cover.
(iii) Galois-descent via Maschke-exact deck-covariant averaging.

Proposed mechanism: Crainic 2004 equivariant HPL (arXiv:math/0403266) with $\mathbb{Z}/d_f$-equivariant strong deformation retract. The averaging operator $P = (1/d_f) \sum_g \sigma_g^*$ commutes with Q_{BRST} via deck-covariant pullback, giving chain-level (not merely cohomological) identities to X.

This closes the stated holography functor only on the scoped proved lanes where the DS/bar-shadow and equivariant HPL hypotheses are verified. Arbitrary good-graded, exotic, and non-principal nilpotents remain outside the unconditional theorem; the first open non-hook test is the $(3, 2)$ nilpotent in $\mathfrak{sl}_5$.

54.5 Six-slot fingerprint classification

The Infinite Fingerprint Classification (Vol II Theorem G) upgrades from five slots to six: $\varphi'(A) = (p_{\max}, r_{\max}, \chi_{\mathrm{VOA}}, n_{\mathrm{strong}}, \mathrm{coset}, \kappa_{\mathrm{ch}})$. The sixth slot is the central-extension / level / curvature class pulled back from Theorem D's universal Arakelov class. Refinement required because the Heisenberg family $\{\mathcal{H}_k\}_{k\in\mathbb{C}^*}$ collides at five slots but has genuinely different R-matrix $\exp(k\hbar/z)$ and bar curvature $d_B^2 = k \cdot \omega_g$ distinguishing different k.

The G/L/C/M quadrichotomy becomes a double coarse projection $\varphi' \twoheadrightarrow \varphi \twoheadrightarrow \{\mathrm{G}, \mathrm{L}, \mathrm{C}, \mathrm{M}\}$. The canonical five-bucket ledger is G/L/C/M/B, where B is the Vol III Mukai-enhanced K3 Heisenberg witness. Feigin–Frenkel critical structure is a separate critical-level regime, not a fifth G/L/C/M archetype.

54.6 Higher Massey odd-weight tower

The chain-level chiral Deligne–Tamarkin statement for Yangian $r(u) = \Omega/u$ is established *associator-dependent* via depth-3 Drinfeld–Kohno analysis: the $\hbar^3$-coefficient of Φ_{KZ} lives in $t_3^{\langle 3\rangle}$ (rank 2 over $\mathbb{Q}$); decomposable-bracket image rank 1; non-decomposable residue $= \zeta(3)\,\sigma_3$ via Brown motivic depth.

Extension to all odd weights: for each $2k+1 \geq 3$, the Higher Massey $\langle r^{\times(2k+1)} \rangle$ is detected by $\zeta(2k+1)\,\sigma_{2k+1}$ in $\mathrm{grt}_1^{\text{weight } 2k+1}$ (Brown motivic). Each odd-weight class is independent, so the *associator-independent* form of chiral Deligne–Tamarkin FAILS –- no single Drinfeld associator absorbs all obstructions. The $\mathrm{GRT}_1(\mathbb{Q})$-torsor of associator-dependent statements is the correct Platonic form.

Relevant to Vol I's Theorem $\mathrm{A}^{\infty,2}$: the R-matrix-twisted Σ_n-descent requires a Drinfeld associator choice; under $\mathrm{GRT}_1(\mathbb{Q})$ action, Theorem A has an orbit of equivalent forms indexed by the associator.

54.7 $\mathcal{W}(p)$ temperedness: the retraction was over-retracted

The 2026-04-17 Beilinson audit retracted "$\mathcal{W}(p)$ tempered, CLOSED" on the grounds that the Zhu-bounded-Massey proof chain fails at step 2 by Gurarie–Flohr: the indecomposable-module sector of $\mathcal{W}(p)$ carries unbounded Massey amplitudes $\langle \Omega, \Omega, \Omega \rangle$ at every weight, so a finite-dimensional Zhu algebra does not imply bounded A_∞ operations.

The *retraction of the proof chain* is correct. The *retraction of the statement* is premature. A second route exists and should be examined before the status of $\mathcal{W}(p)$ tempering is consigned to `OPEN`.

Second route (direct character-amplitude bound). $\mathcal{W}(p) = \mathcal{W}(2, 2p-1)$ is C_2-cofinite (Adamović–Milas 2009). The associated Zhu algebra has dimension $3p-1$ (Flohr–Grabow–Koehn). The rank and C_2-cofiniteness combine to give: *every $\mathcal{W}(p)$-module partition function satisfies a linear ODE of order $3p-1$ with regular singularities.* By the classical Frobenius theorem on regular-singular ODEs, such solutions grow at most polynomially with the modular parameter. Passing to q-expansions via $q = e^{2\pi i \tau}$, the Fourier coefficients $\chi_n = \sum_{\text{modules}} a_n$ satisfy $|\chi_n| \leq C \cdot n^{3p-2}$ for some module-independent $C = C(p)$. The shadow coefficients S_r are extracted from the partition function by the Cauchy contour formula (Bar propagator $d \log E(z,w)$, weight 1); integrating against $(2\pi i)^{-1} \oint q^{-r-1} \cdot (\cdot)\, dq$ gives $|S_r(\mathcal{W}(p))| \leq C' \cdot r^{3p-2}$, polynomial in r. Polynomial growth in r is stronger than tempered growth (sub-exponential) and, in particular, the $\mathcal{W}(p)$ shadow sits in the *polynomial* sub-stratum of the tempered stratum.

Refinement. The two properties–-*temperedness* (growth of the shadow coefficients $|S_r|$) and *Koszul-locus membership* (vanishing of the triple Massey product $\langle \Omega, \Omega, \Omega \rangle$ at weight 3)–-are *independent* for $\mathcal{W}(p)$:

- **Tempered.** Via the direct character-amplitude route: $\mathcal{W}(p)$ sits in the tempered stratum, more specifically in the polynomial sub-stratum of order $3p-2$. This is unaffected by the Gurarie–Flohr Massey unboundedness, which measures a different invariant.
- **Non-Koszul.** The triple Massey product $\langle \Omega, \Omega, \Omega \rangle$ on the indecomposable-module sector is nonzero at weight 3 (Gurarie 1993, Eq. 2.11; Flohr 1996, Fig. 3); by Theorem $\mathrm{A}^{\infty,2}$ the bar-cobar $(\infty, 2)$-adjoint equivalence is obstructed at this Massey class. $\mathcal{W}(p)$ is off the Koszul locus.

The bar-complex shadow tower of $\mathcal{W}(p)$ is thus a polynomial tempered object whose higher $\mathcal{E}_\infty$-structure is obstructed on the indecomposable sector. The programme's class labels must be refined: *tempered* and *Koszul* are orthogonal axes.

Consequence for Theorem F (Universal Holography). The non-logarithmic standard landscape has Universal Holography as an $(\infty, 2)$-adjoint equivalence. On the logarithmic $\mathcal{W}(p)$ stratum the $(\infty, 2)$-equivalence fails (Koszul-locus obstruction), but the $(\infty, 1)$-level bulk-boundary assembly exists: the bulk $\mathsf{SC}^{\mathrm{ch,top}}$-algebra is the derived centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{W}(p))$, and the boundary is $\mathcal{W}(p)$, related by a twist-deformation *functor* rather than an equivalence. The Gurarie–Flohr Massey unboundedness is the explicit twist class obstructing upgrade from $(\infty, 1)$ to $(\infty, 2)$.

This refinement should be inscribed into the residual-boundary piece (a) of the programme overview (Vol I `chapters/frame/programme_overview_platonic.tex`) at the next iteration of this loop, together with the updated CLAUDE.md theorem-status row for Theorem F. Marked `CONJECTURAL` at present because the polynomial-character-growth $\Rightarrow$ tempered S_r-bound chain requires explicit verification of the $(3p-2)$-polynomial-order constant $C(p)$ for $p = 2, 3$ at the compute layer before it can be promoted.

Test protocol. The minimal falsification is: compute the shadow coefficients S_r of $\mathcal{W}(2)$ (the triplet model at $c = -2$) for $r \leq 10$, fit to $S_r \sim A \cdot r^{\alpha}$ at large r, check $\alpha \leq 3 \cdot 2 - 2 = 4$. If $\alpha > 4$, the polynomial route is refuted and the retraction stands. If $\alpha \leq 4$, $\mathcal{W}(p)$ is tempered via the direct character-amplitude route and the Vol I Theorem F scope should be re-examined.

Correction to the polynomial estimate (2026-04-17 post-inscription review). The heuristic chain above conflated two growth indices: *character-weight index* h (via $\chi(q) = \sum c_h q^h$) with *shadow-depth index* r. The Gaberdiel–Kausch graded dimensions of $\mathcal{W}(2)$ at $c = -2$ are $[1, 0, 1, 4, 5, 8, 19, 28, 49, 84, 135, 216, 356, 548, 862, 1296, 1969, 2928, 4389, 6448, 9555]$ (weights $h = 0, 1, \ldots, 20$; source `compute/lib/triplet_wp_character_engine.py` line 264). A log-log fit gives $c_h \sim h^{6.1}$ as the apparent polynomial order; but this is the local slope, and the true asymptotic is sub-exponential $c_h \sim h^{-3/4} \exp(\pi\sqrt{2h/3})$ (Hardy–Ramanujan plus Fuchs correction). Consequently the ODE-order argument yields a sub-exponential bound on c_h, *not* polynomial; and the shadow depth r is related to h by a bar-complex combinatorial factor that needs to be extracted.

The refined claim is therefore: $|S_r(\mathcal{W}(p))|$ grows sub-exponentially (Fuchs regime from the $3p-1$ regular-singular ODE on the partition function), hence $\mathcal{W}(p)$ sits in the *sub-exponential stratum* (one order coarser than polynomial but still bounded by $\exp(C\sqrt{r})$, stronger than exponential). This is stronger than "tempered" in the distributional sense (which is polynomial growth) but weaker than "polynomial" in the strict sense. The programme may need a three-way refinement of growth strata: polynomial $\subset$ sub-exponential $\subset$ tempered (exp), with $\mathcal{W}(p)$ at the sub-exponential level and the non-logarithmic standard landscape at the polynomial level.

The independence of *temperedness* and *Koszul-locus membership* is preserved under this refinement: $\mathcal{W}(p)$ is sub-exponentially tempered via Fuchs + Hardy–Ramanujan, and off the Koszul locus via Gurarie–Flohr. Tempered and Koszul are orthogonal axes; the specific *stratum* within "tempered" gives a third axis that the programme should track (polynomial vs sub-exponential vs exponential).

Numerical refinement 2026-04-17 (after Hardy-Ramanujan naive fit fails). Fitting $\log c_h = a + b\sqrt{h}$ to the Gaberdiel–Kausch sequence for $h \in [8, 20]$ gives $b = 3.2054 \pm 0.02$, not $\pi\sqrt{2/3} \approx 2.5651$. The deviation is 25% and persists across the fit range. Two explanations, both compatible with the sub-exponential stratum:

(N1) *Effective central charge correction.* Cardy's formula gives $c_h \sim \exp(2\pi\sqrt{c_{\text{eff}} \cdot h/6})$ where $c_{\text{eff}} = c - 24h_{\min}$. For $\mathcal{W}(2)$ at $c = -2$ the minimal-weight indecomposable module sits at $h_{\min} = -1/8$ in the symplectic-fermion realization; naive Cardy gives $c_{\text{eff}} = -2 - 24(-1/8) = 1$, hence slope $2\pi\sqrt{1/6} = 2.5651$ (i.e. Hardy–Ramanujan). The fit $b = 3.2054$ corresponds to $c_{\text{eff}} \approx 1.56$. The effective-$c$ value 1.56 is not a clean rational; this flags a *logarithmic enhancement* at finite h (consistent with Gurarie–Flohr non-semisimple Zhu action multiplying the leading asymptotic by $\log h$ factors that shift the naive slope).

(N2) *Finite-h corrections dominate.* The $h \leq 20$ range is below the asymptotic regime; the genuine slope is $\pi\sqrt{2/3}$, recovered only at $h \gtrsim 50$. Extending the Gaberdiel–Kausch sequence (via the FGST theta formula inscribed in `compute/lib/triplet_wp_character_engine.py`) is a test protocol for this explanation.

Either explanation preserves sub-exponential growth: the fit slope 3.2054 still gives $c_h = O(\exp(3.21\sqrt{h}))$, and shadow coefficients inherit sub-exponential growth in r through the bar-complex r-to-h relation. The polynomial stratum (which would have $b = 0$) is firmly ruled out; the sub-exponential stratum stands.

Conclusion: the programme's growth-stratification axis needs three layers, empirically verified.

- *Polynomial.* The non-logarithmic standard landscape (Virasoro, $\mathcal{W}_N$, Schellekens holomorphic) has rational-function character generators, hence polynomial $|S_r|$.
- *Sub-exponential.* The logarithmic stratum (C_2-cofinite with non-semisimple Zhu, e.g. $\mathcal{W}(p)$) has character with Hardy–Ramanujan or faster-Cardy growth, giving $|S_r|$ sub-exponential.
- *Exponential.* Non-C_2-cofinite VOAs (generic $\beta\gamma$ system without conformal restriction, logarithmic free fields) admit exponential character growth; these sit outside the tempered stratum entirely.

The refined Theorem F scope for Universal Holography: $(\infty, 2)$-adjoint equivalence holds on the POLYNOMIAL sub-stratum of the tempered class; $(\infty, 1)$-adjoint morphism holds on the SUB-EXPONENTIAL sub-stratum (including $\mathcal{W}(p)$) with Gurarie–Flohr Massey class as the obstruction to $(\infty, 2)$; status OPEN on the EXPONENTIAL sub-stratum pending C_2-cofinite rescoping.

Second numerical refinement (2026-04-17): pure partition function is also pre-asymptotic at $n = 100$. The partition function $p(n)$ satisfies $\log p(n)/\sqrt{n} \to \pi\sqrt{2/3} = 2.5651$ at $n \to \infty$, but at $n = 100$ the local ratio is only 1.9066, fit over $[80, 100]$ gives slope 2.3589, deviation 8% from HR. Hence the 25% deviation observed for $\mathcal{W}(2)$ vacuum at $h \in [8, 20]$ is not anomalous: it is the same pre-asymptotic regime, amplified by the p-fold generator multiplicity.

This resolves hypothesis (N1) vs (N2) in favor of (N2): the finite-h correction is the dominant source of deviation. The true asymptotic slope for $\mathcal{W}(2)$ vacuum is $\pi\sqrt{2/3}$ (after effective-c correction $c_{\text{eff}} = 1$), but finite-h corrections are $O(1)$ at $h \leq 20$.

For $\mathcal{W}(p)$ more generally, the effective central charge is $c_{\text{eff}}^{(p)} = 1$ (minimal-weight module is the indecomposable logarithmic pair at $h_{\min} = -(p-1)^2/4p$), giving universal leading asymptotic slope $\pi\sqrt{2/3}$. Sub-leading corrections depend on p through the theta-function polynomial prefactors. The sub-exponential stratum is *universal* at leading order across all $\mathcal{W}(p)$; it does not depend on p.

This simplifies the three-layer stratification: polynomial (standard landscape), sub-exponential (logarithmic C_2-cofinite, all p), exponential (non-C_2-cofinite). The middle layer has a universal leading-order asymptotic slope $\pi\sqrt{2/3}$.

The self-correcting Beilinson loop in action. Iteration 1: inscribed naive polynomial bound via ODE order. Iteration 2: caught conflation of h-index with r-index; refined to sub-exponential via Hardy–Ramanujan on η^{-1}. Iteration 3: numerical fit gave slope 3.2054, not 2.5651; refined to "effective-c correction or finite-h". Iteration 4: pure partition function also pre-asymptotic at $n = 100$; hypothesis (N2) wins; W(p) sub-exponential stratum has universal leading slope $\pi\sqrt{2/3}$. Iteration 5 (this entry): re-examined r-index growth using existing Virasoro engine; the "polynomial stratum" label was wrong at r-index. Each iteration takes the previous inscription as input and refines it via first-principles numerical check plus Beilinson audit.

54.8 r-index growth correction: the standard landscape is exponential, not polynomial

The Virasoro shadow leading asymptotic (from `compute/lib/shadow_tower_higher_vir.py` `leading_asymptotic(r)`) is

$$A_r^{\mathrm{Vir}} = \frac{8 \cdot (-6)^{r-4}}{r},$$

i.e. $|A_r| \sim 6^r/r$: linear exponential in r. The h-index sub-exponential growth of the character c_h does not transfer to the r-index shadow coefficient by a simple substitution; the bar-complex combinatorics introduce an explicit 6^r factor reflecting the 6-fold multiplicity of FM_r boundary strata contributing to the Virasoro shadow at each depth.

Corrected four-layer stratification at the r-index:

- *Finite (terminating).* Classes G/L/C with $d_{\mathrm{alg}} \leq 2$ (Heisenberg, affine KM non-critical, $\beta\gamma$): $S_r = 0$ for $r > r_0$. Terminating tower.
- *Exponential.* Class M with $d_{\mathrm{alg}} = \infty$ (Virasoro, $\mathcal{W}_N$ at generic c): $|S_r| \sim C^r/r$, $C = 6$ for Virasoro. Linear exponential; this is the "standard tempered" class.
- *Gevrey-α with $\alpha > 0$.* Class M with non-semisimple Zhu, logarithmic $\mathcal{W}(p)$: bound $|S_r| \leq C \cdot r!^{\alpha} \cdot e^{c\sqrt{r}}$, factorial factor from configuration-space integration combined with Massey-transport indecomposability.
- *Non-Borel-summable.* Pathological; outside tempered; C_2-cofiniteness rules this out by Zhu finiteness.

Consequences requiring propagation:

- **Theorem F scope.** $(\infty, 2)$-adjoint equivalence holds on the *exponential* stratum (standard landscape); $(\infty, 1)$-adjoint morphism only on the *Gevrey-α* stratum ($\mathcal{W}(p)$) with Gurarie–Flohr Massey class as obstruction to $(\infty, 2)$.
- **Theorem H scope.** $\{0, 1, 2\}$ concentration holds on the Koszul locus, which is a subset of the exponential stratum. On the Gevrey stratum, concentration may fail; ChirHoch support spreads in proportion to the Massey-class obstruction.
- **Scalar complementarity $K_{\mathcal{A}} = \kappa + \kappa^!$.** For finite-shadow algebras, $K_{\mathcal{A}}$ reads off as polynomial. For exponential-stratum algebras (Vir, $\mathcal{W}_N$), $K_{\mathcal{A}}$ is the residue-of-leading-asymptotic (for Virasoro: central-charge conductor $K = c + c' = 26$, while the kappa-complement is $K^{\kappa} = \kappa + \kappa^! = 13$). For Gevrey-stratum algebras, $K_{\mathcal{A}}$ is defined by Borel resummation.

This correction is the fifth Beilinson-iteration refinement on the W(p) temperedness thread. Key conceptual move: distinguish *h-index* growth (character dimension) from *r-index* growth (shadow coefficient); bar-complex combinatorics introduce a multiplicative constant-to-r-th-power factor that shifts sub-exponential-h to exponential-r for non-terminating class M.

54.9 The shadow-exponential base: a new invariant of class M

Status after the W_3 W-line computation: the exponential base is lane-dependent. The Virasoro/T-line base is 6; the W_3 W-line base is 12; the supremal base of a multi-line algebra is therefore not determined by the T-line alone. Any earlier universal "$C_{\mathcal{A}} = 6$" statement is to be read only as the T-line statement.

Iteration 6. The geometric ratio $|A_{r+1}^{\mathrm{Vir}}/A_r^{\mathrm{Vir}}| \to 6$ as $r \to \infty$ is a specific algebraic invariant of the Virasoro shadow. Define:

$$C_{\mathcal{A}} := \limsup_{r\to\infty} \left|A_{r+1}^{\mathcal{A}}/A_r^{\mathcal{A}}\right|,$$

the *shadow-exponential base* of a class M chiral algebra $\mathcal{A}$ (on the exponential stratum). The Virasoro engine computation (verified in `compute/lib/shadow_tower_higher_vir.py` `leading_asymptotic` at $r = 5, 6, 7, 8, 9, 10$: ratios $5.00, 5.14, 5.25, 5.33, 5.40, \ldots, \to 6$) confirms $C_{\mathrm{Vir}} = 6$.

The Platonic question: *what is $C_{\mathcal{A}}$ for $\mathcal{A} \in \{\mathcal{W}_N, \mathrm{BP}, \beta\gamma$ extended, $\mathcal{N} = 2$ sca, lattice $V_L\}$?*

First-principles hypothesis: $C_{\mathcal{A}}$ is determined by the highest pole order of the leading-order OPE of the stress-tensor subalgebra. For Virasoro: OPE $T(z)T(w) \sim (c/2)/(z-w)^4 + \ldots$, pole order 4; but $C_{\mathrm{Vir}} = 6$, not 4. So direct pole-order identification fails.

Iteration 7 (2026-04-17): the origin of $C_{\mathrm{Vir}} = 6$ is derived from the leading-order Riccati kernel. Re-reading the engine's proof sketch (`shadow_tower_higher_vir.py` lines 260–272) and tracing the leading-order behaviour of the rescaled recurrence, the factor of 6 is *not* a configuration-space Euler characteristic; it is the product of (initial bar depth $r_0 = 3$) and (Riccati seed coefficient $S_{r_0} = 2$).

Derivation. The rescaled Virasoro recurrence $r \cdot \Phi_r \cdot c = -\sum_{j+k=r+2,\, 3\le j\le k<r} f(j,k)\, jk \Phi_j \Phi_k$ with $\Phi_3 = 2c$ (the seed) and $\Phi_r = A_r + O(1/c)$ for $r \geq 4$ (the leading asymptotic). At leading order in c, the only non-trivial contribution comes from $j = r_0 = 3$ (since Φ_{r_0} is the only one that carries an explicit factor of c; all other Φ_j are $O(1)$ in c). Extracting the $j = 3$, $k = r-1$ term:

$$r \cdot A_r \cdot c \;=\; -f(3, r-1)\cdot 3 \cdot (r-1) \cdot \Phi_3 \cdot A_{r-1} \;=\; -1 \cdot 3(r-1)\cdot 2c \cdot A_{r-1} \;=\; -6(r-1)cA_{r-1},$$

so

$$\frac{A_r}{A_{r-1}} \;=\; -\frac{6(r-1)}{r} \quad\Longrightarrow\quad \left|\frac{A_{r+1}}{A_r}\right| \;=\; \frac{6r}{r+1} \to 6.$$

The factor 6 decomposes as $6 = 3 \cdot 2 = r_0 \cdot S_{r_0}$ (bar-depth seed times Riccati seed coefficient).

T-line identification (candidate beyond Virasoro). For a class M chiral algebra $\mathcal{A}$ with a lowest non-trivial shadow coefficient $S_{r_0}(\mathcal{A}) = s_0$ at bar-depth r_0, the shadow-exponential base is

$$C_{\mathcal{A}} \;=\; r_0 \cdot s_0 \;=\; r_0 \cdot S_{r_0}(\mathcal{A}).$$

This formula is proved for the Virasoro T-line and is a candidate for other isolated scalar lines. For Virasoro: $r_0 = 3$ (first T-line depth), $S_3 = 2$; $C = 6$. For any chiral algebra containing Virasoro as a sub-chiral-algebra with $S_3 = 2$ inherited from the T-line, the T-line base is 6; it need not dominate the full multi-line algebra.

Failed prediction for $\mathcal{W}_3$. The T-line of $\mathcal{W}_3$ inherits $S_3 = 2$ from the Virasoro sub-current, so its own leading base is 6. The later W-line computation shows that this does not dominate the full multi-line algebra: the W-line has base 12.

For algebras without a Virasoro subalgebra (pure $\beta\gamma$ at non-standard weight, bc ghost at specific λ): the lowest non-trivial shadow depth can be 4, the seed coefficient can be different, and $C_{\mathcal{A}} = r_0 \cdot S_{r_0}$ gives a distinct numerical value.

The T-line computation. Using the existing engine `compute/lib/shadow_tower_extended_families.py`:

$$S_3(\mathcal{W}_3, T) = 2, \qquad S_3(\mathcal{W}_3, W) = 0, \qquad S_4(\mathcal{W}_3, T) = \frac{10}{c(5c+22)}, \qquad S_4(\mathcal{W}_3, W) = \frac{2560}{c(5c+22)^3}.$$

At the leading-c asymptotic:

$$\Phi_3^T = c \cdot 2 = 2c, \quad \Phi_3^W = 0, \quad \Phi_4^T \to 2 - \frac{44}{5c}, \quad \Phi_4^W \to 0.$$

The T-line shadow tower of $\mathcal{W}_3$ is *identical* to the Virasoro shadow tower at leading-c order ($A_3 = 2$, $A_4 = 2$); this proves $C^{T\text{-line}}_{\mathcal{W}_3} = 6$, not the total W_3 supremal base.

Generalisation to all $\mathcal{W}_N$ with principal-generator data $(T, W^{(3)}, \ldots, W^{(N)})$:

- T-line inherits Virasoro shadow at leading order ($S_3^T = 2$ universally).
- $W^{(k)}$-line vanishes at depth $< 2k$ by $\mathbb{Z}_2$ (or higher cyclic) parity.
- Therefore the T-line shadow equals the Virasoro shadow at leading order. Higher generator-lines may have larger bases.

The identification $C_{\mathcal{A}} = r_0 \cdot S_{r_0}$ is valid for the Virasoro T-line computation. It is not a total-algebra classification for all $\mathcal{W}_N$.

Differentiation from the polynomial stratum: non-trivial $C_{\mathcal{A}}$ values only arise for algebras where the leading-c asymptotic of Φ_r has a non-vanishing limit. Class G (finite shadow) has $A_r = 0$ for r above the termination depth, hence $C_{\text{Heisenberg}} = 0$ in a degenerate sense (or "undefined"). The Virasoro/T-line projection in class M has base 6; multi-line supremal bases can be larger. Class C ($\beta\gamma$) terminates at finite shadow depth; it has no nonterminating exponential base in the class M sense.

Consolidation. The seven Beilinson iterations in this session's W(p) thread (1: polynomial, 2: sub-exp, 3: fit slope, 4: pre-asymptotic, 5: exp at r-index, 6: invariant $C_{\mathcal{A}}$, 7: formula $r_0 S_{r_0}$, 8: verified $\mathcal{W}_3$) produced a consolidated inscription: the Virasoro/T-line shadow-exponential base is $C^T = r_0 \cdot S_{r_0} = 6$. This is not the total base of a multi-line class-M algebra. Logarithmic $\mathcal{W}(p)$ Gevrey stratum: C undefined (growth is factorial-modified exponential, not pure exponential). The tempered decomposition is by channel: finite tails, exponential lanes such as $C^T = 6$ or $C^{W_3,W} = 12$, and Gevrey-α lanes.

Iteration 9 (2026-04-17): $C^T_{\text{BP}} = 6$. Using the BP block of `compute/lib/shadow_tower_extended_families.py`:

$$S_3(\text{BP}, T) = 2, \quad S_3(\text{BP}, J) = 0, \quad S_3(\text{BP}, G^{\pm}) = 0, \quad S_4(\text{BP}, T) = S_4^{\text{Vir}}(c_{\text{BP}}(k)),$$

so the BP T-line is Virasoro at central charge $c_{\text{BP}}(k) = 2 - 24(k+1)^2/(k+3)$. The J-line is class G (depth 2, abelian); the $G^{\pm}$-line vanishes at $r = 3$ by parity and is subleading at large c. Therefore

$$C^T_{\text{BP}} = r_0 \cdot S^T_{r_0} = 3 \cdot 2 = 6.$$

T-line class-M heuristic. Let $\mathcal{A}$ be a vertex operator algebra in class M and suppose the chosen scalar projection is the Virasoro T-line, with seed $S_3^T = 2$ at bar-depth 3. Then the leading T-line base is

$$C^T_{\mathcal{A}} = 6.$$

This is not a universal class-M base. Principal $\mathcal{W}_3$ has a proved W-line base 12; higher-spin bases are conjectural and channel-dependent. Lattice VOAs are class G: their shadow towers terminate at depth 2 and should not be placed in a non-terminating class-M stratum merely because they contain a Sugawara stress tensor.

Examples for the T-line statement include Virasoro and the Virasoro sub-current of standard principal $\mathcal{W}_N$ or Bershadsky–Polyakov algebras after the central-charge convention is fixed. The full multi-line algebra may have a different leading base.

Algebras *outside* this class (potentially different C):

- Pure Heisenberg $\mathcal{H}_k$: class G (terminating); C undefined in the non-terminating sense, degenerate case.

- Symplectic fermion $\mathrm{SF}_{\mathrm{odd}}$ at the pure fermionic realization: potentially different S_{r_0} if T is induced not primary.
- Logarithmic $\mathcal{W}(p)$: Gevrey stratum; C undefined as pure exponential base (factorial-modified growth).

Third layer refinement. The three-layer stratification at r-index becomes:

(i) *Finite (terminating).* $d_{\mathrm{alg}} \leq 2$; class G/L/C; $C =$ undefined (zero-tail).

(ii) *Exponential, channel-dependent base.* Class M with a Virasoro T-line has $C^T = 6$; other channels can dominate (for $\mathcal{W}_3$, the W-line has base 12).

(iii) *Gevrey-α.* Logarithmic C_2-cofinite; $\mathcal{W}(p)$.

(iv) *Non-Borel-summable.* Non-C_2-cofinite; outside tempered stratum.

The second layer is therefore not a single equivalence class: it is a channel-dependent exponential stratum. The programme's earlier "standard tempered class" language should be read as a Virasoro/T-line statement only.

This positions the shadow-exponential base $C_{\mathcal{A}}$ as a channel-level invariant within the standard landscape. It can distinguish the Virasoro/T-line base 6 from the $\mathcal{W}_3$ W-line base 12, and it is undefined for genuinely Gevrey growth.

Significance for Theorem F. Any link between $(\infty, 2)$-holography and a channel base such as $C^T = 6$ remains conjectural. The Gurarie–Flohr obstruction on $\mathcal{W}(p)$ is evidence that logarithmic/Gevrey growth belongs to a different categorical regime, not a proved equivalence between C-definedness and a holography theorem.

Beilinson attack: is this hypothesis verifiable? The answer requires computing the $\mathcal{W}_N$ shadow leading asymptotic via the programme's engines. The Virasoro engine exists; the $\mathcal{W}_N$ extension is a separate compute task. Inscribed as task #42.

Programme implication. If the hypothesis $C_{\mathcal{A}} = \chi(\partial \mathrm{FM}_{r_0})$ holds, the shadow-exponential base becomes a *bar-complex structural invariant* computable from configuration-space combinatorics alone, without explicit Riccati recurrence. This positions the shadow tower as a *topological* invariant of the algebra's bar resolution — consistent with the programme's E_1-first prose architecture (all shadows are configuration-space invariants of FM_*).

Until the hypothesis is verified for $\mathcal{W}_3, \mathcal{W}_4$, BP, etc., the shadow-exponential base remains conjectural.

54.10 /loop thread summary (iterations 1–29, 2026-04-17)

Consolidated record of the 29-iteration self-correcting Beilinson loop on the shadow-exponential base and super-Yangian complementarity threads.

Shadow-tower thread (iterations 1–28, Vol I):

Iter 1–5 W(p) temperedness retraction refinement: polynomial $\to$ sub-exponential $\to$ three-layer stratification at r-index.

Iter 6–8 Shadow-exponential base $C_{\mathcal{A}} := \limsup |A_{r+1}/A_r|$ defined; $C_{\mathcal{A}} = r_0 \cdot S_{r_0}$ formula derived from leading Riccati; $C_{\mathrm{Vir}} = C^{T\text{-line}}_{\mathcal{W}_3} = C^{T\text{-line}}_{\mathrm{BP}} = 6$ verified.

Iter 9–12 Earlier universal $C_{\mathcal{A}} = 6$ claim retracted to the Virasoro/T-line statement; the inverse-radius identification is channel-dependent ($\mathcal{W}_3$ has a proved W-line base 12); four-stratum analytic decomposition.

Iter 13–15 Closed form $\Sigma_{\mathrm{Vir}}(z) = 4z^3/9 - z^2/9 + z/27 - \log(1+6z)/162$; subleading closed form with $(6z+1)^{-2}$ double pole; pole-doubling pattern verified at $k = 2$.

Iter 16–17 Pole-doubling all-k theorem ($2k$-pole for $\Sigma^{(k)}_{\mathrm{Vir}}$); verified at $k = 3$ with arithmetic $-1/(2^4 \cdot 3^8)$.

Iter 18 Subleading uniformity extension: W_3 T-line asymptotic matches Virasoro at $k = 0, 1$.

Iter 19–21 Arithmetic structure of leading φ_k coefficient: $(-1)^k/(2^{2k-2} \cdot 3^{3k-1})$; per-iteration factor $-1/108 = -1/(2^2 \cdot 3^3)$; sub-leading-$r$ content ramifies through $\{5, 11, 31, \ldots\}$.

Iter 22 Cross-reference from Vol I programme overview Theorem F to shadow-tower theorems.

Iter 23–24 Cross-volume identification $\beta_A = C_{\mathcal{A}}$ (Vol II tempering rate equals Vol I shadow-exp base at leading $1/c$); later W_3 data splits the T-line base 6 from the W-line base 12.

Iter 25 CLAUDE.md Key Constants propagation.

Iter 26–28 Cross-volume coherence with Vol III Gevrey-1 Borel picture; numerical tightening of $|S_k|$ growth (≈ 5 ratios at $c = 1$); Borel radius ≈ 1.6 matches Vol III's shadow-quadratic prediction.

Super-Yangian thread (iteration 29, Vol III):

Iter 29 Super-Yangian complementarity $\kappa_{\mathrm{ch}} + \kappa^!_{\mathrm{ch}} = \max(m, n)$ extended from unbalanced ($m \neq n$) to uniform (all ranks including $m = n$) via limit interpretation: $c_{\mathrm{sub}}(n|n, k) = n$ constant, self-dual $k' = k$ gives $n + n = 2n$, hence $\max(n, n) = n$.

Cumulative Platonic inscription:

- $C^T_{\mathcal{A}} = 6$ uniformly on the Virasoro/T-line projection of the non-logarithmic class M standard landscape (single arithmetic constant governs shadow-series radius of convergence on that lane).
- $|\omega_\pm(\mathrm{Vir})| \to 2$ as $c \to \infty$ via shadow-quadratic $Q_L(\zeta)$; $|\omega|^2(c) = 1 + (15c + 61)/(5c + 22)$ gives $|\omega| \in [1.94, 2)$ across all $c > 0$ — nearly c-independent Borel singularity distance, distinct from $C_{\mathrm{Vir}} = 6$ exponential rate. Ratio $|\omega|/C_{\mathrm{Vir}} \to 1/3$ at large c (iter 33).
- $\kappa_{\mathrm{ch}} + \kappa^!_{\mathrm{ch}} = \max(m, n)$ uniformly for super-Yangian $Y(\mathfrak{sl}(m|n))$ (all (m, n) including balanced via limit interpretation).
- *Pole-doubling pattern* $\Sigma^{(k)}_{\mathrm{Vir}}$ has pole $2k$ at $z = -1/6$ plus log branch; P_k, Q_k polynomials with degrees bounded by $2k + 3$, $2k$.
- *Arithmetic* $(-1)^k/(2^{2k-2} \cdot 3^{3k-1})$ as the leading-r coefficient of φ_k, uniformly $\{2, 3\}$-adic at leading order with higher primes appearing in sub-leading-r content.
- *Uniformity extension* (iter 31): the above $\{2, 3\}$-adic arithmetic is conjectural beyond the Virasoro lane and must be checked channel by channel.
- *Borel geometric identifications* (iters 33–42): $|\omega|^2(c) = (20c + 83)/(5c + 22)$ closed form; $|\omega|(c) = 2 - 1/(4c) + O(c^{-2})$; angle $\to \pi/3 = 2\pi/6$ at $c \to \infty$ (Virasoro-specific coincidence); minimal polynomial $\omega^2 - 2\omega + 4 = 0$; cyclotomic field $\mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{-3})$.
- *Phase transitions* (iters 44–50): formula has pole at $c = -22/5 = -4.4$ (identified as Lee-Yang minimal model $M(2, 5)$ central charge) and zero at $c = -83/20 = -4.15$ (Q_L roots at $\zeta = 0$ and $\zeta = 2$; shadow-asymptotic model fails near origin + positive-real-axis Borel obstruction). Unphysical gap $c \in (-4.4, -4.15)$ where $|\omega|^2 < 0$.

The thread's inner music is the convergence of twenty-nine iterations from a retraction ("$\mathcal{W}(p)$ tempered OPEN") to a rich analytic + arithmetic theory of the Virasoro shadow-series structure, cross-linked across three volumes, with channel-level invariants $C_{\mathcal{A}}$, β_A, $\max(m, n)$ cleanly separated from any total-algebra assertion.

54.11 Iterations 57–58: W_3 W-line closed form and the second lane at $C = 12$

Pivot. After iteration 56 exhausted the Virasoro Borel-geometry self-correction cycle (ratios approaching 12 from below, pole at $c = -22/5$ identified with Lee-Yang, zero at $c = -83/20$ identified with smallest-complexity admissible $\mathfrak{sl}_3$), the programme pivoted to a *second lane*: the W-line of $W_3 = W(\mathfrak{sl}_3)$ at generic central charge.

Finding (iteration 58). The normalised integer sequence $a_n := |N_{2m}|/2560^{m-1}$ on the W-line (where N_{2m} is the numerator of $S_{2m}(W_3, W)$) satisfies the exact closed form

$$a_n \;=\; \frac{2 \cdot 3^{n-2}\,(2n-4)!}{(n-2)!\,n!} \;=\; \frac{2 \cdot 3^{n-2}}{n(n-1)}\binom{2n-4}{n-2}, \qquad n \geq 2,$$

with consecutive ratio

$$\frac{a_{n+1}}{a_n} \;=\; \frac{6(2n-3)}{n+1} \;=\; 12 - \frac{30}{n+1},$$

approaching the shadow-exponential base $C_{W_3}^{W\text{-line}} = 12$ exactly (not 6). Stirling asymptotic $a_n \sim 12^n/(72\sqrt{\pi} \cdot n^{5/2})$. Verified symbolically through $n = 25$ against the Poisson-bracket recurrence in `w3_wline_shadow_tower.py::normalized_recursion`; ten HZ-IV-decorated tests pass (`compute/tests/test_w3_wline_closed_form.py`).

Structural interpretation. The T-line of W_3 inherits the Virasoro T-line base $C_{W_3}^{T\text{-line}} = 6$; the W-line is a *second lane* with its own integer recursion and its own shadow-exponential base $C_{W_3}^{W\text{-line}} = 12 = 2 \cdot 6$. The doubling factor 2 has a clean operadic interpretation (Remark **??** in `chapters/theory/shadow_tower_higher_coefficients.tex`): the W-line Hessian propagator is $3/c$ vs T-line $2/c$ (ratio $3/2$), and the W-line recursion kernel carries an additional factor $8/6 = 4/3$ from the W-generator seed, which combine to $(3/2) \cdot (4/3) = 2$.

Higher-spin boundary. The W_3 W-line proves the second class-M lane with base 12. For $s \geq 4$ there is no all-spin theorem yet. The formula $C_s = 4s$ is a conditional Riccati-template consequence under $(H1)$–$(H4)$; the quadratic guess $s(s+1)$ is only a two-point fit and is not used as evidence until the Fateev–Lukyanov OPE seeds are computed.

Koszul-duality invariance. Because a_n is c-independent (the shadow integer-sequence normalised out $2560^{n-1}/[c^{2m-3}(5c+22)^{3(m-1)}]$), $C_{W_3}^{W\text{-line}} = 12$ is manifestly a Koszul-duality invariant under the trace-form duality $c \mapsto 100 - c$ because the normalised sequence is c-independent. The map $c \mapsto 4 - c$ belongs to a different free-field convention and is not the trace-form conductor.

Reconstitution. The proved class M statement now has two lanes: the T-line base 6, and the W_3 W-line base 12 with its exact Catalan closed form. Higher-spin spin-stratification is conjectural until the $s \geq 4$ lanes are computed.

54.12 Iterations 59–63: rectification to the exact generating function and the linear-vs-quadratic dichotomy

Iter 59 (Beilinson correction). The iter 58 conjecture $C^{(s)} = s(s+1)$ rests on only two data points, fitting multiple functional forms. Three candidate parabolas all match at $s = 2, 3$ but differ at $s = 4$: $s(s+1) = 20$, $6(s-1) = 18$, $2s^2 - 4s + 6 = 22$. The two-point ambiguity is the strictly stronger replacement for the unsupported extrapolation.

Iter 60 (exact generating function). The W_3 W-line sequence a_n admits an ALGEBRAIC closed form for its generating function:

$$F(z) \;=\; \frac{z}{3} + \frac{(1-12z)^{3/2}-1}{54}, \qquad F'(z) \;=\; \frac{1-\sqrt{1-12z}}{3}.$$

F' solves the quadratic $\frac{3}{2}F'^2 - F' + 2z = 0$; the branch $F'(0) = 0$ is the correct one; radius of convergence = 1/12 exactly. Self-consistency evaluation at the radius $F'(1/12) = 1/3$ gives the sharp identity $\sum_{n\geq 2} n\, a_n\, 12^{1-n} = 1/3$, verified numerically through $N = 300$ (residual matches Stirling tail $1/(3\sqrt{\pi N})$ to four significant figures). Via the Catalan identity $(1-\sqrt{1-4u})/2 = \sum C_m u^{m+1}$, the sequence identifies as $a_n = 2C_{n-2}3^{n-2}/n$ where C_m is the m-th Catalan. Structural prediction (under the "Riccati template" with seed $a_2 = 1$, Hessian s/c, and $j, k \geq 2$ summation): $C_s = 4s$. Matches W_3 W-line ($s = 3$, $C = 12$) but not Virasoro T-line ($s = 2$, $C = 6 \neq 8$).

Iter 61 (template hypotheses). Four hypotheses $(H1)$–$(H4)$ isolate when $C = 4s$ applies: $(H1)$ Z_2 parity of the generator, $(H2)$ Hessian s/c, $(H3)$ seed $a_2 = 1$, $(H4)$ weight-4 OPE producing a_3 from a_2^2. W_3 W-line satisfies all four; Virasoro T-line fails $(H1)$; higher-spin $W^{(s)}$ for $s \geq 4$ open pending Fateev–Lukyanov verification.

Iter 62 (linear-vs-quadratic dichotomy). Deeper structural mechanism: how leading-c projection interacts with the quadratic shadow recurrence.

- Virasoro T-line: $S_3 \sim c^0$, $S_r \sim c^{-(r-2)}$ for $r \geq 4$ creates a *cascade of c-orders*. Under $A_r := c^{r-2}S_r$ scaling, only the pair $(3, r-1)$ contributes at leading c; other pairs are sub-leading. Quadratic shadow recurrence collapses to LINEAR: $A_r = -6(r-1)/r \cdot A_{r-1}$, $r \geq 5$. Seeds $A_3 = A_4 = 2$ determine the whole tower $A_r = 8(-6)^{r-4}/r$. GF is TRANSCENDENTAL (polynomial plus $\log(1+6z)$), subexponential $\sim r^{-1}$.
- W_3 W-line: integer sequence a_m normalises out all c-dependence; no cascade, all pairs contribute at the same order. Shadow recurrence remains QUADRATIC on (a_m). GF is ALGEBRAIC (polynomial plus $(1-12z)^{3/2}$), subexponential $\sim m^{-5/2}$.

Both end at finite exponential base but the subexponential power differs by two orders.

Iter 63 (HZ-IV test suite). New test file `compute/tests/test_virasoro_linear_recurrence.py` independently verifies the linear recurrence claim against the closed form, across $r = 4, \ldots, 19$. Four tests pass. Twenty-two tests total across W_3 W-line and Virasoro T-line infrastructure (iter 58: 11; iter 60: 5; iter 63: 4; + 2 higher-spin conjecture tests).

Cumulative structural picture (iters 57–63).

- *W_3 W-line*: exact algebraic GF, Catalan reduction, $C = 12 = 4 \cdot 3$, Koszul-dual invariant. Proved rigorously.
- *Riccati template* $C = 4s$ under $(H1)$–$(H4)$: structural formula, not just curve-fit. Gives unambiguous higher-spin prediction: $W^{(4)}$-line in W_4 should have $C = 16$ (if all four hypotheses hold).
- *Linear-vs-quadratic asymptotic recurrence*: the key structural mechanism distinguishing lanes. Virasoro T-line (no parity, c-cascade) has linear recurrence and log GF; W_3 W-line (parity, integer normalisation) has quadratic recurrence and algebraic GF.
- *Template fails* at Virasoro T-line because $S_2 = \kappa$ is not x-dependent on the T-line (it acts via ∇_H rather than as a shadow coefficient), whereas on the W_3 W-line, $Sh_2 = (c/3)x_W^2$ IS x_W-dependent and enters the recursion as a seed at weight $4 = 2 + 2$.